Ethylene Glycol Plant Design Report

Preliminary Techno-Economic Feasibility Report

Target product: Ethylene Glycol

Design capacity: 200000.00 TPA

Region: India

Issue date: 2026-03-31

Report Basis

Table 1. Report Basis

Columns: Field; Value
Row 1: Field: Target product | Value: Ethylene Glycol
Row 2: Field: Capacity | Value: 200000.00 TPA
Row 3: Field: Purity target | Value: 99.90 wt%
Row 4: Field: Operating mode | Value: continuous
Row 5: Field: Region | Value: India
Row 6: Field: Currency | Value: INR
Row 7: Field: Process template | Value: ethylene_glycol_india

Document Control

Table 2. Document Control

Columns: Field; Value
Row 1: Field: Report type | Value: End-to-end techno-economic feasibility and plant design report
Row 2: Field: Issue date | Value: 2026-03-31
Row 3: Field: Report basis | Value: Solver-backed benchmark-parity report package
Row 4: Field: Document coverage | Value: Chapters, references, appendices, annexures, and datasheet bundle

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1. Executive Summary

Executive Summary

This report develops an India-mode conceptual plant design for Ethylene Glycol at 200000 TPA. The selected basis uses ethylene oxide hydration, India-specific siting and tariff assumptions, traceable calculations for balances and major equipment, and approval gates around route, design basis, reactor/column basis, HAZOP, India economics, and final release.

2. Introduction and Product Profile

Introduction and Product Profile

Ethylene glycol is a high-volume petrochemical intermediate with deep downstream demand in polyester, PET resin, coolants, and solvent applications. For a 200000 TPA India project, the product profile must be tied directly to feedstock integration, high-water hydration chemistry, and energy-intensive purification rather than written as a generic chemistry summary.

The design basis therefore uses cited thermophysical properties, continuous operation, and India-specific market relevance as direct inputs to route, site, and economics decisions.

3. Literature Survey

Literature Survey

The route survey compares industrial EG pathways on maturity, selectivity, safety, utility intensity, and India feedstock fit. Ethylene oxide hydration remains the strongest basis because it is the most defensible large-scale route for a 200000 TPA feasibility-style home paper.

4. Process Selection

Process Selection

Route Comparison

Table 5. Route Comparison

Columns: Route; Family; Total score; Residual hot utility (kW); Heat case; Scenario stability
Row 1: Route: eo_hydration | Family: liquid_hydration_train | Total score: 39.95 | Residual hot utility (kW): 49263.839 | Heat case: eo_hydration_pinch_htm | Scenario stability: reviewed
Row 2: Route: omega_catalytic | Family: liquid_hydration_train | Total score: 60.60 | Residual hot utility (kW): 43800.060 | Heat case: omega_catalytic_pinch_htm | Scenario stability: stable
Row 3: Route: chlorohydrin | Family: chlorinated_hydrolysis_train | Total score: -1.00 | Residual hot utility (kW): 13044.246 | Heat case: chlorohydrin_no_recovery | Scenario stability: reviewed

Selected Route

OMEGA catalytic route via ethylene carbonate is selected because the Liquid Hydration Train route family stays economically competitive after the chosen heat-integration case omega_catalytic_pinch_htm, while retaining credible industrial maturity and India fit.

Route Family Profiles

Table 6. Route Family Profiles

Columns: Route; Family; Reactor Basis; Separation Train; Heat Style; India Blocker
Row 1: Route: eo_hydration | Family: Liquid Hydration Train | Reactor Basis: Tubular plug-flow hydrator | Separation Train: EO flash -> water removal -> vacuum glycol distillation | Heat Style: condenser_reboiler_cluster | India Blocker: none
Row 2: Route: omega_catalytic | Family: Liquid Hydration Train | Reactor Basis: Tubular plug-flow hydrator | Separation Train: EO flash -> water removal -> vacuum glycol distillation | Heat Style: condenser_reboiler_cluster | India Blocker: none
Row 3: Route: chlorohydrin | Family: Chlorinated Hydrolysis Train | Reactor Basis: Agitated hydrolysis CSTR train | Separation Train: Salt removal -> brine handling -> water removal -> purification | Heat Style: utility_led_with_brine_management | India Blocker: Route generates chloride-heavy waste and is not preferred for India deployment under the current policy basis.

Unit-Operation Family Expansion

Route omega_catalytic is expanded into the Liquid Hydration Train unit-operation family.

Table 7. Unit-Operation Family Expansion

Columns: Service Group; Candidate; Status; Score; Rationale
Row 1: Service Group: reactor | Candidate: tubular_plug_flow_hydrator | Status: preferred | Score: 102.0 | Rationale: Tubular hydrator with strong dehydration integration fit
Row 2: Service Group: reactor | Candidate: jacketed_cstr_train | Status: preferred | Score: 78.0 | Rationale: CSTR fallback for liquid hydration service
Row 3: Service Group: reactor | Candidate: high_pressure_carbonylation_loop | Status: blocked | Score: 18.0 | Rationale: High-pressure carbonylation loop
Row 4: Service Group: separation | Candidate: distillation_train | Status: preferred | Score: 100.0 | Rationale: Flash and glycol distillation train
Row 5: Service Group: separation | Candidate: extractive_distillation_train | Status: fallback | Score: 82.0 | Rationale: Extractive/vacuum polishing distillation

Row 6: Service Group: separation | Candidate: packed_absorption_train |
Status: blocked | Score: 20.0 | Rationale: Packed absorption train

- Supporting unit operations: flash, dehydration_column, glycol_polishing_column, heat_exchanger_cluster, reactor, distillation, heat_exchanger
- Applicability critics: dehydration_duty, glycol_split_vle_basis, water_recycle_closure, relative_volatility_claims, reboiler_duty_burden

Chemistry Family Adapter

Adapter continuous_liquid_organic_train supports continuous liquid organic train under a organic chemistry basis.

- Route hints: hydration, carbonylation, hydrogenation, dehydration cleanup
- Property priority: molecular_weight, normal_boiling_point, liquid_density, liquid_viscosity, liquid_heat_capacity, heat_of_vaporization, antoine_constants, binary_interaction_parameters
- Preferred reactor candidates: tubular_plug_flow_hydrator, jacketed_cstr_train
- Preferred separation candidates: distillation_train, extraction_recovery_train
- Common unit operations: reactor, flash, distillation, heat_exchanger, storage
- Corrosion cues: organic_acid_presence, water_content, chloride_trace_monitoring
- Heat-integration patterns: reboiler_condenser_cluster, shared_htm_island_network, moderate_temperature_recovery
- Critic focus: relative_volatility_claims, water_recycle_closure, reboiler_duty_burden
- Sparse-data blockers: missing_vle_basis, missing_density_viscosity_for_column_hydraulics

Sparse-Data Policy

Table 8. Sparse-Data Policy

Columns: Stage; Subject; Artifact Family; Est.; Analogy; Heuristic Fallback;
Min Confidence; Status; Triggered Items
Row 1
Stage: thermodynamic_feasibility
Subject: molecular_weight
Artifact Family: pure_component_identity
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.55
Status: covered
Triggered Items: -
Row 2
Stage: thermodynamic_feasibility
Subject: normal_boiling_point
Artifact Family: pure_component_thermo
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -
Row 3
Stage: energy_balance
Subject: liquid_heat_capacity
Artifact Family: bulk_thermal_properties
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.55
Status: covered
Triggered Items: -
Row 4
Stage: energy_balance
Subject: heat_of_vaporization
Artifact Family: bulk_thermal_properties

Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.55
Status: covered
Triggered Items: -

Row 5
Stage: reactor_design
Subject: liquid_density
Artifact Family: transport_properties
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 6
Stage: reactor_design
Subject: liquid_viscosity
Artifact Family: transport_properties
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 7
Stage: distillation_design
Subject: liquid_density
Artifact Family: separation_transport
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 8
Stage: distillation_design
Subject: liquid_viscosity
Artifact Family: separation_transport
Est.: yes
Analogy: no
Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 9
Stage: distillation_design
Subject: binary_interaction_parameters
Artifact Family: vle_nonideal_basis
Est.: no
Analogy: no
Heuristic Fallback: yes
Min Confidence: 0.00
Status: warning
Triggered Items: ethylene_glycol__ethylene_oxide, ethylene_glycol__water, ethylene_oxide__water

Selected Reactor Basis

Tubular or plug-flow hydrator service selected as the highest-ranked alternative.

Selected Separation Basis

Distillation and flash purification train selected as the highest-ranked alternative.

Selected Heat-Integration Case

Pinch + HTM loop: Pinch-based heat recovery with an indirect hot-oil loop materially reduces purchased steam.

5. Market and Capacity Selection

Market and Capacity Selection

The 200000 TPA basis is framed as a large continuous petrochemical train aligned with India polyester and resin demand rather than a niche batch-chemical facility. The market case depends on large throughput, stable EO integration, and India-based logistics and tariff assumptions, all normalized in INR.

Capacity Decision

200,000 TPA plant basis selected because it best balances the benchmark throughput, market basis, and integration leverage.

6. Site Selection

Site Selection

Dahej, Gujarat is selected because it best combines petrochemical integration, port logistics, utilities, industrial services, and a defendable India cost basis for a 200000 TPA EG plant.

Deterministic Site Decision

Dahej is selected because it provides the strongest combined cluster, logistics, utility, and India execution fit for a 200000 TPA EG plant.

7. Thermodynamic Feasibility

Thermodynamic Feasibility

Selected Property Basis

Use cited public anchors with library-backed calculation and benchmark seeds for missing constants. selected for the active component set of route omega_catalytic.

Table 9. Selected Property Basis

Columns: Property Basis; Feasible; Score
Row 1: Property Basis: direct_public_package | Feasible: no | Score: 16.00
Row 2: Property Basis: hybrid_cited_plus_library | Feasible: yes | Score: 62.00
Row 3: Property Basis: estimation_dominant | Feasible: yes | Score: 40.00
Row 4: Property Basis: analogy_basis | Feasible: yes | Score: 10.00

Separation Thermodynamics Basis

Table 10. Separation Thermodynamics Basis

Columns: Component; Ttop (C); Gamma top; Ktop; Method; Tbottom (C); Gamma bottom; Kbottom; Method
Row 1
Component: Ethylene Glycol
Ttop (C): 108.0
Gamma top: 1.0000
Ktop: 0.0160
Method: ideal_raoult_missing_bip_clausius_clapeyron
Tbottom (C): 203.3
Gamma bottom: 1.0000
Kbottom: 0.3669
Method: ideal_raoult_missing_bip_clausius_clapeyron
Row 2
Component: Ethylene oxide
Ttop (C): 108.0
Gamma top: 1.0000
Ktop: 4.5086
Method: ideal_raoult_clausius_clapeyron
Tbottom (C): 203.3
Gamma bottom: 1.0000
Kbottom: 21.3927
Method: ideal_raoult_clausius_clapeyron
Row 3
Component: Water
Ttop (C): 108.0
Gamma top: 1.0000
Ktop: 0.4138
Method: ideal_raoult_missing_bip_antoine_extrapolated
Tbottom (C): 203.3
Gamma bottom: 1.0000
Kbottom: 5.2836
Method: ideal_raoult_missing_bip_antoine_extrapolated

Light key: Water

Heavy key: Ethylene Glycol

Activity model: ideal_raoult_missing_bip_fallback

Average relative volatility: 19.3148

System pressure: 3.240 bar

Fallback notes:

- Binary interaction parameters for Water / Ethylene Glycol were not resolved from cited/public sources.
- Activity coefficients defaulted to gamma=1.0 and the separation basis fell back to ideal Raoult's law for the key pair.

Selected Thermodynamic Method

Use direct public property and reaction data where available. selected as the primary thermodynamic basis.

Table 11. Selected Thermodynamic Method

Columns: Method; Feasible; Score; Rejected Reasons
Row 1: Method: direct_public_data | Feasible: yes | Score: 91.10 | Rejected Reasons: -
Row 2: Method: correlation_interpolation | Feasible: yes | Score: 88.10 | Rejected Reasons: -
Row 3: Method: estimation_method | Feasible: yes | Score: 65.90 | Rejected Reasons: -
Row 4: Method: analogy_basis | Feasible: yes | Score: 53.50 | Rejected Reasons: -

Thermodynamic Interpretation

The selected reaction is treated as thermodynamically favorable, with a negative Gibbs free energy and exothermic heat release that supports conversion but requires controlled heat removal.

8. Reaction Kinetics

Reaction Kinetics

Selected Kinetic Method

Use cited industrial or literature rate law directly. selected as the primary kinetic basis.

Table 12. Selected Kinetic Method

Columns: Method; Feasible; Score; Rejected Reasons
Row 1: Method: cited_rate_law | Feasible: yes | Score: 92.00 | Rejected Reasons: -
Row 2: Method: arrhenius_fit | Feasible: yes | Score: 88.50 | Rejected Reasons: -
Row 3: Method: apparent_order_fit | Feasible: yes | Score: 72.50 | Rejected Reasons: -
Row 4: Method: conservative_analogy | Feasible: yes | Score: 56.50 | Rejected Reasons: -

Kinetic Interpretation

The seeded kinetics package supports a practical residence-time basis for preliminary equipment sizing.

9. Block Flow Diagram

Block Flow Diagram

Figure 1. Block Flow Diagram

```
mermaid
flowchart LR
A[Feed Storage] --> B[Feed Preparation]
B --> C[Reactor]
C --> D[Separation]
D --> E[Purification]
E --> F[Product Storage]
```

10. Process Description

Process Description

Solver-derived process description for route omega_catalytic built from 5 solved unit packets.

Unit Sequence

Table 13. Unit Sequence

Columns: Unit; Type; Service; Inlet Streams; Outlet Streams; Closure; Coverage

Row 1
- Unit: feed_prep
- Type: feed_preparation
- Service: Feed preparation
- Inlet Streams: S-101, S-102, S-401
- Outlet Streams: S-150
- Closure: converged
- Coverage: complete

Row 2
- Unit: reactor
- Type: reactor
- Service: Reaction zone
- Inlet Streams: S-150
- Outlet Streams: S-201
- Closure: converged
- Coverage: complete

Row 3
- Unit: primary_flash
- Type: flash
- Service: Primary flash and purge recovery
- Inlet Streams: S-201
- Outlet Streams: S-202, S-203
- Closure: converged
- Coverage: complete

Row 4
- Unit: waste_treatment
- Type: waste_handling
- Service: Waste treatment
- Inlet Streams: S-202, S-403
- Outlet Streams: -
- Closure: converged
- Coverage: partial

Row 5
- Unit: recycle_recovery
- Type: recycle
- Service: Recycle recovery
- Inlet Streams: S-203
- Outlet Streams: S-301
- Closure: estimated
- Coverage: partial

Row 6
- Unit: purification
- Type: distillation
- Service: Purification train
- Inlet Streams: S-301
- Outlet Streams: S-401, S-402, S-403
- Closure: converged
- Coverage: partial

Row 7
- Unit: storage
- Type: storage
- Service: Product storage
- Inlet Streams: S-402
- Outlet Streams: -
- Closure: converged
- Coverage: partial

Section Topology

Table 14. Section Topology

Columns: Section; Label; Type; Units; Inlet Streams; Outlet Streams; Side Draws; Status
Row 1
- Section: feed_handling
- Label: Feed handling
- Type: feed_preparation
- Units: feed_prep
- Inlet Streams: S-101, S-102, S-401
- Outlet Streams: S-150
- Side Draws: -
- Status: converged

Row 2
- Section: reaction
- Label: Reaction
- Type: reaction
- Units: reactor
- Inlet Streams: S-150
- Outlet Streams: S-201
- Side Draws: -
- Status: converged

Row 3
- Section: primary_recovery
- Label: Primary recovery
- Type: primary_recovery
- Units: primary_flash
- Inlet Streams: S-201
- Outlet Streams: S-202, S-203
- Side Draws: -
- Status: converged

Row 4
- Section: recycle_recovery_carbonate_loop_cleanup
- Label: Carbonate loop cleanup
- Type: recycle_recovery
- Units: recycle_recovery
- Inlet Streams: S-203
- Outlet Streams: S-301
- Side Draws: -
- Status: estimated

Row 5
- Section: purification
- Label: Purification
- Type: purification
- Units: purification
- Inlet Streams: S-301
- Outlet Streams: S-401, S-402, S-403
- Side Draws: -
- Status: estimated

Separation Architecture

Table 15. Separation Architecture

Columns: Separation; Family; Driving Force; Product Streams; Recycle Streams; Status
Row 1: Separation: primary_flash_separation_packet | Family: flash | Driving Force: temperature/volatility | Product Streams: S-203 | Recycle Streams: - | Status: converged
Row 2: Separation: purification_separation_packet | Family: distillation | Driving Force: temperature/volatility | Product Streams: S-402 | Recycle Streams: S-401 | Status: converged

Recycle Architecture

Table 16. Recycle Architecture

Columns: Loop; Source Unit; Recycle Streams; Purge Streams; Status

Row 1: Loop: recycle_recovery_recycle_loop | Source Unit: recycle_recovery | Recycle Streams: S-301 | Purge Streams: - | Status: converged
Row 2: Loop: purification_recycle_loop | Source Unit: purification | Recycle Streams: S-401 | Purge Streams: S-403 | Status: estimated

Unit Narrative

- feed_prep: Feed preparation. Inlet streams S-101, S-102, S-401 and outlet streams S-150 with converged closure and complete coverage.
- reactor: Reaction zone. Inlet streams S-150 and outlet streams S-201 with converged closure and complete coverage.
- primary_flash: Primary flash and purge recovery. Inlet streams S-201 and outlet streams S-202, S-203 with converged closure and complete coverage.
- waste_treatment: Waste treatment. Inlet streams S-202, S-403 and outlet streams - with converged closure and partial coverage.
- recycle_recovery: Recycle recovery. Inlet streams S-203 and outlet streams S-301 with estimated closure and partial coverage.
- purification: Purification train. Inlet streams S-301 and outlet streams S-401, S-402, S-403 with converged closure and partial coverage.
- storage: Product storage. Inlet streams S-402 and outlet streams - with converged closure and partial coverage.

11. Material Balance

Material Balance

Overall Plant Balance Summary

Table 17. Overall Plant Balance Summary

Columns: Basis; Mass Flow (kg/h); Molar Flow (kmol/h)
Row 1: Basis: Fresh feed | Mass Flow (kg/h): 25952.368 | Molar Flow (kmol/h): 837.633819
Row 2: Basis: External outlet | Mass Flow (kg/h): 25952.368 | Molar Flow (kmol/h): 432.910401
Row 3: Basis: Recycle circulation | Mass Flow (kg/h): 27585.316 | Molar Flow (kmol/h): 492.451084
Row 4: Basis: Side draws | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000
Row 5: Basis: Closure error (%) | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): -

Section Balance Summary

Table 18. Section Balance Summary

Columns: Section; Label; Type; Inlet kg/h; Outlet kg/h; Side Draw kg/h; Recycle kg/h; Status
Row 1
- Section: feed_handling
- Label: Feed handling
- Type: feed_preparation
- Inlet kg/h: 26953.864
- Outlet kg/h: 26953.864
- Side Draw kg/h: 0.000
- Recycle kg/h: 0.000
- Status: converged

Row 2
- Section: reaction
- Label: Reaction
- Type: reaction
- Inlet kg/h: 26953.864
- Outlet kg/h: 26953.863
- Side Draw kg/h: 0.000
- Recycle kg/h: 0.000
- Status: converged

Row 3
- Section: primary_recovery
- Label: Primary recovery
- Type: primary_recovery
- Inlet kg/h: 26953.863
- Outlet kg/h: 26559.945
- Side Draw kg/h: 0.000
- Recycle kg/h: 0.000
- Status: converged

Row 4
- Section: recycle_recovery_carbonate_loop_cleanup
- Label: Carbonate loop cleanup
- Type: recycle_recovery
- Inlet kg/h: 25582.324
- Outlet kg/h: 26583.820
- Side Draw kg/h: 0.000
- Recycle kg/h: 26583.820
- Status: estimated

Row 5
- Section: purification
- Label: Purification
- Type: purification
- Inlet kg/h: 26583.820
- Outlet kg/h: 25976.243
- Side Draw kg/h: 0.000

Recycle kg/h: 1001.496
Status: estimated

Reaction Extent Allocation

Table 19. Reaction Extent Allocation

Columns: Extent; Kind; Representative Component; Fraction of Converted Feed; Status
Row 1: Extent: omega_catalytic_main_extent | Kind: main | Representative Component: Ethylene glycol | Fraction of Converted Feed: 0.990000 | Status: converged
Row 2: Extent: omega_catalytic_side_extent_1 | Kind: byproduct | Representative Component: Trace heavy glycols | Fraction of Converted Feed: 0.010000 | Status: estimated

Byproduct Closure

Table 20. Byproduct Closure

Columns: Component; Basis; Allocation Fraction; Provenance; Status
Row 1: Component: Trace heavy glycols | Basis: Declared route byproducts | Allocation Fraction: 1.000000 | Provenance: declared_trace | Status: estimated

Unit Packet Balance Summary

Table 21. Unit Packet Balance Summary

Columns: Unit; Type; Service; Inlet Streams; Outlet Streams; Inlet kg/h; Outlet kg/h; Closure Error (%); Status; Coverage
Row 1
Unit: feed_prep
Type: feed_preparation
Service: Feed preparation
Inlet Streams: S-101, S-102, S-401
Outlet Streams: S-150
Inlet kg/h: 26953.864
Outlet kg/h: 26953.864
Closure Error (%): 0.000
Status: converged
Coverage: complete
Row 2
Unit: primary_flash
Type: flash
Service: Primary flash and vent recovery
Inlet Streams: S-201
Outlet Streams: S-202, S-203
Inlet kg/h: 26953.863
Outlet kg/h: 26953.863
Closure Error (%): 0.000
Status: converged
Coverage: complete
Row 3
Unit: purification
Type: distillation
Service: Purification train
Inlet Streams: S-301
Outlet Streams: S-401, S-402, S-403
Inlet kg/h: 26583.820
Outlet kg/h: 26583.820
Closure Error (%): 0.000
Status: converged
Coverage: partial
Row 4
Unit: reactor
Type: reactor
Service: Reaction zone
Inlet Streams: S-150
Outlet Streams: S-201

Inlet kg/h: 26953.864
Outlet kg/h: 26953.863
Closure Error (%): 0.000
Status: converged
Coverage: complete

Row 5
Unit: recycle_recovery
Type: recycle
Service: Recycle recovery
Inlet Streams: S-203
Outlet Streams: S-301
Inlet kg/h: 25582.324
Outlet kg/h: 26583.820
Closure Error (%): 3.767
Status: estimated
Coverage: partial

Recycle and Purge Summary

Table 22. Recycle and Purge Summary

Columns: Loop; Source Unit; Target Unit; Recycle Streams; Purge Streams; Max Error (%); Mean Error (%); Status

Row 1
Loop: recycle_recovery_recycle_loop
Source Unit: recycle_recovery
Target Unit: purification
Recycle Streams: S-301
Purge Streams: -
Max Error (%): 0.100
Mean Error (%): 0.100
Status: converged

Row 2
Loop: purification_recycle_loop
Source Unit: purification
Target Unit: feed_prep
Recycle Streams: S-401
Purge Streams: S-403
Max Error (%): 8.650
Mean Error (%): 4.004
Status: estimated

Composition Closure Summary

Table 23. Composition Closure Summary

Columns: Unit; Reactive; Inlet Fraction Sum; Outlet Fraction Sum; New Components; Missing Components; Error (%); Status

Row 1
Unit: feed_prep
Reactive: no
Inlet Fraction Sum: 1.0000
Outlet Fraction Sum: 1.0000
New Components: -
Missing Components: -
Error (%): 0.000
Status: converged

Row 2
Unit: primary_flash
Reactive: no
Inlet Fraction Sum: 1.0000
Outlet Fraction Sum: 1.0000
New Components: -
Missing Components: -
Error (%): 0.000
Status: converged

Row 3
Unit: purification
Reactive: no
Inlet Fraction Sum: 1.0000
Outlet Fraction Sum: 1.0000

New Components: -
Missing Components: Trace heavy glycols
Error (%): 0.000
Status: estimated
Row 4
Unit: reactor
Reactive: yes
Inlet Fraction Sum: 1.0000
Outlet Fraction Sum: 1.0000
New Components: Ethylene glycol, Trace heavy glycols
Missing Components: -
Error (%): 0.000
Status: converged
Row 5
Unit: recycle_recovery
Reactive: no
Inlet Fraction Sum: 1.0000
Outlet Fraction Sum: 1.0000
New Components: -
Missing Components: -
Error (%): 0.000
Status: converged

Unitwise Outlet Composition Snapshot

Table 24. Unitwise Outlet Composition Snapshot

Columns: Unit; Type; Inlet Phase; Outlet Phase; Outlet Mass Fractions; Status
Row 1: Unit: feed_prep | Type: feed_preparation | Inlet Phase: gas | Outlet Phase: gas | Outlet Mass Fractions: Ethylene oxide=0.700, Water=0.300 | Status: converged
Row 2: Unit: primary_flash | Type: flash | Inlet Phase: gas | Outlet Phase: gas | Outlet Mass Fractions: Ethylene oxide=0.028, Trace heavy glycols=0.010, Water=0.011, Ethylene glycol=0.951 | Status: converged
Row 3: Unit: purification | Type: distillation | Inlet Phase: gas | Outlet Phase: gas | Outlet Mass Fractions: Ethylene oxide=0.018, Water=0.020, Ethylene glycol=0.961 | Status: estimated
Row 4: Unit: reactor | Type: reactor | Inlet Phase: gas | Outlet Phase: gas | Outlet Mass Fractions: Ethylene glycol=0.937, Ethylene oxide=0.028, Trace heavy glycols=0.009, Water=0.026 | Status: converged
Row 5: Unit: recycle_recovery | Type: recycle | Inlet Phase: gas | Outlet Phase: gas | Outlet Mass Fractions: Ethylene glycol=0.950, Ethylene oxide=0.020, Trace heavy glycols=0.009, Water=0.021 | Status: converged

Stream Role Summary

Table 25. Stream Role Summary

Columns: Stream; Role; Section; From; To; kg/h; kmol/h
Row 1
Stream: S-101
Role: feed
Section: feed_handling
From: battery_limits
To: feed_prep
kg/h: 18378.663
kmol/h: 417.222770
Row 2
Stream: S-102
Role: feed
Section: feed_handling
From: battery_limits
To: feed_prep
kg/h: 7573.705
kmol/h: 420.411049
Row 3
Stream: S-150
Role: intermediate
Section: reaction
From: feed_prep

To: reactor
kg/h: 26953.864
kmol/h: 877.547235
Row 4
Stream: S-201
Role: intermediate
Section: reaction
From: reactor
To: primary_flash
kg/h: 26953.863
kmol/h: 465.036538
Row 5
Stream: S-202
Role: vent
Section: primary_recovery
From: primary_flash
To: waste_treatment
kg/h: 977.621
kmol/h: 30.546176
Row 6
Stream: S-203
Role: intermediate
Section: primary_recovery
From: primary_flash
To: recycle_recovery
kg/h: 25582.324
kmol/h: 412.624251
Row 7
Stream: S-301
Role: recycle
Section: recycle_recovery_carbonate_loop_cleanup
From: recycle_recovery
To: purification
kg/h: 26583.820
kmol/h: 452.537668
Row 8
Stream: S-401
Role: recycle
Section: purification
From: purification
To: feed_prep
kg/h: 1001.496
kmol/h: 39.913416
Row 9
Stream: S-402
Role: product
Section: purification
From: purification
To: storage
kg/h: 24974.747
kmol/h: 402.364225
Row 10
Stream: S-403
Role: waste
Section: purification
From: purification
To: waste_treatment
kg/h: 0.000
kmol/h: 0.000000

Route-Family Stream Focus

Table 26. Route-Family Stream Focus

Columns: Stream; Role; From; To; Phase; kg/h; Dominant Components
Row 1
Stream: S-101
Role: feed
From: battery_limits
To: feed_prep
Phase: gas
kg/h: 18378.663

Dominant Components: Ethylene oxide=18378.7
Row 2
Stream: S-102
Role: feed
From: battery_limits
To: feed_prep
Phase: gas
kg/h: 7573.705
Dominant Components: Water=7573.7
Row 3
Stream: S-150
Role: intermediate
From: feed_prep
To: reactor
Phase: gas
kg/h: 26953.864
Dominant Components: Ethylene oxide=18856.6, Water=8097.3
Row 4
Stream: S-201
Role: intermediate
From: reactor
To: primary_flash
Phase: gas
kg/h: 26953.863
Dominant Components: Ethylene glycol=25252.5, Ethylene oxide=754.3, Water=694.1, Trace heavy glycols=253.0
Row 5
Stream: S-202
Role: vent
From: primary_flash
To: waste_treatment
Phase: gas
kg/h: 977.621
Dominant Components: Ethylene oxide=709.0, Water=258.5, Trace heavy glycols=10.1
Row 6
Stream: S-203
Role: intermediate
From: primary_flash
To: recycle_recovery
Phase: gas
kg/h: 25582.324
Dominant Components: Ethylene glycol=25252.5, Trace heavy glycols=242.9, Ethylene oxide=45.3, Water=41.6
Row 7
Stream: S-301
Role: recycle
From: recycle_recovery
To: purification
Phase: gas
kg/h: 26583.820
Dominant Components: Ethylene glycol=25252.5, Water=565.2, Ethylene oxide=523.2, Trace heavy glycols=242.9
Row 8
Stream: S-401
Role: recycle
From: purification
To: feed_prep
Phase: gas
kg/h: 1001.496
Dominant Components: Water=523.6, Ethylene oxide=477.9
Row 9
Stream: S-402
Role: product
From: purification
To: storage
Phase: gas
kg/h: 24974.747
Dominant Components: Ethylene glycol=24974.7
Row 10
Stream: S-403
Role: waste
From: purification

To: waste_treatment
Phase: mixed
kg/h: 0.000
Dominant Components: -

Long Stream Ledger

Table 27. Long Stream Ledger

Columns: Stream; Description; From; To; Component; kg/h; kmol/h; T (C); P (bar)

Row 1
Stream: S-101
Description: Ethylene oxide fresh feed
From: battery_limits
To: feed_prep
Component: Ethylene oxide
kg/h: 18378.663
kmol/h: 417.222770
T (C): 25.0
P (bar): 18.00

Row 2
Stream: S-102
Description: Water fresh feed
From: battery_limits
To: feed_prep
Component: Water
kg/h: 7573.705
kmol/h: 420.411049
T (C): 25.0
P (bar): 18.00

Row 3
Stream: S-150
Description: Mixed reactor feed
From: feed_prep
To: reactor
Component: Ethylene oxide
kg/h: 18856.564
kmol/h: 428.071822
T (C): 225.0
P (bar): 18.00

Row 4
Stream: S-150
Description: Mixed reactor feed
From: feed_prep
To: reactor
Component: Water
kg/h: 8097.300
kmol/h: 449.475413
T (C): 225.0
P (bar): 18.00

Row 5
Stream: S-201
Description: Reactor effluent
From: reactor
To: primary_flash
Component: Ethylene glycol
kg/h: 25252.525
kmol/h: 406.839460
T (C): 240.0
P (bar): 18.00

Row 6
Stream: S-201
Description: Reactor effluent
From: reactor
To: primary_flash
Component: Ethylene oxide
kg/h: 754.263
kmol/h: 17.122873
T (C): 240.0
P (bar): 18.00

Row 7

Stream: S-201
Description: Reactor effluent
From: reactor
To: primary_flash
Component: Trace heavy glycols
kg/h: 253.021
kmol/h: 2.547741
T (C): 240.0
P (bar): 18.00

Row 8
Stream: S-201
Description: Reactor effluent
From: reactor
To: primary_flash
Component: Water
kg/h: 694.054
kmol/h: 38.526464
T (C): 240.0
P (bar): 18.00

Row 9
Stream: S-202
Description: Primary separation overhead / vent / purge
From: primary_flash
To: waste_treatment
Component: Ethylene oxide
kg/h: 709.007
kmol/h: 16.095501
T (C): 205.0
P (bar): 16.50

Row 10
Stream: S-202
Description: Primary separation overhead / vent / purge
From: primary_flash
To: waste_treatment
Component: Trace heavy glycols
kg/h: 10.121
kmol/h: 0.101910
T (C): 205.0
P (bar): 16.50

Row 11
Stream: S-202
Description: Primary separation overhead / vent / purge
From: primary_flash
To: waste_treatment
Component: Water
kg/h: 258.493
kmol/h: 14.348765
T (C): 205.0
P (bar): 16.50

Row 12
Stream: S-203
Description: Primary separation bottoms / rich liquid
From: primary_flash
To: recycle_recovery
Component: Ethylene glycol
kg/h: 25252.525
kmol/h: 406.839460
T (C): 230.0
P (bar): 17.20

Row 13
Stream: S-203
Description: Primary separation bottoms / rich liquid
From: primary_flash
To: recycle_recovery
Component: Ethylene oxide
kg/h: 45.256
kmol/h: 1.027372
T (C): 230.0
P (bar): 17.20

Row 14
Stream: S-203
Description: Primary separation bottoms / rich liquid
From: primary_flash

To: recycle_recovery
Component: Trace heavy glycols
kg/h: 242.900
kmol/h: 2.445831
T (C): 230.0
P (bar): 17.20

Row 15
Stream: S-203
Description: Primary separation bottoms / rich liquid
From: primary_flash
To: recycle_recovery
Component: Water
kg/h: 41.643
kmol/h: 2.311588
T (C): 230.0
P (bar): 17.20

Row 16
Stream: S-301
Description: Cleanup / recycle recovery stream
From: recycle_recovery
To: purification
Component: Ethylene glycol
kg/h: 25252.525
kmol/h: 406.839460
T (C): 215.0
P (bar): 17.00

Row 17
Stream: S-301
Description: Cleanup / recycle recovery stream
From: recycle_recovery
To: purification
Component: Ethylene oxide
kg/h: 523.157
kmol/h: 11.876425
T (C): 215.0
P (bar): 17.00

Row 18
Stream: S-301
Description: Cleanup / recycle recovery stream
From: recycle_recovery
To: purification
Component: Trace heavy glycols
kg/h: 242.900
kmol/h: 2.445831
T (C): 215.0
P (bar): 17.00

Row 19
Stream: S-301
Description: Cleanup / recycle recovery stream
From: recycle_recovery
To: purification
Component: Water
kg/h: 565.238
kmol/h: 31.375952
T (C): 215.0
P (bar): 17.00

Row 20
Stream: S-401
Description: Light ends / recoverable recycle
From: purification
To: feed_prep
Component: Ethylene oxide
kg/h: 477.901
kmol/h: 10.849052
T (C): 45.0
P (bar): 1.00

Row 21
Stream: S-401
Description: Light ends / recoverable recycle
From: purification
To: feed_prep
Component: Water
kg/h: 523.595

kmol/h: 29.064364
T (C): 45.0
P (bar): 1.00
Row 22
Stream: S-402
Description: On-spec product
From: purification
To: storage
Component: Ethylene glycol
kg/h: 24974.747
kmol/h: 402.364225
T (C): 45.0
P (bar): 1.10

Stream-Balance Calculation Traces

Table 28. Stream-Balance Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Main reaction extent | Formula: extent = product_kmol / nu_product | Inputs: product_kmol=406.839460; nu_product=1.000 | Result: 406.839460 kmol/h | Notes: -
Row 2: Trace: Feed extent with conversion/selectivity | Formula: feed_extent = (product_kmol / nu_product) / (selectivity * conversion) | Inputs: selectivity=0.9900; conversion=0.9600 | Result: 428.071822 kmol/h | Notes: -
Row 3: Trace: Byproduct closure allocation | Formula: Residual byproduct mass is allocated across explicit byproduct-closure estimates | Inputs: closure_status=estimated; estimate_count=1; residual_mass_kg_hr=253.021 | Result: 1 estimates items | Notes: This replaces the older implicit heavy-ends fallback whenever the reaction system carries a byproduct-closure artifact.
Row 4: Trace: Multi-component recycle solution | Formula: Solve recycle per reactant using target_total, consumed, recovery, and purge fractions | Inputs: Ethylene oxide=fresh=417.223; recycle=10.849; total=428.072; Water=fresh=420.411; recycle=29.064; total=449.475 | Result: 2 components solved components | Notes: This is the first generic convergence slice: reactants are solved through the same recycle/purge loop instead of family-specific inline math.
Row 5: Trace: Unit-sequence expansion | Formula: stream_count = feeds + mixed feed + reactor + separation + recycle + product/waste | Inputs: family=generic | Result: 10 streams | Notes: -
Row 6: Trace: Fresh-feed to external-out closure | Formula: closure = | Inputs: fresh_feed - external_out | Result: / fresh_feed | Notes: fresh_feed=25952.368; external_out=25952.368 | Field 6: 0.000000 % | Field 7: -

12. Energy Balance

Energy Balance

Overall Energy Summary

Table 29. Overall Energy Summary

Columns: Basis; Value; Units
Row 1: Basis: Total heating duty | Value: 5781.464 | Units: kW
Row 2: Basis: Total cooling duty | Value: 8224.582 | Units: kW
Row 3: Basis: Net external duty | Value: -2443.118 | Units: kW
Row 4: Basis: Thermal packets | Value: 3 | Units: count
Row 5: Basis: Recovery candidates | Value: 2 | Units: count
Row 6: Basis: Recoverable packet duty | Value: 624.369 | Units: kW

Unit Duty Summary

Table 30. Unit Duty Summary

Columns: Unit; Section; Duty Type; Heating (kW); Cooling (kW); Notes
Row 1: Unit: E-101 | Section: - | Duty Type: sensible | Heating (kW): 4532.727 | Cooling (kW): 0.000 | Notes: Feed preheat to operating temperature.
Row 2: Unit: R-101 | Section: - | Duty Type: reaction | Heating (kW): 0.000 | Cooling (kW): 7600.213 | Notes: Net reactor thermal duty from the selected thermodynamic basis.
Row 3: Unit: SEP-201 | Section: - | Duty Type: sensible | Heating (kW): 1248.737 | Cooling (kW): 624.369 | Notes: Generic purification duty.

Section Duty Summary

Table 31. Section Duty Summary

Columns: Section; Label; Heating (kW); Cooling (kW); Recoverable (kW); Status
Row 1: Section: feed_handling | Label: Feed handling | Heating (kW): 0.000 | Cooling (kW): 0.000 | Recoverable (kW): 0.000 | Status: converged
Row 2: Section: reaction | Label: Reaction | Heating (kW): 0.000 | Cooling (kW): 0.000 | Recoverable (kW): 0.000 | Status: converged
Row 3: Section: primary_recovery | Label: Primary recovery | Heating (kW): 0.000 | Cooling (kW): 0.000 | Recoverable (kW): 0.000 | Status: converged
Row 4: Section: recycle_recovery_carbonate_loop_cleanup | Label: Carbonate loop cleanup | Heating (kW): 0.000 | Cooling (kW): 0.000 | Recoverable (kW): 0.000 | Status: estimated
Row 5: Section: purification | Label: Purification | Heating (kW): 0.000 | Cooling (kW): 0.000 | Recoverable (kW): 0.000 | Status: estimated

Unit Thermal Packet Summary

Table 32. Unit Thermal Packet Summary

Columns: Packet; Unit; Section; Type; Heating (kW); Cooling (kW); Hot In (C); Hot Out (C); Cold In (C); Cold Out (C); Recoverable (kW); Candidate Media
Row 1
- Packet: E-101_thermal_packet
- Unit: E-101
- Section: -
- Type: sensible
- Heating (kW): 4532.727
- Cooling (kW): 0.000
- Hot In (C): 270.0
- Hot Out (C): 255.0
- Cold In (C): 225.0
- Cold Out (C): 240.0
- Recoverable (kW): 0.000
- Candidate Media: steam, hot oil

Row 2
Packet: R-101_thermal_packet
Unit: R-101
Section: -
Type: reaction
Heating (kW): 0.000
Cooling (kW): 7600.213
Hot In (C): 240.0
Hot Out (C): 195.0
Cold In (C): 30.0
Cold Out (C): 42.0
Recoverable (kW): 0.000
Candidate Media: Dowtherm A, cooling water
Row 3
Packet: SEP-201_thermal_packet
Unit: SEP-201
Section: -
Type: sensible
Heating (kW): 1248.737
Cooling (kW): 624.369
Hot In (C): 270.0
Hot Out (C): 245.0
Cold In (C): 30.0
Cold Out (C): 75.0
Recoverable (kW): 624.369
Candidate Media: cooling water, steam, hot oil

Recovery Candidate Summary

Table 33. Recovery Candidate Summary

Columns: Candidate; Source Unit; Sink Unit; Topology; Recovered Duty (kW); Min Approach (C); Feasible; Notes
Row 1
Candidate: R-101_to_SEP-201
Source Unit: R-101
Sink Unit: SEP-201
Topology: direct
Recovered Duty (kW): 1123.863
Min Approach (C): 20.0
Feasible: yes
Notes: Direct packet-level recovery candidate.
Row 2
Candidate: SEP-201_to_E-101
Source Unit: SEP-201
Sink Unit: E-101
Topology: direct
Recovered Duty (kW): 561.932
Min Approach (C): 20.0
Feasible: yes
Notes: Direct packet-level recovery candidate.

Utility Consumption Basis

Table 34. Utility Consumption Basis

Columns: Unit; Route; Candidate Media; Peak Duty (kW); Recoverable Duty (kW); Service
Row 1: Unit: E-101 | Route: omega_catalytic | Candidate Media: steam, hot oil | Peak Duty (kW): 4532.727 | Recoverable Duty (kW): 0.000 | Service: Feed preheat to operating temperature.
Row 2: Unit: R-101 | Route: omega_catalytic | Candidate Media: Dowtherm A, cooling water | Peak Duty (kW): 7600.213 | Recoverable Duty (kW): 0.000 | Service: Net reactor thermal duty from the selected thermodynamic basis.
Row 3: Unit: SEP-201 | Route: omega_catalytic | Candidate Media: cooling water, steam, hot oil | Peak Duty (kW): 1248.737 | Recoverable Duty (kW): 624.369 | Service: Generic purification duty.

Route-Family Duty Focus

Table 35. Route-Family Duty Focus

Columns: Unit; Duty Type; Heating (kW); Cooling (kW); Section; Notes
Row 1: Unit: E-101 | Duty Type: sensible | Heating (kW): 4532.727 | Cooling (kW): 0.000 | Section: - | Notes: Feed preheat to operating temperature.
Row 2: Unit: R-101 | Duty Type: reaction | Heating (kW): 0.000 | Cooling (kW): 7600.213 | Section: - | Notes: Net reactor thermal duty from the selected thermodynamic basis.

Energy-Balance Calculation Traces

Table 36. Energy-Balance Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Feed preheat duty | Formula: Q = m * Cp * dT / 3600 | Inputs: m=25952.368 kg/h; dT=215.0 K | Result: 4532.727 kW | Notes: -
Row 2: Trace: Reaction duty | Formula: Q = | Inputs: n * dH | Result: / 3600 | Notes: n=402.364225 kmol/h; dH=-68.000 kJ/mol | Field 6: 7600.213 kW | Field 7: -
Row 3: Trace: Unitwise duty expansion | Formula: duty_count = number of explicit unit duties in the solved energy envelope | Inputs: - | Result: 3 duties | Notes: The generic energy solver now emits duty rows for each major process section rather than a single family-level total.
Row 4: Trace: Thermal packet expansion | Formula: packet_count = number of unitwise thermal packets carried into utility and equipment design | Inputs: - | Result: 3 packets | Notes: These packets preserve unit-level hot and cold thermal interfaces downstream.
Row 5: Trace: Packet-derived exchanger candidates | Formula: candidate_count = number of packet-to-packet heat-recovery candidates | Inputs: - | Result: 2 candidates | Notes: These candidates seed utility-architecture selection and detailed exchanger sizing.
Row 6: Trace: Composition-driven thermal basis | Formula: Cp and latent-duty basis are derived from solved unit composition states and mixture-property packages | Inputs: feed_state=feed_prep_composition_state; primary_state=primary_flash_composition_state; purification_state=purification_composition_state; feed_cp=2.816; product_cp=2.452; mixture_packages=5 | Result: generic family | Notes: The energy solver now consumes mixture-property packages first and only retains heuristics as a compatibility fallback.

13. Heat Integration and Utility Optimization

Heat Integration and Utility Optimization

The heat-integration study evaluates no-recovery, multi-effect, and pinch/HTM-loop cases before final route choice. Routes that remain utility-heavy after recovery are penalized in downstream economics.

Table 37. Heat Integration and Utility Optimization

Columns: Route; Selected case; Recovered duty (kW); Residual hot utility (kW); Savings (INR bn/y); Payback (y); Stability

Row 1
- Route: eo_hydration
- Selected case: Pinch + HTM loop
- Recovered duty (kW): 51714.0
- Residual hot utility (kW): 49263.8
- Savings (INR bn/y): 1.45
- Payback (y): 0.40
- Stability: stable

Row 2
- Route: omega_catalytic
- Selected case: Pinch + HTM loop
- Recovered duty (kW): 47256.7
- Residual hot utility (kW): 43800.1
- Savings (INR bn/y): 1.32
- Payback (y): 0.44
- Stability: stable

Row 3
- Route: chlorohydrin
- Selected case: No recovery
- Recovered duty (kW): 0.0
- Residual hot utility (kW): 13044.2
- Savings (INR bn/y): 0.00
- Payback (y): 0.00
- Stability: stable

14. Reactor Design

Reactor Design

Reactor R-101 uses Tubular Plug Flow Hydrator basis in gas_phase_catalytic service with k=14.9632 1/h and Da=14.963. Thermal stability basis gives DeltaTad=414.73 C and heat-removal margin 0.120. Integrated duty basis is 0.000 kW via shared HTM island network (shared_htm) with 1 utility islands [shared_htm], with explicit coupled area 0.000 m2 at 0.000 K.

Governing Equations

- V = tau * volumetric_flow
- k = A * exp(-Ea/RT)
- Da = k * tau
- Q = U * A * LMTD
- Nu = 0.023 * Re^0.8 * Pr^0.4

Route-Family Basis

Table 38. Route-Family Basis

Columns: Parameter; Value
Row 1: Parameter: Route family | Value: Liquid Hydration Train
Row 2: Parameter: Route family id | Value: liquid_hydration_train
Row 3: Parameter: Primary reactor class | Value: Tubular plug-flow hydrator
Row 4: Parameter: Primary separation train | Value: EO flash -> water removal -> vacuum glycol distillation
Row 5: Parameter: Heat recovery style | Value: condenser_reboiler_cluster
Row 6: Parameter: Data anchors | Value: binary_interaction_parameters, liquid_density, liquid_viscosity, missing_vle_basis, missing_density_viscosity_for_column_hydraulics

Solver Packet Basis

Table 39. Solver Packet Basis

Columns: Packet; Primary Value 1; Primary Value 2; Closure / Recoverable; Status / Media
Row 1: Packet: reactor_unit_packet | Primary Value 1: 26953.864 | Primary Value 2: 26953.863 | Closure / Recoverable: 0.000 | Status / Media: converged
Row 2: Packet: R-101_thermal_packet | Primary Value 1: 0.000 | Primary Value 2: 7600.213 | Closure / Recoverable: 0.000 | Status / Media: Dowtherm A, cooling water

Balance Reference Snapshot

Table 40. Balance Reference Snapshot

Columns: Stream; Role; From; To; kg/h; Dominant Components
Row 1: Stream: S-150 | Role: intermediate | From: feed_prep | To: reactor | kg/h: 26953.864 | Dominant Components: Ethylene oxide=18856.6, Water=8097.3
Row 2: Stream: S-201 | Role: intermediate | From: reactor | To: primary_flash | kg/h: 26953.863 | Dominant Components: Ethylene glycol=25252.5, Ethylene oxide=754.3, Water=694.1, Trace heavy glycols=253.0

Reactor Feed / Product / Recycle Summary

Table 41. Reactor Feed / Product / Recycle Summary

Columns: Unit; Service; Stream; Local Role; Stream Role; Section; Phase;

kg/h; kmol/h; Dominant Components
Row 1
Unit: reactor
Service: Reaction zone
Stream: S-150
Local Role: feed
Stream Role: intermediate
Section: reaction
Phase: gas
kg/h: 26953.864
kmol/h: 877.547235
Dominant Components: Ethylene oxide=18856.6, Water=8097.3
Row 2
Unit: reactor
Service: Reaction zone
Stream: S-201
Local Role: product
Stream Role: intermediate
Section: reaction
Phase: gas
kg/h: 26953.863
kmol/h: 465.036538
Dominant Components: Ethylene glycol=25252.5, Ethylene oxide=754.3, Water=694.1, Trace heavy glycols=253.0

Reactor Local Stream Split Summary

Table 42. Reactor Local Stream Split Summary

Columns: Split; Stream Count; Mass Flow (kg/h); Molar Flow (kmol/h)
Row 1: Split: fresh_feed | Stream Count: 1 | Mass Flow (kg/h): 26953.864 | Molar Flow (kmol/h): 877.547235
Row 2: Split: recycle_feed | Stream Count: 0 | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000
Row 3: Split: total_feed | Stream Count: 1 | Mass Flow (kg/h): 26953.864 | Molar Flow (kmol/h): 877.547235
Row 4: Split: product_effluent | Stream Count: 1 | Mass Flow (kg/h): 26953.863 | Molar Flow (kmol/h): 465.036538
Row 5: Split: recycle_effluent | Stream Count: 0 | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000
Row 6: Split: side_draw_purge_vent | Stream Count: 0 | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000

Key Reactor Component Balance

Table 43. Key Reactor Component Balance

Columns: Component; Inlet kg/h; Outlet kg/h; Delta kg/h
Row 1: Component: Ethylene glycol | Inlet kg/h: 0.000 | Outlet kg/h: 25252.525 | Delta kg/h: 25252.525
Row 2: Component: Ethylene oxide | Inlet kg/h: 18856.564 | Outlet kg/h: 754.263 | Delta kg/h: -18102.301
Row 3: Component: Water | Inlet kg/h: 8097.300 | Outlet kg/h: 694.054 | Delta kg/h: -7403.246
Row 4: Component: Trace heavy glycols | Inlet kg/h: 0.000 | Outlet kg/h: 253.021 | Delta kg/h: 253.021

Reactor Design Inputs

Table 44. Reactor Design Inputs

Columns: Parameter; Value
Row 1: Parameter: Residence time hr | Value: 1.000
Row 2: Parameter: Design volume m3 | Value: 27004.165
Row 3: Parameter: Design conversion fraction | Value: 0.9600
Row 4: Parameter: Phase regime | Value: gas_phase_catalytic
Row 5: Parameter: Kinetic rate constant 1 hr | Value: 14.9632
Row 6: Parameter: Kinetic space time hr | Value: 0.6000
Row 7: Parameter: Kinetic damkohler number | Value: 14.9632
Row 8: Parameter: Heat duty kw | Value: 7600.213

Row 9: Parameter: Heat release density kw m3 | Value: 0.281
Row 10: Parameter: Adiabatic temperature rise c | Value: 414.729
Row 11: Parameter: Heat removal capacity kw | Value: 8512.239
Row 12: Parameter: Heat removal margin fraction | Value: 0.1200
Row 13: Parameter: Thermal stability score | Value: 5.00
Row 14: Parameter: Runaway risk label | Value: moderate
Row 15: Parameter: Integrated thermal duty kw | Value: 0.000
Row 16: Parameter: Residual utility duty kw | Value: 7600.213
Row 17: Parameter: Integrated lmtd k | Value: 0.000
Row 18: Parameter: Integrated exchange area m2 | Value: 0.000
Row 19: Parameter: Allocated recovered duty target kw | Value: 0.000

Reactor Operating Envelope

Table 45. Reactor Operating Envelope

Columns: Parameter; Value
Row 1: Parameter: Temperature c | Value: 255.0
Row 2: Parameter: Pressure bar | Value: 20.00
Row 3: Parameter: Cooling medium | Value: Dowtherm A / cooling water
Row 4: Parameter: Utility topology | Value: shared HTM island network (shared_htm) with 1 utility islands
Row 5: Parameter: Utility architecture family | Value: shared_htm
Row 6: Parameter: Coupled service basis | Value: standalone utility service
Row 7: Parameter: Selected utility islands | Value: none
Row 8: Parameter: Selected header levels | Value: none
Row 9: Parameter: Selected cluster ids | Value: none
Row 10: Parameter: Catalyst name | Value: CO2 catalytic loop
Row 11: Parameter: Catalyst inventory kg | Value: 4914758.026
Row 12: Parameter: Catalyst cycle days | Value: 270.0
Row 13: Parameter: Catalyst regeneration days | Value: 5.0
Row 14: Parameter: Catalyst whsv 1 hr | Value: 0.0055

Reactor Sizing Basis

Table 46. Reactor Sizing Basis

Columns: Parameter; Value
Row 1: Parameter: Selected reactor type | Value: Tubular Plug Flow Hydrator
Row 2: Parameter: Design basis | Value: tubular_plug_flow_hydrator selected at 1.00 h residence time and 0.960 conversion basis.
Row 3: Parameter: Phase regime | Value: gas_phase_catalytic
Row 4: Parameter: Residence time (h) | Value: 1.000
Row 5: Parameter: Design volume (m3) | Value: 27004.165
Row 6: Parameter: Design temperature (C) | Value: 255.0
Row 7: Parameter: Design pressure (bar) | Value: 20.00

Reaction and Sizing Derivation Basis

Table 47. Reaction and Sizing Derivation Basis

Columns: Check; Formula / Basis; Result
Row 1: Check: Volumetric throughput | Formula / Basis: Vdot = V / tau | Result: 27004.165 m3/h
Row 2: Check: Kinetic consistency | Formula / Basis: Da = k * tau | Result: 14.9632
Row 3: Check: Heat release density | Formula / Basis: q''' = Q / V | Result: 0.281 kW/m3
Row 4: Check: Heat-transfer area check | Formula / Basis: A = Q / (U * LMTD) | Result: n/a
Row 5: Check: Design conversion basis | Formula / Basis: Target conversion | Result: 0.9600
Row 6: Check: Packet closure basis | Formula / Basis: Unit packet closure | Result: 0.000 %

Reactor Equation-Substitution Sheet

Table 48. Reactor Equation-Substitution Sheet

Columns: Check; Equation; Substitution; Result
Row 1: Check: Residence-time sizing | Equation: Vdot = V / tau | Substitution: V=27004.165 m3; tau=1.000 h | Result: 27004.165 m3/h
Row 2: Check: Damkohler basis | Equation: Da = k * tau | Substitution: k=14.963167 1/h; tau=1.000 h | Result: 14.9632
Row 3: Check: Thermal intensity | Equation: q''' = Q / V | Substitution: Q=7600.213 kW; V=27004.165 m3 | Result: 0.281 kW/m3
Row 4: Check: Heat-transfer area check | Equation: A = Q / (U * LMTD) | Substitution: Integrated LMTD not available | Result: n/a
Row 5: Check: Heat-removal margin | Equation: Margin = Qrem / Qduty - 1 | Substitution: Qrem=8512.239 kW; Qduty=7600.213 kW | Result: 0.1200
Row 6: Check: Residual utility demand | Equation: Qres = Qduty - Qint | Substitution: Qduty=7600.213 kW; Qint=0.000 kW | Result: 7600.213 kW

Kinetic Design Basis

Table 49. Kinetic Design Basis

Columns: Parameter; Value
Row 1: Parameter: Design conversion fraction | Value: 0.9600
Row 2: Parameter: Rate constant (1/h) | Value: 14.963167
Row 3: Parameter: Kinetic space time (h) | Value: 0.600000
Row 4: Parameter: Damkohler number | Value: 14.963167

Reactor Geometry and Internals

Table 50. Reactor Geometry and Internals

Columns: Parameter; Value
Row 1: Parameter: Liquid holdup (m3) | Value: 22884.886
Row 2: Parameter: Shell diameter (m) | Value: 19.016
Row 3: Parameter: Shell length (m) | Value: 95.081
Row 4: Parameter: Tube count | Value: 24
Row 5: Parameter: Tube length (m) | Value: 83.671
Row 6: Parameter: Heat-transfer area (m2) | Value: 50.578
Row 7: Parameter: Cooling medium | Value: Dowtherm A / cooling water
Row 8: Parameter: Utility topology | Value: shared HTM island network (shared_htm) with 1 utility islands

Heat-Transfer Derivation Basis

Table 51. Heat-Transfer Derivation Basis

Columns: Parameter; Value
Row 1: Parameter: Heat duty (kW) | Value: 7600.213
Row 2: Parameter: Heat-release density (kW/m3) | Value: 0.281
Row 3: Parameter: Adiabatic temperature rise (C) | Value: 414.729
Row 4: Parameter: Heat-removal capacity (kW) | Value: 8512.239
Row 5: Parameter: Heat-removal margin | Value: 0.1200
Row 6: Parameter: Thermal stability score | Value: 5.00
Row 7: Parameter: Runaway risk label | Value: moderate
Row 8: Parameter: Heat-transfer area (m2) | Value: 50.578
Row 9: Parameter: Overall U (W/m2-K) | Value: 850.0
Row 10: Parameter: Reynolds number | Value: 1,000.0
Row 11: Parameter: Prandtl number | Value: 122.136
Row 12: Parameter: Nusselt number | Value: 39.49
Row 13: Parameter: Tube count | Value: 24
Row 14: Parameter: Tube length (m) | Value: 83.671

Thermal Stability and Hazard Envelope

Table 52. Thermal Stability and Hazard Envelope

Columns: Parameter; Value
Row 1: Parameter: Adiabatic temperature rise (C) | Value: 414.729
Row 2: Parameter: Heat-removal capacity (kW) | Value: 8512.239
Row 3: Parameter: Heat-removal margin | Value: 0.1200
Row 4: Parameter: Thermal stability score | Value: 5.00
Row 5: Parameter: Runaway risk label | Value: moderate

Row 6: Parameter: Residual utility duty (kW) | Value: 7600.213
Row 7: Parameter: Integrated recovered duty (kW) | Value: 0.000

Catalyst Service Basis

Table 53. Catalyst Service Basis

Columns: Parameter; Value
Row 1: Parameter: Catalyst | Value: CO2 catalytic loop
Row 2: Parameter: Catalyst inventory (kg) | Value: 4914758.026
Row 3: Parameter: Catalyst cycle (days) | Value: 270.0
Row 4: Parameter: Catalyst regeneration (days) | Value: 5.0
Row 5: Parameter: Catalyst void fraction | Value: 0.350
Row 6: Parameter: Catalyst WHSV (1/h) | Value: 0.0055

Integrated Utility Package Basis

Table 54. Integrated Utility Package Basis

Columns: Parameter; Value
Row 1: Parameter: Architecture family | Value: shared_htm
Row 2: Parameter: Coupled service basis | Value: standalone utility service
Row 3: Parameter: Integrated LMTD (K) | Value: 0.000
Row 4: Parameter: Integrated exchange area (m2) | Value: 0.000
Row 5: Parameter: Allocated recovered-duty target (kW) | Value: 0.000
Row 6: Parameter: Selected utility islands | Value: none
Row 7: Parameter: Selected header levels | Value: none
Row 8: Parameter: Selected cluster ids | Value: none
Row 9: Parameter: Selected train steps | Value: none

Utility Coupling

Table 55. Utility Coupling

Columns: Parameter; Value
Row 1: Parameter: Utility topology | Value: shared HTM island network (shared_htm) with 1 utility islands
Row 2: Parameter: Architecture family | Value: shared_htm
Row 3: Parameter: Cooling medium | Value: Dowtherm A / cooling water
Row 4: Parameter: Integrated duty (kW) | Value: 0.000
Row 5: Parameter: Allocated island target (kW) | Value: 0.000
Row 6: Parameter: Residual utility duty (kW) | Value: 7600.213
Row 7: Parameter: Integrated LMTD (K) | Value: 0.000
Row 8: Parameter: Integrated exchange area (m2) | Value: 0.000
Row 9: Parameter: Selected utility islands | Value: none
Row 10: Parameter: Selected header levels | Value: none
Row 11: Parameter: Selected cluster ids | Value: none
Row 12: Parameter: Coupled service basis | Value: standalone utility service
Row 13: Parameter: Selected train steps | Value: none

Reactor Calculation Traces

Table 56. Reactor Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Reactor holdup | Formula: V = Qv * tau * factor | Inputs: Qv=22884.886; tau=1.000 | Result: 22884.886 m3 | Notes: -
Row 2: Trace: Reactor design volume | Formula: Vd = V * design_factor | Inputs: V=22884.886; factor=1.180 | Result: 27004.165 m3 | Notes: -
Row 3: Trace: Kinetics-coupled sizing basis | Formula: k = A*exp(-Ea/RT); tau_design from kinetic space time, route time, and solved conversion target | Inputs: Ea_kJ_mol=58.000; A_1_hr=1.200e+07; T_K=513.15; k_1_hr=14.9632; tau_kin_hr=0.6000; X_target=0.9600 | Result: 1.000 h | Notes: Stage 3 reactor design now ties residence time to the selected kinetic basis instead of only a fixed route heuristic.
Row 4: Trace: Reactor-side Reynolds number | Formula: Re = rho * v * Dh / mu | Inputs: rho=1.2; v=0.200; Dh=0.032; mu=0.01304 | Result: 1000.0 - | Notes: -

Row 5: Trace: Reactor-side Nusselt number | Formula: Nu = 0.023 Re^0.8 Pr^0.4 | Inputs: Re=1000.0; Pr=122.14 | Result: 39.49 - | Notes: -
Row 6: Trace: Reactor property-package basis | Formula: Pr = Cp * mu / k using mixture-property package values when available | Inputs: Cp=2.448; mu=0.013037; k=0.2613; mixture_package=reactor_mixture_properties | Result: 122.14 - | Notes: -
Row 7: Trace: Solved reactor packet basis | Formula: reactor packet -> inlet mass and thermal duty | Inputs: packet_inlet_mass_kg_hr=26953.864; thermal_packet=R-101_thermal_packet; composition_state=reactor_composition_state | Result: 7600.213 kW | Notes: Reactor sizing now reads the solved reactor unit packet, composition state, and thermal packet before falling back to route-level heuristics.
Row 8: Trace: Reactor utility-train coupling | Formula: Integrated reactor duty = sum(selected train steps tied to reactor source/sink) | Inputs: selected_steps=none; topology=shared HTM island network (shared_htm) with 1 utility islands; architecture_family=shared_htm; selected_islands=none; header_levels=none; allocated_target_kw=0.000 | Result: 0.000 kW | Notes: This captures recovered reactor duty exported to or imported from the selected utility train.
Row 9: Trace: Reactor integrated heat-transfer basis | Formula: A_int = Q_int / (U * LMTD_int) | Inputs: Q_int=0.0; U=850.0; LMTD_int=0.00 | Result: 0.000 m2 | Notes: Integrated heat-transfer area is derived from the selected train-step thermal driving force, not only aggregate recovered duty.
Row 10: Trace: Reactor thermal stability basis | Formula: DeltaTad = Q/(m*Cp); margin = (Qcap-Qduty)/Qduty | Inputs: Qduty=7600.213; m_feed=26953.864; Cp_kJ_kg_K=2.448; DeltaTad=414.729; Qcap=8512.239 | Result: 0.1200 fraction margin | Notes: This is a screening runaway and heat-removal check to keep reactor selection tied to thermal controllability.
Row 11: Trace: Catalyst service basis | Formula: mcat = V*(1-eps)*rhobulk ; WHSV = mfeed/mcat | Inputs: catalyst=CO2 catalytic loop; void_fraction=0.350; inventory_kg=4914758.026; WHSV_1_hr=0.0055; cycle_days=270.0 | Result: 4914758.026 kg | Notes: Catalyst inventory and cycle are screening values used to connect reactor selection with service-life assumptions.

15. Distillation / Process-Unit Design

Distillation / Process-Unit Design

Column PU-201 hydraulic basis uses tray spacing 0.450 m and flooding fraction 0.538, with superficial vapor velocity 0.188 m/s against allowable 0.350 m/s. Integrated reboiler duty basis is 0.000 kW via shared HTM island network (shared_htm) with 1 utility islands, family shared_htm, with reboiler area 0.000 m2 at 0.000 K.

Route-Family Basis

Table 57. Route-Family Basis

Columns: Parameter; Value
Row 1: Parameter: Route family | Value: Liquid Hydration Train
Row 2: Parameter: Route family id | Value: liquid_hydration_train
Row 3: Parameter: Primary separation train | Value: EO flash -> water removal -> vacuum glycol distillation
Row 4: Parameter: Heat recovery style | Value: condenser_reboiler_cluster
Row 5: Parameter: Dominant phase pattern | Value: liquid_reactive
Row 6: Parameter: Data anchors | Value: binary_interaction_parameters, liquid_density, liquid_viscosity, missing_vle_basis, missing_density_viscosity_for_column_hydraulics

Governing Equations

- Fenske / Underwood / Gilliland equivalents
- Q = U * A * LMTD
- A = Q / (U * LMTD)
- m_phase = Q / lambda

Solver Packet Basis

Table 58. Solver Packet Basis

Columns: Packet; Unit; Type; Inlet kg/h; Outlet kg/h; Closure Error (%); Status
Row 1
Packet: purification_unit_packet
Unit: purification
Type: distillation
Inlet kg/h: 26583.820
Outlet kg/h: 26583.820
Closure Error (%): 0.000
Status: converged

Balance Reference Snapshot

Table 59. Balance Reference Snapshot

Columns: Stream; Role; From; To; kg/h; Dominant Components
Row 1: Stream: S-301 | Role: recycle | From: recycle_recovery | To: purification | kg/h: 26583.820 | Dominant Components: Ethylene glycol=25252.5, Water=565.2, Ethylene oxide=523.2, Trace heavy glycols=242.9
Row 2: Stream: S-401 | Role: recycle | From: purification | To: feed_prep | kg/h: 1001.496 | Dominant Components: Water=523.6, Ethylene oxide=477.9
Row 3: Stream: S-402 | Role: product | From: purification | To: storage | kg/h: 24974.747 | Dominant Components: Ethylene glycol=24974.7
Row 4: Stream: S-403 | Role: waste | From: purification | To: waste_treatment | kg/h: 0.000 | Dominant Components: -

Unit-by-Unit Feed / Product / Recycle Summary

Table 60. Unit-by-Unit Feed / Product / Recycle Summary

Columns: Unit; Service; Stream; Local Role; Stream Role; Section; Phase; kg/h; kmol/h; Dominant Components

Row 1
- Unit: purification
- Service: Purification train
- Stream: S-301
- Local Role: recycle feed
- Stream Role: recycle
- Section: recycle_recovery_carbonate_loop_cleanup
- Phase: gas
- kg/h: 26583.820
- kmol/h: 452.537668
- Dominant Components: Ethylene glycol=25252.5, Water=565.2, Ethylene oxide=523.2, Trace heavy glycols=242.9

Row 2
- Unit: purification
- Service: Purification train
- Stream: S-401
- Local Role: recycle product
- Stream Role: recycle
- Section: purification
- Phase: gas
- kg/h: 1001.496
- kmol/h: 39.913416
- Dominant Components: Water=523.6, Ethylene oxide=477.9

Row 3
- Unit: purification
- Service: Purification train
- Stream: S-402
- Local Role: product
- Stream Role: product
- Section: purification
- Phase: gas
- kg/h: 24974.747
- kmol/h: 402.364225
- Dominant Components: Ethylene glycol=24974.7

Row 4
- Unit: purification
- Service: Purification train
- Stream: S-403
- Local Role: waste
- Stream Role: waste
- Section: purification
- Phase: mixed
- kg/h: 0.000
- kmol/h: 0.000000
- Dominant Components: -

Process-Unit Local Stream Split Summary

Table 61. Process-Unit Local Stream Split Summary

Columns: Split; Stream Count; Mass Flow (kg/h); Molar Flow (kmol/h)
Row 1: Split: fresh_feed | Stream Count: 0 | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000
Row 2: Split: recycle_feed | Stream Count: 1 | Mass Flow (kg/h): 26583.820 | Molar Flow (kmol/h): 452.537668
Row 3: Split: total_feed | Stream Count: 1 | Mass Flow (kg/h): 26583.820 | Molar Flow (kmol/h): 452.537668
Row 4: Split: product_effluent | Stream Count: 1 | Mass Flow (kg/h): 24974.747 | Molar Flow (kmol/h): 402.364225
Row 5: Split: recycle_effluent | Stream Count: 1 | Mass Flow (kg/h): 1001.496 | Molar Flow (kmol/h): 39.913416
Row 6: Split: side_draw_purge_vent | Stream Count: 1 | Mass Flow (kg/h): 0.000 | Molar Flow (kmol/h): 0.000000

Key Process-Unit Component Balance

Table 62. Key Process-Unit Component Balance

Columns: Unit; Service; Component; Inlet kg/h; Outlet kg/h; Delta kg/h

Row 1: Unit: purification | Service: Purification train | Component: Ethylene glycol | Inlet kg/h: 25252.525 | Outlet kg/h: 24974.747 | Delta kg/h: -277.778
Row 2: Unit: purification | Service: Purification train | Component: Water | Inlet kg/h: 565.238 | Outlet kg/h: 523.595 | Delta kg/h: -41.643
Row 3: Unit: purification | Service: Purification train | Component: Ethylene oxide | Inlet kg/h: 523.157 | Outlet kg/h: 477.901 | Delta kg/h: -45.256
Row 4: Unit: purification | Service: Purification train | Component: Trace heavy glycols | Inlet kg/h: 242.900 | Outlet kg/h: 0.000 | Delta kg/h: -242.900

Separation Design Inputs

Table 63. Separation Design Inputs

Columns: Parameter; Value
Row 1: Parameter: Service | Value: Distillation and purification train
Row 2: Parameter: Equilibrium model | Value: screening_vle
Row 3: Parameter: Equilibrium parameter ids | Value: none
Row 4: Parameter: Light key | Value: Water
Row 5: Parameter: Heavy key | Value: Ethylene Glycol
Row 6: Parameter: Relative volatility | Value: 19.315
Row 7: Parameter: Minimum stages | Value: 3.278
Row 8: Parameter: Theoretical stages | Value: 4.278
Row 9: Parameter: Design stages | Value: 7
Row 10: Parameter: Minimum reflux ratio | Value: 0.150
Row 11: Parameter: Operating reflux ratio | Value: 0.200

Section and Feed Basis

Table 64. Section and Feed Basis

Columns: Parameter; Value
Row 1: Parameter: Feed quality q-factor | Value: 1.106
Row 2: Parameter: Murphree efficiency | Value: 0.634
Row 3: Parameter: Top relative volatility | Value: 25.909
Row 4: Parameter: Bottom relative volatility | Value: 14.399
Row 5: Parameter: Rectifying theoretical stages | Value: 2.444
Row 6: Parameter: Stripping theoretical stages | Value: 1.833
Row 7: Parameter: Rectifying vapor load (kg/h) | Value: 6352.735
Row 8: Parameter: Stripping vapor load (kg/h) | Value: 5683.010
Row 9: Parameter: Rectifying liquid load (m3/h) | Value: 6090.003
Row 10: Parameter: Stripping liquid load (m3/h) | Value: 16500.036

Distillation Equation-Substitution Basis

Table 65. Distillation Equation-Substitution Basis

Columns: Check; Equation; Substitution; Result
Row 1: Check: Fenske minimum stages | Equation: Nmin = Fenske(alpha, LK/HK split) | Substitution: alpha=19.315; alpha_top=25.909; alpha_bottom=14.399 | Result: 3.278
Row 2: Check: Underwood minimum reflux | Equation: Rmin = Underwood(alpha, q, keys) | Substitution: alpha_top=25.909; alpha_bottom=14.399; q=1.106 | Result: 0.150
Row 3: Check: Gilliland operating stages | Equation: N = Gilliland(Nmin, R/Rmin) | Substitution: Nmin=3.278; R/Rmin=1.333 | Result: 4.278
Row 4: Check: Murphree tray conversion | Equation: Nactual = Ntheoretical / Em | Substitution: Ntheoretical=4.278; Em=0.634 | Result: 6.748 equivalent trays

Feed and Internal Flow Derivation

Table 66. Feed and Internal Flow Derivation

Columns: Check; Formula / Basis; Result
Row 1: Check: Minimum stages | Formula / Basis: Fenske screening | Result: 3.278
Row 2: Check: Minimum reflux | Formula / Basis: Underwood screening |

Result: 0.150
Row 3: Check: Operating stages | Formula / Basis: Gilliland screening | Result: 4.278
Row 4: Check: Feed condition | Formula / Basis: q-factor basis | Result: 1.106
Row 5: Check: Rectifying section split | Formula / Basis: Section stage allocation | Result: 2.444 stages
Row 6: Check: Stripping section split | Formula / Basis: Section stage allocation | Result: 1.833 stages
Row 7: Check: Rectifying internal vapor load | Formula / Basis: Top section vapor load | Result: 6352.735 kg/h
Row 8: Check: Stripping internal vapor load | Formula / Basis: Bottom section vapor load | Result: 5683.010 kg/h
Row 9: Check: Rectifying liquid load | Formula / Basis: Top section liquid load | Result: 6090.003 m3/h
Row 10: Check: Stripping liquid load | Formula / Basis: Bottom section liquid load | Result: 16500.036 m3/h

Feed Condition and Internal Flow Substitution Sheet

Table 67. Feed Condition and Internal Flow Substitution Sheet

Columns: Check; Equation; Substitution; Result
Row 1: Check: Feed quality basis | Equation: q = feed thermal condition parameter | Substitution: selected feed stage=4; q=1.106 | Result: 1.106
Row 2: Check: Rectifying section split | Equation: Nrect = f(feed stage, N) | Substitution: feed stage=4; N=7 | Result: 2.444 stages
Row 3: Check: Stripping section split | Equation: Nstrip = N - Nrect | Substitution: N=4.278; Nrect=2.444 | Result: 1.833 stages
Row 4: Check: Rectifying vapor load | Equation: Vrect from section screening | Substitution: R=0.200; Rmin=0.150 | Result: 6352.735 kg/h
Row 5: Check: Stripping vapor load | Equation: Vstrip from bottom section screening | Substitution: Nstrip=1.833; q=1.106 | Result: 5683.010 kg/h
Row 6: Check: Rectifying liquid load | Equation: Lrect from reflux / internal flow basis | Substitution: R=0.200 | Result: 6090.003 m3/h
Row 7: Check: Stripping liquid load | Equation: Lstrip from boilup / internal flow basis | Substitution: Reboiler duty=3179.805 kW | Result: 16500.036 m3/h

Column Operating Envelope

Table 68. Column Operating Envelope

Columns: Parameter; Value
Row 1: Parameter: Column diameter (m) | Value: 1.500
Row 2: Parameter: Column height (m) | Value: 12.000
Row 3: Parameter: Top temperature (C) | Value: 108.0
Row 4: Parameter: Bottom temperature (C) | Value: 203.3
Row 5: Parameter: Liquid density (kg/m3) | Value: 1.158
Row 6: Parameter: Vapor density (kg/m3) | Value: 6.005
Row 7: Parameter: Superficial vapor velocity (m/s) | Value: 0.188
Row 8: Parameter: Allowable vapor velocity (m/s) | Value: 0.350
Row 9: Parameter: Flooding fraction | Value: 0.538
Row 10: Parameter: Pressure drop per stage (kPa) | Value: 0.282

Reboiler and Condenser Package Basis

Table 69. Reboiler and Condenser Package Basis

Columns: Parameter; Value
Row 1: Parameter: Reboiler package type | Value: none
Row 2: Parameter: Reboiler medium | Value: none
Row 3: Parameter: Reboiler integrated duty (kW) | Value: 0.000
Row 4: Parameter: Reboiler LMTD (K) | Value: 0.000
Row 5: Parameter: Reboiler integrated area (m2) | Value: 0.000
Row 6: Parameter: Reboiler phase-change load (kg/h) | Value: 6187.729
Row 7: Parameter: Condenser package type | Value: none
Row 8: Parameter: Condenser recovery medium | Value: none
Row 9: Parameter: Condenser recovery duty (kW) | Value: 0.000
Row 10: Parameter: Condenser recovery LMTD (K) | Value: 0.000
Row 11: Parameter: Condenser recovery area (m2) | Value: 0.000

Row 12: Parameter: Condenser phase-change load (kg/h) | Value: 7049.642

Reboiler and Condenser Thermal Substitution Sheet

Table 70. Reboiler and Condenser Thermal Substitution Sheet

Columns: Check; Equation; Substitution; Result
Row 1: Check: Operating reflux multiple | Equation: R/Rmin = R / Rmin | Substitution: R=0.200; Rmin=0.150 | Result: 1.333
Row 2: Check: Integrated reboiler area | Equation: A = Q / (U * LMTD) | Substitution: Q=0.0 W; U=0.0 W/m2-K; LMTD=0.000 K | Result: 0.000 m2
Row 3: Check: Reboiler phase-change basis | Equation: m = Q / lambda | Substitution: Q=3179.805 kW; lambda~=1850.0 kJ/kg | Result: 6187.729 kg/h
Row 4: Check: Condenser recovery area | Equation: A = Q / (U * LMTD) | Substitution: Q=0.0 W; U=0.0 W/m2-K; LMTD=0.000 K | Result: 0.000 m2
Row 5: Check: Condenser phase-change basis | Equation: m = Q / lambda | Substitution: Q=0.000 kW; lambda~=0.0 kJ/kg | Result: 7049.642 kg/h

Process-Unit Sizing Basis

Table 71. Process-Unit Sizing Basis

Columns: Parameter; Value
Row 1: Parameter: Service | Value: Distillation and purification train
Row 2: Parameter: Light key | Value: Water
Row 3: Parameter: Heavy key | Value: Ethylene Glycol
Row 4: Parameter: Relative volatility | Value: 19.315
Row 5: Parameter: Minimum stages | Value: 3.278
Row 6: Parameter: Theoretical stages | Value: 4.278
Row 7: Parameter: Design stages | Value: 7
Row 8: Parameter: Tray efficiency | Value: 0.660
Row 9: Parameter: Minimum reflux ratio | Value: 0.150
Row 10: Parameter: Reflux ratio | Value: 0.200
Row 11: Parameter: R / Rmin | Value: 1.333
Row 12: Parameter: Diameter (m) | Value: 1.500
Row 13: Parameter: Height (m) | Value: 12.000
Row 14: Parameter: Feed stage | Value: 4

Hydraulics Basis

Table 72. Hydraulics Basis

Columns: Parameter; Value
Row 1: Parameter: Tray spacing (m) | Value: 0.450
Row 2: Parameter: Flooding fraction | Value: 0.538
Row 3: Parameter: Downcomer area fraction | Value: 0.140
Row 4: Parameter: Vapor velocity (m/s) | Value: 0.188
Row 5: Parameter: Allowable vapor velocity (m/s) | Value: 0.350
Row 6: Parameter: Capacity factor (m/s) | Value: 350000.000
Row 7: Parameter: Active area (m2) | Value: 1.520
Row 8: Parameter: Liquid load (m3/h) | Value: 6090.003
Row 9: Parameter: Vapor load (kg/h) | Value: 6187.729
Row 10: Parameter: Liquid density (kg/m3) | Value: 1.158
Row 11: Parameter: Vapor density (kg/m3) | Value: 6.005
Row 12: Parameter: Pressure drop per stage (kPa) | Value: 0.282
Row 13: Parameter: Top temperature (C) | Value: 108.0
Row 14: Parameter: Bottom temperature (C) | Value: 203.3

Heat-Transfer Package Inputs

Table 73. Heat-Transfer Package Inputs

Columns: Parameter; Value
Row 1: Parameter: Heat load kw | Value: 34024.794
Row 2: Parameter: Lmtd k | Value: 8.000
Row 3: Parameter: U w m2 k | Value: 470.000
Row 4: Parameter: Package family | Value: process_exchange
Row 5: Parameter: Circulation flow m3 hr | Value: 1818.329
Row 6: Parameter: Phase change load kg hr | Value: 0.000
Row 7: Parameter: Package holdup m3 | Value: 542.949

Row 8: Parameter: Utility topology | Value: shared HTM island network (shared_htm) with 1 utility islands
Row 9: Parameter: Utility architecture family | Value: shared_htm
Row 10: Parameter: Selected train step id | Value: omega_catalytic_pinch_htm__shared_htm_step_01
Row 11: Parameter: Selected island id | Value: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Row 12: Parameter: Selected header level | Value: 1
Row 13: Parameter: Selected cluster id | Value: none
Row 14: Parameter: Allocated recovered duty target kw | Value: 42530.992
Row 15: Parameter: Boiling side coefficient w m2 k | Value: 1200.000
Row 16: Parameter: Condensing side coefficient w m2 k | Value: 1200.000

Exchanger Package Selection Basis

Table 74. Exchanger Package Selection Basis

Columns: Parameter; Value
Row 1: Parameter: Package family | Value: process_exchange
Row 2: Parameter: Selected package roles | Value: circulation, controls, exchanger, expansion, header, relief
Row 3: Parameter: Selected package items | Value: omega_catalytic_pinch_htm__shared_htm_step_01_exchanger, omega_catalytic_pinch_htm__shared_htm_step_01_controls, omega_catalytic_pinch_htm__shared_htm_step_01_header, omega_catalytic_pinch_htm__shared_htm_step_01_circulation, omega_catalytic_pinch_htm__shared_htm_step_01_expansion, omega_catalytic_pinch_htm__shared_htm_step_01_relief

Utility Coupling

Table 75. Utility Coupling

Columns: Parameter; Value
Row 1: Parameter: Utility topology | Value: shared HTM island network (shared_htm) with 1 utility islands
Row 2: Parameter: Architecture family | Value: shared_htm
Row 3: Parameter: Integrated reboiler duty (kW) | Value: 0.000
Row 4: Parameter: Allocated reboiler target (kW) | Value: 0.000
Row 5: Parameter: Residual reboiler utility (kW) | Value: 3179.805
Row 6: Parameter: Integrated reboiler LMTD (K) | Value: 0.000
Row 7: Parameter: Integrated reboiler area (m2) | Value: 0.000
Row 8: Parameter: Reboiler medium | Value: none
Row 9: Parameter: Reboiler package type | Value: none
Row 10: Parameter: Reboiler circulation ratio | Value: 3.000
Row 11: Parameter: Reboiler phase-change load (kg/h) | Value: 6187.729
Row 12: Parameter: Reboiler package items | Value: none
Row 13: Parameter: Condenser recovery duty (kW) | Value: 0.000
Row 14: Parameter: Allocated condenser target (kW) | Value: 0.000
Row 15: Parameter: Condenser recovery LMTD (K) | Value: 0.000
Row 16: Parameter: Condenser recovery area (m2) | Value: 0.000
Row 17: Parameter: Condenser recovery medium | Value: none
Row 18: Parameter: Condenser package type | Value: none
Row 19: Parameter: Condenser phase-change load (kg/h) | Value: 7049.642
Row 20: Parameter: Condenser circulation flow (m3/h) | Value: 6090.003
Row 21: Parameter: Condenser package items | Value: none
Row 22: Parameter: Selected utility islands | Value: none
Row 23: Parameter: Selected header levels | Value: none
Row 24: Parameter: Selected cluster ids | Value: none
Row 25: Parameter: Selected train steps | Value: none

Heat-Exchanger Thermal Basis

Table 76. Heat-Exchanger Thermal Basis

Columns: Parameter; Value
Row 1: Parameter: Configuration | Value: Shell And Tube
Row 2: Parameter: Heat load (kW) | Value: 34024.794
Row 3: Parameter: LMTD (K) | Value: 8.0
Row 4: Parameter: Overall U (W/m2-K) | Value: 470.0
Row 5: Parameter: Area (m2) | Value: 9049.147

Row 6: Parameter: Package family | Value: process_exchange
Row 7: Parameter: Architecture family | Value: shared_htm
Row 8: Parameter: Selected train step | Value: omega_catalytic_pinch_htm__shared_htm_step_01
Row 9: Parameter: Selected utility island | Value: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Row 10: Parameter: Selected header level | Value: 1
Row 11: Parameter: Selected cluster id | Value: none
Row 12: Parameter: Allocated island target (kW) | Value: 42530.992
Row 13: Parameter: Package roles | Value: circulation, controls, exchanger, expansion, header, relief
Row 14: Parameter: Selected package items | Value: omega_catalytic_pinch_htm__shared_htm_step_01_exchanger, omega_catalytic_pinch_htm__shared_htm_step_01_controls, omega_catalytic_pinch_htm__shared_htm_step_01_header, omega_catalytic_pinch_htm__shared_htm_step_01_circulation, omega_catalytic_pinch_htm__shared_htm_step_01_expansion, omega_catalytic_pinch_htm__shared_htm_step_01_relief
Row 15: Parameter: Boiling-side coefficient (W/m2-K) | Value: 1200.000
Row 16: Parameter: Condensing-side coefficient (W/m2-K) | Value: 1200.000

Process-Unit Calculation Traces

Table 77. Process-Unit Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Process-unit family | Formula: service = selected separation family | Inputs: - | Result: Distillation and purification train | Notes: -
Row 2: Trace: Separation thermodynamics basis | Formula: alpha = sqrt(alpha_top * alpha_bottom) from component K-values when a VLE basis exists | Inputs: light_key=Water; heavy_key=Ethylene Glycol; alpha_top=25.9091; alpha_bottom=14.3988; method=ideal_raoult_missing_bip_fallback | Result: 19.3148 - | Notes: Column volatility basis now prefers the separation-thermodynamics artifact built from Antoine / Clausius-Clapeyron K-values when available.
Row 3: Trace: Minimum theoretical stages | Formula: Nmin = log[(xD,LK/xB,LK)*(xB,HK/xD,HK)] / log(alpha) | Inputs: xD,LK=0.9850; xB,LK=0.0040; xD,HK=0.0150; xB,HK=0.9990; alpha=19.315 | Result: 3.278 stages | Notes: -
Row 4: Trace: Minimum reflux proxy | Formula: Rmin = f(alpha, q, xD,LK) | Inputs: alpha=19.315; q=1.106; xD,LK=0.9850 | Result: 0.150 - | Notes: This is a screening Underwood-style reflux estimate built from the solved property basis and separation severity.
Row 5: Trace: Feed and volatility section basis | Formula: q from Cp and DeltaT; section volatility anchored to top/bottom alpha estimates | Inputs: q=1.106; alpha_top=25.909; alpha_bottom=14.399; murphree_efficiency=0.634 | Result: 1.106 - | Notes: Stage 2 separation design keeps the feed-quality and section-volatility basis explicit instead of burying it inside the reflux proxy.
Row 6: Trace: Actual stage proxy | Formula: Nactual = g(Nmin, R/Rmin, tray efficiency) | Inputs: Nmin=3.278; R=0.200; Rmin=0.150; tray_eff=0.660 | Result: 7.000 actual stages | Notes: -
Row 7: Trace: Rectifying and stripping section basis | Formula: Nrect + Nstrip = Ntheoretical ; section loads derived from reflux and bottoms circulation | Inputs: feed_stage=4; Ntheoretical=4.278; Nrect=2.444; Nstrip=1.833; Vrect=6352.735; Vstrip=5683.010 | Result: 4.278 stages | Notes: This makes the section-level column basis explicit before deeper M9-style tray-by-tray work.
Row 8: Trace: Equivalent diameter | Formula: D = sqrt(4*Aactive/[pi*(1-Adc)]) | Inputs: Aactive=1.105; Adc=0.140 | Result: 1.500 m | Notes: -
Row 9: Trace: Column hydraulic capacity basis | Formula: uallow = C * sqrt[(rhoL-rhoV)/rhoV]; uflood = usuperficial/uallow | Inputs: C=0.300; rhoL=1.158; rhoV=6.005; usuperficial=0.188 | Result: 0.538 - | Notes: -
Row 10: Trace: Process-unit property-package basis | Formula: Hydraulics proxy uses density, viscosity, and Cp from the mixture-property package | Inputs: density=1.158; viscosity=0.014479; Cp=2.452; mixture_package=purification_mixture_properties | Result: 1.158 kg/m3 | Notes: -
Row 11: Trace: Solved process-unit packet basis | Formula: packet basis -> outlet throughput and matched thermal packets | Inputs: process_packet=purification_unit_packet; heating_packet=fallback;

cooling_packet=fallback; composition_state=purification_composition_state | Result: 3179.805 kW | Notes: Process-unit sizing now prefers solved separation packet, composition state, and thermal packet before route-level utility heuristics.
Row 12: Trace: Process-unit utility-train coupling | Formula: Integrated reboiler duty = sum(selected train steps tied to process-unit cold sinks | Inputs: heating_steps=none; condenser_steps=none; topology=shared HTM island network (shared_htm) with 1 utility islands; architecture_family=shared_htm; selected_islands=none; header_levels=none | Result: 0.000 kW | Notes: This captures recovered duty delivered into the reboiler/process unit and heat recovered from condenser-side streams.
Row 13: Trace: Reboiler package basis | Formula: Vboil = Qreb / lambda; circulation from selected train package when available | Inputs: package=fallback; package_type=generic; phase_change_load_kg_hr=6187.729; circulation_ratio=3.000; selected_islands=none; target_recovered_kw=0.000; cluster_ids=none | Result: 3179.805 kW | Notes: -
Row 14: Trace: Condenser package basis | Formula: mcond = Qcond / lambda; circulation from selected train package when available | Inputs: package=fallback; package_type=generic; phase_change_load_kg_hr=7049.642; circulation_flow_m3_hr=6090.003; selected_islands=none; target_recovered_kw=0.000; cluster_ids=none | Result: 4112.291 kW | Notes: -
Row 15: Trace: Integrated reboiler heat-transfer basis | Formula: A_reb,int = Q_reb,int / (U_reb * LMTD_reb,int) | Inputs: Q_reb,int=0.0; U_reb=780.0; LMTD_reb,int=0.00 | Result: 0.000 m2 | Notes: Integrated reboiler area is derived from selected train steps that feed the process-unit heating sink.
Row 16: Trace: Condenser-side recovery basis | Formula: A_cond,rec = Q_cond,rec / (U_cond * LMTD_cond,rec) | Inputs: Q_cond,rec=0.0; U_cond=700.0; LMTD_cond,rec=0.00 | Result: 0.000 m2 | Notes: Recovered condenser duty is tied to selected train steps sourced from the process-unit hot side.
Row 17: Trace: Solved equilibrium packet basis | Formula: selected separation packets provide the active phase-equilibrium model and split basis for this process-unit family | Inputs: packet_ids=purification_separation_packet; models=heuristic; parameter_ids=none; fallback=no | Result: heuristic_split | Notes: This basis is consumed directly from the solved separator packets rather than being inferred from chapter prose.
Row 18: Trace: Exchanger area | Formula: A = Q/(U*dTlm) | Inputs: Q=34024794.0; U=470.0; dTlm=8.0 | Result: 9049.147 m2 | Notes: -
Row 19: Trace: Solved exchanger packet basis | Formula: packet basis -> selected thermal packet | Inputs: thermal_packet=fallback; selected_train_step=omega_catalytic_pinch_htm__shared_htm_step_01; architecture_family=shared_htm; selected_island=omega_catalytic_pinch_htm__shared_htm_shared_htm_01 | Result: 34024.794 kW | Notes: Exchanger sizing now prefers the selected utility-train step and solved thermal packet before aggregate-duty fallback.
Row 20: Trace: Utility package exchanger basis | Formula: Selected exchanger package -> family, area, LMTD, circulation, phase load | Inputs: package_family=process_exchange; exchanger_package=omega_catalytic_pinch_htm__shared_htm_step_01_exchanger; circulation_flow_m3_hr=1818.329; phase_change_load_kg_hr=0.000; header_level=1; cluster_id=none; target_recovered_kw=42530.992 | Result: 542.949 m3 | Notes: Reboiler/condenser package sizing now reads the selected utility package item before applying generic exchanger fallback.

16. Equipment Design and Sizing

Equipment Design and Sizing

Storage Decision

Vertical atmospheric tank farm selected as the highest-ranked alternative.

Material of Construction Decision

Carbon steel construction selected as the highest-ranked alternative.

Storage Basis

Table 78. Storage Basis

Columns: Parameter; Value
Row 1: Parameter: Inventory days | Value: 14.1
Row 2: Parameter: Dispatch buffer days | Value: 9.4
Row 3: Parameter: Operating stock days | Value: 2.5
Row 4: Parameter: Restart buffer days | Value: 1.5
Row 5: Parameter: Turnaround buffer factor | Value: 1.050
Row 6: Parameter: Working volume (m3) | Value: 7661.521
Row 7: Parameter: Total volume (m3) | Value: 8580.903
Row 8: Parameter: Tank diameter (m) | Value: 13.979
Row 9: Parameter: Straight-side height (m) | Value: 55.914

Operations Planning Basis

Table 79. Operations Planning Basis

Columns: Parameter; Value
Row 1: Parameter: Operating mode | Value: continuous
Row 2: Parameter: Service family | Value: continuous_liquid_purification
Row 3: Parameter: Raw-material buffer (d) | Value: 19.6
Row 4: Parameter: Finished-goods buffer (d) | Value: 9.4
Row 5: Parameter: Operating stock (d) | Value: 2.5
Row 6: Parameter: Restart buffer (d) | Value: 1.5
Row 7: Parameter: Startup ramp (d) | Value: 2.0
Row 8: Parameter: Campaign length (d) | Value: 170.0
Row 9: Parameter: Annual restart loss (kg/y) | Value: 16,426.7

Storage Transfer Pump Basis

Table 80. Storage Transfer Pump Basis

Columns: Parameter; Value
Row 1: Parameter: Pump id | Value: TK-301_pump
Row 2: Parameter: Service | Value: Ethylene Glycol storage via vertical tank farm transfer
Row 3: Parameter: Flow (m3/h) | Value: 957.690
Row 4: Parameter: Differential head (m) | Value: 52.0
Row 5: Parameter: Power (kW) | Value: 159.360
Row 6: Parameter: NPSH margin (m) | Value: 2.500

Storage and Inventory Vessel Basis

Table 81. Storage and Inventory Vessel Basis

Columns: ID; Type; Service; Volume (m3); Design T (C); Design P (bar); MoC; Design Basis
Row 1
ID: V-101
Type: Flash drum
Service: Intermediate disengagement
Volume (m3): 4860.750

Design T (C): 85.0
Design P (bar): 3.00
MoC: Carbon Steel
Design Basis: Generic separator hold-up
Row 2
ID: TK-301
Type: Storage tank
Service: Ethylene Glycol storage via vertical tank farm
Volume (m3): 8580.903
Design T (C): 45.0
Design P (bar): 1.20
MoC: SS304
Design Basis: 14.1 days inventory
Row 3
ID: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Volume (m3): 2.835
Design T (C): 210.0
Design P (bar): 6.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Major Process Equipment Basis

Table 82. Major Process Equipment Basis

Columns: ID; Type; Service; Volume (m3); Duty (kW); Design T (C); Design P (bar); MoC
Row 1
ID: R-101
Type: Reactor
Service: Tubular Plug Flow Hydrator
Volume (m3): 27004.165
Duty (kW): 7600.213
Design T (C): 255.0
Design P (bar): 20.00
MoC: Carbon Steel
Row 2
ID: PU-201
Type: Distillation column
Service: Distillation and purification train
Volume (m3): 21.206
Duty (kW): 3179.805
Design T (C): 140.0
Design P (bar): 2.00
MoC: Carbon Steel
Row 3
ID: E-101
Type: Heat exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 723.932
Duty (kW): 34024.794
Design T (C): 180.0
Design P (bar): 8.00
MoC: Carbon Steel
Row 4
ID: TK-301
Type: Storage tank
Service: Ethylene Glycol storage via vertical tank farm
Volume (m3): 8580.903
Duty (kW): 0.000
Design T (C): 45.0
Design P (bar): 1.20
MoC: SS304

Heat Exchanger and Thermal-Service Basis

Table 83. Heat Exchanger and Thermal-Service Basis

Columns: ID; Type; Service; Duty (kW); Design T (C); Design P (bar); MoC; Design Basis
Row 1
ID: E-101
Type: Heat exchanger
Service: R-rough to D-rough heat recovery
Duty (kW): 34024.794
Design T (C): 180.0
Design P (bar): 8.00
MoC: Carbon Steel
Design Basis: LMTD 8.0 K
Row 2
ID: HX-01
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Duty (kW): 34024.794
Design T (C): 213.0
Design P (bar): 11.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 3
ID: HX-01-CTRL
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Duty (kW): 0.000
Design T (C): 203.0
Design P (bar): 9.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 4
ID: HX-01-HDR
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Duty (kW): 34024.794
Design T (C): 205.0
Design P (bar): 14.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 5
ID: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Duty (kW): 15.466
Design T (C): 205.0
Design P (bar): 14.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 6
ID: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Duty (kW): 0.000
Design T (C): 210.0
Design P (bar): 6.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 7

ID: HX-01-RV
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Duty (kW): 0.000
Design T (C): 213.0
Design P (bar): 13.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 8
ID: HX-02
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Duty (kW): 8506.198
Design T (C): 162.0
Design P (bar): 6.50
MoC: SS316
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 9
ID: HX-02-CTRL
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Duty (kW): 0.000
Design T (C): 158.0
Design P (bar): 5.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Rotating and Auxiliary Package Basis

Table 84. Rotating and Auxiliary Package Basis

Columns: ID; Type; Service; Flow/Volume Basis; Head/Pressure Basis; Power/Duty (kW); NPSH Margin; Design Basis
Row 1
ID: TK-301_pump
Type: Transfer pump
Service: Ethylene Glycol storage via vertical tank farm transfer
Flow/Volume Basis: 957.690
Head/Pressure Basis: 52.000
Power/Duty (kW): 159.360
NPSH Margin: 2.500
Design Basis: Dedicated pump design artifact
Row 2
ID: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Flow/Volume Basis: 0.250
Head/Pressure Basis: 14.50
Power/Duty (kW): 15.466
NPSH Margin: n/a
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Utility-Coupled Package Inventory

Table 85. Utility-Coupled Package Inventory

Columns: ID; Type; Service; Volume (m3); Duty (kW); MoC; Design Basis
Row 1
ID: V-101
Type: Flash drum
Service: Intermediate disengagement

Volume (m3): 4860.750
Duty (kW): 0.000
MoC: Carbon Steel
Design Basis: Generic separator hold-up
Row 2
ID: HX-01
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 542.949
Duty (kW): 34024.794
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 3
ID: HX-01-CTRL
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Duty (kW): 0.000
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 4
ID: HX-01-HDR
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Volume (m3): 0.851
Duty (kW): 34024.794
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 5
ID: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Volume (m3): 0.250
Duty (kW): 15.466
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 6
ID: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Volume (m3): 2.835
Duty (kW): 0.000
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 7
ID: HX-01-RV
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Volume (m3): 1.309
Duty (kW): 0.000
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 8
ID: HX-02
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Volume (m3): 30.379

Duty (kW): 8506.198
MoC: SS316
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 9
ID: HX-02-CTRL
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Duty (kW): 0.000
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Datasheet Coverage Matrix

Table 86. Datasheet Coverage Matrix

Columns: Equipment Family; Datasheet Count; Representative IDs
Row 1: Equipment Family: Distillation column | Datasheet Count: 1 | Representative IDs: PU-201
Row 2: Equipment Family: Flash drum | Datasheet Count: 1 | Representative IDs: V-101
Row 3: Equipment Family: HTM circulation skid | Datasheet Count: 1 | Representative IDs: HX-01-PMP
Row 4: Equipment Family: HTM expansion tank | Datasheet Count: 1 | Representative IDs: HX-01-EXP
Row 5: Equipment Family: HTM loop exchanger | Datasheet Count: 1 | Representative IDs: HX-01
Row 6: Equipment Family: HTM relief package | Datasheet Count: 1 | Representative IDs: HX-01-RV
Row 7: Equipment Family: Heat exchanger | Datasheet Count: 1 | Representative IDs: E-101
Row 8: Equipment Family: Heat integration exchanger | Datasheet Count: 1 | Representative IDs: HX-02
Row 9: Equipment Family: Reactor | Datasheet Count: 1 | Representative IDs: R-101
Row 10: Equipment Family: Shared HTM header manifold | Datasheet Count: 1 | Representative IDs: HX-01-HDR
Row 11: Equipment Family: Storage tank | Datasheet Count: 1 | Representative IDs: TK-301
Row 12: Equipment Family: Transfer pump | Datasheet Count: 1 | Representative IDs: TK-301_pump
Row 13: Equipment Family: Utility control package | Datasheet Count: 2 | Representative IDs: HX-01-CTRL, HX-02-CTRL

Equipment-by-Equipment Sizing Summary

Table 87. Equipment-by-Equipment Sizing Summary

Columns: ID; Type; Service; Volume (m3); Duty (kW); Design T (C); Design P (bar); MoC; Design Basis
Row 1
ID: R-101
Type: Reactor
Service: Tubular Plug Flow Hydrator
Volume (m3): 27004.165
Duty (kW): 7600.213
Design T (C): 255.0
Design P (bar): 20.00
MoC: Carbon Steel
Design Basis: tubular_plug_flow_hydrator selected at 1.00 h residence time and 0.960 conversion basis.
Row 2
ID: PU-201
Type: Distillation column
Service: Distillation and purification train
Volume (m3): 21.206

Duty (kW): 3179.805
Design T (C): 140.0
Design P (bar): 2.00
MoC: Carbon Steel
Design Basis: 7 stages equivalent
Row 3
ID: V-101
Type: Flash drum
Service: Intermediate disengagement
Volume (m3): 4860.750
Duty (kW): 0.000
Design T (C): 85.0
Design P (bar): 3.00
MoC: Carbon Steel
Design Basis: Generic separator hold-up
Row 4
ID: E-101
Type: Heat exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 723.932
Duty (kW): 34024.794
Design T (C): 180.0
Design P (bar): 8.00
MoC: Carbon Steel
Design Basis: LMTD 8.0 K
Row 5
ID: TK-301
Type: Storage tank
Service: Ethylene Glycol storage via vertical tank farm
Volume (m3): 8580.903
Duty (kW): 0.000
Design T (C): 45.0
Design P (bar): 1.20
MoC: SS304
Design Basis: 14.1 days inventory
Row 6
ID: HX-01
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 542.949
Duty (kW): 34024.794
Design T (C): 213.0
Design P (bar): 11.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 7
ID: HX-01-CTRL
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Duty (kW): 0.000
Design T (C): 203.0
Design P (bar): 9.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 8
ID: HX-01-HDR
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Volume (m3): 0.851
Duty (kW): 34024.794
Design T (C): 205.0
Design P (bar): 14.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Row 9
ID: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Volume (m3): 0.250
Duty (kW): 15.466
Design T (C): 205.0
Design P (bar): 14.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 10
ID: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Volume (m3): 2.835
Duty (kW): 0.000
Design T (C): 210.0
Design P (bar): 6.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 11
ID: HX-01-RV
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Volume (m3): 1.309
Duty (kW): 0.000
Design T (C): 213.0
Design P (bar): 13.50
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 12
ID: HX-02
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Volume (m3): 30.379
Duty (kW): 8506.198
Design T (C): 162.0
Design P (bar): 6.50
MoC: SS316
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -
Row 13
ID: HX-02-CTRL
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Duty (kW): 0.000
Design T (C): 158.0
Design P (bar): 5.00
MoC: Carbon steel
Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_02; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

17. Mechanical Design and MoC

Mechanical Design and MoC

Table 88. Mechanical Design and MoC

Columns: Equipment; Type; Shell t (mm); Head t (mm); Class; Hydrotest (bar); Nozzle (mm); Support; Load Case; Support t (mm); Load (kN); Thermal Growth (mm); Reinforcement (mm2); Platform

Row 1
- Equipment: R-101
- Type: Reactor
- Shell t (mm): 157.13
- Head t (mm): 144.56
- Class: ASME 600
- Hydrotest (bar): 26.00
- Nozzle (mm): 9336.1
- Support: skirt support
- Load Case: operating + wind + seismic + nozzle loads
- Support t (mm): 109.99
- Load (kN): 259711.84
- Thermal Growth (mm): 148.17
- Reinforcement (mm2): 1941902.0
- Platform: yes

Row 2
- Equipment: PU-201
- Type: Distillation column
- Shell t (mm): 4.41
- Head t (mm): 4.06
- Class: PN16
- Hydrotest (bar): 3.00
- Nozzle (mm): 640.2
- Support: skirt support
- Load Case: operating + wind + seismic + nozzle loads
- Support t (mm): 24.00
- Load (kN): 250.03
- Thermal Growth (mm): 6.83
- Reinforcement (mm2): 13316.9
- Platform: yes

Row 3
- Equipment: V-101
- Type: Flash drum
- Shell t (mm): 15.94
- Head t (mm): 14.66
- Class: PN16
- Hydrotest (bar): 4.00
- Nozzle (mm): 3917.0
- Support: base support
- Load Case: operating + nozzle loads
- Support t (mm): 24.00
- Load (kN): 46738.28
- Thermal Growth (mm): 21.82
- Reinforcement (mm2): 65806.4
- Platform: no

Row 4
- Equipment: E-101
- Type: Heat exchanger
- Shell t (mm): 21.34
- Head t (mm): 19.63
- Class: ASME 150
- Hydrotest (bar): 10.40
- Nozzle (mm): 2021.0
- Support: saddle support
- Load Case: operating + nozzle loads
- Support t (mm): 24.00
- Load (kN): 7376.03
- Thermal Growth (mm): 29.89
- Reinforcement (mm2): 90539.7
- Platform: yes

Row 5
- Equipment: TK-301

Type: Storage tank
Shell t (mm): 9.19
Head t (mm): 8.45
Class: PN16
Hydrotest (bar): 2.20
Nozzle (mm): 5177.3
Support: saddle / leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 75768.79
Thermal Growth (mm): 12.46
Reinforcement (mm2): 34791.6
Platform: no

Row 6
Equipment: HX-01
Type: HTM loop exchanger
Shell t (mm): 25.95
Head t (mm): 23.87
Class: ASME 300
Hydrotest (bar): 14.30
Nozzle (mm): 1822.7
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 5636.10
Thermal Growth (mm): 32.93
Reinforcement (mm2): 152379.0
Platform: yes

Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Hydrotest (bar): 11.70
Nozzle (mm): 25.0
Support: instrument frame
Load Case: skid dead load + piping reaction
Support t (mm): 12.00
Load (kN): 6.00
Thermal Growth (mm): 5.77
Reinforcement (mm2): 1260.0
Platform: no

Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Hydrotest (bar): 18.85
Nozzle (mm): 417.0
Support: pipe rack support
Load Case: rack piping + thermal expansion
Support t (mm): 24.00
Load (kN): 594.50
Thermal Growth (mm): 5.83
Reinforcement (mm2): 45956.1
Platform: no

Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Hydrotest (bar): 18.85
Nozzle (mm): 117.3
Support: base frame
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 10.59
Thermal Growth (mm): 5.83
Reinforcement (mm2): 12930.0
Platform: no

Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Shell t (mm): 5.31
Head t (mm): 4.88
Class: ASME 150
Hydrotest (bar): 7.80
Nozzle (mm): 175.1
Support: leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 33.03
Thermal Growth (mm): 5.99
Reinforcement (mm2): 7984.8
Platform: no
Row 11
Equipment: HX-01-RV
Type: HTM relief package
Shell t (mm): 8.21
Head t (mm): 7.56
Class: ASME 300
Hydrotest (bar): 17.55
Nozzle (mm): 145.4
Support: base support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 18.91
Thermal Growth (mm): 6.09
Reinforcement (mm2): 14920.7
Platform: no
Row 12
Equipment: HX-02
Type: Heat integration exchanger
Shell t (mm): 8.95
Head t (mm): 8.24
Class: ASME 150
Hydrotest (bar): 8.45
Nozzle (mm): 613.7
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 402.13
Thermal Growth (mm): 13.00
Reinforcement (mm2): 22339.3
Platform: yes
Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Hydrotest (bar): 6.50
Nozzle (mm): 25.0
Support: instrument frame
Load Case: skid dead load + piping reaction
Support t (mm): 12.00
Load (kN): 6.00
Thermal Growth (mm): 4.31
Reinforcement (mm2): 700.0
Platform: no

Mechanical Basis

Table 89. Mechanical Basis

Columns: Field; Value
Row 1: Field: Code basis | Value: ASME-style screening logic
Row 2: Field: Design pressure basis | Value: Selected equipment design pressure with feasibility-study margin
Row 3: Field: Design temperature basis | Value: Selected equipment design temperature with preliminary utility envelope
Row 4: Field: Support design basis | Value: Support screening includes

vertical, piping, wind, seismic, and thermal-growth load components by equipment family.
Row 5: Field: Load case basis | Value: Tall process equipment uses operating + wind + seismic + nozzle loads; headers use rack piping + thermal expansion; skids use dead load + piping reaction.
Row 6: Field: Foundation basis | Value: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
Row 7: Field: Nozzle load basis | Value: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.
Row 8: Field: Connection rating basis | Value: Nozzle and header connection classes follow design-pressure screening via ASME/PN envelope mapping.
Row 9: Field: Access platform basis | Value: Columns, absorbers, reactors, and exchanger services above grade include screening access-platform area allowances.

Mechanical Design Input Matrix

Table 90. Mechanical Design Input Matrix

Columns: Equipment; Type; Pdesign (bar); Tdesign (C); Allowable Stress (MPa); Joint Efficiency; CA (mm); Pressure Class

Row 1
- Equipment: R-101
- Type: Reactor
- Pdesign (bar): 20.00
- Tdesign (C): 255.0
- Allowable Stress (MPa): 138.0
- Joint Efficiency: 0.85
- CA (mm): 3.00
- Pressure Class: ASME 600

Row 2
- Equipment: PU-201
- Type: Distillation column
- Pdesign (bar): 2.00
- Tdesign (C): 140.0
- Allowable Stress (MPa): 138.0
- Joint Efficiency: 0.85
- CA (mm): 3.00
- Pressure Class: PN16

Row 3
- Equipment: V-101
- Type: Flash drum
- Pdesign (bar): 3.00
- Tdesign (C): 85.0
- Allowable Stress (MPa): 138.0
- Joint Efficiency: 0.85
- CA (mm): 3.00
- Pressure Class: PN16

Row 4
- Equipment: E-101
- Type: Heat exchanger
- Pdesign (bar): 8.00
- Tdesign (C): 180.0
- Allowable Stress (MPa): 138.0
- Joint Efficiency: 0.85
- CA (mm): 3.00
- Pressure Class: ASME 150

Row 5
- Equipment: TK-301
- Type: Storage tank
- Pdesign (bar): 1.20
- Tdesign (C): 45.0
- Allowable Stress (MPa): 120.0
- Joint Efficiency: 0.85
- CA (mm): 2.00
- Pressure Class: PN16

Row 6
- Equipment: HX-01
- Type: HTM loop exchanger

Pdesign (bar): 11.00
Tdesign (C): 213.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 300

Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Pdesign (bar): 9.00
Tdesign (C): 203.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 150

Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Pdesign (bar): 14.50
Tdesign (C): 205.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 300

Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Pdesign (bar): 14.50
Tdesign (C): 205.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 300

Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Pdesign (bar): 6.00
Tdesign (C): 210.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 150

Row 11
Equipment: HX-01-RV
Type: HTM relief package
Pdesign (bar): 13.50
Tdesign (C): 213.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 300

Row 12
Equipment: HX-02
Type: Heat integration exchanger
Pdesign (bar): 6.50
Tdesign (C): 162.0
Allowable Stress (MPa): 120.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 150

Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Pdesign (bar): 5.00
Tdesign (C): 158.0
Allowable Stress (MPa): 138.0
Joint Efficiency: 0.85
CA (mm): 3.00
Pressure Class: ASME 150

Table 91. Mechanical Design Input Matrix

Columns: Equipment; Load Case; Support Load (kN); Wind (kN); Seismic (kN); Piping (kN); Anchor Bolt (mm); Base Plate (mm); Platform; Platform Area (m2)

Row 1
- Equipment: R-101
- Load Case: operating + wind + seismic + nozzle loads
- Support Load (kN): 328630.148
- Wind (kN): 1152.822
- Seismic (kN): 46748.132
- Piping (kN): 240.402
- Anchor Bolt (mm): 10269.688
- Base Plate (mm): 185.721
- Platform: yes
- Platform Area (m2): 61.737

Row 2
- Equipment: PU-201
- Load Case: operating + wind + seismic + nozzle loads
- Support Load (kN): 341.554
- Wind (kN): 9.813
- Seismic (kN): 45.005
- Piping (kN): 16.706
- Anchor Bolt (mm): 149.391
- Base Plate (mm): 21.185
- Platform: yes
- Platform Area (m2): 5.696

Row 3
- Equipment: V-101
- Load Case: operating + nozzle loads
- Support Load (kN): 55418.590
- Wind (kN): 168.445
- Seismic (kN): 4673.828
- Piping (kN): 98.976
- Anchor Bolt (mm): 2247.216
- Base Plate (mm): 33.876
- Platform: no
- Platform Area (m2): 0.000

Row 4
- Equipment: E-101
- Load Case: operating + nozzle loads
- Support Load (kN): 8804.373
- Wind (kN): 47.328
- Seismic (kN): 737.603
- Piping (kN): 53.324
- Anchor Bolt (mm): 664.724
- Base Plate (mm): 24.186
- Platform: yes
- Platform Area (m2): 3.000

Row 5
- Equipment: TK-301
- Load Case: operating + settlement + thermal growth
- Support Load (kN): 89850.174
- Wind (kN): 313.146
- Seismic (kN): 7576.879
- Piping (kN): 129.853
- Anchor Bolt (mm): 3385.266
- Base Plate (mm): 40.352
- Platform: no
- Platform Area (m2): 0.000

Row 6
- Equipment: HX-01
- Load Case: operating + nozzle loads
- Support Load (kN): 6761.324
- Wind (kN): 39.069
- Seismic (kN): 563.610
- Piping (kN): 71.656
- Anchor Bolt (mm): 570.030
- Base Plate (mm): 23.526
- Platform: yes
- Platform Area (m2): 3.000

Row 7
- Equipment: HX-01-CTRL
- Load Case: skid dead load + piping reaction
- Support Load (kN): 45.600
- Wind (kN): 1.500

Seismic (kN): 1.000
Piping (kN): 3.775
Anchor Bolt (mm): 20.000
Base Plate (mm): 18.000
Platform: no
Platform Area (m2): 0.000

Row 8
Equipment: HX-01-HDR
Load Case: rack piping + thermal expansion
Support Load (kN): 737.938
Wind (kN): 2.066
Seismic (kN): 71.340
Piping (kN): 22.476
Anchor Bolt (mm): 110.194
Base Plate (mm): 20.565
Platform: no
Platform Area (m2): 0.000

Row 9
Equipment: HX-01-PMP
Load Case: operating + nozzle loads
Support Load (kN): 91.200
Wind (kN): 1.500
Seismic (kN): 1.059
Piping (kN): 11.612
Anchor Bolt (mm): 28.093
Base Plate (mm): 20.520
Platform: no
Platform Area (m2): 0.000

Row 10
Equipment: HX-01-EXP
Load Case: operating + settlement + thermal growth
Support Load (kN): 68.400
Wind (kN): 1.701
Seismic (kN): 3.303
Piping (kN): 9.393
Anchor Bolt (mm): 40.233
Base Plate (mm): 18.000
Platform: no
Platform Area (m2): 0.000

Row 11
Equipment: HX-01-RV
Load Case: operating + nozzle loads
Support Load (kN): 91.200
Wind (kN): 1.500
Seismic (kN): 1.892
Piping (kN): 12.123
Anchor Bolt (mm): 34.274
Base Plate (mm): 20.520
Platform: no
Platform Area (m2): 0.000

Row 12
Equipment: HX-02
Load Case: operating + nozzle loads
Support Load (kN): 497.849
Wind (kN): 5.715
Seismic (kN): 40.213
Piping (kN): 17.618
Anchor Bolt (mm): 147.464
Base Plate (mm): 20.857
Platform: yes
Platform Area (m2): 3.000

Row 13
Equipment: HX-02-CTRL
Load Case: skid dead load + piping reaction
Support Load (kN): 45.600
Wind (kN): 1.500
Seismic (kN): 1.000
Piping (kN): 2.375
Anchor Bolt (mm): 20.000
Base Plate (mm): 18.000
Platform: no
Platform Area (m2): 0.000

Foundation and Access Basis

Table 92. Foundation and Access Basis

Columns: Equipment; Support Variant; Anchor Groups; Footprint (m2); Clearance (m); Ladder; Lifting Lugs

Row 1
- Equipment: R-101
- Support Variant: cylindrical skirt with anchor chair
- Anchor Groups: 8
- Footprint (m2): 656.205
- Clearance (m): 1.800
- Ladder: yes
- Lifting Lugs: yes

Row 2
- Equipment: PU-201
- Support Variant: cylindrical skirt with anchor chair
- Anchor Groups: 8
- Footprint (m2): 6.620
- Clearance (m): 1.800
- Ladder: yes
- Lifting Lugs: yes

Row 3
- Equipment: V-101
- Support Variant: grouted base frame
- Anchor Groups: 4
- Footprint (m2): 12.125
- Clearance (m): 1.200
- Ladder: yes
- Lifting Lugs: yes

Row 4
- Equipment: E-101
- Support Variant: dual saddle
- Anchor Groups: 6
- Footprint (m2): 47.328
- Clearance (m): 1.500
- Ladder: yes
- Lifting Lugs: yes

Row 5
- Equipment: TK-301
- Support Variant: dual saddle
- Anchor Groups: 6
- Footprint (m2): 246.043
- Clearance (m): 1.200
- Ladder: yes
- Lifting Lugs: yes

Row 6
- Equipment: HX-01
- Support Variant: dual saddle
- Anchor Groups: 6
- Footprint (m2): 39.069
- Clearance (m): 1.500
- Ladder: yes
- Lifting Lugs: yes

Row 7
- Equipment: HX-01-CTRL
- Support Variant: panel frame
- Anchor Groups: 4
- Footprint (m2): 1.200
- Clearance (m): 1.000
- Ladder: no
- Lifting Lugs: no

Row 8
- Equipment: HX-01-HDR
- Support Variant: guided rack shoe
- Anchor Groups: 2
- Footprint (m2): 2.430
- Clearance (m): 1.200
- Ladder: no
- Lifting Lugs: yes

Row 9
- Equipment: HX-01-PMP

Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.200
Clearance (m): 1.200
Ladder: no
Lifting Lugs: no

Row 10
Equipment: HX-01-EXP
Support Variant: four-leg support
Anchor Groups: 4
Footprint (m2): 1.600
Clearance (m): 1.200
Ladder: no
Lifting Lugs: no

Row 11
Equipment: HX-01-RV
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.200
Clearance (m): 1.200
Ladder: no
Lifting Lugs: no

Row 12
Equipment: HX-02
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 5.715
Clearance (m): 1.500
Ladder: yes
Lifting Lugs: yes

Row 13
Equipment: HX-02-CTRL
Support Variant: panel frame
Anchor Groups: 4
Footprint (m2): 1.200
Clearance (m): 1.000
Ladder: no
Lifting Lugs: no

Shell and Head Thickness Derivation

Table 93. Shell and Head Thickness Derivation

Columns: Equipment; Equation; Substitution; Shell t (mm); Head t (mm); CA (mm); Pressure Class; Hydrotest (bar)

Row 1
Equipment: R-101
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=2.000 MPa; D·17.895 m; CA=3.0 mm
Shell t (mm): 157.134
Head t (mm): 144.563
CA (mm): 3.000
Pressure Class: ASME 600
Hydrotest (bar): 26.000

Row 2
Equipment: PU-201
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.200 MPa; D·1.651 m; CA=3.0 mm
Shell t (mm): 4.409
Head t (mm): 4.056
CA (mm): 3.000
Pressure Class: PN16
Hydrotest (bar): 3.000

Row 3
Equipment: V-101
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.300 MPa; D·10.104 m; CA=3.0 mm
Shell t (mm): 15.940
Head t (mm): 14.665
CA (mm): 3.000
Pressure Class: PN16
Hydrotest (bar): 4.000

Row 4
Equipment: E-101
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.800 MPa; D·5.356 m; CA=3.0 mm
Shell t (mm): 21.338
Head t (mm): 19.631
CA (mm): 3.000
Pressure Class: ASME 150
Hydrotest (bar): 10.400
Row 5
Equipment: TK-301
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.120 MPa; D·12.211 m; CA=2.0 mm
Shell t (mm): 9.188
Head t (mm): 8.453
CA (mm): 2.000
Pressure Class: PN16
Hydrotest (bar): 2.200
Row 6
Equipment: HX-01
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=1.100 MPa; D·4.866 m; CA=3.0 mm
Shell t (mm): 25.945
Head t (mm): 23.869
CA (mm): 3.000
Pressure Class: ASME 300
Hydrotest (bar): 14.300
Row 7
Equipment: HX-01-HDR
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=1.450 MPa; D·0.900 m; CA=3.0 mm
Shell t (mm): 8.604
Head t (mm): 7.916
CA (mm): 3.000
Pressure Class: ASME 300
Hydrotest (bar): 18.850
Row 8
Equipment: HX-01-PMP
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=1.450 MPa; D·0.900 m; CA=3.0 mm
Shell t (mm): 8.604
Head t (mm): 7.916
CA (mm): 3.000
Pressure Class: ASME 300
Hydrotest (bar): 18.850
Row 9
Equipment: HX-01-EXP
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.600 MPa; D·0.900 m; CA=3.0 mm
Shell t (mm): 5.309
Head t (mm): 4.884
CA (mm): 3.000
Pressure Class: ASME 150
Hydrotest (bar): 7.800
Row 10
Equipment: HX-01-RV
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=1.350 MPa; D·0.900 m; CA=3.0 mm
Shell t (mm): 8.215
Head t (mm): 7.558
CA (mm): 3.000
Pressure Class: ASME 300
Hydrotest (bar): 17.550
Row 11
Equipment: HX-02
Equation: t = P*D / (2*S*J - 1.2*P) + CA
Substitution: P=0.650 MPa; D·1.861 m; CA=3.0 mm
Shell t (mm): 8.953
Head t (mm): 8.237
CA (mm): 3.000
Pressure Class: ASME 150
Hydrotest (bar): 8.450

Support and Overturning Derivation

Table 94. Support and Overturning Derivation

Columns: Equipment; Load Case; Equation; Substitution; Support Load (kN); Overturning (kN.m); Anchor Bolt (mm); Base Plate (mm); Foundation Note

Row 1
- Equipment: R-101
- Load Case: operating + wind + seismic + nozzle loads
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=280488.792; Wp=240.402; Ww=1152.822; Ws=46748.132
- Support Load (kN): 328630.148
- Overturning (kN.m): 168812.699
- Anchor Bolt (mm): 10269.688
- Base Plate (mm): 185.721
- Foundation Note: Tall vessel service assumes ring foundation and local anchor-chair reinforcement.

Row 2
- Equipment: PU-201
- Load Case: operating + wind + seismic + nozzle loads
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=270.030; Wp=16.706; Ww=9.813; Ws=45.005
- Support Load (kN): 341.554
- Overturning (kN.m): 162.518
- Anchor Bolt (mm): 149.391
- Base Plate (mm): 21.185
- Foundation Note: Tall vessel service assumes ring foundation and local anchor-chair reinforcement.

Row 3
- Equipment: V-101
- Load Case: operating + nozzle loads
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=50477.341; Wp=98.976; Ww=168.445; Ws=4673.828
- Support Load (kN): 55418.590
- Overturning (kN.m): 30379.881
- Anchor Bolt (mm): 2247.216
- Base Plate (mm): 33.876
- Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 4
- Equipment: E-101
- Load Case: operating + nozzle loads
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=7966.118; Wp=53.324; Ww=47.328; Ws=737.603
- Support Load (kN): 8804.373
- Overturning (kN.m): 4794.423
- Anchor Bolt (mm): 664.724
- Base Plate (mm): 24.186
- Foundation Note: Access platform and ladder loads are included in the support screening basis.

Row 5
- Equipment: TK-301
- Load Case: operating + settlement + thermal growth
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=81830.296; Wp=129.853; Ww=313.146; Ws=7576.879
- Support Load (kN): 89850.174
- Overturning (kN.m): 49249.715
- Anchor Bolt (mm): 3385.266
- Base Plate (mm): 40.352
- Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 6
- Equipment: HX-01
- Load Case: operating + nozzle loads
- Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
- Substitution: Wv=6086.989; Wp=71.656; Ww=39.069; Ws=563.610
- Support Load (kN): 6761.324
- Overturning (kN.m): 3663.466
- Anchor Bolt (mm): 570.030
- Base Plate (mm): 23.526
- Foundation Note: Access platform and ladder loads are included in the support screening basis.

Row 7
Equipment: HX-01-CTRL
Load Case: skid dead load + piping reaction
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=7.000; Wp=3.775; Ww=1.500; Ws=1.000
Support Load (kN): 45.600
Overturning (kN.m): 5.000
Anchor Bolt (mm): 20.000
Base Plate (mm): 18.000
Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 8
Equipment: HX-01-HDR
Load Case: rack piping + thermal expansion
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=642.056; Wp=22.476; Ww=2.066; Ws=71.340
Support Load (kN): 737.938
Overturning (kN.m): 386.423
Anchor Bolt (mm): 110.194
Base Plate (mm): 20.565
Foundation Note: Utility-header service assumes pipe-rack tie-in with local guide/restraint design.

Row 9
Equipment: HX-01-PMP
Load Case: operating + nozzle loads
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=11.589; Wp=11.612; Ww=1.500; Ws=1.059
Support Load (kN): 91.200
Overturning (kN.m): 6.883
Anchor Bolt (mm): 28.093
Base Plate (mm): 20.520
Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 10
Equipment: HX-01-EXP
Load Case: operating + settlement + thermal growth
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=35.673; Wp=9.393; Ww=1.701; Ws=3.303
Support Load (kN): 68.400
Overturning (kN.m): 21.470
Anchor Bolt (mm): 40.233
Base Plate (mm): 18.000
Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 11
Equipment: HX-01-RV
Load Case: operating + nozzle loads
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=20.428; Wp=12.123; Ww=1.500; Ws=1.892
Support Load (kN): 91.200
Overturning (kN.m): 12.295
Anchor Bolt (mm): 34.274
Base Plate (mm): 20.520
Foundation Note: Base/foundation detail remains a preliminary screening basis.

Row 12
Equipment: HX-02
Load Case: operating + nozzle loads
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=434.303; Wp=17.618; Ww=5.715; Ws=40.213
Support Load (kN): 497.849
Overturning (kN.m): 261.386
Anchor Bolt (mm): 147.464
Base Plate (mm): 20.857
Foundation Note: Access platform and ladder loads are included in the support screening basis.

Row 13
Equipment: HX-02-CTRL
Load Case: skid dead load + piping reaction
Equation: Wsupport = Wvertical + Wpiping + Wwind + Wseismic
Substitution: Wv=7.000; Wp=2.375; Ww=1.500; Ws=1.000
Support Load (kN): 45.600
Overturning (kN.m): 5.000

Anchor Bolt (mm): 20.000
Base Plate (mm): 18.000
Foundation Note: Base/foundation detail remains a preliminary screening basis.

Nozzle Reinforcement and Connection Basis

Table 95. Nozzle Reinforcement and Connection Basis

Columns: Equipment; Services; Reinforcement Family; Equation; Substitution; Area (mm2); Shell Factor; Class; Load Cases (kN)

Row 1
- Equipment: R-101
- Services: feed, product, utility, instrument
- Reinforcement Family: integral repad
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=9336.1 mm; P=20.00 bar
- Area (mm2): 1941901.983
- Shell Factor: 2.359
- Class: ASME 600
- Load Cases (kN): 60.10, 60.10, 60.10, 60.10

Row 2
- Equipment: PU-201
- Services: feed, overhead, bottoms, instrument
- Reinforcement Family: integral repad
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=640.2 mm; P=2.00 bar
- Area (mm2): 13316.919
- Shell Factor: 1.848
- Class: PN16
- Load Cases (kN): 4.18, 4.18, 4.18, 4.18

Row 3
- Equipment: V-101
- Services: inlet, outlet, instrument
- Reinforcement Family: shell excess area
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=3917.0 mm; P=3.00 bar
- Area (mm2): 65806.429
- Shell Factor: 1.743
- Class: PN16
- Load Cases (kN): 32.99, 32.99, 32.99

Row 4
- Equipment: E-101
- Services: inlet, outlet, instrument
- Reinforcement Family: shell excess area
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=2021.0 mm; P=8.00 bar
- Area (mm2): 90539.654
- Shell Factor: 1.799
- Class: ASME 150
- Load Cases (kN): 17.77, 17.77, 17.77

Row 5
- Equipment: TK-301
- Services: inlet, outlet, vent
- Reinforcement Family: shell excess area
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=5177.3 mm; P=1.20 bar
- Area (mm2): 34791.617
- Shell Factor: 1.781
- Class: PN16
- Load Cases (kN): 43.28, 43.28, 43.28

Row 6
- Equipment: HX-01
- Services: inlet, outlet, instrument
- Reinforcement Family: set-on repad
- Equation: Areinf = f(dnozzle, Pdesign, equipment family)
- Substitution: d=1822.7 mm; P=11.00 bar
- Area (mm2): 152378.966
- Shell Factor: 1.839
- Class: ASME 300
- Load Cases (kN): 23.89, 23.89, 23.89

Row 7

Equipment: HX-01-CTRL
Services: inlet, outlet, instrument
Reinforcement Family: shell excess area
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=25.0 mm; P=9.00 bar
Area (mm2): 1260.000
Shell Factor: 1.185
Class: ASME 150
Load Cases (kN): 1.26, 1.26, 1.26

Row 8
Equipment: HX-01-HDR
Services: inlet, outlet, branch, drain
Reinforcement Family: set-on repad
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=417.0 mm; P=14.50 bar
Area (mm2): 45956.111
Shell Factor: 2.132
Class: ASME 300
Load Cases (kN): 5.62, 5.62, 5.62, 5.62

Row 9
Equipment: HX-01-PMP
Services: inlet, outlet, instrument
Reinforcement Family: set-on repad
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=117.3 mm; P=14.50 bar
Area (mm2): 12929.954
Shell Factor: 1.452
Class: ASME 300
Load Cases (kN): 3.87, 3.87, 3.87

Row 10
Equipment: HX-01-EXP
Services: inlet, outlet, vent
Reinforcement Family: set-on repad
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=175.1 mm; P=6.00 bar
Area (mm2): 7984.835
Shell Factor: 1.440
Class: ASME 150
Load Cases (kN): 3.13, 3.13, 3.13

Row 11
Equipment: HX-01-RV
Services: inlet, outlet, instrument
Reinforcement Family: set-on repad
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=145.4 mm; P=13.50 bar
Area (mm2): 14920.743
Shell Factor: 1.493
Class: ASME 300
Load Cases (kN): 4.04, 4.04, 4.04

Row 12
Equipment: HX-02
Services: inlet, outlet, instrument
Reinforcement Family: shell excess area
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=613.7 mm; P=6.50 bar
Area (mm2): 22339.297
Shell Factor: 1.691
Class: ASME 150
Load Cases (kN): 5.87, 5.87, 5.87

Row 13
Equipment: HX-02-CTRL
Services: inlet, outlet, instrument
Reinforcement Family: shell excess area
Equation: Areinf = f(dnozzle, Pdesign, equipment family)
Substitution: d=25.0 mm; P=5.00 bar
Area (mm2): 700.000
Shell Factor: 1.125
Class: ASME 150
Load Cases (kN): 0.80, 0.80, 0.80

Connection and Piping Class Basis

Table 96. Connection and Piping Class Basis

Columns: Equipment; Support Variant; Connection Classes; Piping Class Basis; Orientations (deg); Projections (mm); Rack Tie-In

Row 1
- Equipment: R-101
- Support Variant: cylindrical skirt with anchor chair
- Connection Classes: ASME 600, ASME 600, ASME 600, ASME 600
- Piping Class Basis: ASME 600 carbon/alloy steel process class
- Orientations (deg): 0, 90, 180, 270
- Projections (mm): 9802.9, 9802.9, 9802.9, 9802.9
- Rack Tie-In: no

Row 2
- Equipment: PU-201
- Support Variant: cylindrical skirt with anchor chair
- Connection Classes: PN16, PN16, PN16, PN16
- Piping Class Basis: PN16 carbon/alloy steel process class
- Orientations (deg): 0, 90, 180, 270
- Projections (mm): 672.2, 672.2, 672.2, 672.2
- Rack Tie-In: no

Row 3
- Equipment: V-101
- Support Variant: grouted base frame
- Connection Classes: PN16, PN16, PN16
- Piping Class Basis: PN16 carbon/alloy steel process class
- Orientations (deg): 0, 180, 270
- Projections (mm): 4112.9, 4112.9, 4112.9
- Rack Tie-In: no

Row 4
- Equipment: E-101
- Support Variant: dual saddle
- Connection Classes: ASME 150, ASME 150, ASME 150
- Piping Class Basis: ASME 150 thermal-oil / utility piping class
- Orientations (deg): 0, 180, 270
- Projections (mm): 2122.0, 2122.0, 2122.0
- Rack Tie-In: no

Row 5
- Equipment: TK-301
- Support Variant: dual saddle
- Connection Classes: PN16, PN16, PN16
- Piping Class Basis: PN16 tank farm and transfer class
- Orientations (deg): 0, 180, 270
- Projections (mm): 5436.2, 5436.2, 5436.2
- Rack Tie-In: no

Row 6
- Equipment: HX-01
- Support Variant: dual saddle
- Connection Classes: ASME 300, ASME 300, ASME 300
- Piping Class Basis: ASME 300 thermal-oil / utility piping class
- Orientations (deg): 0, 180, 270
- Projections (mm): 2278.4, 2278.4, 2278.4
- Rack Tie-In: no

Row 7
- Equipment: HX-01-CTRL
- Support Variant: panel frame
- Connection Classes: ASME 150, ASME 150, ASME 150
- Piping Class Basis: ASME 150 thermal-oil / utility piping class
- Orientations (deg): 0, 180, 270
- Projections (mm): 90.0, 90.0, 90.0
- Rack Tie-In: no

Row 8
- Equipment: HX-01-HDR
- Support Variant: guided rack shoe
- Connection Classes: ASME 300, ASME 300, ASME 300, ASME 300
- Piping Class Basis: ASME 300 thermal-oil / utility piping class
- Orientations (deg): 0, 180, 270
- Projections (mm): 521.3, 521.3, 521.3, 521.3
- Rack Tie-In: yes

Row 9
- Equipment: HX-01-PMP
- Support Variant: grouted base frame
- Connection Classes: ASME 300, ASME 300, ASME 300

Piping Class Basis: ASME 300 thermal-oil / utility piping class
Orientations (deg): 0, 180, 270
Projections (mm): 146.7, 146.7, 146.7
Rack Tie-In: no
Row 10
Equipment: HX-01-EXP
Support Variant: four-leg support
Connection Classes: ASME 150, ASME 150, ASME 150
Piping Class Basis: ASME 150 thermal-oil / utility piping class
Orientations (deg): 0, 180, 270
Projections (mm): 218.9, 218.9, 218.9
Rack Tie-In: no
Row 11
Equipment: HX-01-RV
Support Variant: grouted base frame
Connection Classes: ASME 300, ASME 300, ASME 300
Piping Class Basis: ASME 300 thermal-oil / utility piping class
Orientations (deg): 0, 180, 270
Projections (mm): 181.8, 181.8, 181.8
Rack Tie-In: no
Row 12
Equipment: HX-02
Support Variant: dual saddle
Connection Classes: ASME 150, ASME 150, ASME 150
Piping Class Basis: ASME 150 thermal-oil / utility piping class
Orientations (deg): 0, 180, 270
Projections (mm): 644.4, 644.4, 644.4
Rack Tie-In: no
Row 13
Equipment: HX-02-CTRL
Support Variant: panel frame
Connection Classes: ASME 150, ASME 150, ASME 150
Piping Class Basis: ASME 150 thermal-oil / utility piping class
Orientations (deg): 0, 180, 270
Projections (mm): 90.0, 90.0, 90.0
Rack Tie-In: no

Material of Construction Matrix

Table 97. Material of Construction Matrix

Columns: Equipment; Type; Service; Selected MoC; Design T (C); Design P (bar); Service Basis
Row 1
Equipment: R-101
Type: Reactor
Service: Tubular Plug Flow Hydrator
Selected MoC: Carbon Steel
Design T (C): 255.0
Design P (bar): 20.00
Service Basis: general process service
Row 2
Equipment: PU-201
Type: Distillation column
Service: Distillation and purification train
Selected MoC: Carbon Steel
Design T (C): 140.0
Design P (bar): 2.00
Service Basis: general process service
Row 3
Equipment: V-101
Type: Flash drum
Service: Intermediate disengagement
Selected MoC: Carbon Steel
Design T (C): 85.0
Design P (bar): 3.00
Service Basis: general process service
Row 4
Equipment: E-101
Type: Heat exchanger
Service: R-rough to D-rough heat recovery
Selected MoC: Carbon Steel

Design T (C): 180.0
Design P (bar): 8.00
Service Basis: general process service
Row 5
Equipment: TK-301
Type: Storage tank
Service: Ethylene Glycol storage via vertical tank farm
Selected MoC: SS304
Design T (C): 45.0
Design P (bar): 1.20
Service Basis: general process service
Row 6
Equipment: HX-01
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Selected MoC: Carbon steel
Design T (C): 213.0
Design P (bar): 11.00
Service Basis: general process service
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Selected MoC: Carbon steel
Design T (C): 203.0
Design P (bar): 9.00
Service Basis: general process service
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Selected MoC: Carbon steel
Design T (C): 205.0
Design P (bar): 14.50
Service Basis: general process service
Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Selected MoC: Carbon steel
Design T (C): 205.0
Design P (bar): 14.50
Service Basis: general process service
Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Selected MoC: Carbon steel
Design T (C): 210.0
Design P (bar): 6.00
Service Basis: general process service
Row 11
Equipment: HX-01-RV
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Selected MoC: Carbon steel
Design T (C): 213.0
Design P (bar): 13.50
Service Basis: general process service
Row 12
Equipment: HX-02
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Selected MoC: SS316
Design T (C): 162.0
Design P (bar): 6.50
Service Basis: general process service
Row 13
Equipment: HX-02-CTRL

Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Selected MoC: Carbon steel
Design T (C): 158.0
Design P (bar): 5.00
Service Basis: general process service

MoC Option Screening

Table 98. MoC Option Screening

Columns: Candidate; Description; Score; Role; Feasible; Screening Notes
Row 1: Candidate: carbon_steel | Description: Carbon steel construction | Score: 100.00 | Role: selected | Feasible: yes | Screening Notes: screening-feasible
Row 2: Candidate: ss316l | Description: SS316L construction | Score: 98.00 | Role: alternate | Feasible: yes | Screening Notes: screening-feasible
Row 3: Candidate: alloy_steel_converter_service | Description: Alloy steel converter service | Score: 52.00 | Role: alternate | Feasible: yes | Screening Notes: screening-feasible
Row 4: Candidate: rubber_lined_cs | Description: Rubber-lined carbon steel | Score: 58.00 | Role: alternate | Feasible: yes | Screening Notes: screening-feasible

Equipment-Wise MoC Justification Matrix

Table 99. Equipment-Wise MoC Justification Matrix

Columns: Equipment; Service Basis; Corrosion Driver; Selected MoC; Alternate MoC; Inspection / Cleaning Basis; Rationale
Row 1
Equipment: R-101
Service Basis: general process liquid service
Corrosion Driver: elevated-temperature metal-loss allowance
Selected MoC: Carbon Steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: thickness survey + nozzle / tray / internals inspection
Rationale: Carbon Steel retained for omega_catalytic because general process liquid service controls the screening basis.
Row 2
Equipment: PU-201
Service Basis: general process liquid service
Corrosion Driver: general aqueous-organic corrosion allowance basis
Selected MoC: Carbon Steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: thickness survey + nozzle / tray / internals inspection
Rationale: Carbon Steel retained for omega_catalytic because general process liquid service controls the screening basis.
Row 3
Equipment: V-101
Service Basis: general process liquid service
Corrosion Driver: general aqueous-organic corrosion allowance basis
Selected MoC: Carbon Steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: periodic shell and connection inspection
Rationale: Carbon Steel retained for omega_catalytic because general process liquid service controls the screening basis.
Row 4
Equipment: E-101
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon Steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon Steel retained for omega_catalytic because thermal / utility service controls the screening basis.
Row 5

Equipment: TK-301
Service Basis: inventory / storage service
Corrosion Driver: general aqueous-organic corrosion allowance basis
Selected MoC: SS304
Alternate MoC: Carbon steel construction
Inspection / Cleaning Basis: shell-floor, vent, and nozzle pad inspection
Rationale: SS304 retained for omega_catalytic because inventory / storage service controls the screening basis.

Row 6

Equipment: HX-01
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 7

Equipment: HX-01-CTRL
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 8

Equipment: HX-01-HDR
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 9

Equipment: HX-01-PMP
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 10

Equipment: HX-01-EXP
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 11

Equipment: HX-01-RV
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 12

Equipment: HX-02
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: SS316

Alternate MoC: Carbon steel construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: SS316 retained for omega_catalytic because thermal / utility service controls the screening basis.

Row 13
Equipment: HX-02-CTRL
Service Basis: thermal / utility service
Corrosion Driver: thermal cycling, condensate, and fouling-side exposure
Selected MoC: Carbon steel
Alternate MoC: SS316L construction
Inspection / Cleaning Basis: tube-side fouling, rack shoe, and support expansion inspection
Rationale: Carbon steel retained for omega_catalytic because thermal / utility service controls the screening basis.

Inspection and Maintainability Basis

Table 100. Inspection and Maintainability Basis

Columns: Equipment; Service Basis; Inspection Basis; Maintainability Basis; Platform; Ladder

Row 1: Equipment: R-101 | Service Basis: general process liquid service | Inspection Basis: thickness survey + nozzle / tray / internals inspection | Maintainability Basis: platform/ladder access with routine shell and nozzle inspection | Platform: yes | Ladder: yes
Row 2: Equipment: PU-201 | Service Basis: general process liquid service | Inspection Basis: thickness survey + nozzle / tray / internals inspection | Maintainability Basis: platform/ladder access with routine shell and nozzle inspection | Platform: yes | Ladder: yes
Row 3: Equipment: V-101 | Service Basis: general process liquid service | Inspection Basis: periodic shell and connection inspection | Maintainability Basis: platform/ladder access with routine shell and nozzle inspection | Platform: no | Ladder: yes
Row 4: Equipment: E-101 | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: yes | Ladder: yes
Row 5: Equipment: TK-301 | Service Basis: inventory / storage service | Inspection Basis: shell-floor, vent, and nozzle pad inspection | Maintainability Basis: shell-floor inspection, vent maintenance, and transfer-line isolation access | Platform: no | Ladder: yes
Row 6: Equipment: HX-01 | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: yes | Ladder: yes
Row 7: Equipment: HX-01-CTRL | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no
Row 8: Equipment: HX-01-HDR | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no
Row 9: Equipment: HX-01-PMP | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no
Row 10: Equipment: HX-01-EXP | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no
Row 11: Equipment: HX-01-RV | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no
Row 12: Equipment: HX-02 | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: yes | Ladder: yes
Row 13: Equipment: HX-02-CTRL | Service Basis: thermal / utility service | Inspection Basis: tube-side fouling, rack shoe, and support expansion

inspection | Maintainability Basis: rack access, expansion restraint checks, and isolation spool maintenance | Platform: no | Ladder: no

Corrosion and Service Basis

Table 101. Corrosion and Service Basis

Columns: Parameter; Value
Row 1: Parameter: Selected MoC decision | Value: carbon_steel
Row 2: Parameter: Decision summary | Value: Carbon steel construction selected as the highest-ranked alternative.
Row 3: Parameter: Route family service | Value: omega_catalytic
Row 4: Parameter: Storage material | Value: SS304
Row 5: Parameter: Reactor material | Value: Carbon Steel
Row 6: Parameter: Main separation material | Value: Carbon Steel

Utility and Storage MoC Basis

Table 102. Utility and Storage MoC Basis

Columns: Service Class; Equipment; Typical MoC; Primary Driver; Piping / Connection Basis
Row 1: Service Class: Storage and inventory systems | Equipment: HX-01-EXP, TK-301 | Typical MoC: Carbon steel, SS304 | Primary Driver: inventory breathing, drainage, and shell-floor exposure | Piping / Connection Basis: tank nozzle and vent classes follow selected vessel pressure class basis
Row 2: Service Class: Thermal and utility packages | Equipment: E-101, HX-01, HX-01-CTRL, HX-01-EXP, HX-01-HDR, HX-01-PMP, HX-01-RV, HX-02, HX-02-CTRL | Typical MoC: Carbon Steel, Carbon steel, SS316 | Primary Driver: thermal cycling, condensation, and fouling-side service | Piping / Connection Basis: utility headers and exchanger nozzles inherit screening connection classes from mechanical basis
Row 3: Service Class: Primary process vessels | Equipment: PU-201, R-101 | Typical MoC: Carbon Steel | Primary Driver: core reaction / separation chemistry and cleanability | Piping / Connection Basis: primary process connections follow equipment pressure class and nozzle reinforcement family

Nozzle and Connection Schedule

Table 103. Nozzle and Connection Schedule

Columns: Equipment; Pressure Class; Hydrotest (bar); Reinforcement Family; Nozzle Services; Connection Classes; Orientations (deg); Projections (mm); Shell Factors; Load Cases (kN)
Row 1
- Equipment: R-101
- Pressure Class: ASME 600
- Hydrotest (bar): 26.000
- Reinforcement Family: integral repad
- Nozzle Services: feed, product, utility, instrument
- Connection Classes: ASME 600, ASME 600, ASME 600, ASME 600
- Orientations (deg): 0, 90, 180, 270
- Projections (mm): 9802.9, 9802.9, 9802.9, 9802.9
- Shell Factors: 2.36, 2.36, 2.36, 2.36
- Load Cases (kN): 60.10, 60.10, 60.10, 60.10

Row 2
- Equipment: PU-201
- Pressure Class: PN16
- Hydrotest (bar): 3.000
- Reinforcement Family: integral repad
- Nozzle Services: feed, overhead, bottoms, instrument
- Connection Classes: PN16, PN16, PN16, PN16
- Orientations (deg): 0, 90, 180, 270
- Projections (mm): 672.2, 672.2, 672.2, 672.2
- Shell Factors: 1.85, 1.85, 1.85, 1.85
- Load Cases (kN): 4.18, 4.18, 4.18, 4.18

Row 3
- Equipment: V-101

Pressure Class: PN16
Hydrotest (bar): 4.000
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, instrument
Connection Classes: PN16, PN16, PN16
Orientations (deg): 0, 180, 270
Projections (mm): 4112.9, 4112.9, 4112.9
Shell Factors: 1.74, 1.74, 1.74
Load Cases (kN): 32.99, 32.99, 32.99

Row 4
Equipment: E-101
Pressure Class: ASME 150
Hydrotest (bar): 10.400
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 150, ASME 150, ASME 150
Orientations (deg): 0, 180, 270
Projections (mm): 2122.0, 2122.0, 2122.0
Shell Factors: 1.80, 1.80, 1.80
Load Cases (kN): 17.77, 17.77, 17.77

Row 5
Equipment: TK-301
Pressure Class: PN16
Hydrotest (bar): 2.200
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, vent
Connection Classes: PN16, PN16, PN16
Orientations (deg): 0, 180, 270
Projections (mm): 5436.2, 5436.2, 5436.2
Shell Factors: 1.78, 1.78, 1.78
Load Cases (kN): 43.28, 43.28, 43.28

Row 6
Equipment: HX-01
Pressure Class: ASME 300
Hydrotest (bar): 14.300
Reinforcement Family: set-on repad
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 300, ASME 300, ASME 300
Orientations (deg): 0, 180, 270
Projections (mm): 2278.4, 2278.4, 2278.4
Shell Factors: 1.84, 1.84, 1.84
Load Cases (kN): 23.89, 23.89, 23.89

Row 7
Equipment: HX-01-CTRL
Pressure Class: ASME 150
Hydrotest (bar): 11.700
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 150, ASME 150, ASME 150
Orientations (deg): 0, 180, 270
Projections (mm): 90.0, 90.0, 90.0
Shell Factors: 1.19, 1.19, 1.19
Load Cases (kN): 1.26, 1.26, 1.26

Row 8
Equipment: HX-01-HDR
Pressure Class: ASME 300
Hydrotest (bar): 18.850
Reinforcement Family: set-on repad
Nozzle Services: inlet, outlet, branch, drain
Connection Classes: ASME 300, ASME 300, ASME 300, ASME 300
Orientations (deg): 0, 180, 270
Projections (mm): 521.3, 521.3, 521.3, 521.3
Shell Factors: 2.13, 2.13, 2.13, 2.13
Load Cases (kN): 5.62, 5.62, 5.62, 5.62

Row 9
Equipment: HX-01-PMP
Pressure Class: ASME 300
Hydrotest (bar): 18.850
Reinforcement Family: set-on repad
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 300, ASME 300, ASME 300
Orientations (deg): 0, 180, 270
Projections (mm): 146.7, 146.7, 146.7

Shell Factors: 1.45, 1.45, 1.45
Load Cases (kN): 3.87, 3.87, 3.87

Row 10

Equipment: HX-01-EXP
Pressure Class: ASME 150
Hydrotest (bar): 7.800
Reinforcement Family: set-on repad
Nozzle Services: inlet, outlet, vent
Connection Classes: ASME 150, ASME 150, ASME 150
Orientations (deg): 0, 180, 270
Projections (mm): 218.9, 218.9, 218.9
Shell Factors: 1.44, 1.44, 1.44
Load Cases (kN): 3.13, 3.13, 3.13

Row 11

Equipment: HX-01-RV
Pressure Class: ASME 300
Hydrotest (bar): 17.550
Reinforcement Family: set-on repad
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 300, ASME 300, ASME 300
Orientations (deg): 0, 180, 270
Projections (mm): 181.8, 181.8, 181.8
Shell Factors: 1.49, 1.49, 1.49
Load Cases (kN): 4.04, 4.04, 4.04

Row 12

Equipment: HX-02
Pressure Class: ASME 150
Hydrotest (bar): 8.450
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 150, ASME 150, ASME 150
Orientations (deg): 0, 180, 270
Projections (mm): 644.4, 644.4, 644.4
Shell Factors: 1.69, 1.69, 1.69
Load Cases (kN): 5.87, 5.87, 5.87

Row 13

Equipment: HX-02-CTRL
Pressure Class: ASME 150
Hydrotest (bar): 6.500
Reinforcement Family: shell excess area
Nozzle Services: inlet, outlet, instrument
Connection Classes: ASME 150, ASME 150, ASME 150
Orientations (deg): 0, 180, 270
Projections (mm): 90.0, 90.0, 90.0
Shell Factors: 1.12, 1.12, 1.12
Load Cases (kN): 0.80, 0.80, 0.80

18. Storage and Utilities

Storage and Utilities

Storage Basis

Table 104. Storage Basis

Columns: Parameter; Value
Row 1: Parameter: Inventory days | Value: 14.1
Row 2: Parameter: Working volume (m3) | Value: 7661.521
Row 3: Parameter: Total volume (m3) | Value: 8580.903
Row 4: Parameter: Material of construction | Value: SS304

Storage Inventory and Buffer Basis

Table 105. Storage Inventory and Buffer Basis

Columns: Parameter; Value
Row 1: Parameter: Recommended operating mode | Value: continuous
Row 2: Parameter: Availability policy | Value: continuous_liquid_train
Row 3: Parameter: Raw-material buffer (d) | Value: 19.6
Row 4: Parameter: Finished-goods buffer (d) | Value: 9.4
Row 5: Parameter: Operating stock (d) | Value: 2.5
Row 6: Parameter: Dispatch buffer (d) | Value: 9.4
Row 7: Parameter: Restart buffer (d) | Value: 1.5
Row 8: Parameter: Campaign length (d) | Value: 170.0
Row 9: Parameter: Startup ramp (d) | Value: 2.0
Row 10: Parameter: Restart loss fraction | Value: 0.0062
Row 11: Parameter: Annual restart loss (kg/y) | Value: 16,426.7
Row 12: Parameter: Turnaround buffer factor | Value: 1.050
Row 13: Parameter: Shared buffer basis | Value: Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.

Storage Service Matrix

Table 106. Storage Service Matrix

Columns: Equipment; Service; Type; MoC; Volume (m3); Design Basis
Row 1: Equipment: TK-301 | Service: Ethylene Glycol storage via vertical tank farm | Type: Storage tank | MoC: SS304 | Volume (m3): 8580.903 | Design Basis: 14.1 days inventory
Row 2: Equipment: HX-01-EXP | Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up | Type: HTM expansion tank | MoC: Carbon steel | Volume (m3): 2.835 | Design Basis: Utility train package item for omega_catalytic_pinch_htm__shared_htm_step_01; island omega_catalytic_pinch_htm__shared_htm_shared_htm_01; header 1; cluster -

Selected Heat Integration

Table 107. Selected Heat Integration

Columns: Parameter; Value
Row 1: Parameter: Selected case | Value: Pinch + HTM loop
Row 2: Parameter: Recovered duty (kW) | Value: 47256.658
Row 3: Parameter: Residual hot utility (kW) | Value: 43800.060
Row 4: Parameter: Residual cold utility (kW) | Value: 32255.698

Utility Basis Decision

Table 108. Utility Basis Decision

Columns: Parameter; Value
Row 1: Parameter: Selected basis | Value: hp_steam_20bar
Row 2: Parameter: Steam pressure (bar) | Value: 20.0

Row 3: Parameter: Steam cost (INR/kg) | Value: 1.80
Row 4: Parameter: Power cost (INR/kWh) | Value: 8.50

Utility Consumption Summary

Table 109. Utility Consumption Summary

Columns: Utility; Load; Units; Annualized Proxy; Basis
Row 1: Utility: Steam | Load: 71672.825 | Units: kg/h | Annualized Proxy: 567648774.0 kg/y | Basis: 20.0 bar saturated steam equivalent
Row 2: Utility: Cooling water | Load: 2778.003 | Units: m3/h | Annualized Proxy: 22001783.8 m3/y | Basis: 10 C cooling-water rise
Row 3: Utility: Electricity | Load: 28732.696 | Units: kW | Annualized Proxy: 227562952.3 kWh/y | Basis: Agitation, pumps, vacuum auxiliaries, and transfer drives
Row 4: Utility: DM water | Load: 2.000 | Units: m3/h | Annualized Proxy: 15840.0 m3/y | Basis: Boiler and wash service allowance
Row 5: Utility: Nitrogen | Load: 3.608 | Units: Nm3/h | Annualized Proxy: 28575.4 Nm3/y | Basis: Inerting and blanketing
Row 6: Utility: Heat-integration auxiliaries | Load: 16.166 | Units: kW | Annualized Proxy: 128034.7 kWh/y | Basis: Selected utility train circulation, HTM pumping, and exchanger-network auxiliaries

Utility Service System Matrix

Table 110. Utility Service System Matrix

Columns: Utility System; Supply Basis; Primary Services; Standby / Design Note
Row 1: Utility System: Steam | Supply Basis: 20.0 bar saturated steam header | Primary Services: reactor heating, reboiler duty, and dryer endpoint polish | Standby / Design Note: 1 x operating + startup margin on steam letdown / control valve basis
Row 2: Utility System: Cooling water | Supply Basis: recirculating cooling-water network with 10 C rise | Primary Services: condenser duty, exchanger cooling, and thermal packet cold-side service | Standby / Design Note: peak summer duty covered by screening CW oversizing factor
Row 3: Utility System: Electricity | Supply Basis: plant electrical distribution / MCC basis | Primary Services: agitation, pumping, fans, controls, and heat-recovery auxiliaries | Standby / Design Note: feasibility standby allowance embedded in peak demand factor
Row 4: Utility System: DM water | Supply Basis: demineralized water and wash-service header | Primary Services: boiler make-up, washdown, and utility support | Standby / Design Note: intermittent demand carried as service allowance rather than dedicated redundancy
Row 5: Utility System: Nitrogen | Supply Basis: nitrogen blanketing / inerting manifold | Primary Services: storage blanketing, shutdown purge, and inert atmosphere protection | Standby / Design Note: peak inerting event handled by conservative surge factor
Row 6: Utility System: Heat-integration auxiliaries | Supply Basis: plant electrical distribution / MCC basis | Primary Services: agitation, pumping, fans, controls, and heat-recovery auxiliaries | Standby / Design Note: feasibility standby allowance embedded in peak demand factor

Utility Peak and Annualized Demand

Table 111. Utility Peak and Annualized Demand

Columns: Utility; Normal Load; Units; Peak Factor; Peak Load; Annualized Usage; Cost Proxy
Row 1
- Utility: Steam
- Normal Load: 71672.825
- Units: kg/h
- Peak Factor: 1.08
- Peak Load: 77406.651
- Annualized Usage: 567648774.0 kg/y
- Cost Proxy: 1021767793 INR/y

Row 2
- Utility: Cooling water

Normal Load: 2778.003
Units: m3/h
Peak Factor: 1.10
Peak Load: 3055.803
Annualized Usage: 22001783.8 m3/y
Cost Proxy: 176014270 INR/y
Row 3
Utility: Electricity
Normal Load: 28732.696
Units: kW
Peak Factor: 1.15
Peak Load: 33042.600
Annualized Usage: 227562952.3 kWh/y
Cost Proxy: 1934285095 INR/y
Row 4
Utility: DM water
Normal Load: 2.000
Units: m3/h
Peak Factor: 1.05
Peak Load: 2.100
Annualized Usage: 15840.0 m3/y
Cost Proxy: screening service allowance
Row 5
Utility: Nitrogen
Normal Load: 3.608
Units: Nm3/h
Peak Factor: 1.20
Peak Load: 4.330
Annualized Usage: 28575.4 Nm3/y
Cost Proxy: screening service allowance
Row 6
Utility: Heat-integration auxiliaries
Normal Load: 16.166
Units: kW
Peak Factor: 1.15
Peak Load: 18.591
Annualized Usage: 128034.7 kWh/y
Cost Proxy: 1088295 INR/y

Utility Demand by Major Unit

Table 112. Utility Demand by Major Unit

Columns: Utility; Contributor; Estimated Demand; Basis
Row 1: Utility: Steam | Contributor: E-101 | Estimated Demand: 4532.727 kW | Basis: Feed preheat to operating temperature.
Row 2: Utility: Steam | Contributor: SEP-201 | Estimated Demand: 1248.737 kW | Basis: Generic purification duty.
Row 3: Utility: Cooling water | Contributor: R-101 | Estimated Demand: 7600.213 kW | Basis: Net reactor thermal duty from the selected thermodynamic basis.
Row 4: Utility: Cooling water | Contributor: SEP-201 | Estimated Demand: 624.369 kW | Basis: Generic purification duty.
Row 5: Utility: Electricity | Contributor: TK-301_pump | Estimated Demand: 159.360 kW | Basis: storage transfer and dispatch pumping basis
Row 6: Utility: Heat-integration auxiliaries | Contributor: omega_catalytic_pinch_htm__shared_htm | Estimated Demand: 42530.992 kW recovered | Basis: selected heat-recovery train circulation, controls, and HTM support
Row 7: Utility: Nitrogen | Contributor: TK-301 | Estimated Demand: 3.608 Nm3/h | Basis: storage blanketing and inerting demand
Row 8: Utility: DM water | Contributor: boiler_makeup_and_wash | Estimated Demand: 2.000 m3/h | Basis: boiler make-up and wash-service allowance

Utilities

Table 113. Utilities

Columns: Utility; Load; Units; Basis
Row 1: Utility: Steam | Load: 71672.825 | Units: kg/h | Basis: 20.0 bar saturated steam equivalent

Row 2: Utility: Cooling water | Load: 2778.003 | Units: m3/h | Basis: 10 C cooling-water rise
Row 3: Utility: Electricity | Load: 28732.696 | Units: kW | Basis: Agitation, pumps, vacuum auxiliaries, and transfer drives
Row 4: Utility: DM water | Load: 2.000 | Units: m3/h | Basis: Boiler and wash service allowance
Row 5: Utility: Nitrogen | Load: 3.608 | Units: Nm3/h | Basis: Inerting and blanketing
Row 6: Utility: Heat-integration auxiliaries | Load: 16.166 | Units: kW | Basis: Selected utility train circulation, HTM pumping, and exchanger-network auxiliaries

Utility Island Service Basis

Table 114. Utility Island Service Basis

Columns: Island; Role; Header Level; Cluster; Recovered Duty (kW); Target Duty (kW); HTM Inventory (m3); Header Pressure (bar); Control Complexity
Row 1
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Role: shared_htm
Header Level: 1
Cluster: -
Recovered Duty (kW): 42530.992
Target Duty (kW): 42530.992
HTM Inventory (m3): 3.936
Header Pressure (bar): 14.500
Control Complexity: 2.500

Header and Thermal-Loop Basis

Table 115. Header and Thermal-Loop Basis

Columns: Step; Island; Header; Cluster; Role; Family; Pressure (bar); Temperature (C); Service
Row 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Header: 1
Cluster: -
Role: controls
Family: process_exchange
Pressure (bar): 9.00
Temperature (C): 203.0
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Row 2
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Header: 1
Cluster: -
Role: header
Family: process_exchange
Pressure (bar): 14.50
Temperature (C): 205.0
Service: R-rough to D-rough heat recovery network header and isolation package
Row 3
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Header: 1
Cluster: -
Role: circulation
Family: process_exchange
Pressure (bar): 14.50
Temperature (C): 205.0
Service: R-rough to D-rough heat recovery circulation loop
Row 4
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Header: 1

Cluster: -
Role: relief
Family: process_exchange
Pressure (bar): 13.50
Temperature (C): 213.0
Service: R-rough to D-rough heat recovery thermal relief and collection package

Row 5
Step: omega_catalytic_pinch_htm__shared_htm_step_02
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Header: 1
Cluster: -
Role: controls
Family: process_exchange
Pressure (bar): 5.00
Temperature (C): 158.0
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station

Utility Architecture

Selected utility architecture: shared HTM island network (shared_htm) with 1 utility islands.

Selected case id: omega_catalytic_pinch_htm__shared_htm

Selected base case id: omega_catalytic_pinch_htm

Recoverable duty target: 59070.823 kW

Selected annual utility cost: INR 1,171,379,913.01/y

Thermal packet count: 3

Packet-derived exchanger candidates: 2

Composite intervals: 11

Selected utility islands: 1

Selected train steps: 2

19. Instrumentation and Process Control

Instrumentation and Process Control

The deterministic control plan derives 10 loops from 13 major equipment items and 6 utility services. Loops are attached to thermal, pressure, level, feed-stability, startup/shutdown, and override risks rather than generated as generic instrumentation text.

Control Philosophy

Table 116. Control Philosophy

Columns: Parameter; Value
Row 1: Parameter: Selected architecture | Value: cascade_override
Row 2: Parameter: Critical units | Value: R-101, PU-201, TK-301, HX-01-EXP
Row 3: Parameter: Loop count | Value: 10
Row 4: Parameter: Utility services linked | Value: 6
Row 5: Parameter: High-criticality loops | Value: 3

Control Architecture Decision

Selected control architecture: Cascade temperature/pressure control with override protection.

Critical units: R-101, PU-201, TK-301, HX-01-EXP.

Existing control loops captured: 10.

Loop Objective Matrix

Table 117. Loop Objective Matrix

Columns: Loop; Unit; Controlled Variable; Manipulated Variable; Objective; Disturbance Basis; Criticality
Row 1
Loop: TIC-R-101
Unit: R-101
Controlled Variable: Tubular Plug Flow Hydrator temperature
Manipulated Variable: Cooling or heating duty
Objective: Maintain reactor temperature within the conversion/selectivity window.
Disturbance Basis: Reaction heat release, feed-ratio drift, and utility temperature fluctuations.
Criticality: high
Row 2
Loop: PIC-R-101
Unit: R-101
Controlled Variable: Tubular Plug Flow Hydrator pressure
Manipulated Variable: Back-pressure control
Objective: Keep the reactor inside the design-pressure and phase-stability envelope.
Disturbance Basis: Gas evolution, condenser instability, and downstream restriction changes.
Criticality: high
Row 3
Loop: FRC-R-101
Unit: R-101
Controlled Variable: Tubular Plug Flow Hydrator feed ratio
Manipulated Variable: Lead/lag feed flow trim
Objective: Maintain stoichiometric or dilution ratio through startup and steady operation.
Disturbance Basis: Upstream feed composition change and pump-speed drift.
Criticality: medium
Row 4
Loop: LIC-PU-201
Unit: PU-201
Controlled Variable: Distillation and purification train level
Manipulated Variable: Bottoms withdrawal

Objective: Maintain bottoms inventory and protect internals from flooding or dry operation.
Disturbance Basis: Feed swings, vapor-rate changes, and downstream withdrawal interruptions.
Criticality: medium

Row 5

Loop: PIC-PU-201
Unit: PU-201
Controlled Variable: Distillation and purification train pressure
Manipulated Variable: Condenser or vent-duty trim
Objective: Hold the main separation unit on its target pressure / vacuum basis.
Disturbance Basis: Condenser duty swings, vent-load changes, and upstream composition disturbance.
Criticality: high

Row 6

Loop: FIC-PU-201
Unit: PU-201
Controlled Variable: Distillation and purification train quality driver
Manipulated Variable: Reflux / solvent / circulation flow
Objective: Hold the separation-driving flow that protects product quality and capture/yield targets.
Disturbance Basis: Feed flow/composition variation and utility availability changes.
Criticality: medium

Row 7

Loop: LIC-TK-301
Unit: TK-301
Controlled Variable: Ethylene Glycol storage via vertical tank farm level
Manipulated Variable: Dispatch pump permissive
Objective: Prevent overflow and dry-running while maintaining dispatch-ready inventory.
Disturbance Basis: Filling / dispatch mismatch and restart buffer drawdown.
Criticality: medium

Row 8

Loop: PIC-TK-301
Unit: TK-301
Controlled Variable: Ethylene Glycol storage via vertical tank farm blanketing pressure
Manipulated Variable: Nitrogen make-up / vent trim
Objective: Maintain safe storage pressure and inert atmosphere for inventory handling.
Disturbance Basis: Ambient temperature swing, filling surge, and dispatch withdrawal.
Criticality: medium

Row 9

Loop: LIC-HX-01-EXP
Unit: HX-01-EXP
Controlled Variable: R-rough to D-rough heat recovery HTM expansion and inventory hold-up level
Manipulated Variable: Dispatch pump permissive
Objective: Prevent overflow and dry-running while maintaining dispatch-ready inventory.
Disturbance Basis: Filling / dispatch mismatch and restart buffer drawdown.
Criticality: medium

Row 10

Loop: PIC-HX-01-EXP
Unit: HX-01-EXP
Controlled Variable: R-rough to D-rough heat recovery HTM expansion and inventory hold-up blanketing pressure
Manipulated Variable: Nitrogen make-up / vent trim
Objective: Maintain safe storage pressure and inert atmosphere for inventory handling.
Disturbance Basis: Ambient temperature swing, filling surge, and dispatch withdrawal.
Criticality: medium

Controlled and Manipulated Variable Register

Table 118. Controlled and Manipulated Variable Register

Columns: Loop; Unit; Family; Controlled Variable; Manipulated Variable; Sensor; Actuator; Criticality

Row 1
- Loop: TIC-R-101
- Unit: R-101
- Family: temperature_cascade
- Controlled Variable: Tubular Plug Flow Hydrator temperature
- Manipulated Variable: Cooling or heating duty
- Sensor: Temperature transmitter
- Actuator: Control valve
- Criticality: high

Row 2
- Loop: PIC-R-101
- Unit: R-101
- Family: pressure_override
- Controlled Variable: Tubular Plug Flow Hydrator pressure
- Manipulated Variable: Back-pressure control
- Sensor: Pressure transmitter
- Actuator: Control valve
- Criticality: high

Row 3
- Loop: FRC-R-101
- Unit: R-101
- Family: feed_ratio
- Controlled Variable: Tubular Plug Flow Hydrator feed ratio
- Manipulated Variable: Lead/lag feed flow trim
- Sensor: Ratio station with flow transmitters
- Actuator: Feed control valve / VFD
- Criticality: medium

Row 4
- Loop: LIC-PU-201
- Unit: PU-201
- Family: inventory_level
- Controlled Variable: Distillation and purification train level
- Manipulated Variable: Bottoms withdrawal
- Sensor: Level transmitter
- Actuator: Control valve
- Criticality: medium

Row 5
- Loop: PIC-PU-201
- Unit: PU-201
- Family: pressure_regulatory
- Controlled Variable: Distillation and purification train pressure
- Manipulated Variable: Condenser or vent-duty trim
- Sensor: Pressure transmitter
- Actuator: Control valve
- Criticality: high

Row 6
- Loop: FIC-PU-201
- Unit: PU-201
- Family: quality_reflux_or_solvent
- Controlled Variable: Distillation and purification train quality driver
- Manipulated Variable: Reflux / solvent / circulation flow
- Sensor: Flow transmitter with composition or ratio bias
- Actuator: Control valve / VFD
- Criticality: medium

Row 7
- Loop: LIC-TK-301
- Unit: TK-301
- Family: inventory_level
- Controlled Variable: Ethylene Glycol storage via vertical tank farm level
- Manipulated Variable: Dispatch pump permissive
- Sensor: Level transmitter
- Actuator: Pump trip
- Criticality: medium

Row 8
- Loop: PIC-TK-301
- Unit: TK-301
- Family: blanketing_pressure
- Controlled Variable: Ethylene Glycol storage via vertical tank farm blanketing pressure
- Manipulated Variable: Nitrogen make-up / vent trim
- Sensor: Pressure transmitter

Actuator: Blanketing valve
Criticality: medium

Row 9

Loop: LIC-HX-01-EXP
Unit: HX-01-EXP
Family: inventory_level
Controlled Variable: R-rough to D-rough heat recovery HTM expansion and inventory hold-up level
Manipulated Variable: Dispatch pump permissive
Sensor: Level transmitter
Actuator: Pump trip
Criticality: medium

Row 10

Loop: PIC-HX-01-EXP
Unit: HX-01-EXP
Family: blanketing_pressure
Controlled Variable: R-rough to D-rough heat recovery HTM expansion and inventory hold-up blanketing pressure
Manipulated Variable: Nitrogen make-up / vent trim
Sensor: Pressure transmitter
Actuator: Blanketing valve
Criticality: medium

Startup, Shutdown, and Override Basis

Table 119. Startup, Shutdown, and Override Basis

Columns: Loop; Startup Logic; Shutdown Logic; Override / Permissive Basis
Row 1: Loop: TIC-R-101 | Startup Logic: Ramp utility service first, then introduce feed under low-rate permissive until reactor temperature stabilizes. | Shutdown Logic: Isolate feeds, hold cooling/inerting until the reactor contents fall below the safe residual-temperature limit. | Override / Permissive Basis: High-high temperature override closes feed and drives maximum cooling / quench response.
Row 2: Loop: PIC-R-101 | Startup Logic: Hold reactor on pressure trim while vent / recycle path is proven open before feed escalation. | Shutdown Logic: Depressure through the controlled path after feed isolation and maintain inert purge if required. | Override / Permissive Basis: High-pressure override opens relief / vent control path and blocks further feed increase.
Row 3: Loop: FRC-R-101 | Startup Logic: Keep secondary feed on ratio-follow mode to the lead feed until circulation and temperature are stable. | Shutdown Logic: Drive secondary feed closed first, then ramp down the lead feed to avoid off-ratio inventory. | Override / Permissive Basis: High-reactor-temperature or low-circulation permissive forces low-feed bias and ratio hold.
Row 4: Loop: LIC-PU-201 | Startup Logic: Establish reflux / circulation first, then release bottoms under minimum-level permissive. | Shutdown Logic: Reduce feed, maintain reflux / boil-up as needed, and drain to the safe low-level setpoint. | Override / Permissive Basis: High-high level overrides bottoms withdrawal or feed cutback depending on service family.
Row 5: Loop: PIC-PU-201 | Startup Logic: Pull pressure into band before advancing feed and enabling automatic reflux or solvent circulation. | Shutdown Logic: Unload feed, keep pressure control active during rundown, then isolate to safe ambient / inert condition. | Override / Permissive Basis: Low-pressure / high-pressure override biases vent or condenser duty to prevent air ingress or overpressure.
Row 6: Loop: FIC-PU-201 | Startup Logic: Run on minimum stable recycle/reflux, then close quality target once pressure and inventory loops settle. | Shutdown Logic: Return manipulated flow to rundown minimum before stopping duty or feed sources. | Override / Permissive Basis: Analyzer or high-differential-pressure override trims reflux / solvent / circulation to a safe fallback value.
Row 7: Loop: LIC-TK-301 | Startup Logic: Keep dispatch permissive blocked until minimum operating inventory is restored. | Shutdown Logic: Stop transfer on low-low level and retain restart inventory if the plant is entering outage mode. | Override / Permissive Basis: High-high level override stops filling and opens alternate routing if configured.
Row 8: Loop: PIC-TK-301 | Startup Logic: Establish blanketing pressure before opening transfer inlets or outlets. | Shutdown Logic: Maintain minimum blanketing setpoint through static storage and isolation. | Override / Permissive Basis: High-pressure / low-pressure override trims

venting or nitrogen make-up ahead of tank relief action.
Row 9: Loop: LIC-HX-01-EXP | Startup Logic: Keep dispatch permissive blocked until minimum operating inventory is restored. | Shutdown Logic: Stop transfer on low-low level and retain restart inventory if the plant is entering outage mode. | Override / Permissive Basis: High-high level override stops filling and opens alternate routing if configured.
Row 10: Loop: PIC-HX-01-EXP | Startup Logic: Establish blanketing pressure before opening transfer inlets or outlets. | Shutdown Logic: Maintain minimum blanketing setpoint through static storage and isolation. | Override / Permissive Basis: High-pressure / low-pressure override trims venting or nitrogen make-up ahead of tank relief action.

Alarm and Interlock Basis

Table 120. Alarm and Interlock Basis

Columns: Loop; Deviation; Protective Layer; Purpose
Row 1: Loop: TIC-R-101 | Deviation: High/low Tubular Plug Flow Hydrator temperature | Protective Layer: alarm + override / interlock response | Purpose: Feeds reactor runaway protection, utility permissives, and emergency feed isolation logic.
Row 2: Loop: PIC-R-101 | Deviation: High/low Tubular Plug Flow Hydrator pressure | Protective Layer: alarm + override / interlock response | Purpose: Supports relief sizing assumptions, pressure alarms, and shutdown permissives.
Row 3: Loop: FRC-R-101 | Deviation: High/low Tubular Plug Flow Hydrator feed ratio | Protective Layer: alarm + operator response | Purpose: Links throughput control to reactor temperature and pressure safeguard logic.
Row 4: Loop: LIC-PU-201 | Deviation: High/low Distillation and purification train level | Protective Layer: alarm + operator response | Purpose: Protects internals, reboiler circulation, and downstream product-quality stability.
Row 5: Loop: PIC-PU-201 | Deviation: High/low Distillation and purification train pressure | Protective Layer: alarm + override / interlock response | Purpose: Feeds column pressure alarms, vacuum protection, and offgas routing logic.
Row 6: Loop: FIC-PU-201 | Deviation: High/low Distillation and purification train quality driver | Protective Layer: alarm + operator response | Purpose: Ties separation quality control to utility duty, flooding risk, and off-spec prevention.
Row 7: Loop: LIC-TK-301 | Deviation: High/low Ethylene Glycol storage via vertical tank farm level | Protective Layer: alarm + operator response | Purpose: Supports storage overflow protection and working-inventory assumptions.
Row 8: Loop: PIC-TK-301 | Deviation: High/low Ethylene Glycol storage via vertical tank farm blanketing pressure | Protective Layer: alarm + operator response | Purpose: Links storage pressure control to inerting and vent-protection assumptions.
Row 9: Loop: LIC-HX-01-EXP | Deviation: High/low R-rough to D-rough heat recovery HTM expansion and inventory hold-up level | Protective Layer: alarm + operator response | Purpose: Supports storage overflow protection and working-inventory assumptions.
Row 10: Loop: PIC-HX-01-EXP | Deviation: High/low R-rough to D-rough heat recovery HTM expansion and inventory hold-up blanketing pressure | Protective Layer: alarm + operator response | Purpose: Links storage pressure control to inerting and vent-protection assumptions.

Utility-Integrated Control Basis

Table 121. Utility-Integrated Control Basis

Columns: Utility; Load; Control Linkage
Row 1: Utility: Steam | Load: 71672.825 kg/h | Control Linkage: temperature / pressure loops depend on steam or cooling-service availability
Row 2: Utility: Cooling water | Load: 2778.003 m3/h | Control Linkage: temperature / pressure loops depend on steam or cooling-service availability
Row 3: Utility: Electricity | Load: 28732.696 kW | Control Linkage: inventory, purge, or protective logic references this plant utility service
Row 4: Utility: DM water | Load: 2.000 m3/h | Control Linkage: inventory,

purge, or protective logic references this plant utility service
Row 5: Utility: Nitrogen | Load: 3.608 Nm3/h | Control Linkage: inventory, purge, or protective logic references this plant utility service
Row 6: Utility: Heat-integration auxiliaries | Load: 16.166 kW | Control Linkage: inventory, purge, or protective logic references this plant utility service

Control Loop Sheets

Table 122. Control Loop Sheets

Columns: Loop; Unit; Family; Sensor; Actuator; Objective; Startup Basis; Override Basis

Row 1
- Loop: TIC-R-101
- Unit: R-101
- Family: temperature_cascade
- Sensor: Temperature transmitter
- Actuator: Control valve
- Objective: Maintain reactor temperature within the conversion/selectivity window.
- Startup Basis: Ramp utility service first, then introduce feed under low-rate permissive until reactor temperature stabilizes.
- Override Basis: High-high temperature override closes feed and drives maximum cooling / quench response.

Row 2
- Loop: PIC-R-101
- Unit: R-101
- Family: pressure_override
- Sensor: Pressure transmitter
- Actuator: Control valve
- Objective: Keep the reactor inside the design-pressure and phase-stability envelope.
- Startup Basis: Hold reactor on pressure trim while vent / recycle path is proven open before feed escalation.
- Override Basis: High-pressure override opens relief / vent control path and blocks further feed increase.

Row 3
- Loop: FRC-R-101
- Unit: R-101
- Family: feed_ratio
- Sensor: Ratio station with flow transmitters
- Actuator: Feed control valve / VFD
- Objective: Maintain stoichiometric or dilution ratio through startup and steady operation.
- Startup Basis: Keep secondary feed on ratio-follow mode to the lead feed until circulation and temperature are stable.
- Override Basis: High-reactor-temperature or low-circulation permissive forces low-feed bias and ratio hold.

Row 4
- Loop: LIC-PU-201
- Unit: PU-201
- Family: inventory_level
- Sensor: Level transmitter
- Actuator: Control valve
- Objective: Maintain bottoms inventory and protect internals from flooding or dry operation.
- Startup Basis: Establish reflux / circulation first, then release bottoms under minimum-level permissive.
- Override Basis: High-high level overrides bottoms withdrawal or feed cutback depending on service family.

Row 5
- Loop: PIC-PU-201
- Unit: PU-201
- Family: pressure_regulatory
- Sensor: Pressure transmitter
- Actuator: Control valve
- Objective: Hold the main separation unit on its target pressure / vacuum basis.
- Startup Basis: Pull pressure into band before advancing feed and enabling automatic reflux or solvent circulation.
- Override Basis: Low-pressure / high-pressure override biases vent or

condenser duty to prevent air ingress or overpressure.

Row 6

Loop: FIC-PU-201
Unit: PU-201
Family: quality_reflux_or_solvent
Sensor: Flow transmitter with composition or ratio bias
Actuator: Control valve / VFD
Objective: Hold the separation-driving flow that protects product quality and capture/yield targets.
Startup Basis: Run on minimum stable recycle/reflux, then close quality target once pressure and inventory loops settle.
Override Basis: Analyzer or high-differential-pressure override trims reflux / solvent / circulation to a safe fallback value.

Row 7

Loop: LIC-TK-301
Unit: TK-301
Family: inventory_level
Sensor: Level transmitter
Actuator: Pump trip
Objective: Prevent overflow and dry-running while maintaining dispatch-ready inventory.
Startup Basis: Keep dispatch permissive blocked until minimum operating inventory is restored.
Override Basis: High-high level override stops filling and opens alternate routing if configured.

Row 8

Loop: PIC-TK-301
Unit: TK-301
Family: blanketing_pressure
Sensor: Pressure transmitter
Actuator: Blanketing valve
Objective: Maintain safe storage pressure and inert atmosphere for inventory handling.
Startup Basis: Establish blanketing pressure before opening transfer inlets or outlets.
Override Basis: High-pressure / low-pressure override trims venting or nitrogen make-up ahead of tank relief action.

Row 9

Loop: LIC-HX-01-EXP
Unit: HX-01-EXP
Family: inventory_level
Sensor: Level transmitter
Actuator: Pump trip
Objective: Prevent overflow and dry-running while maintaining dispatch-ready inventory.
Startup Basis: Keep dispatch permissive blocked until minimum operating inventory is restored.
Override Basis: High-high level override stops filling and opens alternate routing if configured.

Row 10

Loop: PIC-HX-01-EXP
Unit: HX-01-EXP
Family: blanketing_pressure
Sensor: Pressure transmitter
Actuator: Blanketing valve
Objective: Maintain safe storage pressure and inert atmosphere for inventory handling.
Startup Basis: Establish blanketing pressure before opening transfer inlets or outlets.
Override Basis: High-pressure / low-pressure override trims venting or nitrogen make-up ahead of tank relief action.

20. HAZOP

HAZOP

HAZOP Coverage Summary

Table 123. HAZOP Coverage Summary

Columns: Parameter; Value
Row 1: Parameter: Node count | Value: 7
Row 2: Parameter: High-severity route hazards | Value: 0
Row 3: Parameter: Control loops linked | Value: 10
Row 4: Parameter: Coverage summary | Value: 7 HAZOP nodes across 4 node families derived from the selected flowsheet and equipment list.
Row 5: Parameter: Node families | Value: column_or_main_separation, reactor, storage, thermal_exchange

HAZOP Node Basis

Table 124. HAZOP Node Basis

Columns: Node; Family; Design Intent; Parameter; Guide Word; Deviation; Linked Control Loops
Row 1
- Node: R-101
- Family: reactor
- Design Intent: Maintain controlled conversion at safe temperature and pressure.
- Parameter: Temperature
- Guide Word: More
- Deviation: Higher than intended reactor temperature
- Linked Control Loops: TIC-R-101; PIC-R-101; FRC-R-101

Row 2
- Node: PU-201
- Family: column_or_main_separation
- Design Intent: Maintain stable separation pressure, inventory, and quality-driving internal flows.
- Parameter: Pressure
- Guide Word: Less
- Deviation: Lower than intended main-separation pressure
- Linked Control Loops: LIC-PU-201; PIC-PU-201; FIC-PU-201

Row 3
- Node: E-101
- Family: thermal_exchange
- Design Intent: Deliver required thermal duty without tube-side / shell-side upset or loss of containment.
- Parameter: Heat transfer
- Guide Word: Less
- Deviation: Lower than intended thermal duty
- Linked Control Loops: TIC-R-101; PIC-R-101

Row 4
- Node: TK-301
- Family: storage
- Design Intent: Hold safe inventory, containment, and blanketing during receipt, storage, and dispatch.
- Parameter: Level
- Guide Word: More
- Deviation: Higher than intended storage level
- Linked Control Loops: LIC-TK-301; PIC-TK-301

Row 5
- Node: HX-01
- Family: thermal_exchange
- Design Intent: Deliver required thermal duty without tube-side / shell-side upset or loss of containment.
- Parameter: Heat transfer
- Guide Word: Less
- Deviation: Lower than intended thermal duty
- Linked Control Loops: TIC-R-101; PIC-R-101

Row 6

Node: HX-01-EXP
Family: storage
Design Intent: Hold safe inventory, containment, and blanketing during receipt, storage, and dispatch.
Parameter: Level
Guide Word: More
Deviation: Higher than intended storage level
Linked Control Loops: LIC-HX-01-EXP; PIC-HX-01-EXP

Row 7

Node: HX-02
Family: thermal_exchange
Design Intent: Deliver required thermal duty without tube-side / shell-side upset or loss of containment.
Parameter: Heat transfer
Guide Word: Less
Deviation: Lower than intended thermal duty
Linked Control Loops: TIC-R-101; PIC-R-101

Critical Node Summary

Table 125. Critical Node Summary

Columns: Node; Family; Deviation; Parameter; Guide Word; Severity; Consequences; Safeguards

Row 1

Node: R-101
Family: reactor
Deviation: Higher than intended reactor temperature
Parameter: Temperature
Guide Word: More
Severity: high
Consequences: Runaway tendency; Selectivity loss; Pressure rise and emergency shutdown risk
Safeguards: TIC-R-101; PIC-R-101; FRC-R-101

Row 2

Node: PU-201
Family: column_or_main_separation
Deviation: Lower than intended main-separation pressure
Parameter: Pressure
Guide Word: Less
Severity: high
Consequences: Air ingress; Column instability; Off-spec separation performance
Safeguards: LIC-PU-201; PIC-PU-201; FIC-PU-201

Row 3

Node: E-101
Family: thermal_exchange
Deviation: Lower than intended thermal duty
Parameter: Heat transfer
Guide Word: Less
Severity: medium
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit
Safeguards: TIC-R-101; PIC-R-101

Row 4

Node: TK-301
Family: storage
Deviation: Higher than intended storage level
Parameter: Level
Guide Word: More
Severity: medium
Consequences: Overflow; Containment loss; Transfer spill / emissions event
Safeguards: LIC-TK-301; PIC-TK-301

Row 5

Node: HX-01
Family: thermal_exchange
Deviation: Lower than intended thermal duty
Parameter: Heat transfer
Guide Word: Less
Severity: medium
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit

Safeguards: TIC-R-101; PIC-R-101
Row 6
Node: HX-01-EXP
Family: storage
Deviation: Higher than intended storage level
Parameter: Level
Guide Word: More
Severity: medium
Consequences: Overflow; Containment loss; Transfer spill / emissions event
Safeguards: LIC-HX-01-EXP; PIC-HX-01-EXP
Row 7
Node: HX-02
Family: thermal_exchange
Deviation: Lower than intended thermal duty
Parameter: Heat transfer
Guide Word: Less
Severity: medium
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit
Safeguards: TIC-R-101; PIC-R-101

Deviation Cause-Consequence Matrix

Table 126. Deviation Cause-Consequence Matrix

Columns: Node; Deviation; Causes; Consequences; Safeguards
Row 1: Node: R-101 | Deviation: Higher than intended reactor temperature | Causes: Cooling failure; Excess reactant feed; Utility temperature excursion | Consequences: Runaway tendency; Selectivity loss; Pressure rise and emergency shutdown risk | Safeguards: TIC-R-101; PIC-R-101; FRC-R-101
Row 2: Node: PU-201 | Deviation: Lower than intended main-separation pressure | Causes: Vacuum overshoot; Condenser instability; Vent control malfunction | Consequences: Air ingress; Column instability; Off-spec separation performance | Safeguards: LIC-PU-201; PIC-PU-201; FIC-PU-201
Row 3: Node: E-101 | Deviation: Lower than intended thermal duty | Causes: Fouling; Utility interruption; Flow maldistribution | Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit | Safeguards: TIC-R-101; PIC-R-101
Row 4: Node: TK-301 | Deviation: Higher than intended storage level | Causes: Blocked outlet; Dispatch mismatch; Filling valve left open | Consequences: Overflow; Containment loss; Transfer spill / emissions event | Safeguards: LIC-TK-301; PIC-TK-301
Row 5: Node: HX-01 | Deviation: Lower than intended thermal duty | Causes: Fouling; Utility interruption; Flow maldistribution | Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit | Safeguards: TIC-R-101; PIC-R-101
Row 6: Node: HX-01-EXP | Deviation: Higher than intended storage level | Causes: Blocked outlet; Dispatch mismatch; Filling valve left open | Consequences: Overflow; Containment loss; Transfer spill / emissions event | Safeguards: LIC-HX-01-EXP; PIC-HX-01-EXP
Row 7: Node: HX-02 | Deviation: Lower than intended thermal duty | Causes: Fouling; Utility interruption; Flow maldistribution | Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit | Safeguards: TIC-R-101; PIC-R-101

HAZOP Node Register

Table 127. HAZOP Node Register

Columns: Node; Family; Parameter; Guide Word; Deviation; Causes; Consequences; Safeguards; Linked Loops; Recommendation
Row 1
Node: R-101
Family: reactor
Parameter: Temperature
Guide Word: More
Deviation: Higher than intended reactor temperature
Causes: Cooling failure; Excess reactant feed; Utility temperature excursion
Consequences: Runaway tendency; Selectivity loss; Pressure rise and

emergency shutdown risk
Safeguards: TIC-R-101; PIC-R-101; FRC-R-101
Linked Loops: TIC-R-101; PIC-R-101; FRC-R-101
Recommendation: Add high-high temperature trip, emergency feed isolation, and confirm cooling-failure response sequence.

Row 2
Node: PU-201
Family: column_or_main_separation
Parameter: Pressure
Guide Word: Less
Deviation: Lower than intended main-separation pressure
Causes: Vacuum overshoot; Condenser instability; Vent control malfunction
Consequences: Air ingress; Column instability; Off-spec separation performance
Safeguards: LIC-PU-201; PIC-PU-201; FIC-PU-201
Linked Loops: LIC-PU-201; PIC-PU-201; FIC-PU-201
Recommendation: Protect low-pressure operation, verify oxygen ingress response, and confirm safe reflux / circulation fallback.

Row 3
Node: E-101
Family: thermal_exchange
Parameter: Heat transfer
Guide Word: Less
Deviation: Lower than intended thermal duty
Causes: Fouling; Utility interruption; Flow maldistribution
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit
Safeguards: TIC-R-101; PIC-R-101
Linked Loops: TIC-R-101; PIC-R-101
Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.

Row 4
Node: TK-301
Family: storage
Parameter: Level
Guide Word: More
Deviation: Higher than intended storage level
Causes: Blocked outlet; Dispatch mismatch; Filling valve left open
Consequences: Overflow; Containment loss; Transfer spill / emissions event
Safeguards: LIC-TK-301; PIC-TK-301
Linked Loops: LIC-TK-301; PIC-TK-301
Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.

Row 5
Node: HX-01
Family: thermal_exchange
Parameter: Heat transfer
Guide Word: Less
Deviation: Lower than intended thermal duty
Causes: Fouling; Utility interruption; Flow maldistribution
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit
Safeguards: TIC-R-101; PIC-R-101
Linked Loops: TIC-R-101; PIC-R-101
Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.

Row 6
Node: HX-01-EXP
Family: storage
Parameter: Level
Guide Word: More
Deviation: Higher than intended storage level
Causes: Blocked outlet; Dispatch mismatch; Filling valve left open
Consequences: Overflow; Containment loss; Transfer spill / emissions event
Safeguards: LIC-HX-01-EXP; PIC-HX-01-EXP
Linked Loops: LIC-HX-01-EXP; PIC-HX-01-EXP
Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.

Row 7
Node: HX-02
Family: thermal_exchange
Parameter: Heat transfer
Guide Word: Less

Deviation: Lower than intended thermal duty
Causes: Fouling; Utility interruption; Flow maldistribution
Consequences: Poor temperature control; Downstream off-spec operation; Potential pressure upset in linked unit
Safeguards: TIC-R-101; PIC-R-101
Linked Loops: TIC-R-101; PIC-R-101
Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.

Recommendation Register

Table 128. Recommendation Register

Columns: Node; Priority; Status; Recommendation; Primary Causes
Row 1: Node: R-101 | Priority: high | Status: open | Recommendation: Add high-high temperature trip, emergency feed isolation, and confirm cooling-failure response sequence. | Primary Causes: Cooling failure; Excess reactant feed; Utility temperature excursion
Row 2: Node: PU-201 | Priority: high | Status: open | Recommendation: Protect low-pressure operation, verify oxygen ingress response, and confirm safe reflux / circulation fallback. | Primary Causes: Vacuum overshoot; Condenser instability; Vent control malfunction
Row 3: Node: E-101 | Priority: medium | Status: open | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response. | Primary Causes: Fouling; Utility interruption; Flow maldistribution
Row 4: Node: TK-301 | Priority: medium | Status: open | Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic. | Primary Causes: Blocked outlet; Dispatch mismatch; Filling valve left open
Row 5: Node: HX-01 | Priority: medium | Status: open | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response. | Primary Causes: Fouling; Utility interruption; Flow maldistribution
Row 6: Node: HX-01-EXP | Priority: medium | Status: open | Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic. | Primary Causes: Blocked outlet; Dispatch mismatch; Filling valve left open
Row 7: Node: HX-02 | Priority: medium | Status: open | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response. | Primary Causes: Fouling; Utility interruption; Flow maldistribution

21. Safety, Health, Environment, and Waste Management

Safety, Health, Environment, and Waste Management

The SHE basis prioritizes EO containment, inerting, high-integrity relief and vent routing, hot glycol burn protection, wastewater segregation, and recovery of water and heavy glycol side streams before discharge or disposal. Vent, spill, and emergency response provisions are tied directly to the hydrator, flash, and column sections rather than presented as generic safety text.

Safety Basis

Table 129. Safety Basis

Columns: Severity; Hazard; Safeguard
Row 1: Severity: Moderate | Hazard: Catalytic loop and CO2 handling complexity | Safeguard: Closed-loop catalyst management and pressure control

Hazard and Emergency Response Basis

Table 130. Hazard and Emergency Response Basis

Columns: Node; Emergency Trigger; Immediate Response; Protective Systems; Follow-Up Recommendation
Row 1: Node: R-101 | Emergency Trigger: Higher than intended reactor temperature | Immediate Response: isolate feeds, stabilize utilities, and hold inventory in safe containment | Protective Systems: TIC-R-101; PIC-R-101; FRC-R-101 | Follow-Up Recommendation: Add high-high temperature trip, emergency feed isolation, and confirm cooling-failure response sequence.
Row 2: Node: PU-201 | Emergency Trigger: Lower than intended main-separation pressure | Immediate Response: isolate feeds, stabilize utilities, and hold inventory in safe containment | Protective Systems: LIC-PU-201; PIC-PU-201; FIC-PU-201 | Follow-Up Recommendation: Protect low-pressure operation, verify oxygen ingress response, and confirm safe reflux / circulation fallback.
Row 3: Node: E-101 | Emergency Trigger: Lower than intended thermal duty | Immediate Response: stop transfer / circulation and move to contained safe state | Protective Systems: TIC-R-101; PIC-R-101 | Follow-Up Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.
Row 4: Node: TK-301 | Emergency Trigger: Higher than intended storage level | Immediate Response: stop transfer / circulation and move to contained safe state | Protective Systems: LIC-TK-301; PIC-TK-301 | Follow-Up Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.
Row 5: Node: HX-01 | Emergency Trigger: Lower than intended thermal duty | Immediate Response: stop transfer / circulation and move to contained safe state | Protective Systems: TIC-R-101; PIC-R-101 | Follow-Up Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.
Row 6: Node: HX-01-EXP | Emergency Trigger: Higher than intended storage level | Immediate Response: stop transfer / circulation and move to contained safe state | Protective Systems: LIC-HX-01-EXP; PIC-HX-01-EXP | Follow-Up Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.
Row 7: Node: HX-02 | Emergency Trigger: Lower than intended thermal duty | Immediate Response: stop transfer / circulation and move to contained safe state | Protective Systems: TIC-R-101; PIC-R-101 | Follow-Up Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.

Health and Exposure Basis

Table 131. Health and Exposure Basis

Columns: Topic; Exposure Basis; Monitoring / Verification; PPE / Controls;

Operational Linkage
Row 1: Topic: Operator exposure | Exposure Basis: feed / product / intermediate contact and vapor release during sampling, transfer, and upset response | Monitoring / Verification: exposure monitoring at sampling, transfer, and vent-handling points | PPE / Controls: service-specific PPE, chemical-resistant gloves, eye protection, and respiratory precautions where indicated | Operational Linkage: linked to selected route family and HAZOP node families
Row 2: Topic: Maintenance exposure | Exposure Basis: equipment opening, filter / cake handling, exchanger cleaning, and confined-space entry risk | Monitoring / Verification: permit-to-work, isolation verification, and post-cleaning gas test basis | PPE / Controls: maintenance PPE upgraded for corrosive, hot, or solids-bearing service | Operational Linkage: applies during shutdown, cleanup, and turnaround activity
Row 3: Topic: Training and competency | Exposure Basis: routine operating, emergency isolation, inerting, and chemical handling scenarios | Monitoring / Verification: startup, shutdown, and emergency drill coverage tied to critical loops and HAZOP nodes | PPE / Controls: role-specific operating and emergency response training | Operational Linkage: control philosophy and HAZOP outputs are used as the training basis

Environmental and Waste Basis

Table 132. Environmental and Waste Basis

Columns: Area; Release / Load Basis; Control Basis; Monitoring Basis
Row 1: Area: Air emissions | Release / Load Basis: vent, offgas, absorber discharge, dryer exhaust, and utility-release points remain tied to selected separation and utility architecture | Control Basis: route-family vent routing, control-loop-linked upset response, and utility-integrated safeguards | Monitoring Basis: periodic vent / stack monitoring plus upset-event logging
Row 2: Area: Wastewater | Release / Load Basis: aqueous purge, wash, and cleanup streams require segregation by corrosive / solids / organics burden | Control Basis: segregated collection, neutralization / equalization, and load-based downstream treatment | Monitoring Basis: screening effluent characterization and routed discharge control
Row 3: Area: Solid and hazardous waste | Release / Load Basis: filter cake, spent media, catalyst, contaminated solids, and maintenance residues require route-family-specific handling | Control Basis: contained storage, labeled handling, recovery / disposal routing, and turnaround waste segregation | Monitoring Basis: batch / campaign waste tracking and controlled offsite disposition basis

Environmental Control and Monitoring Basis

Table 133. Environmental Control and Monitoring Basis

Columns: Utility / Service; Load; SHE Linkage
Row 1: Utility / Service: Steam | Load: 71672.825 kg/h | SHE Linkage: utility reliability is tied to SHE performance for temperature / pressure / inerting control
Row 2: Utility / Service: Cooling water | Load: 2778.003 m3/h | SHE Linkage: utility reliability is tied to SHE performance for temperature / pressure / inerting control
Row 3: Utility / Service: Electricity | Load: 28732.696 kW | SHE Linkage: utility demand contributes to environmental load and emergency operability basis
Row 4: Utility / Service: DM water | Load: 2.000 m3/h | SHE Linkage: utility demand contributes to environmental load and emergency operability basis
Row 5: Utility / Service: Nitrogen | Load: 3.608 Nm3/h | SHE Linkage: utility reliability is tied to SHE performance for temperature / pressure / inerting control
Row 6: Utility / Service: Heat-integration auxiliaries | Load: 16.166 kW | SHE Linkage: utility demand contributes to environmental load and emergency operability basis

Waste Handling and Disposal Basis

Table 134. Waste Handling and Disposal Basis

Columns: Waste Class; Typical Source; Handling Basis; Disposition Basis
Row 1: Waste Class: Vent / offgas condensate and purge | Typical Source: reactor, separation, and utility upset / normal purge points | Handling Basis: segregate by corrosive / organic / inerted service and route to recovery or controlled treatment | Disposition Basis: selected route-family vent handling basis
Row 2: Waste Class: Aqueous wash and cleanup streams | Typical Source: sampling, washdown, maintenance, and aqueous separation service | Handling Basis: collect separately and treat as route-specific wastewater load | Disposition Basis: neutralization / equalization before discharge screening
Row 3: Waste Class: Solid waste and contaminated media | Typical Source: filters, dryers, catalyst/media replacement, and solids cleanup | Handling Basis: closed handling, labeled storage, and recovery / disposal by waste class | Disposition Basis: solid-waste and hazardous-waste routing basis

Safeguard Linkage

Table 135. Safeguard Linkage

Columns: Node; Safeguards; Linked Loops; Recommendation
Row 1: Node: R-101 | Safeguards: TIC-R-101; PIC-R-101; FRC-R-101 | Linked Loops: TIC-R-101; PIC-R-101; FRC-R-101 | Recommendation: Add high-high temperature trip, emergency feed isolation, and confirm cooling-failure response sequence.
Row 2: Node: PU-201 | Safeguards: LIC-PU-201; PIC-PU-201; FIC-PU-201 | Linked Loops: LIC-PU-201; PIC-PU-201; FIC-PU-201 | Recommendation: Protect low-pressure operation, verify oxygen ingress response, and confirm safe reflux / circulation fallback.
Row 3: Node: E-101 | Safeguards: TIC-R-101; PIC-R-101 | Linked Loops: TIC-R-101; PIC-R-101 | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.
Row 4: Node: TK-301 | Safeguards: LIC-TK-301; PIC-TK-301 | Linked Loops: LIC-TK-301; PIC-TK-301 | Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.
Row 5: Node: HX-01 | Safeguards: TIC-R-101; PIC-R-101 | Linked Loops: TIC-R-101; PIC-R-101 | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.
Row 6: Node: HX-01-EXP | Safeguards: LIC-HX-01-EXP; PIC-HX-01-EXP | Linked Loops: LIC-HX-01-EXP; PIC-HX-01-EXP | Recommendation: Provide level interlocks, overflow routing checks, and confirm dispatch / receipt isolation logic.
Row 7: Node: HX-02 | Safeguards: TIC-R-101; PIC-R-101 | Linked Loops: TIC-R-101; PIC-R-101 | Recommendation: Confirm utility low-flow alarms, bypass logic, and exchanger isolation / cleaning response.

22. Project and Plant Layout

Project and Plant Layout

Selected layout basis: Segregated hazard-zone layout. The scored winner is tied to Dahej, 13 major equipment items, 6 utility services, and 7 solved flowsheet nodes. The layout narrative therefore follows the scored arrangement rather than freewriting a generic plot plan.

Plot Plan Basis

Table 136. Plot Plan Basis

Columns: Parameter; Value
Row 1: Parameter: Selected site | Value: Dahej
Row 2: Parameter: Winning basis | Value: Segregated hazard-zone layout
Row 3: Parameter: Major equipment count | Value: 13
Row 4: Parameter: Flowsheet node count | Value: 7
Row 5: Parameter: Operating mode | Value: continuous
Row 6: Parameter: Campaign length (d) | Value: 170.0
Row 7: Parameter: Dispatch buffer (d) | Value: 9.4

Plot Layout Schematic

Figure 2. Plot Layout Schematic

```
mermaid
flowchart LR
tank_farm["Tank farm / receipt zone\nfeed_prep, storage"]
reaction_zone["Process reaction zone\nreactor"]
separation_zone["Separation and finishing zone\npurification"]
waste_zone["Waste / recovery zone\nprimary_flash, recycle_recovery, waste_treatment"]
general_zone["General process block\nnone"]
tank_farm --> reaction_zone
reaction_zone --> separation_zone
separation_zone --> waste_zone
```

Area Zoning and Separation Basis

Table 137. Area Zoning and Separation Basis

Columns: Zone; Representative Nodes; Primary Layout Driver; Access / Separation Note
Row 1: Zone: Tank farm / receipt zone | Representative Nodes: feed_prep, storage | Primary Layout Driver: hazard segregation and controlled access | Access / Separation Note: truck / service access maintained
Row 2: Zone: Process reaction zone | Representative Nodes: reactor | Primary Layout Driver: hazard segregation and controlled access | Access / Separation Note: buffered from dispatch / occupied areas
Row 3: Zone: Waste / recovery zone | Representative Nodes: primary_flash, recycle_recovery, waste_treatment | Primary Layout Driver: short process transfer and operability grouping | Access / Separation Note: kept accessible for maintenance and utility routing
Row 4: Zone: Separation and finishing zone | Representative Nodes: purification | Primary Layout Driver: short process transfer and operability grouping | Access / Separation Note: kept accessible for maintenance and utility routing

Equipment Placement Matrix

Table 138. Equipment Placement Matrix

Columns: Node; Label; Section; Upstream; Downstream

Row 1: Node: feed_prep | Label: Feed preparation | Section: feed_handling | Upstream: purification | Downstream: reactor
Row 2: Node: reactor | Label: Reaction zone | Section: reaction | Upstream: feed_prep | Downstream: primary_flash
Row 3: Node: primary_flash | Label: Primary flash and purge recovery | Section: primary_recovery | Upstream: reactor | Downstream: waste_treatment, recycle_recovery
Row 4: Node: waste_treatment | Label: Waste treatment | Section: - | Upstream: primary_flash, purification | Downstream: -
Row 5: Node: recycle_recovery | Label: Recycle recovery | Section: recycle_recovery_carbonate_loop_cleanup | Upstream: primary_flash | Downstream: purification
Row 6: Node: purification | Label: Purification train | Section: purification | Upstream: recycle_recovery | Downstream: feed_prep, storage, waste_treatment
Row 7: Node: storage | Label: Product storage | Section: - | Upstream: purification | Downstream: -

Utility Corridor Matrix

Table 139. Utility Corridor Matrix

Columns: Utility; Corridor Type; Routing Basis; Spacing / Access Note
Row 1: Utility: Steam | Corridor Type: header / rack corridor | Routing Basis: parallel to process train with branch take-offs at major units | Spacing / Access Note: kept off main truck path and outside maintenance lift envelope
Row 2: Utility: Cooling water | Corridor Type: header / rack corridor | Routing Basis: parallel to process train with branch take-offs at major units | Spacing / Access Note: kept off main truck path and outside maintenance lift envelope
Row 3: Utility: Electricity | Corridor Type: electrical and local service corridor | Routing Basis: distributed to storage, purge, and utility consumers with isolation access | Spacing / Access Note: routed with safe separation from wet and hot service zones
Row 4: Utility: DM water | Corridor Type: header / rack corridor | Routing Basis: distributed to storage, purge, and utility consumers with isolation access | Spacing / Access Note: kept off main truck path and outside maintenance lift envelope
Row 5: Utility: Nitrogen | Corridor Type: header / rack corridor | Routing Basis: distributed to storage, purge, and utility consumers with isolation access | Spacing / Access Note: kept off main truck path and outside maintenance lift envelope
Row 6: Utility: Heat-integration auxiliaries | Corridor Type: electrical and local service corridor | Routing Basis: distributed to storage, purge, and utility consumers with isolation access | Spacing / Access Note: kept off main truck path and outside maintenance lift envelope

Utility Routing and Access Basis

Table 140. Utility Routing and Access Basis

Columns: Utility; Basis; Routing Strategy; Layout Linkage
Row 1: Utility: Steam | Basis: 20.0 bar saturated steam equivalent | Routing Strategy: process-side corridor to reaction and separation core | Layout Linkage: Segregated hazard-zone layout
Row 2: Utility: Cooling water | Basis: 10 C cooling-water rise | Routing Strategy: process-side corridor to reaction and separation core | Layout Linkage: Segregated hazard-zone layout
Row 3: Utility: Electricity | Basis: Agitation, pumps, vacuum auxiliaries, and transfer drives | Routing Strategy: service corridor routed away from primary hazard cluster | Layout Linkage: Segregated hazard-zone layout
Row 4: Utility: DM water | Basis: Boiler and wash service allowance | Routing Strategy: service corridor routed away from primary hazard cluster | Layout Linkage: Segregated hazard-zone layout
Row 5: Utility: Nitrogen | Basis: Inerting and blanketing | Routing Strategy: tank farm / blanketing header and service access edge | Layout Linkage: Segregated hazard-zone layout
Row 6: Utility: Heat-integration auxiliaries | Basis: Selected utility train circulation, HTM pumping, and exchanger-network auxiliaries | Routing Strategy: process-side corridor to reaction and separation core | Layout Linkage: Segregated hazard-zone layout

Dispatch and Emergency Access Basis

Table 141. Dispatch and Emergency Access Basis

Columns: Topic; Basis; Layout Consequence
Row 1: Topic: Operating mode | Basis: continuous | Layout Consequence: layout keeps routine operator circulation outside the primary hazard core while preserving direct access to critical units
Row 2: Topic: Dispatch and truck movement | Basis: 9.4 d FG buffer | Layout Consequence: storage and loading area placed toward the dispatch edge with maintained truck and emergency vehicle access
Row 3: Topic: Emergency response | Basis: 1 reaction-zone nodes | Layout Consequence: reaction and high-hazard zones remain segregated from occupied / dispatch zones with clear firefighting approach
Row 4: Topic: Maintenance turnaround | Basis: 170.0 d campaign | Layout Consequence: equipment clearance, platform access, and rack tie-ins inform maintenance-side spacing and isolation access

Maintenance and Foundation Basis

Table 142. Maintenance and Foundation Basis

Columns: Equipment; Support Variant; Footprint (m2); Clearance (m); Platform; Rack Tie-In
Row 1: Equipment: R-101 | Support Variant: cylindrical skirt with anchor chair | Footprint (m2): 656.205 | Clearance (m): 1.800 | Platform: yes | Rack Tie-In: no
Row 2: Equipment: PU-201 | Support Variant: cylindrical skirt with anchor chair | Footprint (m2): 6.620 | Clearance (m): 1.800 | Platform: yes | Rack Tie-In: no
Row 3: Equipment: V-101 | Support Variant: grouted base frame | Footprint (m2): 12.125 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: no
Row 4: Equipment: E-101 | Support Variant: dual saddle | Footprint (m2): 47.328 | Clearance (m): 1.500 | Platform: yes | Rack Tie-In: no
Row 5: Equipment: TK-301 | Support Variant: dual saddle | Footprint (m2): 246.043 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: no
Row 6: Equipment: HX-01 | Support Variant: dual saddle | Footprint (m2): 39.069 | Clearance (m): 1.500 | Platform: yes | Rack Tie-In: no
Row 7: Equipment: HX-01-CTRL | Support Variant: panel frame | Footprint (m2): 1.200 | Clearance (m): 1.000 | Platform: no | Rack Tie-In: no
Row 8: Equipment: HX-01-HDR | Support Variant: guided rack shoe | Footprint (m2): 2.430 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: yes
Row 9: Equipment: HX-01-PMP | Support Variant: grouted base frame | Footprint (m2): 1.200 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: no
Row 10: Equipment: HX-01-EXP | Support Variant: four-leg support | Footprint (m2): 1.600 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: no
Row 11: Equipment: HX-01-RV | Support Variant: grouted base frame | Footprint (m2): 1.200 | Clearance (m): 1.200 | Platform: no | Rack Tie-In: no
Row 12: Equipment: HX-02 | Support Variant: dual saddle | Footprint (m2): 5.715 | Clearance (m): 1.500 | Platform: yes | Rack Tie-In: no
Row 13: Equipment: HX-02-CTRL | Support Variant: panel frame | Footprint (m2): 1.200 | Clearance (m): 1.000 | Platform: no | Rack Tie-In: no

23. Project Cost

Project Cost

Procurement Basis

Domestic cluster-led procurement selected as the current economic basis.

Logistics Basis

Coastal cluster dispatch basis selected as the current economic basis.

Project Cost Build-Up Summary

Table 143. Project Cost Build-Up Summary

Columns: Project-cost head; Value (INR)
Row 1: Project-cost head: Purchased equipment | Value (INR): 21,831,240,626.88
Row 2: Project-cost head: Installed equipment and packages | Value (INR): 46,937,167,347.79
Row 3: Project-cost head: Direct plant cost | Value (INR): 54,447,114,123.44
Row 4: Project-cost head: Indirect cost | Value (INR): 13,067,307,389.62
Row 5: Project-cost head: Contingency | Value (INR): 5,444,711,412.34
Row 6: Project-cost head: Total CAPEX | Value (INR): 72,959,132,925.40

Direct Plant Cost Head Allocation

Table 144. Direct Plant Cost Head Allocation

Columns: Direct-cost head; Allocated value (INR)
Row 1: Direct-cost head: Purchased equipment | Allocated value (INR): 21,831,240,626.88
Row 2: Direct-cost head: Installation | Allocated value (INR): 11,333,890,424.07
Row 3: Direct-cost head: Piping | Allocated value (INR): 6,834,337,162.46
Row 4: Direct-cost head: Instrumentation and control | Allocated value (INR): 3,905,335,521.41
Row 5: Direct-cost head: Electrical | Allocated value (INR): 2,929,001,641.05
Row 6: Direct-cost head: Civil and structural | Allocated value (INR): 5,858,003,282.10
Row 7: Direct-cost head: Insulation and painting | Allocated value (INR): 1,952,667,760.71
Row 8: Direct-cost head: Equipment-level contingency allowance | Allocated value (INR): 2,440,834,700.88
Row 9: Direct-cost head: Total spares allowance | Allocated value (INR): 323,316,967.43
Row 10: Direct-cost head: Allocated direct plant cost | Allocated value (INR): 57,085,311,119.53

Indirect and Contingency Basis

Table 145. Indirect and Contingency Basis

Columns: Allowance; Value (INR)
Row 1: Allowance: Installed-cost base | Value (INR): 46,937,167,347.79
Row 2: Allowance: Direct-cost uplift above installed basis | Value (INR): 7,509,946,775.65
Row 3: Allowance: Indirect engineering / procurement / construction | Value (INR): 13,067,307,389.62
Row 4: Allowance: Contingency | Value (INR): 5,444,711,412.34
Row 5: Allowance: Heat-integration CAPEX | Value (INR): 478,104,852.53
Row 6: Allowance: Utility-island project CAPEX burden | Value (INR): 733,501,670.81

Row 7: Allowance: Total import-duty burden | Value (INR): 353,373,129.56
Row 8: Allowance: Total CAPEX | Value (INR): 72,959,132,925.40

Equipment Family Cost Allocation

Table 146. Equipment Family Cost Allocation

Columns: Equipment family; Count; Bare Cost (INR); Installed Cost (INR); Spares (INR); Import Duty (INR)
Row 1: Equipment family: Reactor | Count: 1 | Bare Cost (INR): 19,244,201,057.55 | Installed Cost (INR): 43,299,452,379.48 | Spares (INR): 283,748,938.25 | Import Duty (INR): 327,605,174.15
Row 2: Equipment family: Flash drum | Count: 1 | Bare Cost (INR): 939,536,778.43 | Installed Cost (INR): 2,113,957,751.47 | Spares (INR): 13,853,137.50 | Import Duty (INR): 15,994,278.43
Row 3: Equipment family: Storage tank | Count: 1 | Bare Cost (INR): 1,031,289,992.04 | Installed Cost (INR): 2,062,579,984.08 | Spares (INR): 15,445,625.40 | Import Duty (INR): 1,581,632.04
Row 4: Equipment family: Heat exchanger | Count: 1 | Bare Cost (INR): 370,219,675.74 | Installed Cost (INR): 832,994,270.41 | Spares (INR): 5,479,441.04 | Import Duty (INR): 4,923,606.54
Row 5: Equipment family: HTM loop exchanger | Count: 1 | Bare Cost (INR): 108,622,690.42 | Installed Cost (INR): 238,969,918.92 | Spares (INR): 2,143,562.02 | Import Duty (INR): 1,444,589.32
Row 6: Equipment family: Shared HTM header manifold | Count: 1 | Bare Cost (INR): 84,771,672.00 | Installed Cost (INR): 161,066,176.79 | Spares (INR): 1,672,885.62 | Import Duty (INR): 1,127,391.08
Row 7: Equipment family: Heat integration exchanger | Count: 1 | Bare Cost (INR): 27,026,028.38 | Installed Cost (INR): 55,403,358.19 | Spares (INR): 535,890.47 | Import Duty (INR): 231,504.68
Row 8: Equipment family: Distillation column | Count: 1 | Bare Cost (INR): 13,157,679.80 | Installed Cost (INR): 29,604,779.55 | Spares (INR): 194,005.33 | Import Duty (INR): 223,990.80
Row 9: Equipment family: HTM circulation skid | Count: 1 | Bare Cost (INR): 4,521,339.97 | Installed Cost (INR): 8,364,478.95 | Spares (INR): 89,224.20 | Import Duty (INR): 60,129.97
Row 10: Equipment family: HTM expansion tank | Count: 1 | Bare Cost (INR): 3,507,952.79 | Installed Cost (INR): 7,015,905.57 | Spares (INR): 69,226.00 | Import Duty (INR): 46,652.79
Row 11: Equipment family: Utility control package | Count: 2 | Bare Cost (INR): 2,600,436.48 | Installed Cost (INR): 4,160,698.37 | Spares (INR): 49,800.00 | Import Duty (INR): 110,436.48
Row 12: Equipment family: HTM relief package | Count: 1 | Bare Cost (INR): 1,785,323.28 | Installed Cost (INR): 3,124,315.74 | Spares (INR): 35,231.60 | Import Duty (INR): 23,743.28

Procurement Timing Basis

Table 147. Procurement Timing Basis

Columns: Field; Value
Row 1: Field: Profile | Value: domestic_cluster_timing
Row 2: Field: Construction months | Value: 19
Row 3: Field: Imported equipment fraction | Value: 0.140
Row 4: Field: Long-lead equipment fraction | Value: 0.254
Row 5: Field: Total import duty | Value: INR 353,373,129.56
Row 6: Field: Advance fraction | Value: 0.150
Row 7: Field: Progress fraction | Value: 0.750
Row 8: Field: Retention fraction | Value: 0.100

Table 148. Procurement Timing Basis

Columns: Package Family; Milestone; Month; Draw Fraction; CAPEX Draw (INR)
Row 1: Package Family: pressure_vessel_package | Milestone: advance_release | Month: 6.7 | Draw Fraction: 0.186 | CAPEX Draw (INR): 13,583,398,188.32
Row 2: Package Family: thermal_exchange_package | Milestone: advance_release | Month: 10.1 | Draw Fraction: 0.005 | CAPEX Draw (INR): 336,687,371.39
Row 3: Package Family: general_fabrication | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 14,904,585.21

Row 4: Package Family: instrumented_skid_package | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,119,309.11
Row 5: Package Family: pressure_vessel_package | Milestone: fabrication_progress | Month: 11.8 | Draw Fraction: 0.428 | CAPEX Draw (INR): 31,241,815,833.14
Row 6: Package Family: tankage_and_balance_of_plant | Milestone: advance_release | Month: 13.5 | Draw Fraction: 0.005 | CAPEX Draw (INR): 369,916,194.26
Row 7: Package Family: thermal_exchange_package | Milestone: fabrication_progress | Month: 13.6 | Draw Fraction: 0.011 | CAPEX Draw (INR): 785,603,866.57
Row 8: Package Family: general_fabrication | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 34,777,365.48
Row 9: Package Family: instrumented_skid_package | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 2,611,721.25
Row 10: Package Family: tankage_and_balance_of_plant | Milestone: fabrication_progress | Month: 15.5 | Draw Fraction: 0.020 | CAPEX Draw (INR): 1,479,664,777.04
Row 11: Package Family: pressure_vessel_package | Milestone: delivery_release | Month: 16.0 | Draw Fraction: 0.177 | CAPEX Draw (INR): 12,904,228,278.90
Row 12: Package Family: thermal_exchange_package | Milestone: delivery_release | Month: 16.6 | Draw Fraction: 0.005 | CAPEX Draw (INR): 374,097,079.32
Row 13: Package Family: general_fabrication | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 16,560,650.23
Row 14: Package Family: instrumented_skid_package | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,243,676.78
Row 15: Package Family: tankage_and_balance_of_plant | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.008 | CAPEX Draw (INR): 616,526,990.43
Row 16: Package Family: pressure_vessel_package | Milestone: site_erection | Month: 17.3 | Draw Fraction: 0.047 | CAPEX Draw (INR): 3,395,849,547.08
Row 17: Package Family: thermal_exchange_package | Milestone: site_erection | Month: 17.5 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65
Row 18: Package Family: general_fabrication | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 19: Package Family: instrumented_skid_package | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 20: Package Family: tankage_and_balance_of_plant | Milestone: site_erection | Month: 17.7 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 21: Package Family: general_fabrication | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 22: Package Family: instrumented_skid_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 23: Package Family: pressure_vessel_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.093 | CAPEX Draw (INR): 6,791,699,094.16
Row 24: Package Family: tankage_and_balance_of_plant | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 25: Package Family: thermal_exchange_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65

Procurement Package Timing

Table 149. Procurement Package Timing

Columns: Equipment; Type; Package Family; Lead (mo); Award Month; Delivery Month; Import Content; Import Duty; CAPEX Burden (INR)
Row 1
Equipment: R-101

Type: Reactor
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211
Import Duty: 327,605,174.15
CAPEX Burden (INR): 64,713,323,286.86
Row 2
Equipment: PU-201
Type: Distillation column
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211
Import Duty: 223,990.80
CAPEX Burden (INR): 44,245,909.93
Row 3
Equipment: V-101
Type: Flash drum
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211
Import Duty: 15,994,278.43
CAPEX Burden (INR): 3,159,421,744.80
Row 4
Equipment: E-101
Type: Heat exchanger
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 4,923,606.54
CAPEX Burden (INR): 1,244,954,024.93
Row 5
Equipment: TK-301
Type: Storage tank
Package Family: tankage_and_balance_of_plant
Lead (mo): 4.54
Award Month: 13.46
Delivery Month: 16.72
Import Content: 0.038
Import Duty: 1,581,632.04
CAPEX Burden (INR): 3,082,634,952.17
Row 6
Equipment: HX-01
Type: HTM loop exchanger
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 1,444,589.32
CAPEX Burden (INR): 357,153,191.76
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Package Family: instrumented_skid_package
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.403
Import Duty: 62,758.08
CAPEX Burden (INR): 3,533,740.26
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Package Family: thermal_exchange_package
Lead (mo): 7.94

Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 1,127,391.08
CAPEX Burden (INR): 240,721,925.95

Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 60,129.97
CAPEX Burden (INR): 12,501,156.50

Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 46,652.79
CAPEX Burden (INR): 10,485,642.20

Row 11
Equipment: HX-01-RV
Type: HTM relief package
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Import Duty: 23,743.28
CAPEX Burden (INR): 4,669,455.23

Row 12
Equipment: HX-02
Type: Heat integration exchanger
Package Family: general_fabrication
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.115
Import Duty: 231,504.68
CAPEX Burden (INR): 82,803,251.14

Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Package Family: instrumented_skid_package
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.403
Import Duty: 47,678.40
CAPEX Burden (INR): 2,684,643.66

Installed Equipment Cost Matrix

Table 150. Installed Equipment Cost Matrix

Columns: Equipment; Type; Bare Cost (INR); Installed Cost (INR); Spares (INR); Package Family; Lead (mo); Import Duty (INR); Basis

Row 1
Equipment: R-101
Type: Reactor
Bare Cost (INR): 19,244,201,057.55
Installed Cost (INR): 43,299,452,379.48
Spares (INR): 283,748,938.25
Package Family: pressure_vessel_package
Lead (mo): 11.34
Import Duty (INR): 327,605,174.15
Basis: tubular_plug_flow_hydrator selected at 1.00 h residence time and

0.960 conversion basis.
Row 2
Equipment: PU-201
Type: Distillation column
Bare Cost (INR): 13,157,679.80
Installed Cost (INR): 29,604,779.55
Spares (INR): 194,005.33
Package Family: pressure_vessel_package
Lead (mo): 11.34
Import Duty (INR): 223,990.80
Basis: 7 stages equivalent
Row 3
Equipment: V-101
Type: Flash drum
Bare Cost (INR): 939,536,778.43
Installed Cost (INR): 2,113,957,751.47
Spares (INR): 13,853,137.50
Package Family: pressure_vessel_package
Lead (mo): 11.34
Import Duty (INR): 15,994,278.43
Basis: Generic separator hold-up
Row 4
Equipment: E-101
Type: Heat exchanger
Bare Cost (INR): 370,219,675.74
Installed Cost (INR): 832,994,270.41
Spares (INR): 5,479,441.04
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 4,923,606.54
Basis: LMTD 8.0 K
Row 5
Equipment: TK-301
Type: Storage tank
Bare Cost (INR): 1,031,289,992.04
Installed Cost (INR): 2,062,579,984.08
Spares (INR): 15,445,625.40
Package Family: tankage_and_balance_of_plant
Lead (mo): 4.54
Import Duty (INR): 1,581,632.04
Basis: 14.1 days inventory
Row 6
Equipment: HX-01
Type: HTM loop exchanger
Bare Cost (INR): 108,622,690.42
Installed Cost (INR): 238,969,918.92
Spares (INR): 2,143,562.02
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 1,444,589.32
Basis: role=exchanger; duty=34024.794 kW; power=0.000 kW; volume=542.949 m3
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Bare Cost (INR): 1,477,758.08
Installed Cost (INR): 2,364,412.93
Spares (INR): 28,300.00
Package Family: instrumented_skid_package
Lead (mo): 6.80
Import Duty (INR): 62,758.08
Basis: role=controls; duty=0.000 kW; power=0.000 kW; volume=0.150 m3
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Bare Cost (INR): 84,771,672.00
Installed Cost (INR): 161,066,176.79
Spares (INR): 1,672,885.62
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 1,127,391.08
Basis: role=header; duty=34024.794 kW; power=0.000 kW; volume=0.851 m3
Row 9

Equipment: HX-01-PMP
Type: HTM circulation skid
Bare Cost (INR): 4,521,339.97
Installed Cost (INR): 8,364,478.95
Spares (INR): 89,224.20
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 60,129.97
Basis: role=circulation; duty=0.000 kW; power=15.466 kW; volume=0.250 m3

Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Bare Cost (INR): 3,507,952.79
Installed Cost (INR): 7,015,905.57
Spares (INR): 69,226.00
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 46,652.79
Basis: role=expansion; duty=0.000 kW; power=0.000 kW; volume=2.835 m3

Row 11
Equipment: HX-01-RV
Type: HTM relief package
Bare Cost (INR): 1,785,323.28
Installed Cost (INR): 3,124,315.74
Spares (INR): 35,231.60
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 23,743.28
Basis: role=relief; duty=0.000 kW; power=0.000 kW; volume=1.309 m3

Row 12
Equipment: HX-02
Type: Heat integration exchanger
Bare Cost (INR): 27,026,028.38
Installed Cost (INR): 55,403,358.19
Spares (INR): 535,890.47
Package Family: general_fabrication
Lead (mo): 6.80
Import Duty (INR): 231,504.68
Basis: role=exchanger; duty=8506.198 kW; power=0.000 kW; volume=30.379 m3

Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Bare Cost (INR): 1,122,678.40
Installed Cost (INR): 1,796,285.44
Spares (INR): 21,500.00
Package Family: instrumented_skid_package
Lead (mo): 6.80
Import Duty (INR): 47,678.40
Basis: role=controls; duty=0.000 kW; power=0.000 kW; volume=0.150 m3

Utility Island Economics

Table 151. Utility Island Economics

Columns: Island; Topology; HTM Inventory (m3); Header Pressure (bar); Pair Score; Cycle (y); Turnaround (d); Project CAPEX Burden (INR); Allocated Utility (INR/y); Service (INR/y); Replacement (INR/y); Operating Burden (INR/y)

Row 1
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Topology: shared HTM island network
HTM Inventory (m3): 3.936
Header Pressure (bar): 14.50
Pair Score: 0.000
Cycle (y): 4.26
Turnaround (d): 2.57
Project CAPEX Burden (INR): 733,501,670.81
Allocated Utility (INR/y): 3,132,067,158.00
Service (INR/y): 2,952,253.22
Replacement (INR/y): 15,494,813.83
Operating Burden (INR/y): 3,150,514,225.04

Recurring Transport / Service Submodels

Table 152. Recurring Transport / Service Submodels

Columns: Recurring submodel; Value (INR/y)
Row 1: Recurring submodel: Utility-island service | Value (INR/y): 2,952,253.22
Row 2: Recurring submodel: Utility-island replacement | Value (INR/y): 15,494,813.83
Row 3: Recurring submodel: Packing replacement | Value (INR/y): 0.00
Row 4: Recurring submodel: Classifier service | Value (INR/y): 0.00
Row 5: Recurring submodel: Filter media replacement | Value (INR/y): 0.00
Row 6: Recurring submodel: Dryer exhaust treatment | Value (INR/y): 0.00
Row 7: Recurring submodel: Total transport/service burden | Value (INR/y): 18,447,067.05

Scenario Cost Snapshot

Table 153. Scenario Cost Snapshot

Columns: Scenario; Utility Cost (INR/y); Transport/Service (INR/y); Utility-Island Burden (INR/y); Operating Cost (INR/y); Revenue (INR/y); Gross Margin (INR/y)
Row 1
- Scenario: base
- Utility Cost (INR/y): 3,132,067,158.00
- Transport/Service (INR/y): 14,583,091.39
- Utility-Island Burden (INR/y): 3,146,650,249.39
- Operating Cost (INR/y): 13,273,756,525.48
- Revenue (INR/y): 12,600,000,000.00
- Gross Margin (INR/y): -673,756,525.48

Row 2
- Scenario: conservative
- Utility Cost (INR/y): 3,775,393,752.25
- Transport/Service (INR/y): 16,167,728.07
- Utility-Island Burden (INR/y): 3,791,561,480.32
- Operating Cost (INR/y): 14,912,994,804.01
- Revenue (INR/y): 11,970,000,000.00
- Gross Margin (INR/y): -2,942,994,804.01

Row 3
- Scenario: upside
- Utility Cost (INR/y): 2,888,705,539.82
- Transport/Service (INR/y): 14,088,928.39
- Utility-Island Burden (INR/y): 2,902,794,468.22
- Operating Cost (INR/y): 12,731,575,600.67
- Revenue (INR/y): 13,104,000,000.00
- Gross Margin (INR/y): 372,424,399.33

24. Cost of Production

Cost of Production

Cost of Production Summary

Table 154. Cost of Production Summary

Columns: Metric; Value
Row 1: Metric: Annual production | Value: 200,000,000.00 kg/y
Row 2: Metric: Annual operating cost | Value: INR 13,281,189,728.49/y
Row 3: Metric: Unit cost of production | Value: 66.41 INR/kg
Row 4: Metric: Selling price basis | Value: 63.00 INR/kg
Row 5: Metric: Gross margin basis | Value: -3.41 INR/kg

Manufacturing Cost Build-Up

Table 155. Manufacturing Cost Build-Up

Columns: Opex bucket; Value (INR/y); Share of OPEX (%)
Row 1: Opex bucket: Raw materials | Value (INR/y): 7,510,532,251.62 | Share of OPEX (%): 56.55
Row 2: Opex bucket: Utilities | Value (INR/y): 3,132,067,158.00 | Share of OPEX (%): 23.58
Row 3: Opex bucket: Labor | Value (INR/y): 156,000,000.00 | Share of OPEX (%): 1.17
Row 4: Opex bucket: Base maintenance | Value (INR/y): 2,042,855,721.91 | Share of OPEX (%): 15.38
Row 5: Opex bucket: Utility-island service | Value (INR/y): 2,952,253.22 | Share of OPEX (%): 0.02
Row 6: Opex bucket: Utility-island replacement | Value (INR/y): 15,494,813.83 | Share of OPEX (%): 0.12
Row 7: Opex bucket: Packing replacement | Value (INR/y): 0.00 | Share of OPEX (%): 0.00
Row 8: Opex bucket: Classifier service | Value (INR/y): 0.00 | Share of OPEX (%): 0.00
Row 9: Opex bucket: Filter media replacement | Value (INR/y): 0.00 | Share of OPEX (%): 0.00
Row 10: Opex bucket: Dryer exhaust treatment | Value (INR/y): 0.00 | Share of OPEX (%): 0.00
Row 11: Opex bucket: Transport/service penalties | Value (INR/y): 18,447,067.05 | Share of OPEX (%): 0.14
Row 12: Opex bucket: Total maintenance | Value (INR/y): 2,061,302,788.96 | Share of OPEX (%): 15.52
Row 13: Opex bucket: Overheads | Value (INR/y): 421,287,529.90 | Share of OPEX (%): 3.17
Row 14: Opex bucket: Total OPEX | Value (INR/y): 13,281,189,728.49 | Share of OPEX (%): 100.00
Row 15: Opex bucket: Unit cost | Value (INR/y): 66.41 INR/kg | Share of OPEX (%): -

Utility and Raw-Material Cost Basis

Table 156. Utility and Raw-Material Cost Basis

Columns: Basis item; Value
Row 1: Basis item: Annual raw-material cost | Value: INR 7,510,532,251.62/y
Row 2: Basis item: Annual utility cost | Value: INR 3,132,067,158.00/y
Row 3: Basis item: Annual labor cost | Value: INR 156,000,000.00/y
Row 4: Basis item: Annual output | Value: 200,000,000.00 kg/y
Row 5: Basis item: Selling price basis | Value: 63.00 INR/kg

Recurring Service and Maintenance Basis

Table 157. Recurring Service and Maintenance Basis

Columns: Service component; Value (INR/y)

Row 1: Service component: Base maintenance | Value (INR/y): 2,042,855,721.91
Row 2: Service component: Utility-island service | Value (INR/y): 2,952,253.22
Row 3: Service component: Utility-island replacement | Value (INR/y): 15,494,813.83
Row 4: Service component: Packing replacement | Value (INR/y): 0.00
Row 5: Service component: Classifier service | Value (INR/y): 0.00
Row 6: Service component: Filter media replacement | Value (INR/y): 0.00
Row 7: Service component: Dryer exhaust treatment | Value (INR/y): 0.00
Row 8: Service component: Total transport/service penalties | Value (INR/y): 18,447,067.05
Row 9: Service component: Total maintenance | Value (INR/y): 2,061,302,788.96

Unit Cost and Selling Basis

Table 158. Unit Cost and Selling Basis

Columns: Metric; Value
Row 1: Metric: Unit cost of production | Value: 66.41 INR/kg
Row 2: Metric: Selling price basis | Value: 63.00 INR/kg
Row 3: Metric: Gross margin basis | Value: -3.41 INR/kg
Row 4: Metric: Operating margin on selling basis | Value: -5.41%

25. Working Capital

Working Capital

Working-Capital Parameter Basis

Table 159. Working-Capital Parameter Basis

Columns: Parameter; Value
Row 1: Parameter: Raw-material inventory days | Value: 20.5
Row 2: Parameter: Product inventory days | Value: 10.9
Row 3: Parameter: Cash buffer days | Value: 7.0
Row 4: Parameter: Operating stock days | Value: 2.5
Row 5: Parameter: Procurement timing factor | Value: 1.197
Row 6: Parameter: Pre-commissioning inventory days | Value: 13.4
Row 7: Parameter: Receivable days | Value: 32.0
Row 8: Parameter: Payable days | Value: 24.0
Row 9: Parameter: Pre-commissioning inventory | Value: INR 275,688,431.02
Row 10: Parameter: Restart loss inventory | Value: INR 1,034,880.00
Row 11: Parameter: Outage buffer inventory | Value: INR 95,515,405.58
Row 12: Parameter: Working capital | Value: INR 1,759,656,318.22
Row 13: Parameter: Peak working capital | Value: INR 1,759,656,318.22

Working-Capital Breakdown

Table 160. Working-Capital Breakdown

Columns: Component; Value
Row 1: Component: Raw-material inventory | Value: INR 421,309,994.11
Row 2: Component: Product inventory | Value: INR 376,273,972.60
Row 3: Component: Receivables | Value: INR 1,104,657,534.25
Row 4: Component: Cash buffer | Value: INR 254,707,748.22
Row 5: Component: Pre-commissioning inventory | Value: INR 275,688,431.02
Row 6: Component: Restart loss inventory | Value: INR 1,034,880.00
Row 7: Component: Outage buffer inventory | Value: INR 95,515,405.58
Row 8: Component: Payables | Value: INR 493,843,216.54

Inventory, Receivable, and Payable Basis

Table 161. Inventory, Receivable, and Payable Basis

Columns: Cash-cycle component; Days; Value
Row 1: Cash-cycle component: Raw-material inventory | Days: 20.5 | Value: INR 421,309,994.11
Row 2: Cash-cycle component: Product inventory | Days: 10.9 | Value: INR 376,273,972.60
Row 3: Cash-cycle component: Receivables | Days: 32.0 | Value: INR 1,104,657,534.25
Row 4: Cash-cycle component: Payables | Days: 24.0 | Value: INR 493,843,216.54
Row 5: Cash-cycle component: Cash buffer | Days: 7.0 | Value: INR 254,707,748.22
Row 6: Cash-cycle component: Operating stock | Days: 2.5 | Value: INR 95,515,405.58

Procurement-Linked Working-Capital Timing

Table 162. Procurement-Linked Working-Capital Timing

Columns: Parameter; Value
Row 1: Parameter: Procurement timing factor | Value: 1.197
Row 2: Parameter: Pre-commissioning inventory month | Value: 17.95
Row 3: Parameter: Peak working-capital month | Value: 18.15
Row 4: Parameter: Peak working capital | Value: INR 1,759,656,318.22

Operations Planning Basis

Table 163. Operations Planning Basis

Columns: Parameter; Value
Row 1: Parameter: Availability policy | Value: continuous_liquid_train
Row 2: Parameter: Operating mode | Value: continuous
Row 3: Parameter: Raw-material buffer (d) | Value: 19.6
Row 4: Parameter: Finished-goods buffer (d) | Value: 9.4
Row 5: Parameter: Operating stock (d) | Value: 2.5
Row 6: Parameter: Restart buffer (d) | Value: 1.5
Row 7: Parameter: Annual restart loss (kg/y) | Value: 16,426.7

26. Financial Analysis

Financial Analysis

Financing Basis

70:30 debt-equity basis selected after lender-coverage reranking (min DSCR 0.050, LLCR 0.000, PLCR 0.000). Higher-leverage options were downgraded because of covenant pressure. The selected basis still carries covenant pressure and requires approval.

Financial Performance Summary

Table 164. Financial Performance Summary

Columns: Metric; Value
Row 1: Metric: Annual revenue | Value: INR 12,600,000,000.00
Row 2: Metric: Annual operating cost | Value: INR 13,281,189,728.49
Row 3: Metric: Gross profit | Value: INR -681,189,728.49
Row 4: Metric: Working capital | Value: INR 1,759,656,318.22
Row 5: Metric: Peak working capital | Value: INR 1,759,656,318.22
Row 6: Metric: Peak working-capital month | Value: 18.15
Row 7: Metric: Total project funding | Value: INR 76,261,521,411.19
Row 8: Metric: Interest during construction | Value: INR 1,542,732,167.57
Row 9: Metric: Payback | Value: 76261521411.190 y
Row 10: Metric: NPV | Value: INR -98,309,630,442.49
Row 11: Metric: IRR | Value: 1.00%
Row 12: Metric: Profitability index | Value: -0.289
Row 13: Metric: Break-even fraction | Value: 1.054
Row 14: Metric: Minimum DSCR | Value: 0.050
Row 15: Metric: Average DSCR | Value: 0.050
Row 16: Metric: LLCR | Value: 0.000
Row 17: Metric: PLCR | Value: 0.000
Row 18: Metric: Selected financing option | Value: debt_equity_70_30
Row 19: Metric: Downside scenario | Value: conservative
Row 20: Metric: Downside-preferred financing option | Value: debt_equity_70_30
Row 21: Metric: Scenario reversal | Value: no
Row 22: Metric: Covenant breaches | Value: minimum_dscr_breach, average_dscr_breach, llcr_breach, plcr_breach

Profitability and Return Summary

Table 165. Profitability and Return Summary

Columns: Metric; Value
Row 1: Metric: Payback | Value: 76261521411.190 y
Row 2: Metric: NPV | Value: INR -98,309,630,442.49
Row 3: Metric: IRR | Value: 1.00%
Row 4: Metric: Profitability index | Value: -0.289
Row 5: Metric: Break-even fraction | Value: 1.054
Row 6: Metric: Minimum DSCR | Value: 0.050
Row 7: Metric: Average DSCR | Value: 0.050
Row 8: Metric: LLCR | Value: 0.000
Row 9: Metric: PLCR | Value: 0.000

Funding and Capital Structure Basis

Table 166. Funding and Capital Structure Basis

Columns: Funding item; Value
Row 1: Funding item: Total project funding | Value: INR 76,261,521,411.19
Row 2: Funding item: Interest during construction | Value: INR 1,542,732,167.57
Row 3: Funding item: Working capital | Value: INR 1,759,656,318.22
Row 4: Funding item: Peak working capital | Value: INR 1,759,656,318.22
Row 5: Funding item: Selected financing option | Value: debt_equity_70_30

Row 6: Funding item: Downside-preferred financing option | Value: debt_equity_70_30

Procurement and Construction Funding Basis

Table 167. Procurement and Construction Funding Basis

Columns: Package Family; Milestone; Month; Draw Fraction; CAPEX Draw (INR)
Row 1: Package Family: pressure_vessel_package | Milestone: advance_release | Month: 6.7 | Draw Fraction: 0.186 | CAPEX Draw (INR): 13,583,398,188.32
Row 2: Package Family: thermal_exchange_package | Milestone: advance_release | Month: 10.1 | Draw Fraction: 0.005 | CAPEX Draw (INR): 336,687,371.39
Row 3: Package Family: general_fabrication | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 14,904,585.21
Row 4: Package Family: instrumented_skid_package | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,119,309.11
Row 5: Package Family: pressure_vessel_package | Milestone: fabrication_progress | Month: 11.8 | Draw Fraction: 0.428 | CAPEX Draw (INR): 31,241,815,833.14
Row 6: Package Family: tankage_and_balance_of_plant | Milestone: advance_release | Month: 13.5 | Draw Fraction: 0.005 | CAPEX Draw (INR): 369,916,194.26
Row 7: Package Family: thermal_exchange_package | Milestone: fabrication_progress | Month: 13.6 | Draw Fraction: 0.011 | CAPEX Draw (INR): 785,603,866.57
Row 8: Package Family: general_fabrication | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 34,777,365.48
Row 9: Package Family: instrumented_skid_package | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 2,611,721.25
Row 10: Package Family: tankage_and_balance_of_plant | Milestone: fabrication_progress | Month: 15.5 | Draw Fraction: 0.020 | CAPEX Draw (INR): 1,479,664,777.04
Row 11: Package Family: pressure_vessel_package | Milestone: delivery_release | Month: 16.0 | Draw Fraction: 0.177 | CAPEX Draw (INR): 12,904,228,278.90
Row 12: Package Family: thermal_exchange_package | Milestone: delivery_release | Month: 16.6 | Draw Fraction: 0.005 | CAPEX Draw (INR): 374,097,079.32
Row 13: Package Family: general_fabrication | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 16,560,650.23
Row 14: Package Family: instrumented_skid_package | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,243,676.78
Row 15: Package Family: tankage_and_balance_of_plant | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.008 | CAPEX Draw (INR): 616,526,990.43
Row 16: Package Family: pressure_vessel_package | Milestone: site_erection | Month: 17.3 | Draw Fraction: 0.047 | CAPEX Draw (INR): 3,395,849,547.08
Row 17: Package Family: thermal_exchange_package | Milestone: site_erection | Month: 17.5 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65
Row 18: Package Family: general_fabrication | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 19: Package Family: instrumented_skid_package | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 20: Package Family: tankage_and_balance_of_plant | Milestone: site_erection | Month: 17.7 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 21: Package Family: general_fabrication | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 22: Package Family: instrumented_skid_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 23: Package Family: pressure_vessel_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.093 | CAPEX Draw (INR): 6,791,699,094.16

Row 24: Package Family: tankage_and_balance_of_plant | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 25: Package Family: thermal_exchange_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65

Procurement-Linked Working-Capital Basis

Table 168. Procurement-Linked Working-Capital Basis

Columns: Metric; Value
Row 1: Metric: Procurement timing factor | Value: 1.197
Row 2: Metric: Pre-commissioning inventory days | Value: 13.4
Row 3: Metric: Pre-commissioning inventory month | Value: 17.95
Row 4: Metric: Pre-commissioning inventory | Value: INR 275,688,431.02
Row 5: Metric: Peak working-capital month | Value: 18.15
Row 6: Metric: Peak working capital | Value: INR 1,759,656,318.22

Availability and Outage Calendar

Table 169. Availability and Outage Calendar

Columns: Year; Availability (%); Minor Outage (d); Turnaround (d); Startup Loss (d); Revenue Loss (INR); Available Days; Calendar Basis
Row 1
Year: 1.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 152,727,272.73
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.
Row 2
Year: 2.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 162,272,727.27
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.
Row 3
Year: 3.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.
Row 4
Year: 4.0
Availability (%): 92.71
Minor Outage (d): 5.00
Turnaround (d): 15.57
Startup Loss (d): 3.50
Revenue Loss (INR): 827,029,636.36
Available Days: 305.93
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. A major process-unit turnaround is assumed in a low-demand maintenance window.
Row 5
Year: 5.0

Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.

Row 6
Year: 6.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.

Row 7
Year: 7.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.

Row 8
Year: 8.0
Availability (%): 92.71
Minor Outage (d): 5.00
Turnaround (d): 15.57
Startup Loss (d): 3.50
Revenue Loss (INR): 827,029,636.36
Available Days: 305.93
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. A major process-unit turnaround is assumed in a low-demand maintenance window.

Row 9
Year: 9.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.

Row 10
Year: 10.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. No major turnaround scheduled this year.

Row 11
Year: 11.0
Availability (%): 98.48
Minor Outage (d): 5.00
Turnaround (d): 0.00
Startup Loss (d): 0.00
Revenue Loss (INR): 171,818,181.82
Available Days: 325.00
Calendar Basis: Routine exchanger cleaning and column internals inspection

are represented as annual minor outages. No major turnaround scheduled this year.

Row 12

Year: 12.0
Availability (%): 92.71
Minor Outage (d): 5.00
Turnaround (d): 15.57
Startup Loss (d): 3.50
Revenue Loss (INR): 827,029,636.36
Available Days: 305.93
Calendar Basis: Routine exchanger cleaning and column internals inspection are represented as annual minor outages. A major process-unit turnaround is assumed in a low-demand maintenance window.

Debt Service Coverage Schedule

Table 170. Debt Service Coverage Schedule

Columns: Year; Principal (INR); Interest (INR); Debt Service (INR); CFADS (INR); DSCR

Row 1: Year: 1.0 | Principal (INR): 10,522,825,043.07 | Interest (INR): 5,524,483,147.61 | Debt Service (INR): 16,047,308,190.68 | CFADS (INR): 0.00 | DSCR: 0.050
Row 2: Year: 2.0 | Principal (INR): 10,522,825,043.07 | Interest (INR): 4,419,586,518.09 | Debt Service (INR): 14,942,411,561.16 | CFADS (INR): 0.00 | DSCR: 0.050
Row 3: Year: 3.0 | Principal (INR): 10,522,825,043.07 | Interest (INR): 3,314,689,888.57 | Debt Service (INR): 13,837,514,931.64 | CFADS (INR): 0.00 | DSCR: 0.050
Row 4: Year: 4.0 | Principal (INR): 10,522,825,043.07 | Interest (INR): 2,209,793,259.04 | Debt Service (INR): 12,732,618,302.11 | CFADS (INR): 0.00 | DSCR: 0.050
Row 5: Year: 5.0 | Principal (INR): 10,522,825,043.07 | Interest (INR): 1,104,896,629.52 | Debt Service (INR): 11,627,721,672.59 | CFADS (INR): 0.00 | DSCR: 0.050
Row 6: Year: 6.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 7: Year: 7.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 8: Year: 8.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 9: Year: 9.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 10: Year: 10.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 11: Year: 11.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000
Row 12: Year: 12.0 | Principal (INR): 0.00 | Interest (INR): 0.00 | Debt Service (INR): 0.00 | CFADS (INR): 0.00 | DSCR: 0.000

Lender Coverage Screening

Table 171. Lender Coverage Screening

Columns: Metric; Value

Row 1: Metric: Minimum DSCR | Value: 0.050
Row 2: Metric: Average DSCR | Value: 0.050
Row 3: Metric: LLCR | Value: 0.000
Row 4: Metric: PLCR | Value: 0.000
Row 5: Metric: Downside scenario | Value: conservative
Row 6: Metric: Downside-preferred financing option | Value: debt_equity_70_30
Row 7: Metric: Scenario reversal | Value: no

Financing Option Ranking

Table 172. Financing Option Ranking

Columns: Option; Score; Min DSCR; LLCR; PLCR; Downside DSCR; Downside LLCR; Downside PLCR; IRR (%); IDC (INR); Coverage; Downside; Rejected / Pressure

Row 1
Option: 70:30 debt-equity basis
Score: 32.248
Min DSCR: 0.050
LLCR: 0.000
PLCR: 0.000
Downside DSCR: 0.000
Downside LLCR: 0.000
Downside PLCR: 0.000
IRR (%): 1.00
IDC (INR): 1542732167.57
Coverage: warning
Downside: warning
Rejected / Pressure: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 2
Option: 60:40 debt-equity basis
Score: 26.418
Min DSCR: 0.050
LLCR: 0.000
PLCR: 0.000
Downside DSCR: 0.000
Downside LLCR: 0.000
Downside PLCR: 0.000
IRR (%): 1.00
IDC (INR): 1385310517.82
Coverage: warning
Downside: warning
Rejected / Pressure: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 3
Option: 50:50 conservative financing basis
Score: 17.013
Min DSCR: 0.050
LLCR: 0.000
PLCR: 0.000
Downside DSCR: 0.000
Downside LLCR: 0.000
Downside PLCR: 0.000
IRR (%): 1.00
IDC (INR): 1259373198.01
Coverage: warning
Downside: warning
Rejected / Pressure: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.

Covenant Warnings

- Minimum DSCR 0.050 is below the screening covenant threshold of 1.10.
- Average DSCR 0.050 is below the screening covenant threshold of 1.25.
- LLCR 0.000 is below the screening covenant threshold of 1.30.
- PLCR 0.000 is below the screening covenant threshold of 1.45.

Scenario Margin Snapshot

Table 173. Scenario Margin Snapshot

Columns: Scenario; Revenue (INR/y); Transport/Service (INR/y); Utility-Island Burden (INR/y); Operating Cost (INR/y); Gross Margin (INR/y)
Row 1: Scenario: base | Revenue (INR/y): 12,600,000,000.00 | Transport/Service (INR/y): 14,583,091.39 | Utility-Island Burden (INR/y): 3,146,650,249.39 | Operating Cost (INR/y): 13,273,756,525.48 | Gross Margin (INR/y): -673,756,525.48
Row 2: Scenario: conservative | Revenue (INR/y): 11,970,000,000.00 | Transport/Service (INR/y): 16,167,728.07 | Utility-Island Burden (INR/y): 3,791,561,480.32 | Operating Cost (INR/y): 14,912,994,804.01 | Gross Margin (INR/y): -2,942,994,804.01
Row 3: Scenario: upside | Revenue (INR/y): 13,104,000,000.00 | Transport/Service (INR/y): 14,088,928.39 | Utility-Island Burden (INR/y): 2,902,794,468.22 | Operating Cost (INR/y): 12,731,575,600.67 | Gross Margin (INR/y): 372,424,399.33

Scenario Recurring Service Breakdown

Table 174. Scenario Recurring Service Breakdown

Columns: Scenario; Utility-Island Service (INR/y); Utility-Island Replacement (INR/y); Packing (INR/y); Classifier (INR/y); Filter Media (INR/y); Dryer Exhaust (INR/y)
Row 1
- Scenario: base
- Utility-Island Service (INR/y): 885,675.97
- Utility-Island Replacement (INR/y): 13,697,415.43
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Row 2
- Scenario: conservative
- Utility-Island Service (INR/y): 1,001,404.29
- Utility-Island Replacement (INR/y): 15,166,323.78
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Row 3
- Scenario: upside
- Utility-Island Service (INR/y): 847,060.49
- Utility-Island Replacement (INR/y): 13,241,867.90
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Scenario Utility Island Breakdown

Table 175. Scenario Utility Island Breakdown

Columns: Scenario; Island; Capex Burden (INR); Allocated Utility (INR/y); Service (INR/y); Replacement (INR/y); Operating Burden (INR/y)
Row 1
- Scenario: base
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Capex Burden (INR): 648,415,477.00
- Allocated Utility (INR/y): 3,132,067,158.00
- Service (INR/y): 885,675.97
- Replacement (INR/y): 13,697,415.43
- Operating Burden (INR/y): 3,146,650,249.39

Row 2
- Scenario: conservative
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Capex Burden (INR): 717,951,435.39
- Allocated Utility (INR/y): 3,775,393,752.25

Service (INR/y): 1,001,404.29
Replacement (INR/y): 15,166,323.78
Operating Burden (INR/y): 3,791,561,480.32
Row 3
Scenario: upside
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Capex Burden (INR): 626,850,527.87
Allocated Utility (INR/y): 2,888,705,539.82
Service (INR/y): 847,060.49
Replacement (INR/y): 13,241,867.90
Operating Burden (INR/y): 2,902,794,468.22

Multi-Year Financial Schedule

Table 176. Multi-Year Financial Schedule

Columns: Year; Capacity Utilization (%); Availability (%); Revenue Loss (INR); Revenue (INR); Operating Cost (INR); Transport/Service (INR); Utility-Island Service (INR); Utility-Island Replacement (INR); Packing (INR); Turnaround (INR); Utility-Island Turnaround (INR); Principal (INR); Interest (INR); Debt Service (INR); CFADS (INR); DSCR; Depreciation (INR); PBT (INR); Tax (INR); PAT (INR); Cash Accrual (INR)
Row 1
Year: 1.0
Capacity Utilization (%): 78.79
Availability (%): 98.48
Revenue Loss (INR): 152,727,272.73
Revenue (INR): 9,927,272,727.27
Operating Cost (INR): 11,486,378,287.07
Transport/Service (INR): 2,795,694.34
Utility-Island Service (INR): 2,795,694.34
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 5,524,483,147.61
Debt Service (INR): 16,047,308,190.68
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -13,163,516,451.19
Tax (INR): 0.00
PAT (INR): -13,163,516,451.19
Cash Accrual (INR): -7,083,588,707.41
Row 2
Year: 2.0
Capacity Utilization (%): 83.71
Availability (%): 98.48
Revenue Loss (INR): 162,272,727.27
Revenue (INR): 10,547,727,272.73
Operating Cost (INR): 11,898,750,500.79
Transport/Service (INR): 2,832,038.36
Utility-Island Service (INR): 2,832,038.36
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 4,419,586,518.09
Debt Service (INR): 14,942,411,561.16
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -11,850,537,489.94
Tax (INR): 0.00
PAT (INR): -11,850,537,489.94
Cash Accrual (INR): -5,770,609,746.16
Row 3
Year: 3.0
Capacity Utilization (%): 88.64
Availability (%): 98.48

Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 3,314,689,888.57
Debt Service (INR): 13,837,514,931.64
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -10,537,558,528.69
Tax (INR): 0.00
PAT (INR): -10,537,558,528.69
Cash Accrual (INR): -4,457,630,784.90

Row 4

Year: 4.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Transport/Service (INR): 68,894,229.09
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Packing (INR): 0.00
Turnaround (INR): 768,606,535.25
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 10,522,825,043.07
Interest (INR): 2,209,793,259.04
Debt Service (INR): 12,732,618,302.11
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -10,655,092,861.23
Tax (INR): 0.00
PAT (INR): -10,655,092,861.23
Cash Accrual (INR): -4,575,165,117.44

Row 5

Year: 5.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 1,104,896,629.52
Debt Service (INR): 11,627,721,672.59
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,327,765,269.64
Tax (INR): 0.00
PAT (INR): -8,327,765,269.64
Cash Accrual (INR): -2,247,837,525.85

Row 6

Year: 6.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39

Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33

Row 7

Year: 7.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33

Row 8

Year: 8.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Transport/Service (INR): 68,894,229.09
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Packing (INR): 0.00
Turnaround (INR): 768,606,535.25
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Cash Accrual (INR): -2,365,371,858.40

Row 9

Year: 9.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00

Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 10
Year: 10.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 11
Year: 11.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Transport/Service (INR): 2,868,382.39
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Packing (INR): 0.00
Turnaround (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 12
Year: 12.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Transport/Service (INR): 68,894,229.09
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Packing (INR): 0.00
Turnaround (INR): 768,606,535.25
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00

CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Cash Accrual (INR): -2,365,371,858.40

Profit and Loss Schedule

Table 177. Profit and Loss Schedule

Columns: Year; Revenue (INR); Operating Cost (INR); Depreciation (INR); Interest (INR); PBT (INR); Tax (INR); PAT (INR)

Row 1
Year: 1.0
Revenue (INR): 9,927,272,727.27
Operating Cost (INR): 11,486,378,287.07
Depreciation (INR): 6,079,927,743.78
Interest (INR): 5,524,483,147.61
PBT (INR): -13,163,516,451.19
Tax (INR): 0.00
PAT (INR): -13,163,516,451.19

Row 2
Year: 2.0
Revenue (INR): 10,547,727,272.73
Operating Cost (INR): 11,898,750,500.79
Depreciation (INR): 6,079,927,743.78
Interest (INR): 4,419,586,518.09
PBT (INR): -11,850,537,489.94
Tax (INR): 0.00
PAT (INR): -11,850,537,489.94

Row 3
Year: 3.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 3,314,689,888.57
PBT (INR): -10,537,558,528.69
Tax (INR): 0.00
PAT (INR): -10,537,558,528.69

Row 4
Year: 4.0
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Depreciation (INR): 6,079,927,743.78
Interest (INR): 2,209,793,259.04
PBT (INR): -10,655,092,861.23
Tax (INR): 0.00
PAT (INR): -10,655,092,861.23

Row 5
Year: 5.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 1,104,896,629.52
PBT (INR): -8,327,765,269.64
Tax (INR): 0.00
PAT (INR): -8,327,765,269.64

Row 6
Year: 6.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12

Row 7
Year: 7.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52

Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 8
Year: 8.0
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Row 9
Year: 9.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 10
Year: 10.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 11
Year: 11.0
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 12
Year: 12.0
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Depreciation (INR): 6,079,927,743.78
Interest (INR): 0.00
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19

Cash Accrual and Funding Schedule

Table 178. Cash Accrual and Funding Schedule

Columns: Year; CAPEX Draw (INR); Debt Draw (INR); Equity Draw (INR); IDC (INR); CFADS (INR); Debt Service (INR); Cash Accrual (INR)
Row 1
Year: 1.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 16,047,308,190.68
Cash Accrual (INR): -7,083,588,707.41
Row 2
Year: 2.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00

CFADS (INR): 0.00
Debt Service (INR): 14,942,411,561.16
Cash Accrual (INR): -5,770,609,746.16
Row 3
Year: 3.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 13,837,514,931.64
Cash Accrual (INR): -4,457,630,784.90
Row 4
Year: 4.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 12,732,618,302.11
Cash Accrual (INR): -4,575,165,117.44
Row 5
Year: 5.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 11,627,721,672.59
Cash Accrual (INR): -2,247,837,525.85
Row 6
Year: 6.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -1,142,940,896.33
Row 7
Year: 7.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -1,142,940,896.33
Row 8
Year: 8.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -2,365,371,858.40
Row 9
Year: 9.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -1,142,940,896.33
Row 10
Year: 10.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00

Debt Service (INR): 0.00
Cash Accrual (INR): -1,142,940,896.33
Row 11
Year: 11.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -1,142,940,896.33
Row 12
Year: 12.0
CAPEX Draw (INR): 0.00
Debt Draw (INR): 0.00
Equity Draw (INR): 0.00
IDC (INR): 0.00
CFADS (INR): 0.00
Debt Service (INR): 0.00
Cash Accrual (INR): -2,365,371,858.40

Debt Schedule

Table 179. Debt Schedule

Columns: Year; Opening Debt; Principal; Interest; Closing Debt
Row 1: Year: 1 | Opening Debt: 52,614,125,215.35 | Principal: 10,522,825,043.07 | Interest: 5,524,483,147.61 | Closing Debt: 42,091,300,172.28
Row 2: Year: 2 | Opening Debt: 42,091,300,172.28 | Principal: 10,522,825,043.07 | Interest: 4,419,586,518.09 | Closing Debt: 31,568,475,129.21
Row 3: Year: 3 | Opening Debt: 31,568,475,129.21 | Principal: 10,522,825,043.07 | Interest: 3,314,689,888.57 | Closing Debt: 21,045,650,086.14
Row 4: Year: 4 | Opening Debt: 21,045,650,086.14 | Principal: 10,522,825,043.07 | Interest: 2,209,793,259.04 | Closing Debt: 10,522,825,043.07
Row 5: Year: 5 | Opening Debt: 10,522,825,043.07 | Principal: 10,522,825,043.07 | Interest: 1,104,896,629.52 | Closing Debt: 0.00

Tax and Depreciation Basis

Depreciation uses a 12-year straight-line screening basis on total CAPEX of INR 72,959,132,925.40; corporate tax assumed at 25%.

27. Conclusion

Conclusion

The India-mode conceptual study indicates that Ethylene Glycol production can be framed credibly around the selected EO hydration route, Dahej-style cluster siting logic, and a high-capacity continuous plant basis. With headline payback of 76261521411.19 years and IRR of 1.00%, the current feasibility case supports progression to deeper engineering, provided live-cited tariffs, market references, and detailed mechanical code design are carried into the next phase.

28. References

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- user_doc_2: India site note (user supplied document). /Users/ayzaman/.gemini/antigravity/scratch/AoC/examples/eg_sources/india_site_note.md [site; India; ref n/a]
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29. Appendices and Annexures

Appendix Navigation

The appendix package is organized into report-style appendices followed by grouped annexure registers for evidence, properties, flowsheet state, and design backup.

- Appendix A: Material Safety Data Sheet
- Appendix B: Python Code and Reproducibility Bundle
- Appendix C: Process Design Data Sheet
- Annexure I: Evidence and Source Basis
- Annexure II: Property and Thermodynamic Registers
- Annexure III: Process, Decision, and Critic Registers
- Annexure IV: Streams, Flowsheet, and Solver Registers
- Annexure V: Equipment, Utility, and Financial Registers

Appendix A: Material Safety Data Sheet

Screening MSDS-style sheets derived from the selected-route hazard basis, product safety notes, and the report's SHE basis.

Material Safety Data Sheet Summary

Table 180. Material Safety Data Sheet Summary

Columns: Material; Role; Formula; Hazard Summary; Handling / Storage Basis; Emergency / PPE Basis
Row 1: Material: Ethylene oxide | Role: Reactant | Formula: C_2H_4O | Hazard Summary: Catalytic loop and CO_2 handling complexity | Handling / Storage Basis: Closed-loop catalyst management and pressure control | Emergency / PPE Basis: Contained transfer, routine PPE, emergency isolation, and plant-response procedures per the SHE chapter.
Row 2: Material: Water | Role: Reactant | Formula: H_2O | Hazard Summary: Catalytic loop and CO_2 handling complexity | Handling / Storage Basis: Closed-loop catalyst management and pressure control | Emergency / PPE Basis: Contained transfer, routine PPE, emergency isolation, and plant-response procedures per the SHE chapter.
Row 3: Material: Ethylene glycol | Role: Product | Formula: $C_2H_6O_2$ | Hazard Summary: Ethylene oxide feed handling dominates the process hazard envelope because of flammability, toxicity, and polymerization risk.; Ethylene glycol product handling is materially easier, but hot glycol service still requires burn and leak control. | Handling / Storage Basis: Ethylene oxide feed handling dominates the process hazard envelope because of flammability, toxicity, and polymerization risk. | Emergency / PPE Basis: Contained transfer, routine PPE, emergency isolation, and plant-response procedures per the SHE chapter.

Appendix B: Python Code and Reproducibility Bundle

Core Module Register

Table 181. Core Module Register

Columns: Bundle Item; Path; Purpose
Row 1: Bundle Item: Pipeline entry | Path: aoc/pipeline.py | Purpose: Stage orchestration, chapter generation, final report assembly
Row 2: Bundle Item: Engineering calculators | Path: aoc/calculators.py | Purpose: Core process calculations and artifact builders
Row 3: Bundle Item: Unit-operation solvers | Path: aoc/solvers/units.py | Purpose: Reactor, separation, hydraulic, solids, and package design logic
Row 4: Bundle Item: Economics engine | Path: aoc/economics_v2.py | Purpose: Project cost, working capital, financing, and schedules
Row 5: Bundle Item: Property engine | Path: aoc/properties/engine.py | Purpose: Pure-component and phase-equilibrium property routing
Row 6: Bundle Item: Report publisher | Path: aoc/publish.py | Purpose: Markdown assembly, annexures, and PDF rendering support

Reproducibility Artifact Register

Table 182. Reproducibility Artifact Register

Columns: Artifact; Value / Count; Purpose
Row 1: Artifact: Benchmark manifest | Value / Count: ethylene_glycol | Purpose: Benchmark/report parity contract basis
Row 2: Artifact: Resolved values | Value / Count: 121 | Purpose: Cited and estimated value basis carried into the report
Row 3: Artifact: Stream table | Value / Count: 10 | Purpose: Material-balance stream ledger and composition basis
Row 4: Artifact: Equipment list | Value / Count: 13 | Purpose: Equipment inventory for sizing, cost, and datasheets
Row 5: Artifact: Calc trace sections | Value / Count: 12 | Purpose: Trace bundle captured for engineering auditability
Row 6: Artifact: Financial schedule rows | Value / Count: 12 | Purpose: Typed multi-year finance schedule

Appendix C: Process Design Data Sheet

Process Design Data Sheet Summary

Table 183. Process Design Data Sheet Summary

Columns: Equipment; Service; Design Temp (C); Design Pressure (bar); Volume (m3); MoC
Row 1: Equipment: R-101 | Service: Tubular Plug Flow Hydrator | Design Temp (C): 255.0 | Design Pressure (bar): 20.00 | Volume (m3): 27004.165 | MoC: Carbon Steel
Row 2: Equipment: PU-201 | Service: Distillation and purification train | Design Temp (C): 140.0 | Design Pressure (bar): 2.00 | Volume (m3): 21.206 | MoC: Carbon Steel
Row 3: Equipment: V-101 | Service: Intermediate disengagement | Design Temp (C): 85.0 | Design Pressure (bar): 3.00 | Volume (m3): 4860.750 | MoC: Carbon Steel
Row 4: Equipment: E-101 | Service: R-rough to D-rough heat recovery | Design Temp (C): 180.0 | Design Pressure (bar): 8.00 | Volume (m3): 723.932 | MoC: Carbon Steel
Row 5: Equipment: TK-301 | Service: Ethylene Glycol storage via vertical tank farm | Design Temp (C): 45.0 | Design Pressure (bar): 1.20 | Volume (m3): 8580.903 | MoC: SS304
Row 6: Equipment: HX-01 | Service: R-rough to D-rough heat recovery | Design Temp (C): 213.0 | Design Pressure (bar): 11.00 | Volume (m3): 542.949 | MoC: Carbon steel
Row 7: Equipment: HX-01-CTRL | Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station | Design Temp (C): 203.0 | Design Pressure (bar): 9.00 | Volume (m3): 0.150 | MoC: Carbon steel
Row 8: Equipment: HX-01-HDR | Service: R-rough to D-rough heat recovery network header and isolation package | Design Temp (C): 205.0 | Design Pressure (bar): 14.50 | Volume (m3): 0.851 | MoC: Carbon steel
Row 9: Equipment: HX-01-PMP | Service: R-rough to D-rough heat recovery circulation loop | Design Temp (C): 205.0 | Design Pressure (bar): 14.50 | Volume (m3): 0.250 | MoC: Carbon steel
Row 10: Equipment: HX-01-EXP | Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up | Design Temp (C): 210.0 | Design Pressure (bar): 6.00 | Volume (m3): 2.835 | MoC: Carbon steel
Row 11: Equipment: HX-01-RV | Service: R-rough to D-rough heat recovery thermal relief and collection package | Design Temp (C): 213.0 | Design Pressure (bar): 13.50 | Volume (m3): 1.309 | MoC: Carbon steel
Row 12: Equipment: HX-02 | Service: D-rough to E-rough heat recovery | Design Temp (C): 162.0 | Design Pressure (bar): 6.50 | Volume (m3): 30.379 | MoC: SS316
Row 13: Equipment: HX-02-CTRL | Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station | Design Temp (C): 158.0 | Design Pressure (bar): 5.00 | Volume (m3): 0.150 | MoC: Carbon steel

Process Design Data Sheet Bundle

R-101 Datasheet

Service: Tubular Plug Flow Hydrator

Design Data

- design_temperature_c: 255.0
- design_pressure_bar: 20.00
- volume_m3: 27004.165
- material_of_construction: Carbon Steel
- phase_regime: gas_phase_catalytic
- design_conversion_fraction: 0.9600
- kinetic_rate_constant_1_hr: 14.9632
- kinetic_space_time_hr: 0.6000
- kinetic_damkohler_number: 14.9632
- heat_release_density_kw_m3: 0.281
- adiabatic_temperature_rise_c: 414.729

- heat_removal_capacity_kw: 8512.239
- heat_removal_margin_fraction: 0.1200
- thermal_stability_score: 5.00
- runaway_risk_label: moderate
- catalyst_name: CO2 catalytic loop
- catalyst_inventory_kg: 4914758.026
- catalyst_cycle_days: 270.0
- catalyst_regeneration_days: 5.0
- catalyst_whsv_1_hr: 0.0055
- heat_transfer_area_m2: 50.578
- integrated_thermal_duty_kw: 0.000
- integrated_exchange_area_m2: 0.000
- reactor_utility_topology: shared HTM island network (shared_htm) with 1 utility islands
- reactor_utility_architecture_family: shared_htm
- reactor_selected_utility_islands: none
- pressure_class: ASME 600
- hydrotest_pressure_bar: 26.000
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 157.134
- head_thickness_mm: 144.563
- support_type: skirt support
- support_variant: cylindrical skirt with anchor chair
- support_load_case: operating + wind + seismic + nozzle loads
- wind_load_kn: 1152.822
- seismic_load_kn: 46748.132
- piping_load_kn: 240.402
- foundation_footprint_m2: 656.205
- maintenance_clearance_m: 1.800
- platform_required: yes
- platform_area_m2: 61.737
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 8
- nozzle_reinforcement_family: integral repad
- local_shell_load_interaction_factor: 2.359
- nozzle_services: feed, product, utility, instrument
- nozzle_connection_classes: ASME 600, ASME 600, ASME 600, ASME 600
- nozzle_orientations_deg: 0, 90, 180, 270
- nozzle_projection_mm: 9802.9, 9802.9, 9802.9, 9802.9
- local_shell_load_factors: 2.359, 2.359, 2.359, 2.359
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Package Derivation Basis

- volumetric_throughput_basis: 27004.165 m3/h from V/tau
- damkohler_basis: 14.9632 from k*tau
- heat_transfer_area_check: n/a

- residual_utility_basis: 7600.213 kW after integrated duty 0.000 kW

Mechanical Basis

- pressure_class: ASME 600
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 157.134
- head_thickness_mm: 144.563
- support_variant: cylindrical skirt with anchor chair
- support_load_case: operating + wind + seismic + nozzle loads
- foundation_footprint_m2: 656.205
- anchor_group_count: 8
- pipe_rack_tie_in_required: no

Notes

- Selector-driven reactor basis.
- Mechanical basis uses cylindrical skirt with anchor chair support with integral repad nozzle reinforcement.

PU-201 Datasheet

Service: Distillation and purification train

Design Data

- design_temperature_c: 140.0
- design_pressure_bar: 2.00
- volume_m3: 21.206
- material_of_construction: Carbon Steel
- design_stages: 7
- theoretical_stages: 4.278
- minimum_reflux_ratio: 0.150
- reflux_ratio: 0.200
- tray_efficiency: 0.660
- feed_quality_q_factor: 1.106
- murphree_efficiency: 0.634
- top_relative_volatility: 25.909
- bottom_relative_volatility: 14.399
- rectifying_theoretical_stages: 2.444
- stripping_theoretical_stages: 1.833
- rectifying_vapor_load_kg_hr: 6352.735
- stripping_vapor_load_kg_hr: 5683.010
- rectifying_liquid_load_m3_hr: 6090.003
- stripping_liquid_load_m3_hr: 16500.036
- integrated_reboiler_duty_kw: 0.000
- integrated_reboiler_area_m2: 0.000
- allocated_reboiler_recovery_target_kw: 0.000
- allocated_condenser_recovery_target_kw: 0.000
- reboiler_package_type: none
- condenser_package_type: none
- column_utility_topology: shared HTM island network (shared_htm) with 1 utility islands
- column_utility_architecture_family: shared_htm
- column_selected_utility_islands: none
- equilibrium_model: none

- equilibrium_parameter_ids: none
- pressure_class: PN16
- hydrotest_pressure_bar: 3.000
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 4.409
- head_thickness_mm: 4.056
- support_type: skirt support
- support_variant: cylindrical skirt with anchor chair
- support_load_case: operating + wind + seismic + nozzle loads
- wind_load_kn: 9.813
- seismic_load_kn: 45.005
- piping_load_kn: 16.706
- foundation_footprint_m2: 6.620
- maintenance_clearance_m: 1.800
- platform_required: yes
- platform_area_m2: 5.696
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 8
- nozzle_reinforcement_family: integral repad
- local_shell_load_interaction_factor: 1.848
- nozzle_services: feed, overhead, bottoms, instrument
- nozzle_connection_classes: PN16, PN16, PN16, PN16
- nozzle_orientations_deg: 0, 90, 180, 270
- nozzle_projection_mm: 672.2, 672.2, 672.2, 672.2
- local_shell_load_factors: 1.848, 1.848, 1.848, 1.848
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Package Derivation Basis

- fenske_min_stage_basis: 3.278
- underwood_min_reflux_basis: 0.150
- gilliland_stage_basis: 4.278 at R/Rmin 1.333
- rectifying_split_basis: 2.444 stages / 6352.735 kg/h vapor
- stripping_split_basis: 1.833 stages / 5683.010 kg/h vapor

Mechanical Basis

- pressure_class: PN16
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 4.409
- head_thickness_mm: 4.056
- support_variant: cylindrical skirt with anchor chair
- support_load_case: operating + wind + seismic + nozzle loads
- foundation_footprint_m2: 6.620
- anchor_group_count: 8

- pipe_rack_tie_in_required: no

Notes

- Selector-driven process-unit envelope.
- Mechanical basis uses cylindrical skirt with anchor chair support with integral repad nozzle reinforcement.

V-101 Datasheet

Service: Intermediate disengagement

Design Data

- design_temperature_c: 85.0
- design_pressure_bar: 3.00
- volume_m3: 4860.750
- material_of_construction: Carbon Steel
- pressure_class: PN16
- hydrotest_pressure_bar: 4.000
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 15.940
- head_thickness_mm: 14.665
- support_type: base support
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- wind_load_kn: 168.445
- seismic_load_kn: 4673.828
- piping_load_kn: 98.976
- foundation_footprint_m2: 12.125
- maintenance_clearance_m: 1.200
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: shell excess area
- local_shell_load_interaction_factor: 1.743
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: PN16, PN16, PN16
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 4112.9, 4112.9, 4112.9
- local_shell_load_factors: 1.743, 1.743, 1.743
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: PN16
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850

- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 15.940
- head_thickness_mm: 14.665
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 12.125
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- Generic separation hold-up vessel.
- Mechanical basis uses grouted base frame support with shell excess area nozzle reinforcement.

E-101 Datasheet

Service: R-rough to D-rough heat recovery

Design Data

- design_temperature_c: 180.0
- design_pressure_bar: 8.00
- volume_m3: 723.932
- material_of_construction: Carbon Steel
- heat_load_kw: 34024.794
- area_m2: 9049.147
- package_family: process_exchange
- circulation_flow_m3_hr: 1818.329
- phase_change_load_kg_hr: 0.000
- boiling_side_coefficient_w_m2_k: 1200.000
- condensing_side_coefficient_w_m2_k: 1200.000
- selected_package_roles: circulation, controls, exchanger, expansion, header, relief
- selected_train_step_id: omega_catalytic_pinch_htm__shared_htm_step_01
- pressure_class: ASME 150
- hydrotest_pressure_bar: 10.400
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 21.338
- head_thickness_mm: 19.631
- support_type: saddle support
- support_variant: dual saddle
- support_load_case: operating + nozzle loads
- wind_load_kn: 47.328
- seismic_load_kn: 737.603
- piping_load_kn: 53.324
- foundation_footprint_m2: 47.328
- maintenance_clearance_m: 1.500
- platform_required: yes
- platform_area_m2: 3.000
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 6
- nozzle_reinforcement_family: shell excess area

- local_shell_load_interaction_factor: 1.799
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 150, ASME 150, ASME 150
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 2122.0, 2122.0, 2122.0
- local_shell_load_factors: 1.799, 1.799, 1.799
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Package Derivation Basis

- thermal_area_basis: 9049.147 m2 from Q/(U*LMTD)
- package_family: process_exchange
- phase_change_load_basis: 0.000 kg/h

Mechanical Basis

- pressure_class: ASME 150
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 21.338
- head_thickness_mm: 19.631
- support_variant: dual saddle
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 47.328
- anchor_group_count: 6
- pipe_rack_tie_in_required: no

Notes

- Selector-driven exchanger basis.
- Mechanical basis uses dual saddle support with shell excess area nozzle reinforcement.

TK-301 Datasheet

Service: Ethylene Glycol storage via vertical tank farm

Design Data

- design_temperature_c: 45.0
- design_pressure_bar: 1.20
- volume_m3: 8580.903
- material_of_construction: SS304
- inventory_days: 14.1
- working_volume_m3: 7661.521
- total_volume_m3: 8580.903
- dispatch_buffer_days: 9.4
- operating_stock_days: 2.5
- restart_buffer_days: 1.5
- tank_diameter_m: 13.979
- straight_side_height_m: 55.914
- pressure_class: PN16
- hydrotest_pressure_bar: 2.200
- allowable_stress_mpa: 120.000

- joint_efficiency: 0.850
- corrosion_allowance_mm: 2.000
- shell_thickness_mm: 9.188
- head_thickness_mm: 8.453
- support_type: saddle / leg support
- support_variant: dual saddle
- support_load_case: operating + settlement + thermal growth
- wind_load_kn: 313.146
- seismic_load_kn: 7576.879
- piping_load_kn: 129.853
- foundation_footprint_m2: 246.043
- maintenance_clearance_m: 1.200
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 6
- nozzle_reinforcement_family: shell excess area
- local_shell_load_interaction_factor: 1.781
- nozzle_services: inlet, outlet, vent
- nozzle_connection_classes: PN16, PN16, PN16
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 5436.2, 5436.2, 5436.2
- local_shell_load_factors: 1.781, 1.781, 1.781
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Package Derivation Basis

- working_to_total_volume_ratio: 0.892857
- inventory_basis: 14.1 d with dispatch 9.4 d and restart 1.5 d buffers
- tank_slenderness_basis: 3.999857 H/D

Mechanical Basis

- pressure_class: PN16
- allowable_stress_mpa: 120.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 2.000
- shell_thickness_mm: 9.188
- head_thickness_mm: 8.453
- support_variant: dual saddle
- support_load_case: operating + settlement + thermal growth
- foundation_footprint_m2: 246.043
- anchor_group_count: 6
- pipe_rack_tie_in_required: no

Notes

- Selector-driven storage basis.
- Mechanical basis uses dual saddle support with shell excess area nozzle reinforcement.

HX-01 Datasheet

Service: R-rough to D-rough heat recovery

Design Data

- design_temperature_c: 213.0
- design_pressure_bar: 11.00
- volume_m3: 542.949
- material_of_construction: Carbon steel
- pressure_class: ASME 300
- hydrotest_pressure_bar: 14.300
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 25.945
- head_thickness_mm: 23.869
- support_type: saddle support
- support_variant: dual saddle
- support_load_case: operating + nozzle loads
- wind_load_kn: 39.069
- seismic_load_kn: 563.610
- piping_load_kn: 71.656
- foundation_footprint_m2: 39.069
- maintenance_clearance_m: 1.500
- platform_required: yes
- platform_area_m2: 3.000
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 6
- nozzle_reinforcement_family: set-on repad
- local_shell_load_interaction_factor: 1.839
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 300, ASME 300, ASME 300
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 2278.4, 2278.4, 2278.4
- local_shell_load_factors: 1.839, 1.839, 1.839
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 300
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 25.945
- head_thickness_mm: 23.869
- support_variant: dual saddle
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 39.069
- anchor_group_count: 6
- pipe_rack_tie_in_required: no

Notes

- Primary exchanger item derived from selected utility-train step.
- Mechanical basis uses dual saddle support with set-on repad nozzle reinforcement.

HX-01-CTRL Datasheet

Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station

Design Data

- design_temperature_c: 203.0
- design_pressure_bar: 9.00
- volume_m3: 0.150
- material_of_construction: Carbon steel
- pressure_class: ASME 150
- hydrotest_pressure_bar: 11.700
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 0.000
- head_thickness_mm: 0.000
- support_type: instrument frame
- support_variant: panel frame
- support_load_case: skid dead load + piping reaction
- wind_load_kn: 1.500
- seismic_load_kn: 1.000
- piping_load_kn: 3.775
- foundation_footprint_m2: 1.200
- maintenance_clearance_m: 1.000
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: no
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: shell excess area
- local_shell_load_interaction_factor: 1.185
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 150, ASME 150, ASME 150
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 90.0, 90.0, 90.0
- local_shell_load_factors: 1.185, 1.185, 1.185
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 150
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 0.000

- head_thickness_mm: 0.000
- support_variant: panel frame
- support_load_case: skid dead load + piping reaction
- foundation_footprint_m2: 1.200
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- Includes control valve station, bypass, and instrumentation package for the selected train step.
- Mechanical basis uses panel frame support with shell excess area nozzle reinforcement.

HX-01-HDR Datasheet

Service: R-rough to D-rough heat recovery network header and isolation package

Design Data

- design_temperature_c: 205.0
- design_pressure_bar: 14.50
- volume_m3: 0.851
- material_of_construction: Carbon steel
- pressure_class: ASME 300
- hydrotest_pressure_bar: 18.850
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.604
- head_thickness_mm: 7.916
- support_type: pipe rack support
- support_variant: guided rack shoe
- support_load_case: rack piping + thermal expansion
- wind_load_kn: 2.066
- seismic_load_kn: 71.340
- piping_load_kn: 22.476
- foundation_footprint_m2: 2.430
- maintenance_clearance_m: 1.200
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: yes
- pipe_rack_tie_in_required: yes
- anchor_group_count: 2
- nozzle_reinforcement_family: set-on repad
- local_shell_load_interaction_factor: 2.132
- nozzle_services: inlet, outlet, branch, drain
- nozzle_connection_classes: ASME 300, ASME 300, ASME 300, ASME 300
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 521.3, 521.3, 521.3, 521.3
- local_shell_load_factors: 2.132, 2.132, 2.132, 2.132
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 300
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.604
- head_thickness_mm: 7.916
- support_variant: guided rack shoe
- support_load_case: rack piping + thermal expansion
- foundation_footprint_m2: 2.430
- anchor_group_count: 2
- pipe_rack_tie_in_required: yes

Notes

- Network-level header/manifold package added for richer heat-network architecture.
- Mechanical basis uses guided rack shoe support with set-on repad nozzle reinforcement.

HX-01-PMP Datasheet

Service: R-rough to D-rough heat recovery circulation loop

Design Data

- design_temperature_c: 205.0
- design_pressure_bar: 14.50
- volume_m3: 0.250
- material_of_construction: Carbon steel
- pressure_class: ASME 300
- hydrotest_pressure_bar: 18.850
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.604
- head_thickness_mm: 7.916
- support_type: base frame
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- wind_load_kn: 1.500
- seismic_load_kn: 1.059
- piping_load_kn: 11.612
- foundation_footprint_m2: 1.200
- maintenance_clearance_m: 1.200
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: no
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: set-on repad
- local_shell_load_interaction_factor: 1.452
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 300, ASME 300, ASME 300
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 146.7, 146.7, 146.7
- local_shell_load_factors: 1.452, 1.452, 1.452

- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 300
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.604
- head_thickness_mm: 7.916
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 1.200
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- HTM circulation skid sized from recovered duty and selected train topology.
- Mechanical basis uses grouted base frame support with set-on repad nozzle reinforcement.

HX-01-EXP Datasheet

Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up

Design Data

- design_temperature_c: 210.0
- design_pressure_bar: 6.00
- volume_m3: 2.835
- material_of_construction: Carbon steel
- pressure_class: ASME 150
- hydrotest_pressure_bar: 7.800
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 5.309
- head_thickness_mm: 4.884
- support_type: leg support
- support_variant: four-leg support
- support_load_case: operating + settlement + thermal growth
- wind_load_kn: 1.701
- seismic_load_kn: 3.303
- piping_load_kn: 9.393
- foundation_footprint_m2: 1.600
- maintenance_clearance_m: 1.200
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: no
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: set-on repad

- local_shell_load_interaction_factor: 1.440
- nozzle_services: inlet, outlet, vent
- nozzle_connection_classes: ASME 150, ASME 150, ASME 150
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 218.9, 218.9, 218.9
- local_shell_load_factors: 1.440, 1.440, 1.440
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 150
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 5.309
- head_thickness_mm: 4.884
- support_variant: four-leg support
- support_load_case: operating + settlement + thermal growth
- foundation_footprint_m2: 1.600
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- Expansion volume estimated from recovered duty and HTM loop inventory basis.
- Mechanical basis uses four-leg support support with set-on repad nozzle reinforcement.

HX-01-RV Datasheet

Service: R-rough to D-rough heat recovery thermal relief and collection package

Design Data

- design_temperature_c: 213.0
- design_pressure_bar: 13.50
- volume_m3: 1.309
- material_of_construction: Carbon steel
- pressure_class: ASME 300
- hydrotest_pressure_bar: 17.550
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.215
- head_thickness_mm: 7.558
- support_type: base support
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- wind_load_kn: 1.500
- seismic_load_kn: 1.892
- piping_load_kn: 12.123
- foundation_footprint_m2: 1.200
- maintenance_clearance_m: 1.200
- platform_required: no

- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: no
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: set-on repad
- local_shell_load_interaction_factor: 1.493
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 300, ASME 300, ASME 300
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 181.8, 181.8, 181.8
- local_shell_load_factors: 1.493, 1.493, 1.493
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 300
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.215
- head_thickness_mm: 7.558
- support_variant: grouted base frame
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 1.200
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- Relief package includes relief device and small knock-out volume for HTM overpressure screening.
- Mechanical basis uses grouted base frame support with set-on repad nozzle reinforcement.

HX-02 Datasheet

Service: D-rough to E-rough heat recovery

Design Data

- design_temperature_c: 162.0
- design_pressure_bar: 6.50
- volume_m3: 30.379
- material_of_construction: SS316
- pressure_class: ASME 150
- hydrotest_pressure_bar: 8.450
- allowable_stress_mpa: 120.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.953
- head_thickness_mm: 8.237
- support_type: saddle support
- support_variant: dual saddle

- support_load_case: operating + nozzle loads
- wind_load_kn: 5.715
- seismic_load_kn: 40.213
- piping_load_kn: 17.618
- foundation_footprint_m2: 5.715
- maintenance_clearance_m: 1.500
- platform_required: yes
- platform_area_m2: 3.000
- access_ladder_required: yes
- lifting_lug_required: yes
- pipe_rack_tie_in_required: no
- anchor_group_count: 6
- nozzle_reinforcement_family: shell excess area
- local_shell_load_interaction_factor: 1.691
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 150, ASME 150, ASME 150
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 644.4, 644.4, 644.4
- local_shell_load_factors: 1.691, 1.691, 1.691
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 150
- allowable_stress_mpa: 120.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 8.953
- head_thickness_mm: 8.237
- support_variant: dual saddle
- support_load_case: operating + nozzle loads
- foundation_footprint_m2: 5.715
- anchor_group_count: 6
- pipe_rack_tie_in_required: no

Notes

- Primary exchanger item derived from selected utility-train step.
- Mechanical basis uses dual saddle support with shell excess area nozzle reinforcement.

HX-02-CTRL Datasheet

Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station

Design Data

- design_temperature_c: 158.0
- design_pressure_bar: 5.00
- volume_m3: 0.150
- material_of_construction: Carbon steel
- pressure_class: ASME 150
- hydrotest_pressure_bar: 6.500

- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 0.000
- head_thickness_mm: 0.000
- support_type: instrument frame
- support_variant: panel frame
- support_load_case: skid dead load + piping reaction
- wind_load_kn: 1.500
- seismic_load_kn: 1.000
- piping_load_kn: 2.375
- foundation_footprint_m2: 1.200
- maintenance_clearance_m: 1.000
- platform_required: no
- platform_area_m2: 0.000
- access_ladder_required: no
- lifting_lug_required: no
- pipe_rack_tie_in_required: no
- anchor_group_count: 4
- nozzle_reinforcement_family: shell excess area
- local_shell_load_interaction_factor: 1.125
- nozzle_services: inlet, outlet, instrument
- nozzle_connection_classes: ASME 150, ASME 150, ASME 150
- nozzle_orientations_deg: 0, 180, 270
- nozzle_projection_mm: 90.0, 90.0, 90.0
- local_shell_load_factors: 1.125, 1.125, 1.125
- mechanical_code_basis: ASME-style screening logic
- foundation_basis: Support variants differentiate skirt, saddle, leg, rack, and skid foundations with screening footprint and anchor-group logic.
- nozzle_load_basis: Nozzle schedules include reinforcement family, local shell interaction, orientation, projection, and screening piping load allocation.

Mechanical Basis

- pressure_class: ASME 150
- allowable_stress_mpa: 138.000
- joint_efficiency: 0.850
- corrosion_allowance_mm: 3.000
- shell_thickness_mm: 0.000
- head_thickness_mm: 0.000
- support_variant: panel frame
- support_load_case: skid dead load + piping reaction
- foundation_footprint_m2: 1.200
- anchor_group_count: 4
- pipe_rack_tie_in_required: no

Notes

- Includes control valve station, bypass, and instrumentation package for the selected train step.
- Mechanical basis uses panel frame support with shell excess area nozzle reinforcement.

TK-301_pump Datasheet

Service: Ethylene Glycol storage via vertical tank farm transfer

Design Data

- flow_m3_hr: 957.690
- differential_head_m: 52.000
- power_kw: 159.360
- npsh_margin_m: 2.500
- material_of_construction: SS304
- service_basis: storage transfer and dispatch support

Package Derivation Basis

- specific_head_basis: 0.054297 m per m3/h
- power_density_basis: 0.166400 kW per m3/h
- dispatch_turnover_basis: 0.125000 1/h against working storage volume

Notes

- Pump datasheet emitted separately because the pump is a designed auxiliary, not a vessel-family equipment item in equipment_list.

Annexure I: Evidence and Source Basis

Benchmark Manifest

Table 184. Benchmark Manifest

Columns: Field; Value
Row 1: Field: Benchmark ID | Value: ethylene_glycol
Row 2: Field: Target product | Value: Ethylene glycol
Row 3: Field: Archetype family | Value: continuous_liquid_organic_heat_integrated
Row 4: Field: Expected decisions | Value: capacity_case, operating_mode, route_selection, site_selection, reactor_choice, separation_choice, utility_basis, economic_basis
Row 5: Field: Required source domains | Value: technical, safety, market, site, utilities, logistics, regulatory, economics

Source Ranking Table

Table 185. Source Ranking Table

Columns: Domain; Source ID; Score; Selected; Conflict
Row 1: Domain: technical | Source ID: eg_handbook | Score: 78.6 | Selected: yes | Conflict: no
Row 2: Domain: technical | Source ID: eg_kinetics | Score: 73.7 | Selected: yes | Conflict: no
Row 3: Domain: technical | Source ID: user_doc_1 | Score: 60.1 | Selected: no | Conflict: no
Row 4: Domain: safety | Source ID: eg_safety | Score: 74.4 | Selected: yes | Conflict: no
Row 5: Domain: market | Source ID: eg_market_india | Score: 92.0 | Selected: yes | Conflict: no
Row 6: Domain: market | Source ID: india_sheet_1 | Score: 82.1 | Selected: yes | Conflict: no
Row 7: Domain: site | Source ID: eg_sites_india | Score: 96.8 | Selected: yes | Conflict: no
Row 8: Domain: site | Source ID: user_doc_2 | Score: 79.9 | Selected: yes | Conflict: no
Row 9: Domain: utilities | Source ID: eg_utilities_gujarat | Score: 94.2 | Selected: yes | Conflict: no
Row 10: Domain: economics | Source ID: eg_market_india | Score: 92.0 | Selected: yes | Conflict: no
Row 11: Domain: economics | Source ID: india_sheet_1 | Score: 82.1 | Selected: yes | Conflict: no

Source Conflict Register

Table 186. Source Conflict Register

Columns: Domain; Selected; Competing; Score Gap; Blocking; Recommended

Resolution
Row 1: Domain: technical | Selected: eg_handbook | Competing: eg_kinetics, user_doc_1 | Score Gap: 4.94 | Blocking: no | Recommended Resolution: Prefer the highest-ranked India-grounded source or request analyst confirmation if the score gap remains narrow.

Annexure II: Property and Thermodynamic Registers

Property Table

Table 187. Property Table

Columns: Property; Value; Units; Method; Sources
Row 1: Property: Molecular weight | Value: 62.07 | Units: g/mol | Method: sourced | Sources: user_doc_1, user_doc_2, india_sheet_1
Row 2: Property: Melting point | Value: -12.9 | Units: C | Method: sourced | Sources: user_doc_1, user_doc_2, india_sheet_1
Row 3: Property: Boiling point | Value: 197.3 | Units: C | Method: sourced | Sources: user_doc_1, user_doc_2, india_sheet_1
Row 4: Property: Density | Value: 1.113 | Units: g/cm3 | Method: sourced | Sources: user_doc_1, user_doc_2, india_sheet_1
Row 5: Property: Water solubility | Value: Miscible | Units: - | Method: sourced | Sources: user_doc_1, user_doc_2, india_sheet_1

Component Identifier Table

Table 188. Component Identifier Table

Columns: Identifier; Canonical Name; Formula; CAS; Route IDs; Status
Row 1: Identifier: ethylene_glycol | Canonical Name: Ethylene Glycol | Formula: C2H6O2 | CAS: 107-21-1 | Route IDs: chlorohydrin, eo_hydration, omega_catalytic | Status: resolved
Row 2: Identifier: ethylene_oxide | Canonical Name: Ethylene oxide | Formula: C2H4O | CAS: 75-21-8 | Route IDs: eo_hydration, omega_catalytic | Status: resolved
Row 3: Identifier: water | Canonical Name: Water | Formula: H2O | CAS: 7732-18-5 | Route IDs: eo_hydration, omega_catalytic | Status: resolved
Row 4: Identifier: ethylene_chlorohydrin | Canonical Name: Ethylene chlorohydrin | Formula: C2H5ClO | CAS: 107-07-3 | Route IDs: chlorohydrin | Status: resolved
Row 5: Identifier: sodium_hydroxide | Canonical Name: Sodium hydroxide | Formula: NaOH | CAS: 1310-73-2 | Route IDs: chlorohydrin | Status: resolved
Row 6: Identifier: sodium_chloride | Canonical Name: Sodium chloride | Formula: NaCl | CAS: 7647-14-5 | Route IDs: chlorohydrin | Status: resolved

Pure-Component Property Table

Table 189. Pure-Component Property Table

Columns: Component; Property; Value; Units; Ref Temp (C); Method; Status; Sources
Row 1
- Component: Ethylene Glycol
- Property: molecular_weight
- Value: 62.07
- Units: g/mol
- Ref Temp (C): -
- Method: sourced
- Status: resolved
- Sources: user_doc_1, user_doc_2, india_sheet_1

Row 2
- Component: Ethylene Glycol
- Property: normal_boiling_point
- Value: 197.3
- Units: C
- Ref Temp (C): -
- Method: sourced
- Status: resolved

Sources: user_doc_1, user_doc_2, india_sheet_1
Row 3
Component: Ethylene Glycol
Property: melting_point
Value: -12.9
Units: C
Ref Temp (C): -
Method: sourced
Status: resolved
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 4
Component: Ethylene Glycol
Property: liquid_density
Value: 1.113
Units: g/cm3
Ref Temp (C): 25.0
Method: sourced
Status: resolved
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 5
Component: Ethylene Glycol
Property: liquid_viscosity
Value: 0.0161
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 6
Component: Ethylene Glycol
Property: liquid_heat_capacity
Value: 2.42
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 7
Component: Ethylene Glycol
Property: heat_of_vaporization
Value: 800
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 8
Component: Ethylene Glycol
Property: thermal_conductivity
Value: 0.258
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 9
Component: Ethylene oxide
Property: molecular_weight
Value: 44.05
Units: g/mol
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 10
Component: Ethylene oxide
Property: normal_boiling_point
Value: 10.7
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 11
Component: Ethylene oxide
Property: melting_point
Value: -111
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 12
Component: Ethylene oxide
Property: liquid_density
Value: 882
Units: kg/m3
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 13
Component: Ethylene oxide
Property: liquid_viscosity
Value: 0.0012
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 14
Component: Ethylene oxide
Property: liquid_heat_capacity
Value: 2.23
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 15
Component: Ethylene oxide
Property: heat_of_vaporization
Value: 560
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 16
Component: Ethylene oxide
Property: thermal_conductivity
Value: 0.14
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 17
Component: Water
Property: molecular_weight
Value: 18.015
Units: g/mol
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 18
Component: Water
Property: normal_boiling_point
Value: 100
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 19

Component: Water
Property: melting_point
Value: 0
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 20
Component: Water
Property: liquid_density
Value: 997
Units: kg/m3
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 21
Component: Water
Property: liquid_viscosity
Value: 0.00089
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 22
Component: Water
Property: liquid_heat_capacity
Value: 4.18
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 23
Component: Water
Property: heat_of_vaporization
Value: 2257
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 24
Component: Water
Property: thermal_conductivity
Value: 0.598
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 25
Component: Ethylene chlorohydrin
Property: molecular_weight
Value: 80.51
Units: g/mol
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 26
Component: Ethylene chlorohydrin
Property: normal_boiling_point
Value: 128
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 27
Component: Ethylene chlorohydrin

Property: melting_point
Value: -67
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 28
Component: Ethylene chlorohydrin
Property: liquid_density
Value: 1200
Units: kg/m3
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 29
Component: Ethylene chlorohydrin
Property: liquid_viscosity
Value: 0.0031
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 30
Component: Ethylene chlorohydrin
Property: liquid_heat_capacity
Value: 2.1
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 31
Component: Ethylene chlorohydrin
Property: heat_of_vaporization
Value: 430
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 32
Component: Ethylene chlorohydrin
Property: thermal_conductivity
Value: 0.16
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 33
Component: Sodium hydroxide
Property: molecular_weight
Value: 40
Units: g/mol
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 34
Component: Sodium hydroxide
Property: normal_boiling_point
Value: 1388
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 35
Component: Sodium hydroxide
Property: melting_point

Value: 318
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 36
Component: Sodium hydroxide
Property: liquid_density
Value: 2130
Units: kg/m3
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 37
Component: Sodium hydroxide
Property: liquid_viscosity
Value: 0.012
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 38
Component: Sodium hydroxide
Property: liquid_heat_capacity
Value: 1.5
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 39
Component: Sodium hydroxide
Property: heat_of_vaporization
Value: 0
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 40
Component: Sodium hydroxide
Property: thermal_conductivity
Value: 0.5
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 41
Component: Sodium chloride
Property: molecular_weight
Value: 58.44
Units: g/mol
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 42
Component: Sodium chloride
Property: normal_boiling_point
Value: 1465
Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook
Row 43
Component: Sodium chloride
Property: melting_point
Value: 801

Units: C
Ref Temp (C): -
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 44
Component: Sodium chloride
Property: liquid_density
Value: 2160
Units: kg/m3
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 45
Component: Sodium chloride
Property: liquid_viscosity
Value: 0.0045
Units: Pa.s
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 46
Component: Sodium chloride
Property: liquid_heat_capacity
Value: 0.86
Units: kJ/kg-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 47
Component: Sodium chloride
Property: heat_of_vaporization
Value: 0
Units: kJ/kg
Ref Temp (C): 100.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Row 48
Component: Sodium chloride
Property: thermal_conductivity
Value: 6.5
Units: W/m-K
Ref Temp (C): 25.0
Method: calculated
Status: estimated
Sources: user_doc_1, eg_handbook

Property Correlation Register

Table 190. Property Correlation Register

Columns: Identifier; Property; Equation; Parameters; Tmin (C); Tmax (C); Method; Sources

Row 1
Identifier: water
Property: vapor_pressure
Equation: Antoine
Parameters: A=8.07131, B=1730.63, C=233.426
Tmin (C): 1.0
Tmax (C): 100.0
Method: calculated
Sources: user_doc_1, eg_handbook

Binary Interaction Parameter Register

Table 191. Binary Interaction Parameter Register

Columns: Component A; Component B; Model; Tau12; Tau21; Alpha12; Status;

Sources
Row 1
Component A: n/a
Component B: n/a
Model: n/a
Tau12: n/a
Tau21: n/a
Alpha12: n/a
Status: n/a
Sources: n/a

Henry-Law Constant Register

Table 192. Henry-Law Constant Register

Columns: Gas; Solvent; Value; Units; Ref Temp (C); Equation; Status; Sources
Row 1
Gas: n/a
Solvent: n/a
Value: n/a
Units: n/a
Ref Temp (C): n/a
Equation: n/a
Status: n/a
Sources: n/a

Solubility Curve Register

Table 193. Solubility Curve Register

Columns: Solute; Solvent; Equation; Parameters; Tmin (C); Tmax (C); Units; Status; Sources
Row 1
Solute: n/a
Solvent: n/a
Equation: n/a
Parameters: n/a
Tmin (C): n/a
Tmax (C): n/a
Units: n/a
Status: n/a
Sources: n/a

Value Provenance Table

Table 194. Value Provenance Table

Columns: Name; Value; Units; Method; Sensitivity; Blocking; Sources
Row 1
Name: Molecular weight
Value: 62.07
Units: g/mol
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 2
Name: Melting point
Value: -12.9
Units: C
Method: sourced
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 3
Name: Boiling point
Value: 197.3
Units: C
Method: sourced
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1

Row 4
Name: Density
Value: 1.113
Units: g/cm3
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 5
Name: Water solubility
Value: Miscible
Units: -
Method: sourced
Sensitivity: low
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 6
Name: Ethylene Glycol Molecular weight
Value: 62.07
Units: g/mol
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 7
Name: Ethylene Glycol Normal boiling point
Value: 197.3
Units: C
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 8
Name: Ethylene Glycol Melting point
Value: -12.9
Units: C
Method: sourced
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 9
Name: Ethylene Glycol Liquid density
Value: 1.113
Units: g/cm3
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 10
Name: Ethylene Glycol Liquid viscosity
Value: 0.0161
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 11
Name: Ethylene Glycol Liquid heat capacity
Value: 2.42
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 12
Name: Ethylene Glycol Heat of vaporization
Value: 800
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 13

Name: Ethylene Glycol Thermal conductivity
Value: 0.258
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 14
Name: Ethylene oxide Molecular weight
Value: 44.05
Units: g/mol
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 15
Name: Ethylene oxide Normal boiling point
Value: 10.7
Units: C
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 16
Name: Ethylene oxide Melting point
Value: -111
Units: C
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 17
Name: Ethylene oxide Liquid density
Value: 882
Units: kg/m3
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 18
Name: Ethylene oxide Liquid viscosity
Value: 0.0012
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 19
Name: Ethylene oxide Liquid heat capacity
Value: 2.23
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 20
Name: Ethylene oxide Heat of vaporization
Value: 560
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 21
Name: Ethylene oxide Thermal conductivity
Value: 0.14
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 22
Name: Water Molecular weight

Value: 18.015
Units: g/mol
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 23
Name: Water Normal boiling point
Value: 100
Units: C
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 24
Name: Water Melting point
Value: 0
Units: C
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 25
Name: Water Liquid density
Value: 997
Units: kg/m3
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 26
Name: Water Liquid viscosity
Value: 0.00089
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 27
Name: Water Liquid heat capacity
Value: 4.18
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 28
Name: Water Heat of vaporization
Value: 2257
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 29
Name: Water Thermal conductivity
Value: 0.598
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 30
Name: Ethylene chlorohydrin Molecular weight
Value: 80.51
Units: g/mol
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 31
Name: Ethylene chlorohydrin Normal boiling point
Value: 128

Units: C
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 32
Name: Ethylene chlorohydrin Melting point
Value: -67
Units: C
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 33
Name: Ethylene chlorohydrin Liquid density
Value: 1200
Units: kg/m3
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 34
Name: Ethylene chlorohydrin Liquid viscosity
Value: 0.0031
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 35
Name: Ethylene chlorohydrin Liquid heat capacity
Value: 2.1
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 36
Name: Ethylene chlorohydrin Heat of vaporization
Value: 430
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 37
Name: Ethylene chlorohydrin Thermal conductivity
Value: 0.16
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 38
Name: Sodium hydroxide Molecular weight
Value: 40
Units: g/mol
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 39
Name: Sodium hydroxide Normal boiling point
Value: 1388
Units: C
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 40
Name: Sodium hydroxide Melting point
Value: 318
Units: C

Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 41
Name: Sodium hydroxide Liquid density
Value: 2130
Units: kg/m3
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 42
Name: Sodium hydroxide Liquid viscosity
Value: 0.012
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 43
Name: Sodium hydroxide Liquid heat capacity
Value: 1.5
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 44
Name: Sodium hydroxide Heat of vaporization
Value: 0
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 45
Name: Sodium hydroxide Thermal conductivity
Value: 0.5
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 46
Name: Sodium chloride Molecular weight
Value: 58.44
Units: g/mol
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 47
Name: Sodium chloride Normal boiling point
Value: 1465
Units: C
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 48
Name: Sodium chloride Melting point
Value: 801
Units: C
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 49
Name: Sodium chloride Liquid density
Value: 2160
Units: kg/m3
Method: calculated

Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 50
Name: Sodium chloride Liquid viscosity
Value: 0.0045
Units: Pa.s
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 51
Name: Sodium chloride Liquid heat capacity
Value: 0.86
Units: kJ/kg-K
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 52
Name: Sodium chloride Heat of vaporization
Value: 0
Units: kJ/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, eg_handbook
Row 53
Name: Sodium chloride Thermal conductivity
Value: 6.5
Units: W/m-K
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, eg_handbook
Row 54
Name: Reaction enthalpy
Value: -68
Units: kJ/mol
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 55
Name: Reaction Gibbs free energy
Value: -24
Units: kJ/mol
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 56
Name: Activation energy
Value: 58
Units: kJ/mol
Method: sourced
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 57
Name: Pre-exponential factor
Value: 12000000
Units: 1/h
Method: sourced
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 58
Name: Design residence time
Value: 1
Units: h
Method: sourced
Sensitivity: high

Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 59
Name: Reaction conversion
Value: 0.96
Units: fraction
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 60
Name: Reaction selectivity
Value: 0.99
Units: fraction
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 61
Name: Excess ratio
Value: 1.05
Units: mol/mol
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 62
Name: Byproduct selectivity gap
Value: 0.01
Units: fraction
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 63
Name: Product mass flow
Value: 25252.525253
Units: kg/h
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 64
Name: Total fresh-feed mass flow
Value: 25952.368042
Units: kg/h
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 65
Name: Total external outlet mass flow
Value: 25952.368
Units: kg/h
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1, user_doc_2, india_sheet_1
Row 66
Name: Mass-balance closure error
Value: 0
Units: %
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook, user_doc_1,

user_doc_2, india_sheet_1
Row 67
Name: Total heating duty
Value: 5781.464
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 68
Name: Total cooling duty
Value: 8224.582
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 69
Name: Reaction duty
Value: 7600.213139
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 70
Name: Reactor design volume
Value: 27004.164978
Units: m3
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 71
Name: Reactor heat duty
Value: 7600.213
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 72
Name: Reactor rate constant
Value: 14.963167
Units: 1/h
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 73
Name: Reactor heat-removal margin
Value: 0.12
Units: -
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 74
Name: Catalyst inventory
Value: 4914758.026086
Units: kg
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 75
Name: Catalyst cycle length
Value: 270
Units: days
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook

Row 76
Name: Process-unit relative volatility
Value: 19.314777
Units: -
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 77
Name: Process-unit design stages
Value: 7
Units: stages
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 78
Name: Process-unit heating duty
Value: 3179.8052
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 79
Name: Exchanger area
Value: 9049.147
Units: m2
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 80
Name: Inventory days
Value: 14.07
Units: days
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 81
Name: Storage total volume
Value: 8580.903373
Units: m3
Method: calculated
Sensitivity: medium
Blocking: no
Sources: user_doc_1, user_doc_2, india_sheet_1
Row 82
Name: Steam pressure
Value: 20
Units: bar
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 83
Name: Steam cost
Value: 1.8
Units: INR/kg
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 84
Name: Cooling-water cost
Value: 8
Units: INR/m3
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 85

Name: Power cost
Value: 8.5
Units: INR/kWh
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 86
Name: Steam load
Value: 71672.825455
Units: kg/h
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 87
Name: Cooling-water load
Value: 2778.002699
Units: m3/h
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 88
Name: Electrical load
Value: 28732.696447
Units: kW
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 89
Name: Recovered duty
Value: 47256.658
Units: kW
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 90
Name: Total CAPEX
Value: 72959132925.40358
Units: INR
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 91
Name: Total import duty
Value: 353373129.56
Units: INR
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 92
Name: Annual OPEX
Value: 13281189728.486349
Units: INR/y
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 93
Name: Logistics penalty factor
Value: 0.02
Units: fraction
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 94
Name: Annual transport/service penalty cost

Value: 18447067.05
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 95
Name: Annual utility-island service cost
Value: 2952253.22
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 96
Name: Annualized utility-island replacement cost
Value: 15494813.83
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 97
Name: Annual packing replacement cost
Value: 0
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 98
Name: Annual filter media replacement cost
Value: 0
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 99
Name: Annual dryer exhaust treatment cost
Value: 0
Units: INR/y
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 100
Name: Working capital
Value: 1759656318.218509
Units: INR
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 101
Name: Restart loss inventory
Value: 1034880
Units: INR
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 102
Name: Outage buffer inventory
Value: 95515405.581606
Units: INR
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 103
Name: Pre-commissioning inventory
Value: 275688431.02021

Units: INR
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 104
Name: Peak working capital
Value: 1759656318.218509
Units: INR
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 105
Name: Annual revenue
Value: 12600000000
Units: INR/y
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 106
Name: Payback
Value: 76261521411.190002
Units: y
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 107
Name: IRR
Value: 1
Units: %
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 108
Name: Interest during construction
Value: 1542732167.57
Units: INR
Method: calculated
Sensitivity: medium
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 109
Name: Peak working capital
Value: 1759656318.22
Units: INR
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 110
Name: Minimum DSCR
Value: 0.05
Units: ratio
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 111
Name: LLCR
Value: 0
Units: ratio
Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2
Row 112
Name: PLCR
Value: 0
Units: ratio

Method: calculated
Sensitivity: high
Blocking: no
Sources: eg_handbook, india_sheet_1, user_doc_1, user_doc_2

Resolved Value Table

Table 195. Resolved Value Table

Columns: Name; Resolution Level; Status; Sensitivity; Selected Source; Justification
Row 1: Name: Activation energy | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: kinetic_feasibility: matched ranked source.
Row 2: Name: Annual dryer exhaust treatment cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 3: Name: Annual filter media replacement cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 4: Name: Annual OPEX | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 5: Name: Annual packing replacement cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 6: Name: Annual restart loss | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: process_synthesis: matched ranked source.
Row 7: Name: Annual revenue | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 8: Name: Annual transport/service penalty cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 9: Name: Annual utility-island service cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 10: Name: Annualized utility-island replacement cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 11: Name: Boiling point | Resolution Level: 1 | Status: resolved | Sensitivity: medium | Selected Source: user_doc_2 | Justification: Resolved from ranked public sources.
Row 12: Name: Byproduct selectivity gap | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_reaction_system: matched ranked source.
Row 13: Name: Catalyst cycle length | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 14: Name: Catalyst inventory | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 15: Name: Cooling-water cost | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: utility_basis: matched ranked source.
Row 16: Name: Cooling-water load | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: utility_summary: matched ranked source.
Row 17: Name: Density | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: Resolved from ranked public sources.
Row 18: Name: Design residence time | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: kinetic_feasibility: matched ranked source.
Row 19: Name: Electrical load | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: utility_summary: matched ranked source.
Row 20: Name: Electricity price basis | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: india_sheet_1 | Justification: market_capacity: matched ranked source.
Row 21: Name: Ethylene chlorohydrin Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source:

eg_handbook | Justification: property_engine: matched ranked source.
Row 22: Name: Ethylene chlorohydrin Liquid density | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 23: Name: Ethylene chlorohydrin Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 24: Name: Ethylene chlorohydrin Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 25: Name: Ethylene chlorohydrin Melting point | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 26: Name: Ethylene chlorohydrin Molecular weight | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 27: Name: Ethylene chlorohydrin Normal boiling point | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 28: Name: Ethylene chlorohydrin Thermal conductivity | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 29: Name: Ethylene Glycol Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 30: Name: Ethylene Glycol Liquid density | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: property_engine: matched ranked source.
Row 31: Name: Ethylene Glycol Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 32: Name: Ethylene Glycol Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 33: Name: Ethylene Glycol Melting point | Resolution Level: 1 | Status: resolved | Sensitivity: medium | Selected Source: user_doc_2 | Justification: property_engine: matched ranked source.
Row 34: Name: Ethylene Glycol Molecular weight | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: property_engine: matched ranked source.
Row 35: Name: Ethylene Glycol Normal boiling point | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: property_engine: matched ranked source.
Row 36: Name: Ethylene Glycol Thermal conductivity | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 37: Name: Ethylene oxide Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 38: Name: Ethylene oxide Liquid density | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 39: Name: Ethylene oxide Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 40: Name: Ethylene oxide Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 41: Name: Ethylene oxide Melting point | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 42: Name: Ethylene oxide Molecular weight | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 43: Name: Ethylene oxide Normal boiling point | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 44: Name: Ethylene oxide price basis | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: india_sheet_1 | Justification: market_capacity: matched ranked source.
Row 45: Name: Ethylene oxide Thermal conductivity | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 46: Name: Excess ratio | Resolution Level: 2 | Status: estimated |

Sensitivity: medium | Selected Source: user_doc_2 | Justification: material_balance_reaction_system: matched ranked source.
Row 47: Name: Exchanger area | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: heat_exchanger_design: matched ranked source.
Row 48: Name: Finished-goods buffer days | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: process_synthesis: matched ranked source.
Row 49: Name: Interest during construction | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 50: Name: Inventory days | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: equipment_sizing: matched ranked source.
Row 51: Name: IRR | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 52: Name: LLCR | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 53: Name: Logistics penalty factor | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 54: Name: Mass-balance closure error | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_stream_table: matched ranked source.
Row 55: Name: Melting point | Resolution Level: 1 | Status: resolved | Sensitivity: medium | Selected Source: user_doc_2 | Justification: Resolved from ranked public sources.
Row 56: Name: Minimum DSCR | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 57: Name: Molecular weight | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: Resolved from ranked public sources.
Row 58: Name: Monoethylene glycol price basis | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: india_sheet_1 | Justification: market_capacity: matched ranked source.
Row 59: Name: Operating labour price basis | Resolution Level: 1 | Status: resolved | Sensitivity: medium | Selected Source: india_sheet_1 | Justification: market_capacity: matched ranked source.
Row 60: Name: Operating stock days | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: process_synthesis: matched ranked source.
Row 61: Name: Outage buffer inventory | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: working_capital: matched ranked source.
Row 62: Name: Payback | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 63: Name: Peak working capital | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: working_capital: matched ranked source.
Row 64: Name: Peak working capital | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 65: Name: PLCR | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: financial_analysis: matched ranked source.
Row 66: Name: Power cost | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: utility_basis: matched ranked source.
Row 67: Name: Pre-commissioning inventory | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: working_capital: matched ranked source.
Row 68: Name: Pre-exponential factor | Resolution Level: 1 | Status: resolved | Sensitivity: medium | Selected Source: user_doc_2 | Justification: kinetic_feasibility: matched ranked source.
Row 69: Name: Process-unit design stages | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: distillation_design: matched ranked source.
Row 70: Name: Process-unit heating duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: distillation_design: matched ranked source.

Row 71: Name: Process-unit relative volatility | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: distillation_design: matched ranked source.
Row 72: Name: Product mass flow | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_stream_table: matched ranked source.
Row 73: Name: Raw-material buffer days | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: process_synthesis: matched ranked source.
Row 74: Name: Reaction conversion | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_reaction_system: matched ranked source.
Row 75: Name: Reaction duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: energy_balance: matched ranked source.
Row 76: Name: Reaction enthalpy | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: thermodynamic_feasibility: matched ranked source.
Row 77: Name: Reaction Gibbs free energy | Resolution Level: 1 | Status: resolved | Sensitivity: high | Selected Source: user_doc_2 | Justification: thermodynamic_feasibility: matched ranked source.
Row 78: Name: Reaction selectivity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_reaction_system: matched ranked source.
Row 79: Name: Reactor design volume | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 80: Name: Reactor heat duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 81: Name: Reactor heat-removal margin | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 82: Name: Reactor rate constant | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: reactor_design: matched ranked source.
Row 83: Name: Recovered duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: utility_summary: matched ranked source.
Row 84: Name: Restart loss fraction | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: process_synthesis: matched ranked source.
Row 85: Name: Restart loss inventory | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: working_capital: matched ranked source.
Row 86: Name: Sodium chloride Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 87: Name: Sodium chloride Liquid density | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 88: Name: Sodium chloride Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 89: Name: Sodium chloride Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 90: Name: Sodium chloride Melting point | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 91: Name: Sodium chloride Molecular weight | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 92: Name: Sodium chloride Normal boiling point | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 93: Name: Sodium chloride Thermal conductivity | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 94: Name: Sodium hydroxide Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 95: Name: Sodium hydroxide Liquid density | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook |

Justification: property_engine: matched ranked source.
Row 96: Name: Sodium hydroxide Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 97: Name: Sodium hydroxide Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 98: Name: Sodium hydroxide Melting point | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 99: Name: Sodium hydroxide Molecular weight | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 100: Name: Sodium hydroxide Normal boiling point | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 101: Name: Sodium hydroxide Thermal conductivity | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 102: Name: Steam cost | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: utility_basis: matched ranked source.
Row 103: Name: Steam load | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: utility_summary: matched ranked source.
Row 104: Name: Steam pressure | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: utility_basis: matched ranked source.
Row 105: Name: Storage total volume | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: user_doc_2 | Justification: equipment_sizing: matched ranked source.
Row 106: Name: Total CAPEX | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 107: Name: Total cooling duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: energy_balance: matched ranked source.
Row 108: Name: Total external outlet mass flow | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_stream_table: matched ranked source.
Row 109: Name: Total fresh-feed mass flow | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: material_balance_stream_table: matched ranked source.
Row 110: Name: Total heating duty | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: user_doc_2 | Justification: energy_balance: matched ranked source.
Row 111: Name: Total import duty | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: project_cost: matched ranked source.
Row 112: Name: Water Heat of vaporization | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 113: Name: Water Liquid density | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 114: Name: Water Liquid heat capacity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 115: Name: Water Liquid viscosity | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 116: Name: Water Melting point | Resolution Level: 2 | Status: estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 117: Name: Water Molecular weight | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 118: Name: Water Normal boiling point | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 119: Name: Water solubility | Resolution Level: 1 | Status: resolved | Sensitivity: low | Selected Source: user_doc_2 | Justification: Resolved from ranked public sources.
Row 120: Name: Water Thermal conductivity | Resolution Level: 2 | Status:

estimated | Sensitivity: medium | Selected Source: eg_handbook | Justification: property_engine: matched ranked source.
Row 121: Name: Working capital | Resolution Level: 2 | Status: estimated | Sensitivity: high | Selected Source: eg_handbook | Justification: working_capital: matched ranked source.

Property Estimate Register

Table 196. Property Estimate Register

Columns: Property; Selected Method; Candidate Methods; Selected Source; Sensitivity; Blocking
Row 1: Property: Annual dryer exhaust treatment cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 2: Property: Annual filter media replacement cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 3: Property: Annual OPEX | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 4: Property: Annual packing replacement cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 5: Property: Annual restart loss | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 6: Property: Annual revenue | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 7: Property: Annual transport/service penalty cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 8: Property: Annual utility-island service cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 9: Property: Annualized utility-island replacement cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 10: Property: Binary interaction parameters for Ethylene Glycol / Ethylene oxide | Selected Method: estimated | Candidate Methods: sourced, calculated, estimated | Selected Source: - | Sensitivity: high | Blocking: no
Row 11: Property: Binary interaction parameters for Ethylene Glycol / Water | Selected Method: estimated | Candidate Methods: sourced, calculated, estimated | Selected Source: - | Sensitivity: high | Blocking: no
Row 12: Property: Binary interaction parameters for Ethylene oxide / Water | Selected Method: estimated | Candidate Methods: sourced, calculated, estimated | Selected Source: - | Sensitivity: high | Blocking: no
Row 13: Property: Byproduct selectivity gap | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 14: Property: Catalyst cycle length | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 15: Property: Catalyst inventory | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 16: Property: Cooling-water cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 17: Property: Cooling-water load | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 18: Property: Electrical load | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 19: Property: Ethylene chlorohydrin Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 20: Property: Ethylene chlorohydrin Liquid density | Selected Method:

calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 21: Property: Ethylene chlorohydrin Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 22: Property: Ethylene chlorohydrin Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 23: Property: Ethylene chlorohydrin Melting point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 24: Property: Ethylene chlorohydrin Molecular weight | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 25: Property: Ethylene chlorohydrin Normal boiling point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 26: Property: Ethylene chlorohydrin Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 27: Property: Ethylene Glycol Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 28: Property: Ethylene Glycol Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 29: Property: Ethylene Glycol Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 30: Property: Ethylene Glycol Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 31: Property: Ethylene oxide Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 32: Property: Ethylene oxide Liquid density | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 33: Property: Ethylene oxide Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 34: Property: Ethylene oxide Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 35: Property: Ethylene oxide Melting point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 36: Property: Ethylene oxide Molecular weight | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 37: Property: Ethylene oxide Normal boiling point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 38: Property: Ethylene oxide Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 39: Property: Excess ratio | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 40: Property: Exchanger area | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 41: Property: Finished-goods buffer days | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 42: Property: Interest during construction | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 43: Property: Inventory days | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 44: Property: IRR | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook |

Sensitivity: high | Blocking: no
Row 45: Property: LLCR | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 46: Property: Logistics penalty factor | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 47: Property: Mass-balance closure error | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 48: Property: Minimum DSCR | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 49: Property: Operating stock days | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 50: Property: Outage buffer inventory | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 51: Property: Payback | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 52: Property: Peak working capital | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 53: Property: Peak working capital | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 54: Property: PLCR | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 55: Property: Power cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 56: Property: Pre-commissioning inventory | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 57: Property: Process-unit design stages | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 58: Property: Process-unit heating duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 59: Property: Process-unit relative volatility | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 60: Property: Product mass flow | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 61: Property: Raw-material buffer days | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 62: Property: Reaction conversion | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 63: Property: Reaction duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 64: Property: Reaction selectivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 65: Property: Reactor design volume | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 66: Property: Reactor heat duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 67: Property: Reactor heat-removal margin | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 68: Property: Reactor rate constant | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 69: Property: Recovered duty | Selected Method: calculated | Candidate

Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 70: Property: Restart loss fraction | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 71: Property: Restart loss inventory | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 72: Property: Sodium chloride Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 73: Property: Sodium chloride Liquid density | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 74: Property: Sodium chloride Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 75: Property: Sodium chloride Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 76: Property: Sodium chloride Melting point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 77: Property: Sodium chloride Molecular weight | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 78: Property: Sodium chloride Normal boiling point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 79: Property: Sodium chloride Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 80: Property: Sodium hydroxide Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 81: Property: Sodium hydroxide Liquid density | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 82: Property: Sodium hydroxide Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 83: Property: Sodium hydroxide Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 84: Property: Sodium hydroxide Melting point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 85: Property: Sodium hydroxide Molecular weight | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 86: Property: Sodium hydroxide Normal boiling point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 87: Property: Sodium hydroxide Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 88: Property: Steam cost | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 89: Property: Steam load | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 90: Property: Steam pressure | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 91: Property: Storage total volume | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: medium | Blocking: no
Row 92: Property: Total CAPEX | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 93: Property: Total cooling duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no

Row 94: Property: Total external outlet mass flow | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 95: Property: Total fresh-feed mass flow | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 96: Property: Total heating duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: user_doc_2 | Sensitivity: high | Blocking: no
Row 97: Property: Total import duty | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 98: Property: Water Heat of vaporization | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 99: Property: Water Liquid density | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 100: Property: Water Liquid heat capacity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 101: Property: Water Liquid viscosity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 102: Property: Water Melting point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 103: Property: Water Molecular weight | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 104: Property: Water Normal boiling point | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no
Row 105: Property: Water Thermal conductivity | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: medium | Blocking: no
Row 106: Property: Working capital | Selected Method: calculated | Candidate Methods: sourced, calculated, estimated, analogy | Selected Source: eg_handbook | Sensitivity: high | Blocking: no

Property Requirement Coverage

Table 197. Property Requirement Coverage

Columns: Stage; Identifier; Property; Status; Allow Estimated; Blocking
Row 1: Stage: thermodynamic_feasibility | Identifier: ethylene_chlorohydrin | Property: molecular_weight | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 2: Stage: thermodynamic_feasibility | Identifier: ethylene_glycol | Property: molecular_weight | Status: covered | Allow Estimated: yes | Blocking: yes
Row 3: Stage: thermodynamic_feasibility | Identifier: ethylene_oxide | Property: molecular_weight | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 4: Stage: thermodynamic_feasibility | Identifier: sodium_chloride | Property: molecular_weight | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 5: Stage: thermodynamic_feasibility | Identifier: sodium_hydroxide | Property: molecular_weight | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 6: Stage: thermodynamic_feasibility | Identifier: water | Property: molecular_weight | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 7: Stage: thermodynamic_feasibility | Identifier: ethylene_chlorohydrin | Property: normal_boiling_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 8: Stage: thermodynamic_feasibility | Identifier: ethylene_glycol | Property: normal_boiling_point | Status: covered | Allow Estimated: yes | Blocking: yes
Row 9: Stage: thermodynamic_feasibility | Identifier: ethylene_oxide | Property: normal_boiling_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 10: Stage: thermodynamic_feasibility | Identifier: sodium_chloride |

Property: normal_boiling_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 11: Stage: thermodynamic_feasibility | Identifier: sodium_hydroxide | Property: normal_boiling_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 12: Stage: thermodynamic_feasibility | Identifier: water | Property: normal_boiling_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 13: Stage: thermodynamic_feasibility | Identifier: ethylene_chlorohydrin | Property: melting_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 14: Stage: thermodynamic_feasibility | Identifier: ethylene_glycol | Property: melting_point | Status: covered | Allow Estimated: yes | Blocking: yes
Row 15: Stage: thermodynamic_feasibility | Identifier: ethylene_oxide | Property: melting_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 16: Stage: thermodynamic_feasibility | Identifier: sodium_chloride | Property: melting_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 17: Stage: thermodynamic_feasibility | Identifier: sodium_hydroxide | Property: melting_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 18: Stage: thermodynamic_feasibility | Identifier: water | Property: melting_point | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 19: Stage: energy_balance | Identifier: ethylene_chlorohydrin | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 20: Stage: energy_balance | Identifier: ethylene_glycol | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 21: Stage: energy_balance | Identifier: ethylene_oxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 22: Stage: energy_balance | Identifier: sodium_chloride | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 23: Stage: energy_balance | Identifier: sodium_hydroxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 24: Stage: energy_balance | Identifier: water | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 25: Stage: energy_balance | Identifier: ethylene_chlorohydrin | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 26: Stage: energy_balance | Identifier: ethylene_glycol | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 27: Stage: energy_balance | Identifier: ethylene_oxide | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 28: Stage: energy_balance | Identifier: sodium_chloride | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 29: Stage: energy_balance | Identifier: sodium_hydroxide | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 30: Stage: energy_balance | Identifier: water | Property: heat_of_vaporization | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 31: Stage: energy_balance | Identifier: ethylene_chlorohydrin | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 32: Stage: energy_balance | Identifier: ethylene_glycol | Property: liquid_density | Status: covered | Allow Estimated: yes | Blocking: yes
Row 33: Stage: energy_balance | Identifier: ethylene_oxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 34: Stage: energy_balance | Identifier: sodium_chloride | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 35: Stage: energy_balance | Identifier: sodium_hydroxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 36: Stage: energy_balance | Identifier: water | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes

Row 37: Stage: reactor_design | Identifier: ethylene_chlorohydrin | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 38: Stage: reactor_design | Identifier: ethylene_glycol | Property: liquid_density | Status: covered | Allow Estimated: yes | Blocking: yes
Row 39: Stage: reactor_design | Identifier: ethylene_oxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 40: Stage: reactor_design | Identifier: sodium_chloride | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 41: Stage: reactor_design | Identifier: sodium_hydroxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 42: Stage: reactor_design | Identifier: water | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 43: Stage: reactor_design | Identifier: ethylene_chlorohydrin | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 44: Stage: reactor_design | Identifier: ethylene_glycol | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 45: Stage: reactor_design | Identifier: ethylene_oxide | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 46: Stage: reactor_design | Identifier: sodium_chloride | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 47: Stage: reactor_design | Identifier: sodium_hydroxide | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 48: Stage: reactor_design | Identifier: water | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 49: Stage: reactor_design | Identifier: ethylene_chlorohydrin | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 50: Stage: reactor_design | Identifier: ethylene_glycol | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 51: Stage: reactor_design | Identifier: ethylene_oxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 52: Stage: reactor_design | Identifier: sodium_chloride | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 53: Stage: reactor_design | Identifier: sodium_hydroxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 54: Stage: reactor_design | Identifier: water | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 55: Stage: reactor_design | Identifier: ethylene_chlorohydrin | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 56: Stage: reactor_design | Identifier: ethylene_glycol | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 57: Stage: reactor_design | Identifier: ethylene_oxide | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 58: Stage: reactor_design | Identifier: sodium_chloride | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 59: Stage: reactor_design | Identifier: sodium_hydroxide | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 60: Stage: reactor_design | Identifier: water | Property: thermal_conductivity | Status: estimated | Allow Estimated: yes | Blocking: no
Row 61: Stage: distillation_design | Identifier: ethylene_chlorohydrin | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 62: Stage: distillation_design | Identifier: ethylene_glycol | Property: liquid_density | Status: covered | Allow Estimated: yes | Blocking: yes
Row 63: Stage: distillation_design | Identifier: ethylene_oxide |

Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 64: Stage: distillation_design | Identifier: sodium_chloride | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 65: Stage: distillation_design | Identifier: sodium_hydroxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 66: Stage: distillation_design | Identifier: water | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 67: Stage: distillation_design | Identifier: ethylene_chlorohydrin | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 68: Stage: distillation_design | Identifier: ethylene_glycol | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 69: Stage: distillation_design | Identifier: ethylene_oxide | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 70: Stage: distillation_design | Identifier: sodium_chloride | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 71: Stage: distillation_design | Identifier: sodium_hydroxide | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 72: Stage: distillation_design | Identifier: water | Property: liquid_viscosity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 73: Stage: distillation_design | Identifier: ethylene_chlorohydrin | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 74: Stage: distillation_design | Identifier: ethylene_glycol | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 75: Stage: distillation_design | Identifier: ethylene_oxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 76: Stage: distillation_design | Identifier: sodium_chloride | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 77: Stage: distillation_design | Identifier: sodium_hydroxide | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 78: Stage: distillation_design | Identifier: water | Property: liquid_heat_capacity | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 79: Stage: mechanical_design_moc | Identifier: ethylene_chlorohydrin | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 80: Stage: mechanical_design_moc | Identifier: ethylene_glycol | Property: liquid_density | Status: covered | Allow Estimated: yes | Blocking: yes
Row 81: Stage: mechanical_design_moc | Identifier: ethylene_oxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 82: Stage: mechanical_design_moc | Identifier: sodium_chloride | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 83: Stage: mechanical_design_moc | Identifier: sodium_hydroxide | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes
Row 84: Stage: mechanical_design_moc | Identifier: water | Property: liquid_density | Status: estimated | Allow Estimated: yes | Blocking: yes

Separation Thermodynamics Basis

Table 198. Separation Thermodynamics Basis

Columns: Field; Value
Row 1: Field: Family | Value: distillation
Row 2: Field: Pressure (bar) | Value: 3.240
Row 3: Field: Light key | Value: Water
Row 4: Field: Heavy key | Value: Ethylene Glycol

Row 5: Field: Activity model | Value: ideal_raoult_missing_bip_fallback
Row 6: Field: Top temperature (C) | Value: 108.000
Row 7: Field: Bottom temperature (C) | Value: 203.300
Row 8: Field: Average alpha | Value: 19.314777
Row 9: Field: Method | Value: ideal_raoult_missing_bip_fallback
Row 10: Field: Missing BIP pairs | Value: ethylene_glycol__water

Component K-Value Register

Table 199. Component K-Value Register

Columns: Component; Top T (C); Top Gamma; Top K; Top Method; Bottom T (C); Bottom Gamma; Bottom K; Bottom Method
Row 1
Component: Ethylene Glycol
Top T (C): 108.0
Top Gamma: 1.000000
Top K: 0.015973
Top Method: ideal_raoult_missing_bip_clausius_clapeyron
Bottom T (C): 203.3
Bottom Gamma: 1.000000
Bottom K: 0.366948
Bottom Method: ideal_raoult_missing_bip_clausius_clapeyron
Row 2
Component: Ethylene oxide
Top T (C): 108.0
Top Gamma: 1.000000
Top K: 4.508641
Top Method: ideal_raoult_clausius_clapeyron
Bottom T (C): 203.3
Bottom Gamma: 1.000000
Bottom K: 21.392694
Bottom Method: ideal_raoult_clausius_clapeyron
Row 3
Component: Water
Top T (C): 108.0
Top Gamma: 1.000000
Top K: 0.413846
Top Method: ideal_raoult_missing_bip_antoine_extrapolated
Bottom T (C): 203.3
Bottom Gamma: 1.000000
Bottom K: 5.283621
Bottom Method: ideal_raoult_missing_bip_antoine_extrapolated

Blocked / Unresolved Property Register

Table 200. Blocked / Unresolved Property Register

Columns: Component; Property; Package Status; Blocking
Row 1: Component: n/a | Property: n/a | Package Status: n/a | Blocking: n/a

Annexure III: Process, Decision, and Critic Registers

Reaction Extent Set

Table 201. Reaction Extent Set

Columns: Extent; Kind; Representative Component; Fraction of Converted Feed; Status
Row 1: Extent: omega_catalytic_main_extent | Kind: main | Representative Component: Ethylene glycol | Fraction of Converted Feed: 0.990000 | Status: converged
Row 2: Extent: omega_catalytic_side_extent_1 | Kind: byproduct | Representative Component: Trace heavy glycols | Fraction of Converted Feed: 0.010000 | Status: estimated

Byproduct Closure Register

Table 202. Byproduct Closure Register

Columns: Component; Basis; Allocation Fraction; Provenance; Status
Row 1: Component: Trace heavy glycols | Basis: Declared route byproducts | Allocation Fraction: 1.000000 | Provenance: declared_trace | Status: estimated

Process Archetype

Table 203. Process Archetype

Columns: Field; Value
Row 1: Field: Archetype ID | Value: organic_liquid_distillation_flash
Row 2: Field: Compound family | Value: organic
Row 3: Field: Product phase | Value: liquid
Row 4: Field: Feed phase | Value: mixed
Row 5: Field: Separation family | Value: distillation_flash
Row 6: Field: Heat profile | Value: moderate_temperature_integrated
Row 7: Field: Hazard intensity | Value: high
Row 8: Field: Benchmark profile | Value: ethylene_glycol

Route Family Profiles

Table 204. Route Family Profiles

Columns: Route; Family; Reactor Basis; Separation Train; Heat Style; Maturity; India Fit; Critic Flags; India Blocker
Row 1
- Route: eo_hydration
- Family: Liquid Hydration Train
- Reactor Basis: Tubular plug-flow hydrator
- Separation Train: EO flash -> water removal -> vacuum glycol distillation
- Heat Style: condenser_reboiler_cluster
- Maturity: 92.00
- India Fit: 90.00
- Critic Flags: dehydration_duty, glycol_split_vle_basis, water_recycle_closure
- India Blocker: none

Row 2
- Route: omega_catalytic
- Family: Liquid Hydration Train
- Reactor Basis: Tubular plug-flow hydrator
- Separation Train: EO flash -> water removal -> vacuum glycol distillation
- Heat Style: condenser_reboiler_cluster
- Maturity: 92.00
- India Fit: 90.00
- Critic Flags: dehydration_duty, glycol_split_vle_basis, water_recycle_closure
- India Blocker: none

Row 3
- Route: chlorohydrin
- Family: Chlorinated Hydrolysis Train
- Reactor Basis: Agitated hydrolysis CSTR train
- Separation Train: Salt removal -> brine handling -> water removal -> purification
- Heat Style: utility_led_with_brine_management
- Maturity: 30.00
- India Fit: 22.00
- Critic Flags: chloride_waste_burden, brine_handling_penalty, corrosion_material_risk
- India Blocker: Route generates chloride-heavy waste and is not preferred for India deployment under the current policy basis.

Unit-Operation Family Expansion

Table 205. Unit-Operation Family Expansion

Columns: Service Group; Candidate; Status; Score; Description
Row 1: Service Group: reactor | Candidate: tubular_plug_flow_hydrator | Status: preferred | Score: 102.00 | Description: Tubular hydrator with strong dehydration integration fit
Row 2: Service Group: reactor | Candidate: jacketed_cstr_train | Status: preferred | Score: 78.00 | Description: CSTR fallback for liquid hydration service

Row 3: Service Group: reactor | Candidate: high_pressure_carbonylation_loop | Status: blocked | Score: 18.00 | Description: High-pressure carbonylation loop
Row 4: Service Group: separation | Candidate: distillation_train | Status: preferred | Score: 100.00 | Description: Flash and glycol distillation train
Row 5: Service Group: separation | Candidate: extractive_distillation_train | Status: fallback | Score: 82.00 | Description: Extractive/vacuum polishing distillation
Row 6: Service Group: separation | Candidate: packed_absorption_train | Status: blocked | Score: 20.00 | Description: Packed absorption train

Table 206. Unit-Operation Family Expansion

Columns: Field; Value
Row 1: Field: Route | Value: omega_catalytic
Row 2: Field: Route family | Value: Liquid Hydration Train
Row 3: Field: Dominant phase pattern | Value: liquid_reactive
Row 4: Field: Supporting operations | Value: flash, dehydration_column, glycol_polishing_column, heat_exchanger_cluster, reactor, distillation, heat_exchanger
Row 5: Field: Applicability critics | Value: dehydration_duty, glycol_split_vle_basis, water_recycle_closure, relative_volatility_claims, reboiler_duty_burden

Sparse-Data Policy

Table 207. Sparse-Data Policy

Columns: Stage; Subject; Artifact Family; Allow Estimated; Allow Analogy; Allow Heuristic Fallback; Min Confidence; Status; Triggered Items
Row 1
- Stage: thermodynamic_feasibility
- Subject: molecular_weight
- Artifact Family: pure_component_identity
- Allow Estimated: yes
- Allow Analogy: no
- Allow Heuristic Fallback: no
- Min Confidence: 0.55
- Status: covered
- Triggered Items: -

Row 2
- Stage: thermodynamic_feasibility
- Subject: normal_boiling_point
- Artifact Family: pure_component_thermo
- Allow Estimated: yes
- Allow Analogy: no
- Allow Heuristic Fallback: no
- Min Confidence: 0.60
- Status: covered
- Triggered Items: -

Row 3
- Stage: energy_balance
- Subject: liquid_heat_capacity
- Artifact Family: bulk_thermal_properties
- Allow Estimated: yes
- Allow Analogy: no
- Allow Heuristic Fallback: no
- Min Confidence: 0.55
- Status: covered
- Triggered Items: -

Row 4
- Stage: energy_balance
- Subject: heat_of_vaporization
- Artifact Family: bulk_thermal_properties
- Allow Estimated: yes
- Allow Analogy: no
- Allow Heuristic Fallback: no
- Min Confidence: 0.55
- Status: covered
- Triggered Items: -

Row 5

Stage: reactor_design
Subject: liquid_density
Artifact Family: transport_properties
Allow Estimated: yes
Allow Analogy: no
Allow Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 6
Stage: reactor_design
Subject: liquid_viscosity
Artifact Family: transport_properties
Allow Estimated: yes
Allow Analogy: no
Allow Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 7
Stage: distillation_design
Subject: liquid_density
Artifact Family: separation_transport
Allow Estimated: yes
Allow Analogy: no
Allow Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 8
Stage: distillation_design
Subject: liquid_viscosity
Artifact Family: separation_transport
Allow Estimated: yes
Allow Analogy: no
Allow Heuristic Fallback: no
Min Confidence: 0.60
Status: covered
Triggered Items: -

Row 9
Stage: distillation_design
Subject: binary_interaction_parameters
Artifact Family: vle_nonideal_basis
Allow Estimated: no
Allow Analogy: no
Allow Heuristic Fallback: yes
Min Confidence: 0.00
Status: warning
Triggered Items: ethylene_glycol__ethylene_oxide, ethylene_glycol__water, ethylene_oxide__water

Operations Planning Basis

Table 208. Operations Planning Basis

Columns: Field; Value

Row 1: Field: Service family | Value: continuous_liquid_purification
Row 2: Field: Recommended mode | Value: continuous
Row 3: Field: Availability policy | Value: continuous_liquid_train
Row 4: Field: Raw-material buffer (d) | Value: 19.600
Row 5: Field: Finished-goods buffer (d) | Value: 9.400
Row 6: Field: Operating stock (d) | Value: 2.500
Row 7: Field: Restart buffer (d) | Value: 1.500
Row 8: Field: Startup ramp (d) | Value: 2.000
Row 9: Field: Campaign length (d) | Value: 170.000
Row 10: Field: Cleaning cycle (d) | Value: 150.000
Row 11: Field: Cleaning downtime (d) | Value: 2.000
Row 12: Field: Throughput loss fraction | Value: 0.012344
Row 13: Field: Restart loss fraction | Value: 0.006160
Row 14: Field: Annual restart loss (kg/y) | Value: 16,426.667

Specialist Decision Fabric

Table 209. Specialist Decision Fabric

Columns: Packet; Specialist; Selected; Critic Status
Row 1: Packet: technical_sources_packet | Specialist: SourceScout | Selected: eg_handbook | Critic Status: pass
Row 2: Packet: safety_sources_packet | Specialist: SourceScout | Selected: eg_safety | Critic Status: pass
Row 3: Packet: market_sources_packet | Specialist: SourceScout | Selected: eg_market_india | Critic Status: pass
Row 4: Packet: site_sources_packet | Specialist: SourceScout | Selected: eg_sites_india | Critic Status: pass
Row 5: Packet: utilities_sources_packet | Specialist: SourceScout | Selected: eg_utilities_gujarat | Critic Status: pass
Row 6: Packet: logistics_sources_packet | Specialist: SourceScout | Selected: - | Critic Status: pass
Row 7: Packet: regulatory_sources_packet | Specialist: SourceScout | Selected: - | Critic Status: pass
Row 8: Packet: economics_sources_packet | Specialist: SourceScout | Selected: eg_market_india | Critic Status: pass
Row 9: Packet: cost_dryer_exhaust_treatment_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 10: Packet: cost_filter_media_replacement_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 11: Packet: cost_annual_opex_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 12: Packet: cost_packing_replacement_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 13: Packet: ops_annual_restart_loss_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 14: Packet: financial_revenue_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 15: Packet: cost_transport_service_penalty_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 16: Packet: cost_utility_island_service_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 17: Packet: cost_utility_island_replacement_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 18: Packet: ethylene_glycol__ethylene_oxide_bip_estimate_packet | Specialist: PropertyResolver | Selected: estimated | Critic Status: warning
Row 19: Packet: ethylene_glycol__water_bip_estimate_packet | Specialist: PropertyResolver | Selected: estimated | Critic Status: warning
Row 20: Packet: ethylene_oxide__water_bip_estimate_packet | Specialist: PropertyResolver | Selected: estimated | Critic Status: warning
Row 21: Packet: rxn_byproduct_gap_fraction_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 22: Packet: reactor_catalyst_cycle_days_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 23: Packet: reactor_catalyst_inventory_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 24: Packet: utility_cw_cost_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 25: Packet: utility_cw_load_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 26: Packet: utility_power_load_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 27: Packet: ethylene_chlorohydrin_heat_of_vaporization_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 28: Packet: ethylene_chlorohydrin_liquid_density_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 29: Packet: ethylene_chlorohydrin_liquid_heat_capacity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 30: Packet: ethylene_chlorohydrin_liquid_viscosity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 31: Packet: ethylene_chlorohydrin_melting_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 32: Packet: ethylene_chlorohydrin_molecular_weight_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 33: Packet: ethylene_chlorohydrin_normal_boiling_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass

Row 34: Packet: ethylene_chlorohydrin_thermal_conductivity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 35: Packet: ethylene_glycol_heat_of_vaporization_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 36: Packet: ethylene_glycol_liquid_heat_capacity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 37: Packet: ethylene_glycol_liquid_viscosity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 38: Packet: ethylene_glycol_thermal_conductivity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 39: Packet: ethylene_oxide_heat_of_vaporization_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 40: Packet: ethylene_oxide_liquid_density_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 41: Packet: ethylene_oxide_liquid_heat_capacity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 42: Packet: ethylene_oxide_liquid_viscosity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 43: Packet: ethylene_oxide_melting_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 44: Packet: ethylene_oxide_molecular_weight_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 45: Packet: ethylene_oxide_normal_boiling_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 46: Packet: ethylene_oxide_thermal_conductivity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 47: Packet: rxn_excess_ratio_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 48: Packet: exchanger_area_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 49: Packet: ops_finished_goods_buffer_days_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 50: Packet: financial_idc_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 51: Packet: storage_inventory_days_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 52: Packet: financial_irr_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 53: Packet: financial_llcr_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 54: Packet: cost_logistics_penalty_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 55: Packet: stream_closure_error_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 56: Packet: financial_minimum_dscr_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 57: Packet: ops_operating_stock_days_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 58: Packet: wc_outage_buffer_inventory_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 59: Packet: financial_payback_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 60: Packet: wc_peak_working_capital_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 61: Packet: financial_peak_working_capital_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 62: Packet: financial_plcr_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 63: Packet: utility_power_cost_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 64: Packet: wc_precommissioning_inventory_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 65: Packet: process_unit_stages_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 66: Packet: process_unit_reboiler_duty_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 67: Packet: process_unit_relative_volatility_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 68: Packet: stream_product_mass_flow_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 69: Packet: ops_raw_material_buffer_days_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 70: Packet: rxn_conversion_estimate_packet | Specialist:

PropertyResolver | Selected: calculated | Critic Status: pass
Row 71: Packet: energy_reaction_duty_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 72: Packet: rxn_selectivity_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 73: Packet: reactor_design_volume_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 74: Packet: reactor_heat_duty_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 75: Packet: reactor_heat_removal_margin_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 76: Packet: reactor_kinetic_rate_constant_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 77: Packet: utility_recovered_duty_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 78: Packet: ops_restart_loss_fraction_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 79: Packet: wc_restart_loss_inventory_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 80: Packet: sodium_chloride_heat_of_vaporization_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 81: Packet: sodium_chloride_liquid_density_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 82: Packet: sodium_chloride_liquid_heat_capacity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 83: Packet: sodium_chloride_liquid_viscosity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 84: Packet: sodium_chloride_melting_point_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 85: Packet: sodium_chloride_molecular_weight_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 86: Packet: sodium_chloride_normal_boiling_point_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 87: Packet: sodium_chloride_thermal_conductivity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 88: Packet: sodium_hydroxide_heat_of_vaporization_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 89: Packet: sodium_hydroxide_liquid_density_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 90: Packet: sodium_hydroxide_liquid_heat_capacity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 91: Packet: sodium_hydroxide_liquid_viscosity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 92: Packet: sodium_hydroxide_melting_point_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 93: Packet: sodium_hydroxide_molecular_weight_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 94: Packet: sodium_hydroxide_normal_boiling_point_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 95: Packet: sodium_hydroxide_thermal_conductivity_estimate_packet |
Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 96: Packet: utility_steam_cost_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 97: Packet: utility_steam_load_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 98: Packet: utility_steam_pressure_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 99: Packet: storage_total_volume_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 100: Packet: cost_total_capex_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 101: Packet: energy_total_cooling_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 102: Packet: stream_total_external_out_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 103: Packet: stream_total_feed_mass_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 104: Packet: energy_total_heating_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 105: Packet: cost_import_duty_total_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 106: Packet: water_heat_of_vaporization_estimate_packet | Specialist:
PropertyResolver | Selected: calculated | Critic Status: pass
Row 107: Packet: water_liquid_density_estimate_packet | Specialist:

PropertyResolver | Selected: calculated | Critic Status: pass
Row 108: Packet: water_liquid_heat_capacity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 109: Packet: water_liquid_viscosity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 110: Packet: water_melting_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 111: Packet: water_molecular_weight_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 112: Packet: water_normal_boiling_point_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 113: Packet: water_thermal_conductivity_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 114: Packet: wc_working_capital_estimate_packet | Specialist: PropertyResolver | Selected: calculated | Critic Status: pass
Row 115: Packet: capacity_case_packet | Specialist: SourceScout | Selected: 200000_tpa | Critic Status: pass
Row 116: Packet: site_selection_packet | Specialist: SourceScout | Selected: Dahej | Critic Status: warning
Row 117: Packet: route_selection_packet | Specialist: RouteSynthesizer | Selected: omega_catalytic | Critic Status: warning
Row 118: Packet: reactor_choice_packet | Specialist: UnitDesignEngineer | Selected: tubular_plug_flow_hydrator | Critic Status: warning
Row 119: Packet: separation_choice_packet | Specialist: UnitDesignEngineer | Selected: distillation_train | Critic Status: warning
Row 120: Packet: exchanger_choice_packet | Specialist: UtilityIntegrator | Selected: shell_and_tube | Critic Status: pass
Row 121: Packet: storage_choice_packet | Specialist: UnitDesignEngineer | Selected: vertical_tank_farm | Critic Status: pass
Row 122: Packet: moc_choice_packet | Specialist: MechanicalDesigner | Selected: carbon_steel | Critic Status: warning
Row 123: Packet: utility_basis_packet | Specialist: UtilityIntegrator | Selected: hp_steam_20bar | Critic Status: pass
Row 124: Packet: procurement_basis_packet | Specialist: CostEngineer | Selected: domestic_cluster_procurement | Critic Status: pass
Row 125: Packet: logistics_basis_packet | Specialist: CostEngineer | Selected: coastal_cluster_dispatch | Critic Status: pass
Row 126: Packet: financing_basis_packet | Specialist: CostEngineer | Selected: debt_equity_70_30 | Critic Status: warning
Row 127: Packet: economic_basis_packet | Specialist: CostEngineer | Selected: no_recovery_counterfactual | Critic Status: warning
Row 128: Packet: operating_mode_packet | Specialist: Critic | Selected: continuous | Critic Status: warning
Row 129: Packet: heat_integration_omega_catalytic_packet | Specialist: Critic | Selected: omega_catalytic_pinch_htm | Critic Status: warning
Row 130: Packet: thermo_method_packet | Specialist: PropertyResolver | Selected: direct_public_data | Critic Status: pass
Row 131: Packet: kinetics_method_packet | Specialist: PropertyResolver | Selected: cited_rate_law | Critic Status: pass
Row 132: Packet: control_architecture_packet | Specialist: Critic | Selected: cascade_override | Critic Status: pass
Row 133: Packet: layout_choice_packet | Specialist: Critic | Selected: segregated_hazard_zones | Critic Status: pass
Row 134: Packet: heat_integration_omega_catalytic_packet | Specialist: Critic | Selected: omega_catalytic_pinch_htm | Critic Status: warning
Row 135: Packet: distillation_design:sparse_data_policy_warning:sparse_data_policy_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 136: Packet: energy_balance:solve_result:0_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 137: Packet: energy_balance:solve_result:1_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 138: Packet: energy_balance:solve_result:2_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 139: Packet: equipment_sizing:reactor_choice_family_critic_flags_prese nt:reactor_choice_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 140: Packet: equipment_sizing:separation_choice_family_critic_flags_pr esent:separation_choice_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 141: Packet: financial_analysis:economic_basis_counterfactual_selected :economic_basis_decision_packet | Specialist: Critic | Selected: - | Critic Status: warning
Row 142: Packet: financial_analysis:economic_basis_rejects_selected_recove

ry:economic_basis_decision_packet | Specialist: Critic | Selected: - |
Critic Status: warning
Row 143: Packet: financial_analysis:financing_operability_tension:financin
g_basis_decision_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 144: Packet: financial_analysis:heat_recovery_without_economic_benefit
:cost_model_packet | Specialist: Critic | Selected: - | Critic Status:
warning
Row 145: Packet: financial_analysis:operating_mode_integrated_economics_te
nsion:operating_mode_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 146: Packet:
financial_analysis:weak_dscr_screening:financial_model_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 147: Packet:
financial_analysis:weak_llcr_screening:financial_model_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 148: Packet:
financial_analysis:weak_plcr_screening:financial_model_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 149: Packet: kinetic_feasibility:architecture_package_fallback_thermo:
separation_thermo_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 150: Packet: kinetic_feasibility:architecture_package_sparse_data_comp
ounded:sparse_data_policy_packet | Specialist: Critic | Selected: - |
Critic Status: warning
Row 151: Packet:
material_balance:composition_closure_estimated:stream_table_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 152: Packet:
material_balance:flowsheet_unit_unresolved_basis:flowsheet_case_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 153: Packet: route_selection:distillation_family_property_coverage_wea
k:property_packages_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 154: Packet:
route_selection:route_family_critic_flags_present:route_selection_packet |
Specialist: Critic | Selected: - | Critic Status: warning
Row 155: Packet: thermodynamic_feasibility:architecture_package_fallback_t
hermo:separation_thermo_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 156: Packet: thermodynamic_feasibility:architecture_package_sparse_dat
a_compounded:sparse_data_policy_packet | Specialist: Critic | Selected: -
| Critic Status: warning
Row 157: Packet: thermodynamic_feasibility:missing_binary_interaction_cove
rage:separation_thermo_packet | Specialist: Critic | Selected: - | Critic
Status: warning
Row 158: Packet: thermodynamic_feasibility:separation_thermo_fallback_acti
ve:separation_thermo_packet | Specialist: Critic | Selected: - | Critic
Status: warning

Critic Registry

Table 210. Critic Registry

Columns: Stage; Family; Severity; Code; Artifact; Recommended Action
Row 1: Stage: distillation_design | Family: general | Severity: warning |
Code: sparse_data_policy_warning | Artifact: sparse_data_policy |
Recommended Action: Review the critic finding and either improve the basis
or require analyst approval.
Row 2: Stage: energy_balance | Family: flowsheet_solver | Severity:
warning | Code: solve_result_critic_message | Artifact: solve_result |
Recommended Action: Review the solved flowsheet packets and close the
critic message before treating the design as robust.
Row 3: Stage: energy_balance | Family: flowsheet_solver | Severity:
warning | Code: solve_result_critic_message | Artifact: solve_result |
Recommended Action: Review the solved flowsheet packets and close the
critic message before treating the design as robust.
Row 4: Stage: energy_balance | Family: flowsheet_solver | Severity:
warning | Code: solve_result_critic_message | Artifact: solve_result |
Recommended Action: Review the solved flowsheet packets and close the
critic message before treating the design as robust.
Row 5: Stage: equipment_sizing | Family: unit_applicability | Severity:

warning | Code: reactor_choice_family_critic_flags_present | Artifact: reactor_choice | Recommended Action: Re-rank the unit family or revise the equipment basis to match the route family.
Row 6: Stage: equipment_sizing | Family: separation_design | Severity: warning | Code: separation_choice_family_critic_flags_present | Artifact: separation_choice | Recommended Action: Recheck the selected separation family and design basis against the solved process.
Row 7: Stage: financial_analysis | Family: economics_finance | Severity: warning | Code: economic_basis_counterfactual_selected | Artifact: economic_basis_decision | Recommended Action: Align the economic and financing basis with the technical architecture and scenarios.
Row 8: Stage: financial_analysis | Family: economics_finance | Severity: warning | Code: economic_basis_rejects_selected_recovery | Artifact: economic_basis_decision | Recommended Action: Align the economic and financing basis with the technical architecture and scenarios.
Row 9: Stage: financial_analysis | Family: general | Severity: warning | Code: financing_operability_tension | Artifact: financing_basis_decision | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 10: Stage: financial_analysis | Family: economics_finance | Severity: warning | Code: heat_recovery_without_economic_benefit | Artifact: cost_model | Recommended Action: Align the economic and financing basis with the technical architecture and scenarios.
Row 11: Stage: financial_analysis | Family: economics_finance | Severity: warning | Code: operating_mode_integrated_economics_tension | Artifact: operating_mode | Recommended Action: Align the economic and financing basis with the technical architecture and scenarios.
Row 12: Stage: financial_analysis | Family: cross_chapter | Severity: warning | Code: weak_dscr_screening | Artifact: financial_model | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 13: Stage: financial_analysis | Family: cross_chapter | Severity: warning | Code: weak_llcr_screening | Artifact: financial_model | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 14: Stage: financial_analysis | Family: cross_chapter | Severity: warning | Code: weak_plcr_screening | Artifact: financial_model | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 15: Stage: kinetic_feasibility | Family: thermo_property | Severity: warning | Code: architecture_package_fallback_thermo | Artifact: separation_thermo | Recommended Action: Resolve the missing thermodynamic/property basis or explicitly downgrade the design claim.
Row 16: Stage: kinetic_feasibility | Family: general | Severity: warning | Code: architecture_package_sparse_data_compounded | Artifact: sparse_data_policy | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 17: Stage: material_balance | Family: general | Severity: warning | Code: composition_closure_estimated | Artifact: stream_table | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 18: Stage: material_balance | Family: evidence | Severity: warning | Code: flowsheet_unit_unresolved_basis | Artifact: flowsheet_case | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 19: Stage: route_selection | Family: thermo_property | Severity: warning | Code: distillation_family_property_coverage_weak | Artifact: property_packages | Recommended Action: Resolve the missing thermodynamic/property basis or explicitly downgrade the design claim.
Row 20: Stage: route_selection | Family: cross_chapter | Severity: warning | Code: route_family_critic_flags_present | Artifact: route_selection | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 21: Stage: thermodynamic_feasibility | Family: thermo_property | Severity: warning | Code: architecture_package_fallback_thermo | Artifact: separation_thermo | Recommended Action: Resolve the missing thermodynamic/property basis or explicitly downgrade the design claim.
Row 22: Stage: thermodynamic_feasibility | Family: general | Severity: warning | Code: architecture_package_sparse_data_compounded | Artifact: sparse_data_policy | Recommended Action: Review the critic finding and either improve the basis or require analyst approval.
Row 23: Stage: thermodynamic_feasibility | Family: thermo_property | Severity: warning | Code: missing_binary_interaction_coverage | Artifact: separation_thermo | Recommended Action: Resolve the missing

thermodynamic/property basis or explicitly downgrade the design claim.
Row 24: Stage: thermodynamic_feasibility | Family: thermo_property | Severity: warning | Code: separation_thermo_fallback_active | Artifact: separation_thermo | Recommended Action: Resolve the missing thermodynamic/property basis or explicitly downgrade the design claim.

Alternative Set Summary

Table 211. Alternative Set Summary

Columns: Set; Candidate; Description; Score; Selected
Row 1: Set: capacity_case | Candidate: 150000_tpa | Description: 150,000 TPA design basis | Score: 82.50 | Selected: no
Row 2: Set: capacity_case | Candidate: 200000_tpa | Description: 200,000 TPA design basis | Score: 100.00 | Selected: yes
Row 3: Set: capacity_case | Candidate: 250000_tpa | Description: 250,000 TPA design basis | Score: 82.50 | Selected: no
Row 4: Set: operating_mode | Candidate: continuous | Description: Continuous operation | Score: 77.75 | Selected: yes
Row 5: Set: route_screen | Candidate: eo_hydration | Description: Ethylene oxide hydration | Score: 95.00 | Selected: yes
Row 6: Set: route_screen | Candidate: omega_catalytic | Description: OMEGA catalytic route via ethylene carbonate | Score: 89.00 | Selected: no
Row 7: Set: route_screen | Candidate: chlorohydrin | Description: Ethylene chlorohydrin hydrolysis | Score: 59.00 | Selected: no
Row 8: Set: reactor_family | Candidate: cstr_train | Description: CSTR train | Score: 72.00 | Selected: no
Row 9: Set: reactor_family | Candidate: pfr_tubular | Description: Tubular PFR | Score: 88.00 | Selected: yes
Row 10: Set: reactor_family | Candidate: batch_agitated | Description: Batch agitated vessel | Score: 58.00 | Selected: no
Row 11: Set: separation_family | Candidate: distillation_flash | Description: distillation flash | Score: 86.00 | Selected: yes
Row 12: Set: separation_family | Candidate: mixed_purification | Description: Mixed purification train | Score: 68.00 | Selected: no

Annexure IV: Streams, Flowsheet, and Solver Registers

Stream Table

Table 212. Stream Table

Columns: Stream; Description; Component; kg/h; kmol/h
Row 1: Stream: S-101 | Description: Ethylene oxide fresh feed | Component: Ethylene oxide | kg/h: 18378.663 | kmol/h: 417.222770
Row 2: Stream: S-102 | Description: Water fresh feed | Component: Water | kg/h: 7573.705 | kmol/h: 420.411049
Row 3: Stream: S-150 | Description: Mixed reactor feed | Component: Ethylene oxide | kg/h: 18856.564 | kmol/h: 428.071822
Row 4: Stream: S-150 | Description: Mixed reactor feed | Component: Water | kg/h: 8097.300 | kmol/h: 449.475413
Row 5: Stream: S-201 | Description: Reactor effluent | Component: Ethylene glycol | kg/h: 25252.525 | kmol/h: 406.839460
Row 6: Stream: S-201 | Description: Reactor effluent | Component: Ethylene oxide | kg/h: 754.263 | kmol/h: 17.122873
Row 7: Stream: S-201 | Description: Reactor effluent | Component: Trace heavy glycols | kg/h: 253.021 | kmol/h: 2.547741
Row 8: Stream: S-201 | Description: Reactor effluent | Component: Water | kg/h: 694.054 | kmol/h: 38.526464
Row 9: Stream: S-202 | Description: Primary separation overhead / vent / purge | Component: Ethylene oxide | kg/h: 709.007 | kmol/h: 16.095501
Row 10: Stream: S-202 | Description: Primary separation overhead / vent / purge | Component: Trace heavy glycols | kg/h: 10.121 | kmol/h: 0.101910
Row 11: Stream: S-202 | Description: Primary separation overhead / vent / purge | Component: Water | kg/h: 258.493 | kmol/h: 14.348765
Row 12: Stream: S-203 | Description: Primary separation bottoms / rich liquid | Component: Ethylene glycol | kg/h: 25252.525 | kmol/h: 406.839460
Row 13: Stream: S-203 | Description: Primary separation bottoms / rich liquid | Component: Ethylene oxide | kg/h: 45.256 | kmol/h: 1.027372
Row 14: Stream: S-203 | Description: Primary separation bottoms / rich liquid | Component: Trace heavy glycols | kg/h: 242.900 | kmol/h: 2.445831

Row 15: Stream: S-203 | Description: Primary separation bottoms / rich liquid | Component: Water | kg/h: 41.643 | kmol/h: 2.311588
Row 16: Stream: S-301 | Description: Cleanup / recycle recovery stream | Component: Ethylene glycol | kg/h: 25252.525 | kmol/h: 406.839460
Row 17: Stream: S-301 | Description: Cleanup / recycle recovery stream | Component: Ethylene oxide | kg/h: 523.157 | kmol/h: 11.876425
Row 18: Stream: S-301 | Description: Cleanup / recycle recovery stream | Component: Trace heavy glycols | kg/h: 242.900 | kmol/h: 2.445831
Row 19: Stream: S-301 | Description: Cleanup / recycle recovery stream | Component: Water | kg/h: 565.238 | kmol/h: 31.375952
Row 20: Stream: S-401 | Description: Light ends / recoverable recycle | Component: Ethylene oxide | kg/h: 477.901 | kmol/h: 10.849052
Row 21: Stream: S-401 | Description: Light ends / recoverable recycle | Component: Water | kg/h: 523.595 | kmol/h: 29.064364
Row 22: Stream: S-402 | Description: On-spec product | Component: Ethylene glycol | kg/h: 24974.747 | kmol/h: 402.364225

Flowsheet Graph

Table 213. Flowsheet Graph

Columns: Node; Unit Type; Upstream; Downstream; Representative Streams
Row 1: Node: feed_prep | Unit Type: feed_preparation | Upstream: purification | Downstream: reactor | Representative Streams: S-150, S-401
Row 2: Node: reactor | Unit Type: reactor | Upstream: feed_prep | Downstream: primary_flash | Representative Streams: S-150, S-201
Row 3: Node: primary_flash | Unit Type: flash | Upstream: reactor | Downstream: waste_treatment, recycle_recovery | Representative Streams: S-201, S-202, S-203
Row 4: Node: waste_treatment | Unit Type: waste_handling | Upstream: primary_flash, purification | Downstream: - | Representative Streams: S-202, S-403
Row 5: Node: recycle_recovery | Unit Type: recycle | Upstream: primary_flash | Downstream: purification | Representative Streams: S-203, S-301
Row 6: Node: purification | Unit Type: distillation | Upstream: recycle_recovery | Downstream: feed_prep, storage, waste_treatment | Representative Streams: S-301, S-401, S-402, S-403
Row 7: Node: storage | Unit Type: storage | Upstream: purification | Downstream: - | Representative Streams: S-402

Flowsheet Case

Table 214. Flowsheet Case

Columns: Unit; Type; Inlet Streams; Outlet Streams; Closure Status; Coverage; Unresolved Sensitivities
Row 1
- Unit: feed_prep
- Type: feed_preparation
- Inlet Streams: S-101, S-102, S-401
- Outlet Streams: S-150
- Closure Status: converged
- Coverage: complete
- Unresolved Sensitivities: -

Row 2
- Unit: reactor
- Type: reactor
- Inlet Streams: S-150
- Outlet Streams: S-201
- Closure Status: converged
- Coverage: complete
- Unresolved Sensitivities: -

Row 3
- Unit: primary_flash
- Type: flash
- Inlet Streams: S-201
- Outlet Streams: S-202, S-203
- Closure Status: converged
- Coverage: complete
- Unresolved Sensitivities: -

Row 4
- Unit: waste_treatment

Type: waste_handling
Inlet Streams: S-202, S-403
Outlet Streams: -
Closure Status: converged
Coverage: partial
Unresolved Sensitivities: -

Row 5
Unit: recycle_recovery
Type: recycle
Inlet Streams: S-203
Outlet Streams: S-301
Closure Status: estimated
Coverage: partial
Unresolved Sensitivities: material closure above converged range

Row 6
Unit: purification
Type: distillation
Inlet Streams: S-301
Outlet Streams: S-401, S-402, S-403
Closure Status: converged
Coverage: partial
Unresolved Sensitivities: mixed or unresolved phase basis remains in solved streams

Row 7
Unit: storage
Type: storage
Inlet Streams: S-402
Outlet Streams: -
Closure Status: converged
Coverage: partial
Unresolved Sensitivities: -

Unit Composition States

Table 215. Unit Composition States

Columns: Unit; Type; Inlet Phase; Outlet Phase; Top Outlet Components; Status

Row 1: Unit: feed_prep | Type: feed_preparation | Inlet Phase: gas | Outlet Phase: gas | Top Outlet Components: Water, Ethylene oxide | Status: converged
Row 2: Unit: primary_flash | Type: flash | Inlet Phase: gas | Outlet Phase: gas | Top Outlet Components: Ethylene glycol, Ethylene oxide, Water | Status: converged
Row 3: Unit: purification | Type: distillation | Inlet Phase: gas | Outlet Phase: gas | Top Outlet Components: Ethylene glycol, Water, Ethylene oxide | Status: estimated
Row 4: Unit: reactor | Type: reactor | Inlet Phase: gas | Outlet Phase: gas | Top Outlet Components: Ethylene glycol, Water, Ethylene oxide | Status: converged
Row 5: Unit: recycle_recovery | Type: recycle | Inlet Phase: gas | Outlet Phase: gas | Top Outlet Components: Ethylene glycol, Water, Ethylene oxide | Status: converged

Composition Closures

Table 216. Composition Closures

Columns: Unit; Reactive; Inlet Fraction Sum; Outlet Fraction Sum; Composition Error (%); Status

Row 1: Unit: feed_prep | Reactive: no | Inlet Fraction Sum: 1.000000 | Outlet Fraction Sum: 1.000000 | Composition Error (%): 0.000000 | Status: converged
Row 2: Unit: primary_flash | Reactive: no | Inlet Fraction Sum: 1.000000 | Outlet Fraction Sum: 1.000000 | Composition Error (%): 0.000000 | Status: converged
Row 3: Unit: purification | Reactive: no | Inlet Fraction Sum: 1.000000 | Outlet Fraction Sum: 1.000000 | Composition Error (%): 0.000000 | Status: estimated
Row 4: Unit: reactor | Reactive: yes | Inlet Fraction Sum: 1.000000 | Outlet Fraction Sum: 1.000000 | Composition Error (%): 0.000000 | Status: converged
Row 5: Unit: recycle_recovery | Reactive: no | Inlet Fraction Sum:

1.000000 | Outlet Fraction Sum: 1.000000 | Composition Error (%): 0.000000 | Status: converged

Mixture Property Packages

Table 217. Mixture Property Packages

Columns: Unit; Phase; Cp (kJ/kg-K); Density (kg/m3); Viscosity (Pa.s); Status; Estimated Properties

Row 1
- Unit: feed_prep
- Phase: gas
- Cp (kJ/kg-K): 2.815805
- Density (kg/m3): 913.659630
- Viscosity (Pa.s): 0.00109696
- Status: estimated
- Estimated Properties: heat_of_vaporization, liquid_density, liquid_heat_capacity, liquid_viscosity, thermal_conductivity

Row 2
- Unit: primary_flash
- Phase: gas
- Cp (kJ/kg-K): 2.421918
- Density (kg/m3): 1.160609
- Viscosity (Pa.s): 0.01356677
- Status: estimated
- Estimated Properties: heat_of_vaporization, liquid_density, liquid_heat_capacity, liquid_viscosity, thermal_conductivity

Row 3
- Unit: purification
- Phase: gas
- Cp (kJ/kg-K): 2.451983
- Density (kg/m3): 1.157576
- Viscosity (Pa.s): 0.01447883
- Status: estimated
- Estimated Properties: heat_of_vaporization, liquid_density, liquid_heat_capacity, liquid_viscosity, thermal_conductivity

Row 4
- Unit: reactor
- Phase: gas
- Cp (kJ/kg-K): 2.447612
- Density (kg/m3): 1.177802
- Viscosity (Pa.s): 0.01303724
- Status: estimated
- Estimated Properties: heat_of_vaporization, liquid_density, liquid_heat_capacity, liquid_viscosity, thermal_conductivity

Row 5
- Unit: recycle_recovery
- Phase: gas
- Cp (kJ/kg-K): 2.441621
- Density (kg/m3): 1.162047
- Viscosity (Pa.s): 0.01351854
- Status: estimated
- Estimated Properties: heat_of_vaporization, liquid_density, liquid_heat_capacity, liquid_viscosity, thermal_conductivity

Solve Result

Table 218. Solve Result

Columns: Unit; Closure Error (%); Solve Status; Coverage; Unresolved Sensitivities

Row 1: Unit: feed_prep | Closure Error (%): 0.000000 | Solve Status: converged | Coverage: complete | Unresolved Sensitivities: -
Row 2: Unit: reactor | Closure Error (%): 0.000004 | Solve Status: converged | Coverage: complete | Unresolved Sensitivities: -
Row 3: Unit: primary_flash | Closure Error (%): 0.000000 | Solve Status: converged | Coverage: complete | Unresolved Sensitivities: -
Row 4: Unit: waste_treatment | Closure Error (%): 0.000000 | Solve Status: converged | Coverage: partial | Unresolved Sensitivities: -
Row 5: Unit: recycle_recovery | Closure Error (%): 3.767314 | Solve Status: estimated | Coverage: partial | Unresolved Sensitivities: material closure above converged range

Row 6: Unit: purification | Closure Error (%): 0.000000 | Solve Status: converged | Coverage: partial | Unresolved Sensitivities: mixed or unresolved phase basis remains in solved streams
Row 7: Unit: storage | Closure Error (%): 0.000000 | Solve Status: converged | Coverage: partial | Unresolved Sensitivities: -

Phase Split Specs

Table 219. Phase Split Specs

Columns: Unit; Family; Mechanism; Inlet Phases; Product Phase; Waste Phase; Recycle Phase; Status
Row 1
- Unit: primary_flash
- Family: flash
- Mechanism: vapor_liquid_equilibrium
- Inlet Phases: gas
- Product Phase: gas
- Waste Phase: gas
- Recycle Phase: -
- Status: complete

Row 2
- Unit: purification
- Family: distillation
- Mechanism: vapor_liquid_equilibrium
- Inlet Phases: gas
- Product Phase: gas
- Waste Phase: mixed
- Recycle Phase: gas
- Status: complete

Separator Performance

Table 220. Separator Performance

Columns: Unit; Family; Status; Product Mass Frac; Waste Mass Frac; Recycle Mass Frac; Split Closure (%)
Row 1
- Unit: primary_flash
- Family: flash
- Status: converged
- Product Mass Frac: 0.949115
- Waste Mass Frac: 0.050885
- Recycle Mass Frac: 0.000000
- Split Closure (%): 0.000000

Row 2
- Unit: purification
- Family: distillation
- Status: converged
- Product Mass Frac: 0.939472
- Waste Mass Frac: 0.022855
- Recycle Mass Frac: 0.037673
- Split Closure (%): 0.000000

Recycle Convergence Summaries

Table 221. Recycle Convergence Summaries

Columns: Loop; Source Unit; Status; Max Component Error (%); Mean Component Error (%); Max Iterations; Purge Policy
Row 1
- Loop: recycle_recovery_recycle_loop
- Source Unit: recycle_recovery
- Status: converged
- Max Component Error (%): 0.100007
- Mean Component Error (%): 0.100001
- Max Iterations: 2
- Purge Policy: aqueous_diluent=0.001, heavy_byproducts=0.001, recoverable_reactants=0.001

Row 2
- Loop: purification_recycle_loop
- Source Unit: purification

Status: estimated
Max Component Error (%): 8.650479
Mean Component Error (%): 4.004473
Max Iterations: 2
Purge Policy: aqueous_diluent=0.142, heavy_byproducts=1.000, recoverable_reactants=0.166

Annexure V: Equipment, Utility, and Financial Registers

Equipment Table

Table 222. Equipment Table

Columns: ID; Type; Service; Volume (m3); Design Temp (C); Design Pressure (bar); MoC

Row 1
ID: R-101
Type: Reactor
Service: Tubular Plug Flow Hydrator
Volume (m3): 27004.165
Design Temp (C): 255.0
Design Pressure (bar): 20.0
MoC: Carbon Steel

Row 2
ID: PU-201
Type: Distillation column
Service: Distillation and purification train
Volume (m3): 21.206
Design Temp (C): 140.0
Design Pressure (bar): 2.0
MoC: Carbon Steel

Row 3
ID: V-101
Type: Flash drum
Service: Intermediate disengagement
Volume (m3): 4860.750
Design Temp (C): 85.0
Design Pressure (bar): 3.0
MoC: Carbon Steel

Row 4
ID: E-101
Type: Heat exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 723.932
Design Temp (C): 180.0
Design Pressure (bar): 8.0
MoC: Carbon Steel

Row 5
ID: TK-301
Type: Storage tank
Service: Ethylene Glycol storage via vertical tank farm
Volume (m3): 8580.903
Design Temp (C): 45.0
Design Pressure (bar): 1.2
MoC: SS304

Row 6
ID: HX-01
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Volume (m3): 542.949
Design Temp (C): 213.0
Design Pressure (bar): 11.0
MoC: Carbon steel

Row 7
ID: HX-01-CTRL
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Design Temp (C): 203.0

Design Pressure (bar): 9.0
MoC: Carbon steel
Row 8
ID: HX-01-HDR
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Volume (m3): 0.851
Design Temp (C): 205.0
Design Pressure (bar): 14.5
MoC: Carbon steel
Row 9
ID: HX-01-PMP
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Volume (m3): 0.250
Design Temp (C): 205.0
Design Pressure (bar): 14.5
MoC: Carbon steel
Row 10
ID: HX-01-EXP
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Volume (m3): 2.835
Design Temp (C): 210.0
Design Pressure (bar): 6.0
MoC: Carbon steel
Row 11
ID: HX-01-RV
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Volume (m3): 1.309
Design Temp (C): 213.0
Design Pressure (bar): 13.5
MoC: Carbon steel
Row 12
ID: HX-02
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Volume (m3): 30.379
Design Temp (C): 162.0
Design Pressure (bar): 6.5
MoC: SS316
Row 13
ID: HX-02-CTRL
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Volume (m3): 0.150
Design Temp (C): 158.0
Design Pressure (bar): 5.0
MoC: Carbon steel

Mechanical Design Summary View

Table 223. Mechanical Design Summary View

Columns: Equipment; Type; Shell t (mm); Head t (mm); Class; Support; Load Case; Platform
Row 1
Equipment: R-101
Type: Reactor
Shell t (mm): 157.13
Head t (mm): 144.56
Class: ASME 600
Support: skirt support
Load Case: operating + wind + seismic + nozzle loads
Platform: yes
Row 2
Equipment: PU-201

Type: Distillation column
Shell t (mm): 4.41
Head t (mm): 4.06
Class: PN16
Support: skirt support
Load Case: operating + wind + seismic + nozzle loads
Platform: yes
Row 3
Equipment: V-101
Type: Flash drum
Shell t (mm): 15.94
Head t (mm): 14.66
Class: PN16
Support: base support
Load Case: operating + nozzle loads
Platform: no
Row 4
Equipment: E-101
Type: Heat exchanger
Shell t (mm): 21.34
Head t (mm): 19.63
Class: ASME 150
Support: saddle support
Load Case: operating + nozzle loads
Platform: yes
Row 5
Equipment: TK-301
Type: Storage tank
Shell t (mm): 9.19
Head t (mm): 8.45
Class: PN16
Support: saddle / leg support
Load Case: operating + settlement + thermal growth
Platform: no
Row 6
Equipment: HX-01
Type: HTM loop exchanger
Shell t (mm): 25.95
Head t (mm): 23.87
Class: ASME 300
Support: saddle support
Load Case: operating + nozzle loads
Platform: yes
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Support: instrument frame
Load Case: skid dead load + piping reaction
Platform: no
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Support: pipe rack support
Load Case: rack piping + thermal expansion
Platform: no
Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Support: base frame
Load Case: operating + nozzle loads
Platform: no
Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank

Shell t (mm): 5.31
Head t (mm): 4.88
Class: ASME 150
Support: leg support
Load Case: operating + settlement + thermal growth
Platform: no
Row 11
Equipment: HX-01-RV
Type: HTM relief package
Shell t (mm): 8.21
Head t (mm): 7.56
Class: ASME 300
Support: base support
Load Case: operating + nozzle loads
Platform: no
Row 12
Equipment: HX-02
Type: Heat integration exchanger
Shell t (mm): 8.95
Head t (mm): 8.24
Class: ASME 150
Support: saddle support
Load Case: operating + nozzle loads
Platform: yes
Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Support: instrument frame
Load Case: skid dead load + piping reaction
Platform: no

Mechanical Load and Foundation View

Table 224. Mechanical Load and Foundation View

Columns: Equipment; Load (kN); Wind (kN); Seismic (kN); Piping (kN); Anchor Groups; Footprint (m2); Clearance (m)
Row 1
Equipment: R-101
Load (kN): 259711.84
Wind (kN): 1152.82
Seismic (kN): 46748.13
Piping (kN): 240.40
Anchor Groups: 8
Footprint (m2): 656.21
Clearance (m): 1.80
Row 2
Equipment: PU-201
Load (kN): 250.03
Wind (kN): 9.81
Seismic (kN): 45.01
Piping (kN): 16.71
Anchor Groups: 8
Footprint (m2): 6.62
Clearance (m): 1.80
Row 3
Equipment: V-101
Load (kN): 46738.28
Wind (kN): 168.44
Seismic (kN): 4673.83
Piping (kN): 98.98
Anchor Groups: 4
Footprint (m2): 12.12
Clearance (m): 1.20
Row 4
Equipment: E-101
Load (kN): 7376.03
Wind (kN): 47.33

Seismic (kN): 737.60
Piping (kN): 53.32
Anchor Groups: 6
Footprint (m2): 47.33
Clearance (m): 1.50

Row 5
Equipment: TK-301
Load (kN): 75768.79
Wind (kN): 313.15
Seismic (kN): 7576.88
Piping (kN): 129.85
Anchor Groups: 6
Footprint (m2): 246.04
Clearance (m): 1.20

Row 6
Equipment: HX-01
Load (kN): 5636.10
Wind (kN): 39.07
Seismic (kN): 563.61
Piping (kN): 71.66
Anchor Groups: 6
Footprint (m2): 39.07
Clearance (m): 1.50

Row 7
Equipment: HX-01-CTRL
Load (kN): 6.00
Wind (kN): 1.50
Seismic (kN): 1.00
Piping (kN): 3.77
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.00

Row 8
Equipment: HX-01-HDR
Load (kN): 594.50
Wind (kN): 2.07
Seismic (kN): 71.34
Piping (kN): 22.48
Anchor Groups: 2
Footprint (m2): 2.43
Clearance (m): 1.20

Row 9
Equipment: HX-01-PMP
Load (kN): 10.59
Wind (kN): 1.50
Seismic (kN): 1.06
Piping (kN): 11.61
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20

Row 10
Equipment: HX-01-EXP
Load (kN): 33.03
Wind (kN): 1.70
Seismic (kN): 3.30
Piping (kN): 9.39
Anchor Groups: 4
Footprint (m2): 1.60
Clearance (m): 1.20

Row 11
Equipment: HX-01-RV
Load (kN): 18.91
Wind (kN): 1.50
Seismic (kN): 1.89
Piping (kN): 12.12
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20

Row 12
Equipment: HX-02
Load (kN): 402.13
Wind (kN): 5.71
Seismic (kN): 40.21

Piping (kN): 17.62
Anchor Groups: 6
Footprint (m2): 5.71
Clearance (m): 1.50
Row 13
Equipment: HX-02-CTRL
Load (kN): 6.00
Wind (kN): 1.50
Seismic (kN): 1.00
Piping (kN): 2.38
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.00

Mechanical Design Table

Table 225. Mechanical Design Table

Columns: Equipment; Type; Shell t (mm); Head t (mm); Class; Hydrotest (bar); Nozzle (mm); Support; Load Case; Support t (mm); Load (kN); Wind (kN); Seismic (kN); Piping (kN); Thermal Growth (mm); Reinforcement (mm2); Support Variant; Anchor Groups; Footprint (m2); Clearance (m); Ladder; Lifting Lugs; Reinf. Family; Shell Factor; Platform; Platform Area (m2)
Row 1
Equipment: R-101
Type: Reactor
Shell t (mm): 157.13
Head t (mm): 144.56
Class: ASME 600
Hydrotest (bar): 26.00
Nozzle (mm): 9336.1
Support: skirt support
Load Case: operating + wind + seismic + nozzle loads
Support t (mm): 109.99
Load (kN): 259711.84
Wind (kN): 1152.82
Seismic (kN): 46748.13
Piping (kN): 240.40
Thermal Growth (mm): 148.17
Reinforcement (mm2): 1941902.0
Support Variant: cylindrical skirt with anchor chair
Anchor Groups: 8
Footprint (m2): 656.21
Clearance (m): 1.80
Ladder: yes
Lifting Lugs: yes
Reinf. Family: integral repad
Shell Factor: 2.36
Platform: yes
Platform Area (m2): 61.74
Row 2
Equipment: PU-201
Type: Distillation column
Shell t (mm): 4.41
Head t (mm): 4.06
Class: PN16
Hydrotest (bar): 3.00
Nozzle (mm): 640.2
Support: skirt support
Load Case: operating + wind + seismic + nozzle loads
Support t (mm): 24.00
Load (kN): 250.03
Wind (kN): 9.81
Seismic (kN): 45.01
Piping (kN): 16.71
Thermal Growth (mm): 6.83
Reinforcement (mm2): 13316.9
Support Variant: cylindrical skirt with anchor chair
Anchor Groups: 8
Footprint (m2): 6.62
Clearance (m): 1.80
Ladder: yes

Lifting Lugs: yes
Reinf. Family: integral repad
Shell Factor: 1.85
Platform: yes
Platform Area (m2): 5.70

Row 3

Equipment: V-101
Type: Flash drum
Shell t (mm): 15.94
Head t (mm): 14.66
Class: PN16
Hydrotest (bar): 4.00
Nozzle (mm): 3917.0
Support: base support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 46738.28
Wind (kN): 168.44
Seismic (kN): 4673.83
Piping (kN): 98.98
Thermal Growth (mm): 21.82
Reinforcement (mm2): 65806.4
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 12.12
Clearance (m): 1.20
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.74
Platform: no
Platform Area (m2): 0.00

Row 4

Equipment: E-101
Type: Heat exchanger
Shell t (mm): 21.34
Head t (mm): 19.63
Class: ASME 150
Hydrotest (bar): 10.40
Nozzle (mm): 2021.0
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 7376.03
Wind (kN): 47.33
Seismic (kN): 737.60
Piping (kN): 53.32
Thermal Growth (mm): 29.89
Reinforcement (mm2): 90539.7
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 47.33
Clearance (m): 1.50
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.80
Platform: yes
Platform Area (m2): 3.00

Row 5

Equipment: TK-301
Type: Storage tank
Shell t (mm): 9.19
Head t (mm): 8.45
Class: PN16
Hydrotest (bar): 2.20
Nozzle (mm): 5177.3
Support: saddle / leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 75768.79
Wind (kN): 313.15
Seismic (kN): 7576.88

Piping (kN): 129.85
Thermal Growth (mm): 12.46
Reinforcement (mm2): 34791.6
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 246.04
Clearance (m): 1.20
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.78
Platform: no
Platform Area (m2): 0.00

Row 6
Equipment: HX-01
Type: HTM loop exchanger
Shell t (mm): 25.95
Head t (mm): 23.87
Class: ASME 300
Hydrotest (bar): 14.30
Nozzle (mm): 1822.7
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 5636.10
Wind (kN): 39.07
Seismic (kN): 563.61
Piping (kN): 71.66
Thermal Growth (mm): 32.93
Reinforcement (mm2): 152379.0
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 39.07
Clearance (m): 1.50
Ladder: yes
Lifting Lugs: yes
Reinf. Family: set-on repad
Shell Factor: 1.84
Platform: yes
Platform Area (m2): 3.00

Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Hydrotest (bar): 11.70
Nozzle (mm): 25.0
Support: instrument frame
Load Case: skid dead load + piping reaction
Support t (mm): 12.00
Load (kN): 6.00
Wind (kN): 1.50
Seismic (kN): 1.00
Piping (kN): 3.77
Thermal Growth (mm): 5.77
Reinforcement (mm2): 1260.0
Support Variant: panel frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.00
Ladder: no
Lifting Lugs: no
Reinf. Family: shell excess area
Shell Factor: 1.19
Platform: no
Platform Area (m2): 0.00

Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300

Hydrotest (bar): 18.85
Nozzle (mm): 417.0
Support: pipe rack support
Load Case: rack piping + thermal expansion
Support t (mm): 24.00
Load (kN): 594.50
Wind (kN): 2.07
Seismic (kN): 71.34
Piping (kN): 22.48
Thermal Growth (mm): 5.83
Reinforcement (mm2): 45956.1
Support Variant: guided rack shoe
Anchor Groups: 2
Footprint (m2): 2.43
Clearance (m): 1.20
Ladder: no
Lifting Lugs: yes
Reinf. Family: set-on repad
Shell Factor: 2.13
Platform: no
Platform Area (m2): 0.00

Row 9

Equipment: HX-01-PMP
Type: HTM circulation skid
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Hydrotest (bar): 18.85
Nozzle (mm): 117.3
Support: base frame
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 10.59
Wind (kN): 1.50
Seismic (kN): 1.06
Piping (kN): 11.61
Thermal Growth (mm): 5.83
Reinforcement (mm2): 12930.0
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.45
Platform: no
Platform Area (m2): 0.00

Row 10

Equipment: HX-01-EXP
Type: HTM expansion tank
Shell t (mm): 5.31
Head t (mm): 4.88
Class: ASME 150
Hydrotest (bar): 7.80
Nozzle (mm): 175.1
Support: leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 33.03
Wind (kN): 1.70
Seismic (kN): 3.30
Piping (kN): 9.39
Thermal Growth (mm): 5.99
Reinforcement (mm2): 7984.8
Support Variant: four-leg support
Anchor Groups: 4
Footprint (m2): 1.60
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.44

Platform: no
Platform Area (m2): 0.00

Row 11

Equipment: HX-01-RV
Type: HTM relief package
Shell t (mm): 8.21
Head t (mm): 7.56
Class: ASME 300
Hydrotest (bar): 17.55
Nozzle (mm): 145.4
Support: base support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 18.91
Wind (kN): 1.50
Seismic (kN): 1.89
Piping (kN): 12.12
Thermal Growth (mm): 6.09
Reinforcement (mm2): 14920.7
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.49
Platform: no
Platform Area (m2): 0.00

Row 12

Equipment: HX-02
Type: Heat integration exchanger
Shell t (mm): 8.95
Head t (mm): 8.24
Class: ASME 150
Hydrotest (bar): 8.45
Nozzle (mm): 613.7
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 402.13
Wind (kN): 5.71
Seismic (kN): 40.21
Piping (kN): 17.62
Thermal Growth (mm): 13.00
Reinforcement (mm2): 22339.3
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 5.71
Clearance (m): 1.50
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.69
Platform: yes
Platform Area (m2): 3.00

Row 13

Equipment: HX-02-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Hydrotest (bar): 6.50
Nozzle (mm): 25.0
Support: instrument frame
Load Case: skid dead load + piping reaction
Support t (mm): 12.00
Load (kN): 6.00
Wind (kN): 1.50
Seismic (kN): 1.00
Piping (kN): 2.38
Thermal Growth (mm): 4.31
Reinforcement (mm2): 700.0

Support Variant: panel frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.00
Ladder: no
Lifting Lugs: no
Reinf. Family: shell excess area
Shell Factor: 1.12
Platform: no
Platform Area (m2): 0.00

Utility Table

Table 226. Utility Table

Columns: Utility; Load; Units; Basis
Row 1: Utility: Steam | Load: 71672.825 | Units: kg/h | Basis: 20.0 bar saturated steam equivalent
Row 2: Utility: Cooling water | Load: 2778.003 | Units: m3/h | Basis: 10 C cooling-water rise
Row 3: Utility: Electricity | Load: 28732.696 | Units: kW | Basis: Agitation, pumps, vacuum auxiliaries, and transfer drives
Row 4: Utility: DM water | Load: 2.000 | Units: m3/h | Basis: Boiler and wash service allowance
Row 5: Utility: Nitrogen | Load: 3.608 | Units: Nm3/h | Basis: Inerting and blanketing
Row 6: Utility: Heat-integration auxiliaries | Load: 16.166 | Units: kW | Basis: Selected utility train circulation, HTM pumping, and exchanger-network auxiliaries

India Price Evidence Log

Table 227. India Price Evidence Log

Columns: Item; Category; Region; Value (INR); Units; Ref year; Norm year; Sources
Row 1
Item: Monoethylene glycol
Category: product
Region: India
Value (INR): 63.00
Units: INR/kg
Ref year: 2025
Norm year: 2025
Sources: india_sheet_1
Row 2
Item: Ethylene oxide
Category: raw_material
Region: India
Value (INR): 62.00
Units: INR/kg
Ref year: 2025
Norm year: 2025
Sources: india_sheet_1
Row 3
Item: Electricity
Category: utility
Region: Gujarat
Value (INR): 8.50
Units: INR/kWh
Ref year: 2025
Norm year: 2025
Sources: india_sheet_1
Row 4
Item: Operating labour
Category: labor
Region: India
Value (INR): 650,000.00
Units: INR/person-year
Ref year: 2025
Norm year: 2025
Sources: india_sheet_1

India Location Evidence Log

Table 228. India Location Evidence Log

Columns: Site; State; Port access; Utility note; Logistics note; Regulatory note; Ref year; Sources
Row 1
Site: Dahej
State: Gujarat
Port access: Dahej/Hazira marine access
Utility note: Mature industrial utility ecosystem and steam/power availability.
Logistics note: Strong chemical-cluster logistics and export connectivity.
Regulatory note: Established chemical-zone operating context with mature compliance ecosystem.
Ref year: 2025
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 2
Site: Jamnagar
State: Gujarat
Port access: West coast port connectivity
Utility note: Integrated refining-petrochemical utility environment.
Logistics note: Good feedstock and product dispatch linkages.
Regulatory note: Industrial cluster advantage but site-specific compliance envelope still required.
Ref year: 2025
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook
Row 3
Site: Paradip
State: Odisha
Port access: Deep-water east-coast port access
Utility note: Industrial utility support available with project-specific development.
Logistics note: Useful for eastern dispatch but less dense chemical services.
Regulatory note: Would need stronger project-specific ecosystem build-out.
Ref year: 2025
Sources: user_doc_1, user_doc_2, india_sheet_1, eg_handbook

Financial Snapshot

Table 229. Financial Snapshot

Columns: Metric; Value
Row 1: Metric: Annual Revenue | Value: INR 12,600,000,000.00
Row 2: Metric: Annual Opex | Value: INR 13,281,189,728.49
Row 3: Metric: Gross Profit | Value: INR -681,189,728.49
Row 4: Metric: Working Capital | Value: INR 1,759,656,318.22
Row 5: Metric: Peak Working Capital | Value: INR 1,759,656,318.22
Row 6: Metric: Peak Working-Capital Month | Value: 18.15
Row 7: Metric: Total Import Duty | Value: INR 353,373,129.56
Row 8: Metric: IDC | Value: INR 1,542,732,167.57
Row 9: Metric: Minimum DSCR | Value: 0.050
Row 10: Metric: Average DSCR | Value: 0.050
Row 11: Metric: LLCR | Value: 0.000
Row 12: Metric: PLCR | Value: 0.000
Row 13: Metric: Selected Financing Option | Value: debt_equity_70_30
Row 14: Metric: Downside Scenario | Value: conservative
Row 15: Metric: Downside-Preferred Financing Option | Value: debt_equity_70_30
Row 16: Metric: Scenario Reversal | Value: no
Row 17: Metric: Covenant Breaches | Value: minimum_dscr_breach, average_dscr_breach, llcr_breach, plcr_breach
Row 18: Metric: Payback (y) | Value: 76261521411.190
Row 19: Metric: NPV | Value: INR -98,309,630,442.49
Row 20: Metric: IRR (%) | Value: 1.00

Lender Coverage Screening

Table 230. Lender Coverage Screening

Columns: Metric; Value
Row 1: Metric: Minimum DSCR | Value: 0.050

Row 2: Metric: Average DSCR | Value: 0.050
Row 3: Metric: LLCR | Value: 0.000
Row 4: Metric: PLCR | Value: 0.000
Row 5: Metric: Downside Scenario | Value: conservative
Row 6: Metric: Downside-Preferred Financing Option | Value: debt_equity_70_30
Row 7: Metric: Scenario Reversal | Value: no

Covenant Warnings

- Minimum DSCR 0.050 is below the screening covenant threshold of 1.10.
- Average DSCR 0.050 is below the screening covenant threshold of 1.25.
- LLCR 0.000 is below the screening covenant threshold of 1.30.
- PLCR 0.000 is below the screening covenant threshold of 1.45.

Equipment Cost Breakdown

Table 231. Equipment Cost Breakdown

Columns: Equipment; Type; Bare Cost (INR); Installed Cost (INR); Spares (INR); Package Family; Lead (mo); Import Duty (INR); Basis

Row 1
- Equipment: R-101
- Type: Reactor
- Bare Cost (INR): 19,244,201,057.55
- Installed Cost (INR): 43,299,452,379.48
- Spares (INR): 283,748,938.25
- Package Family: pressure_vessel_package
- Lead (mo): 11.34
- Import Duty (INR): 327,605,174.15
- Basis: tubular_plug_flow_hydrator selected at 1.00 h residence time and 0.960 conversion basis.

Row 2
- Equipment: PU-201
- Type: Distillation column
- Bare Cost (INR): 13,157,679.80
- Installed Cost (INR): 29,604,779.55
- Spares (INR): 194,005.33
- Package Family: pressure_vessel_package
- Lead (mo): 11.34
- Import Duty (INR): 223,990.80
- Basis: 7 stages equivalent

Row 3
- Equipment: V-101
- Type: Flash drum
- Bare Cost (INR): 939,536,778.43
- Installed Cost (INR): 2,113,957,751.47
- Spares (INR): 13,853,137.50
- Package Family: pressure_vessel_package
- Lead (mo): 11.34
- Import Duty (INR): 15,994,278.43
- Basis: Generic separator hold-up

Row 4
- Equipment: E-101
- Type: Heat exchanger
- Bare Cost (INR): 370,219,675.74
- Installed Cost (INR): 832,994,270.41
- Spares (INR): 5,479,441.04
- Package Family: thermal_exchange_package
- Lead (mo): 7.94
- Import Duty (INR): 4,923,606.54
- Basis: LMTD 8.0 K

Row 5
- Equipment: TK-301
- Type: Storage tank
- Bare Cost (INR): 1,031,289,992.04
- Installed Cost (INR): 2,062,579,984.08
- Spares (INR): 15,445,625.40
- Package Family: tankage_and_balance_of_plant
- Lead (mo): 4.54
- Import Duty (INR): 1,581,632.04
- Basis: 14.1 days inventory

Row 6
Equipment: HX-01
Type: HTM loop exchanger
Bare Cost (INR): 108,622,690.42
Installed Cost (INR): 238,969,918.92
Spares (INR): 2,143,562.02
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 1,444,589.32
Basis: role=exchanger; duty=34024.794 kW; power=0.000 kW; volume=542.949 m3
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Bare Cost (INR): 1,477,758.08
Installed Cost (INR): 2,364,412.93
Spares (INR): 28,300.00
Package Family: instrumented_skid_package
Lead (mo): 6.80
Import Duty (INR): 62,758.08
Basis: role=controls; duty=0.000 kW; power=0.000 kW; volume=0.150 m3
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Bare Cost (INR): 84,771,672.00
Installed Cost (INR): 161,066,176.79
Spares (INR): 1,672,885.62
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 1,127,391.08
Basis: role=header; duty=34024.794 kW; power=0.000 kW; volume=0.851 m3
Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Bare Cost (INR): 4,521,339.97
Installed Cost (INR): 8,364,478.95
Spares (INR): 89,224.20
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 60,129.97
Basis: role=circulation; duty=0.000 kW; power=15.466 kW; volume=0.250 m3
Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Bare Cost (INR): 3,507,952.79
Installed Cost (INR): 7,015,905.57
Spares (INR): 69,226.00
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 46,652.79
Basis: role=expansion; duty=0.000 kW; power=0.000 kW; volume=2.835 m3
Row 11
Equipment: HX-01-RV
Type: HTM relief package
Bare Cost (INR): 1,785,323.28
Installed Cost (INR): 3,124,315.74
Spares (INR): 35,231.60
Package Family: thermal_exchange_package
Lead (mo): 7.94
Import Duty (INR): 23,743.28
Basis: role=relief; duty=0.000 kW; power=0.000 kW; volume=1.309 m3
Row 12
Equipment: HX-02
Type: Heat integration exchanger
Bare Cost (INR): 27,026,028.38
Installed Cost (INR): 55,403,358.19
Spares (INR): 535,890.47
Package Family: general_fabrication
Lead (mo): 6.80
Import Duty (INR): 231,504.68
Basis: role=exchanger; duty=8506.198 kW; power=0.000 kW; volume=30.379 m3
Row 13
Equipment: HX-02-CTRL

Type: Utility control package
Bare Cost (INR): 1,122,678.40
Installed Cost (INR): 1,796,285.44
Spares (INR): 21,500.00
Package Family: instrumented_skid_package
Lead (mo): 6.80
Import Duty (INR): 47,678.40
Basis: role=controls; duty=0.000 kW; power=0.000 kW; volume=0.150 m3

Plant Cost Summary

Table 232. Plant Cost Summary

Columns: Equipment; Bare; Installation; Piping; Instrumentation; Total Installed
Row 1: Equipment: R-101 | Bare: 19,244,201,057.55 | Installation: 10,103,205,555.21 | Piping: 6,061,923,333.13 | Instrumentation: 3,463,956,190.36 | Total Installed: 50,564,138,278.71
Row 2: Equipment: PU-201 | Bare: 13,157,679.80 | Installation: 6,907,781.89 | Piping: 4,144,669.14 | Instrumentation: 2,368,382.36 | Total Installed: 34,571,803.67
Row 3: Equipment: V-101 | Bare: 939,536,778.43 | Installation: 493,256,808.68 | Piping: 295,954,085.21 | Instrumentation: 169,116,620.12 | Total Installed: 2,468,632,885.33
Row 4: Equipment: E-101 | Bare: 370,219,675.74 | Installation: 194,365,329.76 | Piping: 116,619,197.86 | Instrumentation: 66,639,541.63 | Total Installed: 972,752,198.00
Row 5: Equipment: TK-301 | Bare: 1,031,289,992.04 | Installation: 433,141,796.66 | Piping: 288,761,197.77 | Instrumentation: 165,006,398.73 | Total Installed: 2,475,095,980.90
Row 6: Equipment: HX-01 | Bare: 108,622,690.42 | Installation: 54,745,835.97 | Piping: 33,455,788.65 | Instrumentation: 19,117,593.51 | Total Installed: 280,463,786.66
Row 7: Equipment: HX-01-CTRL | Bare: 1,477,758.08 | Installation: 372,395.04 | Piping: 331,017.81 | Instrumentation: 189,153.03 | Total Installed: 3,008,715.45
Row 8: Equipment: HX-01-HDR | Bare: 84,771,672.00 | Installation: 32,043,692.01 | Piping: 22,549,264.75 | Instrumentation: 12,885,294.14 | Total Installed: 195,737,790.64
Row 9: Equipment: HX-01-PMP | Bare: 4,521,339.97 | Installation: 1,614,118.37 | Piping: 1,171,027.05 | Instrumentation: 669,158.32 | Total Installed: 10,234,053.03
Row 10: Equipment: HX-01-EXP | Bare: 3,507,952.79 | Installation: 1,473,340.17 | Piping: 982,226.78 | Instrumentation: 561,272.45 | Total Installed: 8,419,086.69
Row 11: Equipment: HX-01-RV | Bare: 1,785,323.28 | Installation: 562,376.83 | Piping: 437,404.20 | Instrumentation: 249,945.26 | Total Installed: 3,878,614.83
Row 12: Equipment: HX-02 | Bare: 27,026,028.38 | Installation: 11,918,478.52 | Piping: 7,756,470.15 | Instrumentation: 4,432,268.66 | Total Installed: 66,092,152.41
Row 13: Equipment: HX-02-CTRL | Bare: 1,122,678.40 | Installation: 282,914.96 | Piping: 251,479.96 | Instrumentation: 143,702.84 | Total Installed: 2,285,773.22

Procurement Timing Schedule

Table 233. Procurement Timing Schedule

Columns: Package Family; Milestone; Month; Draw Fraction; CAPEX Draw (INR)
Row 1: Package Family: pressure_vessel_package | Milestone: advance_release | Month: 6.7 | Draw Fraction: 0.186 | CAPEX Draw (INR): 13,583,398,188.32
Row 2: Package Family: thermal_exchange_package | Milestone: advance_release | Month: 10.1 | Draw Fraction: 0.005 | CAPEX Draw (INR): 336,687,371.39
Row 3: Package Family: general_fabrication | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 14,904,585.21
Row 4: Package Family: instrumented_skid_package | Milestone: advance_release | Month: 11.2 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,119,309.11
Row 5: Package Family: pressure_vessel_package | Milestone: fabrication_progress | Month: 11.8 | Draw Fraction: 0.428 | CAPEX Draw

(INR): 31,241,815,833.14
Row 6: Package Family: tankage_and_balance_of_plant | Milestone: advance_release | Month: 13.5 | Draw Fraction: 0.005 | CAPEX Draw (INR): 369,916,194.26
Row 7: Package Family: thermal_exchange_package | Milestone: fabrication_progress | Month: 13.6 | Draw Fraction: 0.011 | CAPEX Draw (INR): 785,603,866.57
Row 8: Package Family: general_fabrication | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 34,777,365.48
Row 9: Package Family: instrumented_skid_package | Milestone: fabrication_progress | Month: 14.3 | Draw Fraction: 0.000 | CAPEX Draw (INR): 2,611,721.25
Row 10: Package Family: tankage_and_balance_of_plant | Milestone: fabrication_progress | Month: 15.5 | Draw Fraction: 0.020 | CAPEX Draw (INR): 1,479,664,777.04
Row 11: Package Family: pressure_vessel_package | Milestone: delivery_release | Month: 16.0 | Draw Fraction: 0.177 | CAPEX Draw (INR): 12,904,228,278.90
Row 12: Package Family: thermal_exchange_package | Milestone: delivery_release | Month: 16.6 | Draw Fraction: 0.005 | CAPEX Draw (INR): 374,097,079.32
Row 13: Package Family: general_fabrication | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 16,560,650.23
Row 14: Package Family: instrumented_skid_package | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.000 | CAPEX Draw (INR): 1,243,676.78
Row 15: Package Family: tankage_and_balance_of_plant | Milestone: delivery_release | Month: 16.7 | Draw Fraction: 0.008 | CAPEX Draw (INR): 616,526,990.43
Row 16: Package Family: pressure_vessel_package | Milestone: site_erection | Month: 17.3 | Draw Fraction: 0.047 | CAPEX Draw (INR): 3,395,849,547.08
Row 17: Package Family: thermal_exchange_package | Milestone: site_erection | Month: 17.5 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65
Row 18: Package Family: general_fabrication | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 19: Package Family: instrumented_skid_package | Milestone: site_erection | Month: 17.6 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 20: Package Family: tankage_and_balance_of_plant | Milestone: site_erection | Month: 17.7 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 21: Package Family: general_fabrication | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 8,280,325.11
Row 22: Package Family: instrumented_skid_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.000 | CAPEX Draw (INR): 621,838.40
Row 23: Package Family: pressure_vessel_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.093 | CAPEX Draw (INR): 6,791,699,094.16
Row 24: Package Family: tankage_and_balance_of_plant | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.004 | CAPEX Draw (INR): 308,263,495.22
Row 25: Package Family: thermal_exchange_package | Milestone: performance_retention | Month: 19.0 | Draw Fraction: 0.003 | CAPEX Draw (INR): 187,048,539.65

Procurement Package Detail

Table 234. Procurement Package Detail

Columns: Equipment; Type; Package Family; Lead (mo); Award Month; Delivery Month; Import Content; Duty Fraction; Import Duty (INR); CAPEX Burden (INR)
Row 1
Equipment: R-101
Type: Reactor
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211

Duty Fraction: 0.082
Import Duty (INR): 327,605,174.15
CAPEX Burden (INR): 64,713,323,286.86
Row 2
Equipment: PU-201
Type: Distillation column
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211
Duty Fraction: 0.082
Import Duty (INR): 223,990.80
CAPEX Burden (INR): 44,245,909.93
Row 3
Equipment: V-101
Type: Flash drum
Package Family: pressure_vessel_package
Lead (mo): 11.34
Award Month: 6.66
Delivery Month: 15.96
Import Content: 0.211
Duty Fraction: 0.082
Import Duty (INR): 15,994,278.43
CAPEX Burden (INR): 3,159,421,744.80
Row 4
Equipment: E-101
Type: Heat exchanger
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 4,923,606.54
CAPEX Burden (INR): 1,244,954,024.93
Row 5
Equipment: TK-301
Type: Storage tank
Package Family: tankage_and_balance_of_plant
Lead (mo): 4.54
Award Month: 13.46
Delivery Month: 16.72
Import Content: 0.038
Duty Fraction: 0.040
Import Duty (INR): 1,581,632.04
CAPEX Burden (INR): 3,082,634,952.17
Row 6
Equipment: HX-01
Type: HTM loop exchanger
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 1,444,589.32
CAPEX Burden (INR): 357,153,191.76
Row 7
Equipment: HX-01-CTRL
Type: Utility control package
Package Family: instrumented_skid_package
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.403
Duty Fraction: 0.110
Import Duty (INR): 62,758.08
CAPEX Burden (INR): 3,533,740.26
Row 8
Equipment: HX-01-HDR
Type: Shared HTM header manifold
Package Family: thermal_exchange_package

Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 1,127,391.08
CAPEX Burden (INR): 240,721,925.95

Row 9
Equipment: HX-01-PMP
Type: HTM circulation skid
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 60,129.97
CAPEX Burden (INR): 12,501,156.50

Row 10
Equipment: HX-01-EXP
Type: HTM expansion tank
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 46,652.79
CAPEX Burden (INR): 10,485,642.20

Row 11
Equipment: HX-01-RV
Type: HTM relief package
Package Family: thermal_exchange_package
Lead (mo): 7.94
Award Month: 10.06
Delivery Month: 16.57
Import Content: 0.173
Duty Fraction: 0.078
Import Duty (INR): 23,743.28
CAPEX Burden (INR): 4,669,455.23

Row 12
Equipment: HX-02
Type: Heat integration exchanger
Package Family: general_fabrication
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.115
Duty Fraction: 0.075
Import Duty (INR): 231,504.68
CAPEX Burden (INR): 82,803,251.14

Row 13
Equipment: HX-02-CTRL
Type: Utility control package
Package Family: instrumented_skid_package
Lead (mo): 6.80
Award Month: 11.20
Delivery Month: 16.72
Import Content: 0.403
Duty Fraction: 0.110
Import Duty (INR): 47,678.40
CAPEX Burden (INR): 2,684,643.66

Working-Capital Timing Basis

Table 235. Working-Capital Timing Basis

Columns: Metric; Value

Row 1: Metric: Procurement Timing Factor | Value: 1.197
Row 2: Metric: Pre-commissioning Inventory Days | Value: 13.40
Row 3: Metric: Pre-commissioning Inventory Month | Value: 17.95
Row 4: Metric: Pre-commissioning Inventory | Value: INR 275,688,431.02
Row 5: Metric: Peak Working-Capital Month | Value: 18.15

Row 6: Metric: Peak Working Capital | Value: INR 1,759,656,318.22

Mechanical Screening Table

Table 236. Mechanical Screening Table

Columns: Equipment; Type; Shell t (mm); Head t (mm); Class; Hydrotest (bar); Nozzle (mm); Support; Load Case; Support t (mm); Load (kN); Wind (kN); Seismic (kN); Piping (kN); Thermal Growth (mm); Reinforcement (mm2); Support Variant; Anchor Groups; Footprint (m2); Clearance (m); Ladder; Lifting Lugs; Reinf. Family; Shell Factor; Platform; Platform Area (m2)

Row 1
- Equipment: R-101
- Type: Reactor
- Shell t (mm): 157.13
- Head t (mm): 144.56
- Class: ASME 600
- Hydrotest (bar): 26.00
- Nozzle (mm): 9336.1
- Support: skirt support
- Load Case: operating + wind + seismic + nozzle loads
- Support t (mm): 109.99
- Load (kN): 259711.84
- Wind (kN): 1152.82
- Seismic (kN): 46748.13
- Piping (kN): 240.40
- Thermal Growth (mm): 148.17
- Reinforcement (mm2): 1941902.0
- Support Variant: cylindrical skirt with anchor chair
- Anchor Groups: 8
- Footprint (m2): 656.21
- Clearance (m): 1.80
- Ladder: yes
- Lifting Lugs: yes
- Reinf. Family: integral repad
- Shell Factor: 2.36
- Platform: yes
- Platform Area (m2): 61.74

Row 2
- Equipment: PU-201
- Type: Distillation column
- Shell t (mm): 4.41
- Head t (mm): 4.06
- Class: PN16
- Hydrotest (bar): 3.00
- Nozzle (mm): 640.2
- Support: skirt support
- Load Case: operating + wind + seismic + nozzle loads
- Support t (mm): 24.00
- Load (kN): 250.03
- Wind (kN): 9.81
- Seismic (kN): 45.01
- Piping (kN): 16.71
- Thermal Growth (mm): 6.83
- Reinforcement (mm2): 13316.9
- Support Variant: cylindrical skirt with anchor chair
- Anchor Groups: 8
- Footprint (m2): 6.62
- Clearance (m): 1.80
- Ladder: yes
- Lifting Lugs: yes
- Reinf. Family: integral repad
- Shell Factor: 1.85
- Platform: yes
- Platform Area (m2): 5.70

Row 3
- Equipment: V-101
- Type: Flash drum
- Shell t (mm): 15.94
- Head t (mm): 14.66
- Class: PN16
- Hydrotest (bar): 4.00
- Nozzle (mm): 3917.0

Support: base support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 46738.28
Wind (kN): 168.44
Seismic (kN): 4673.83
Piping (kN): 98.98
Thermal Growth (mm): 21.82
Reinforcement (mm2): 65806.4
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 12.12
Clearance (m): 1.20
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.74
Platform: no
Platform Area (m2): 0.00

Row 4
Equipment: E-101
Type: Heat exchanger
Shell t (mm): 21.34
Head t (mm): 19.63
Class: ASME 150
Hydrotest (bar): 10.40
Nozzle (mm): 2021.0
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 7376.03
Wind (kN): 47.33
Seismic (kN): 737.60
Piping (kN): 53.32
Thermal Growth (mm): 29.89
Reinforcement (mm2): 90539.7
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 47.33
Clearance (m): 1.50
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.80
Platform: yes
Platform Area (m2): 3.00

Row 5
Equipment: TK-301
Type: Storage tank
Shell t (mm): 9.19
Head t (mm): 8.45
Class: PN16
Hydrotest (bar): 2.20
Nozzle (mm): 5177.3
Support: saddle / leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 75768.79
Wind (kN): 313.15
Seismic (kN): 7576.88
Piping (kN): 129.85
Thermal Growth (mm): 12.46
Reinforcement (mm2): 34791.6
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 246.04
Clearance (m): 1.20
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.78
Platform: no
Platform Area (m2): 0.00

Row 6
- Equipment: HX-01
- Type: HTM loop exchanger
- Shell t (mm): 25.95
- Head t (mm): 23.87
- Class: ASME 300
- Hydrotest (bar): 14.30
- Nozzle (mm): 1822.7
- Support: saddle support
- Load Case: operating + nozzle loads
- Support t (mm): 24.00
- Load (kN): 5636.10
- Wind (kN): 39.07
- Seismic (kN): 563.61
- Piping (kN): 71.66
- Thermal Growth (mm): 32.93
- Reinforcement (mm2): 152379.0
- Support Variant: dual saddle
- Anchor Groups: 6
- Footprint (m2): 39.07
- Clearance (m): 1.50
- Ladder: yes
- Lifting Lugs: yes
- Reinf. Family: set-on repad
- Shell Factor: 1.84
- Platform: yes
- Platform Area (m2): 3.00

Row 7
- Equipment: HX-01-CTRL
- Type: Utility control package
- Shell t (mm): 0.00
- Head t (mm): 0.00
- Class: ASME 150
- Hydrotest (bar): 11.70
- Nozzle (mm): 25.0
- Support: instrument frame
- Load Case: skid dead load + piping reaction
- Support t (mm): 12.00
- Load (kN): 6.00
- Wind (kN): 1.50
- Seismic (kN): 1.00
- Piping (kN): 3.77
- Thermal Growth (mm): 5.77
- Reinforcement (mm2): 1260.0
- Support Variant: panel frame
- Anchor Groups: 4
- Footprint (m2): 1.20
- Clearance (m): 1.00
- Ladder: no
- Lifting Lugs: no
- Reinf. Family: shell excess area
- Shell Factor: 1.19
- Platform: no
- Platform Area (m2): 0.00

Row 8
- Equipment: HX-01-HDR
- Type: Shared HTM header manifold
- Shell t (mm): 8.60
- Head t (mm): 7.92
- Class: ASME 300
- Hydrotest (bar): 18.85
- Nozzle (mm): 417.0
- Support: pipe rack support
- Load Case: rack piping + thermal expansion
- Support t (mm): 24.00
- Load (kN): 594.50
- Wind (kN): 2.07
- Seismic (kN): 71.34
- Piping (kN): 22.48
- Thermal Growth (mm): 5.83
- Reinforcement (mm2): 45956.1
- Support Variant: guided rack shoe
- Anchor Groups: 2

Footprint (m2): 2.43
Clearance (m): 1.20
Ladder: no
Lifting Lugs: yes
Reinf. Family: set-on repad
Shell Factor: 2.13
Platform: no
Platform Area (m2): 0.00

Row 9

Equipment: HX-01-PMP
Type: HTM circulation skid
Shell t (mm): 8.60
Head t (mm): 7.92
Class: ASME 300
Hydrotest (bar): 18.85
Nozzle (mm): 117.3
Support: base frame
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 10.59
Wind (kN): 1.50
Seismic (kN): 1.06
Piping (kN): 11.61
Thermal Growth (mm): 5.83
Reinforcement (mm2): 12930.0
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.45
Platform: no
Platform Area (m2): 0.00

Row 10

Equipment: HX-01-EXP
Type: HTM expansion tank
Shell t (mm): 5.31
Head t (mm): 4.88
Class: ASME 150
Hydrotest (bar): 7.80
Nozzle (mm): 175.1
Support: leg support
Load Case: operating + settlement + thermal growth
Support t (mm): 18.00
Load (kN): 33.03
Wind (kN): 1.70
Seismic (kN): 3.30
Piping (kN): 9.39
Thermal Growth (mm): 5.99
Reinforcement (mm2): 7984.8
Support Variant: four-leg support
Anchor Groups: 4
Footprint (m2): 1.60
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.44
Platform: no
Platform Area (m2): 0.00

Row 11

Equipment: HX-01-RV
Type: HTM relief package
Shell t (mm): 8.21
Head t (mm): 7.56
Class: ASME 300
Hydrotest (bar): 17.55
Nozzle (mm): 145.4
Support: base support
Load Case: operating + nozzle loads
Support t (mm): 24.00

Load (kN): 18.91
Wind (kN): 1.50
Seismic (kN): 1.89
Piping (kN): 12.12
Thermal Growth (mm): 6.09
Reinforcement (mm2): 14920.7
Support Variant: grouted base frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.20
Ladder: no
Lifting Lugs: no
Reinf. Family: set-on repad
Shell Factor: 1.49
Platform: no
Platform Area (m2): 0.00

Row 12

Equipment: HX-02
Type: Heat integration exchanger
Shell t (mm): 8.95
Head t (mm): 8.24
Class: ASME 150
Hydrotest (bar): 8.45
Nozzle (mm): 613.7
Support: saddle support
Load Case: operating + nozzle loads
Support t (mm): 24.00
Load (kN): 402.13
Wind (kN): 5.71
Seismic (kN): 40.21
Piping (kN): 17.62
Thermal Growth (mm): 13.00
Reinforcement (mm2): 22339.3
Support Variant: dual saddle
Anchor Groups: 6
Footprint (m2): 5.71
Clearance (m): 1.50
Ladder: yes
Lifting Lugs: yes
Reinf. Family: shell excess area
Shell Factor: 1.69
Platform: yes
Platform Area (m2): 3.00

Row 13

Equipment: HX-02-CTRL
Type: Utility control package
Shell t (mm): 0.00
Head t (mm): 0.00
Class: ASME 150
Hydrotest (bar): 6.50
Nozzle (mm): 25.0
Support: instrument frame
Load Case: skid dead load + piping reaction
Support t (mm): 12.00
Load (kN): 6.00
Wind (kN): 1.50
Seismic (kN): 1.00
Piping (kN): 2.38
Thermal Growth (mm): 4.31
Reinforcement (mm2): 700.0
Support Variant: panel frame
Anchor Groups: 4
Footprint (m2): 1.20
Clearance (m): 1.00
Ladder: no
Lifting Lugs: no
Reinf. Family: shell excess area
Shell Factor: 1.12
Platform: no
Platform Area (m2): 0.00

Utility Island Economics

Table 237. Utility Island Economics

Columns: Island; Topology; HTM Inventory (m3); Header Pressure (bar); Pair Score; Control Factor; Cycle (y); Replacement Event (INR); Replacement Avg (INR/y); Turnaround (d); Project CAPEX Burden (INR); Allocated Utility (INR/y); Service (INR/y); Operating Burden (INR/y); Utility Share; CAPEX Share

Row 1
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Topology: shared HTM island network
- HTM Inventory (m3): 3.936
- Header Pressure (bar): 14.50
- Pair Score: 0.000
- Control Factor: 2.500
- Cycle (y): 4.26
- Replacement Event (INR): 66,064,226.66
- Replacement Avg (INR/y): 15,494,813.83
- Turnaround (d): 2.57
- Project CAPEX Burden (INR): 733,501,670.81
- Allocated Utility (INR/y): 3,132,067,158.00
- Service (INR/y): 2,952,253.22
- Operating Burden (INR/y): 3,150,514,225.04
- Utility Share: 1.000
- CAPEX Share: 0.010

Utility Island Scenario Breakdown

Table 238. Utility Island Scenario Breakdown

Columns: Scenario; Island; Capex Burden (INR); Allocated Utility (INR/y); Service (INR/y); Replacement (INR/y); Operating Burden (INR/y)

Row 1
- Scenario: base
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Capex Burden (INR): 648,415,477.00
- Allocated Utility (INR/y): 3,132,067,158.00
- Service (INR/y): 885,675.97
- Replacement (INR/y): 13,697,415.43
- Operating Burden (INR/y): 3,146,650,249.39

Row 2
- Scenario: conservative
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Capex Burden (INR): 717,951,435.39
- Allocated Utility (INR/y): 3,775,393,752.25
- Service (INR/y): 1,001,404.29
- Replacement (INR/y): 15,166,323.78
- Operating Burden (INR/y): 3,791,561,480.32

Row 3
- Scenario: upside
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Capex Burden (INR): 626,850,527.87
- Allocated Utility (INR/y): 2,888,705,539.82
- Service (INR/y): 847,060.49
- Replacement (INR/y): 13,241,867.90
- Operating Burden (INR/y): 2,902,794,468.22

Scenario Comparison Table

Table 239. Scenario Comparison Table

Columns: Scenario; Utility Cost (INR/y); Transport/Service (INR/y); Utility-Island Burden (INR/y); Operating Cost (INR/y); Revenue (INR/y); Gross Margin (INR/y)

Row 1
- Scenario: base
- Utility Cost (INR/y): 3,132,067,158.00
- Transport/Service (INR/y): 14,583,091.39
- Utility-Island Burden (INR/y): 3,146,650,249.39
- Operating Cost (INR/y): 13,273,756,525.48
- Revenue (INR/y): 12,600,000,000.00
- Gross Margin (INR/y): -673,756,525.48

Row 2

Scenario: conservative
Utility Cost (INR/y): 3,775,393,752.25
Transport/Service (INR/y): 16,167,728.07
Utility-Island Burden (INR/y): 3,791,561,480.32
Operating Cost (INR/y): 14,912,994,804.01
Revenue (INR/y): 11,970,000,000.00
Gross Margin (INR/y): -2,942,994,804.01

Row 3
Scenario: upside
Utility Cost (INR/y): 2,888,705,539.82
Transport/Service (INR/y): 14,088,928.39
Utility-Island Burden (INR/y): 2,902,794,468.22
Operating Cost (INR/y): 12,731,575,600.67
Revenue (INR/y): 13,104,000,000.00
Gross Margin (INR/y): 372,424,399.33

Multi-Year Financial Schedule

Financial Schedule Summary View

Table 240. Financial Schedule Summary View

Columns: Year; Capacity Utilization (%); Availability (%); Revenue (INR); Operating Cost (INR); CFADS (INR); DSCR; Cash Accrual (INR)

Row 1
Year: 1.0
Capacity Utilization (%): 78.79
Availability (%): 98.48
Revenue (INR): 9,927,272,727.27
Operating Cost (INR): 11,486,378,287.07
CFADS (INR): 0.00
DSCR: 0.050
Cash Accrual (INR): -7,083,588,707.41

Row 2
Year: 2.0
Capacity Utilization (%): 83.71
Availability (%): 98.48
Revenue (INR): 10,547,727,272.73
Operating Cost (INR): 11,898,750,500.79
CFADS (INR): 0.00
DSCR: 0.050
Cash Accrual (INR): -5,770,609,746.16

Row 3
Year: 3.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.050
Cash Accrual (INR): -4,457,630,784.90

Row 4
Year: 4.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
CFADS (INR): 0.00
DSCR: 0.050
Cash Accrual (INR): -4,575,165,117.44

Row 5
Year: 5.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.050
Cash Accrual (INR): -2,247,837,525.85

Row 6
Year: 6.0
Capacity Utilization (%): 88.64

Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -1,142,940,896.33
Row 7
Year: 7.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -1,142,940,896.33
Row 8
Year: 8.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -2,365,371,858.40
Row 9
Year: 9.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -1,142,940,896.33
Row 10
Year: 10.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -1,142,940,896.33
Row 11
Year: 11.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -1,142,940,896.33
Row 12
Year: 12.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
CFADS (INR): 0.00
DSCR: 0.000
Cash Accrual (INR): -2,365,371,858.40

Financial Schedule Debt and Profit View

Table 241. Financial Schedule Debt and Profit View

Columns: Year; Principal (INR); Interest (INR); Debt Service (INR); PBT (INR); Tax (INR); PAT (INR)
Row 1
Year: 1.0
Principal (INR): 10,522,825,043.07
Interest (INR): 5,524,483,147.61

Debt Service (INR): 16,047,308,190.68
PBT (INR): -13,163,516,451.19
Tax (INR): 0.00
PAT (INR): -13,163,516,451.19
Row 2
Year: 2.0
Principal (INR): 10,522,825,043.07
Interest (INR): 4,419,586,518.09
Debt Service (INR): 14,942,411,561.16
PBT (INR): -11,850,537,489.94
Tax (INR): 0.00
PAT (INR): -11,850,537,489.94
Row 3
Year: 3.0
Principal (INR): 10,522,825,043.07
Interest (INR): 3,314,689,888.57
Debt Service (INR): 13,837,514,931.64
PBT (INR): -10,537,558,528.69
Tax (INR): 0.00
PAT (INR): -10,537,558,528.69
Row 4
Year: 4.0
Principal (INR): 10,522,825,043.07
Interest (INR): 2,209,793,259.04
Debt Service (INR): 12,732,618,302.11
PBT (INR): -10,655,092,861.23
Tax (INR): 0.00
PAT (INR): -10,655,092,861.23
Row 5
Year: 5.0
Principal (INR): 10,522,825,043.07
Interest (INR): 1,104,896,629.52
Debt Service (INR): 11,627,721,672.59
PBT (INR): -8,327,765,269.64
Tax (INR): 0.00
PAT (INR): -8,327,765,269.64
Row 6
Year: 6.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 7
Year: 7.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 8
Year: 8.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Row 9
Year: 9.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 10
Year: 10.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00

PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 11
Year: 11.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Row 12
Year: 12.0
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19

Table 242. Financial Schedule Debt and Profit View

Columns: Year; Capacity Utilization (%); Availability (%); Revenue Loss (INR); Revenue (INR); Operating Cost (INR); Utility-Island Service (INR); Utility-Island Replacement (INR); Utility-Island Turnaround (INR); Principal (INR); Interest (INR); Debt Service (INR); CFADS (INR); DSCR; Depreciation (INR); PBT (INR); Tax (INR); PAT (INR); Cash Accrual (INR)
Row 1
Year: 1.0
Capacity Utilization (%): 78.79
Availability (%): 98.48
Revenue Loss (INR): 152,727,272.73
Revenue (INR): 9,927,272,727.27
Operating Cost (INR): 11,486,378,287.07
Utility-Island Service (INR): 2,795,694.34
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 5,524,483,147.61
Debt Service (INR): 16,047,308,190.68
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -13,163,516,451.19
Tax (INR): 0.00
PAT (INR): -13,163,516,451.19
Cash Accrual (INR): -7,083,588,707.41
Row 2
Year: 2.0
Capacity Utilization (%): 83.71
Availability (%): 98.48
Revenue Loss (INR): 162,272,727.27
Revenue (INR): 10,547,727,272.73
Operating Cost (INR): 11,898,750,500.79
Utility-Island Service (INR): 2,832,038.36
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 4,419,586,518.09
Debt Service (INR): 14,942,411,561.16
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -11,850,537,489.94
Tax (INR): 0.00
PAT (INR): -11,850,537,489.94
Cash Accrual (INR): -5,770,609,746.16
Row 3
Year: 3.0
Capacity Utilization (%): 88.64
Availability (%): 98.48

Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 3,314,689,888.57
Debt Service (INR): 13,837,514,931.64
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -10,537,558,528.69
Tax (INR): 0.00
PAT (INR): -10,537,558,528.69
Cash Accrual (INR): -4,457,630,784.90

Row 4

Year: 4.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 10,522,825,043.07
Interest (INR): 2,209,793,259.04
Debt Service (INR): 12,732,618,302.11
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -10,655,092,861.23
Tax (INR): 0.00
PAT (INR): -10,655,092,861.23
Cash Accrual (INR): -4,575,165,117.44

Row 5

Year: 5.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 10,522,825,043.07
Interest (INR): 1,104,896,629.52
Debt Service (INR): 11,627,721,672.59
CFADS (INR): 0.00
DSCR: 0.050
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,327,765,269.64
Tax (INR): 0.00
PAT (INR): -8,327,765,269.64
Cash Accrual (INR): -2,247,837,525.85

Row 6

Year: 6.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12

Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 7
Year: 7.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 8
Year: 8.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Cash Accrual (INR): -2,365,371,858.40
Row 9
Year: 9.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33
Row 10
Year: 10.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00

Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33

Row 11
Year: 11.0
Capacity Utilization (%): 88.64
Availability (%): 98.48
Revenue Loss (INR): 171,818,181.82
Revenue (INR): 11,168,181,818.18
Operating Cost (INR): 12,311,122,714.52
Utility-Island Service (INR): 2,868,382.39
Utility-Island Replacement (INR): 0.00
Utility-Island Turnaround (INR): 0.00
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -7,222,868,640.12
Tax (INR): 0.00
PAT (INR): -7,222,868,640.12
Cash Accrual (INR): -1,142,940,896.33

Row 12
Year: 12.0
Capacity Utilization (%): 83.44
Availability (%): 92.71
Revenue Loss (INR): 827,029,636.36
Revenue (INR): 10,512,970,363.64
Operating Cost (INR): 12,878,342,222.04
Utility-Island Service (INR): 2,830,002.43
Utility-Island Replacement (INR): 66,064,226.66
Utility-Island Turnaround (INR): 7,927,707.20
Principal (INR): 0.00
Interest (INR): 0.00
Debt Service (INR): 0.00
CFADS (INR): 0.00
DSCR: 0.000
Depreciation (INR): 6,079,927,743.78
PBT (INR): -8,445,299,602.19
Tax (INR): 0.00
PAT (INR): -8,445,299,602.19
Cash Accrual (INR): -2,365,371,858.40

Typed Financial Schedule

Table 243. Typed Financial Schedule

Columns: Year; Capacity Utilization (%); Availability (%); DSCR; CFADS; Revenue; Opex; PBT; Cash Accrual

Row 1
Year: 1
Capacity Utilization (%): 78.79
Availability (%): 98.48
DSCR: 0.050
CFADS: 0.00
Revenue: 9,927,272,727.27
Opex: 11,486,378,287.07
PBT: -13,163,516,451.19
Cash Accrual: -7,083,588,707.41

Row 2
Year: 2
Capacity Utilization (%): 83.71
Availability (%): 98.48
DSCR: 0.050
CFADS: 0.00

Revenue: 10,547,727,272.73
Opex: 11,898,750,500.79
PBT: -11,850,537,489.94
Cash Accrual: -5,770,609,746.16
Row 3
Year: 3
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.050
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -10,537,558,528.69
Cash Accrual: -4,457,630,784.90
Row 4
Year: 4
Capacity Utilization (%): 83.44
Availability (%): 92.71
DSCR: 0.050
CFADS: 0.00
Revenue: 10,512,970,363.64
Opex: 12,878,342,222.04
PBT: -10,655,092,861.23
Cash Accrual: -4,575,165,117.44
Row 5
Year: 5
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.050
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -8,327,765,269.64
Cash Accrual: -2,247,837,525.85
Row 6
Year: 6
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.000
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -7,222,868,640.12
Cash Accrual: -1,142,940,896.33
Row 7
Year: 7
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.000
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -7,222,868,640.12
Cash Accrual: -1,142,940,896.33
Row 8
Year: 8
Capacity Utilization (%): 83.44
Availability (%): 92.71
DSCR: 0.000
CFADS: 0.00
Revenue: 10,512,970,363.64
Opex: 12,878,342,222.04
PBT: -8,445,299,602.19
Cash Accrual: -2,365,371,858.40
Row 9
Year: 9
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.000
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -7,222,868,640.12

Cash Accrual: -1,142,940,896.33
Row 10
Year: 10
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.000
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -7,222,868,640.12
Cash Accrual: -1,142,940,896.33
Row 11
Year: 11
Capacity Utilization (%): 88.64
Availability (%): 98.48
DSCR: 0.000
CFADS: 0.00
Revenue: 11,168,181,818.18
Opex: 12,311,122,714.52
PBT: -7,222,868,640.12
Cash Accrual: -1,142,940,896.33
Row 12
Year: 12
Capacity Utilization (%): 83.44
Availability (%): 92.71
DSCR: 0.000
CFADS: 0.00
Revenue: 10,512,970,363.64
Opex: 12,878,342,222.04
PBT: -8,445,299,602.19
Cash Accrual: -2,365,371,858.40

Debt Schedule

Table 244. Debt Schedule

Columns: Year; Opening Debt; Principal; Interest; Closing Debt
Row 1: Year: 1 | Opening Debt: 52,614,125,215.35 | Principal: 10,522,825,043.07 | Interest: 5,524,483,147.61 | Closing Debt: 42,091,300,172.28
Row 2: Year: 2 | Opening Debt: 42,091,300,172.28 | Principal: 10,522,825,043.07 | Interest: 4,419,586,518.09 | Closing Debt: 31,568,475,129.21
Row 3: Year: 3 | Opening Debt: 31,568,475,129.21 | Principal: 10,522,825,043.07 | Interest: 3,314,689,888.57 | Closing Debt: 21,045,650,086.14
Row 4: Year: 4 | Opening Debt: 21,045,650,086.14 | Principal: 10,522,825,043.07 | Interest: 2,209,793,259.04 | Closing Debt: 10,522,825,043.07
Row 5: Year: 5 | Opening Debt: 10,522,825,043.07 | Principal: 10,522,825,043.07 | Interest: 1,104,896,629.52 | Closing Debt: 0.00

Heat Integration Cases

Table 245. Heat Integration Cases

Columns: Route; Case ID; Title; Recovered Duty (kW); Residual Hot Utility (kW); Annual Savings (INR); Payback (y); Feasible
Row 1
Route: eo_hydration
Case ID: eo_hydration_no_recovery
Title: No recovery
Recovered Duty (kW): 0.000
Residual Hot Utility (kW): 100977.873
Annual Savings (INR): 0.00
Payback (y): 0.000
Feasible: yes
Row 2
Route: eo_hydration
Case ID: eo_hydration_multi_effect
Title: Multi-effect only
Recovered Duty (kW): 28442.719
Residual Hot Utility (kW): 72535.154

Annual Savings (INR): 795,438,058.00
Payback (y): 0.277
Feasible: yes
Row 3
Route: eo_hydration
Case ID: eo_hydration_pinch_htm
Title: Pinch + HTM loop
Recovered Duty (kW): 51714.034
Residual Hot Utility (kW): 49263.839
Annual Savings (INR): 1,446,251,014.54
Payback (y): 0.401
Feasible: yes
Row 4
Route: omega_catalytic
Case ID: omega_catalytic_no_recovery
Title: No recovery
Recovered Duty (kW): 0.000
Residual Hot Utility (kW): 91056.718
Annual Savings (INR): 0.00
Payback (y): 0.000
Feasible: yes
Row 5
Route: omega_catalytic
Case ID: omega_catalytic_multi_effect
Title: Multi-effect only
Recovered Duty (kW): 25991.162
Residual Hot Utility (kW): 65065.556
Annual Savings (INR): 726,877,044.14
Payback (y): 0.303
Feasible: yes
Row 6
Route: omega_catalytic
Case ID: omega_catalytic_pinch_htm
Title: Pinch + HTM loop
Recovered Duty (kW): 47256.658
Residual Hot Utility (kW): 43800.060
Annual Savings (INR): 1,321,594,625.71
Payback (y): 0.439
Feasible: yes
Row 7
Route: chlorohydrin
Case ID: chlorohydrin_no_recovery
Title: No recovery
Recovered Duty (kW): 0.000
Residual Hot Utility (kW): 13044.246
Annual Savings (INR): 0.00
Payback (y): 0.000
Feasible: yes
Row 8
Route: chlorohydrin
Case ID: chlorohydrin_multi_effect
Title: Multi-effect only
Recovered Duty (kW): 503.065
Residual Hot Utility (kW): 12541.181
Annual Savings (INR): 14,068,881.48
Payback (y): 7.819
Feasible: no
Row 9
Route: chlorohydrin
Case ID: chlorohydrin_pinch_htm
Title: Pinch + HTM loop
Recovered Duty (kW): 411.599
Residual Hot Utility (kW): 12632.647
Annual Savings (INR): 11,510,903.03
Payback (y): 19.981
Feasible: no

Utility Architecture

Table 246. Utility Architecture

Columns: Case; Base Case; Topology; Family; Recovered Duty (kW); Residual Hot Utility (kW); Header Levels; Shared HTM Islands; Cond-Reb Clusters;

Exchanger Count
Row 1
Case: omega_catalytic_no_recovery
Base Case: omega_catalytic_no_recovery
Topology: utility-only architecture
Family: base
Recovered Duty (kW): 0.000
Residual Hot Utility (kW): 91056.718
Header Levels: 0
Shared HTM Islands: 0
Cond-Reb Clusters: 0
Exchanger Count: 0
Row 2
Case: omega_catalytic_multi_effect
Base Case: omega_catalytic_multi_effect
Topology: multi-effect recovery network
Family: base
Recovered Duty (kW): 25991.162
Residual Hot Utility (kW): 65065.556
Header Levels: 0
Shared HTM Islands: 0
Cond-Reb Clusters: 0
Exchanger Count: 3
Row 3
Case: omega_catalytic_multi_effect__staged_headers
Base Case: omega_catalytic_multi_effect
Topology: staged utility header network
Family: staged_headers
Recovered Duty (kW): 25991.162
Residual Hot Utility (kW): 65065.556
Header Levels: 2
Shared HTM Islands: 0
Cond-Reb Clusters: 0
Exchanger Count: 3
Row 4
Case: omega_catalytic_pinch_htm
Base Case: omega_catalytic_pinch_htm
Topology: reactor-to-reboiler HTM loop
Family: base
Recovered Duty (kW): 47256.658
Residual Hot Utility (kW): 43800.060
Header Levels: 0
Shared HTM Islands: 0
Cond-Reb Clusters: 0
Exchanger Count: 2
Row 5
Case: omega_catalytic_pinch_htm__shared_htm
Base Case: omega_catalytic_pinch_htm
Topology: shared HTM island network
Family: shared_htm
Recovered Duty (kW): 47256.658
Residual Hot Utility (kW): 43800.060
Header Levels: 1
Shared HTM Islands: 1
Cond-Reb Clusters: 0
Exchanger Count: 2
Row 6
Case: omega_catalytic_pinch_htm__staged_headers
Base Case: omega_catalytic_pinch_htm
Topology: staged utility header network
Family: staged_headers
Recovered Duty (kW): 47256.658
Residual Hot Utility (kW): 43800.060
Header Levels: 2
Shared HTM Islands: 0
Cond-Reb Clusters: 0
Exchanger Count: 4

Composite Thermal Intervals

Table 247. Composite Thermal Intervals

Columns: Interval; Shifted Upper (C); Shifted Lower (C); Hot Duty (kW); Cold

Duty (kW); Net Duty (kW)
Row 1: Interval: interval_01 | Shifted Upper (C): 260.0 | Shifted Lower (C): 250.0 | Hot Duty (kW): 249.748 | Cold Duty (kW): 0.000 | Net Duty (kW): 249.748
Row 2: Interval: interval_02 | Shifted Upper (C): 250.0 | Shifted Lower (C): 235.0 | Hot Duty (kW): 374.621 | Cold Duty (kW): 4532.727 | Net Duty (kW): -4158.106
Row 3: Interval: interval_03 | Shifted Upper (C): 235.0 | Shifted Lower (C): 230.0 | Hot Duty (kW): 0.000 | Cold Duty (kW): 0.000 | Net Duty (kW): 0.000
Row 4: Interval: interval_04 | Shifted Upper (C): 230.0 | Shifted Lower (C): 185.0 | Hot Duty (kW): 7600.213 | Cold Duty (kW): 0.000 | Net Duty (kW): 7600.213
Row 5: Interval: interval_05 | Shifted Upper (C): 185.0 | Shifted Lower (C): 140.0 | Hot Duty (kW): 35215.298 | Cold Duty (kW): 49170.628 | Net Duty (kW): -13955.329
Row 6: Interval: interval_06 | Shifted Upper (C): 140.0 | Shifted Lower (C): 130.0 | Hot Duty (kW): 11023.188 | Cold Duty (kW): 10926.806 | Net Duty (kW): 96.382
Row 7: Interval: interval_07 | Shifted Upper (C): 130.0 | Shifted Lower (C): 125.0 | Hot Duty (kW): 5511.594 | Cold Duty (kW): 6576.319 | Net Duty (kW): -1064.725
Row 8: Interval: interval_08 | Shifted Upper (C): 125.0 | Shifted Lower (C): 120.0 | Hot Duty (kW): 1598.783 | Cold Duty (kW): 6576.319 | Net Duty (kW): -4977.536
Row 9: Interval: interval_09 | Shifted Upper (C): 120.0 | Shifted Lower (C): 85.0 | Hot Duty (kW): 11191.481 | Cold Duty (kW): 7790.408 | Net Duty (kW): 3401.073
Row 10: Interval: interval_10 | Shifted Upper (C): 85.0 | Shifted Lower (C): 50.0 | Hot Duty (kW): 11191.481 | Cold Duty (kW): 8761.648 | Net Duty (kW): 2429.833
Row 11: Interval: interval_11 | Shifted Upper (C): 50.0 | Shifted Lower (C): 40.0 | Hot Duty (kW): 0.000 | Cold Duty (kW): 2503.328 | Net Duty (kW): -2503.328

Utility Islands

Utility Island Summary View

Table 248. Utility Island Summary View

Columns: Island; Topology; Role; Header; Units; Target Duty (kW); Recovered Duty (kW); Residual Hot Utility (kW)
Row 1
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Topology: shared HTM island network
- Role: shared_htm
- Header: 1
- Units: D-rough, E-rough, R-rough
- Target Duty (kW): 42530.992
- Recovered Duty (kW): 42530.992
- Residual Hot Utility (kW): 33200.832

Utility Island Operating Window View

Table 249. Utility Island Operating Window View

Columns: Island; Candidates; Direct Matches; Indirect Matches; HTM Inventory (m3); Header Pressure (bar); Pair Score; Control Factor
Row 1
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Candidates: 2
- Direct Matches: 1
- Indirect Matches: 1
- HTM Inventory (m3): 3.936
- Header Pressure (bar): 14.50
- Pair Score: 0.000
- Control Factor: 2.500

Table 250. Utility Island Operating Window View

Columns: Island; Topology; Role; Header; Cluster; Units; Target Duty (kW); Potential (kW); Recovered Duty (kW); Residual Hot Utility (kW); Residual Cold Utility (kW); Candidates; Direct Matches; Indirect Matches; HTM Inventory (m3); Header Pressure (bar); Pair Score; Control Factor

Row 1
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Topology: shared HTM island network
- Role: shared_htm
- Header: 1
- Cluster: -
- Units: D-rough, E-rough, R-rough
- Target Duty (kW): 42530.992
- Potential (kW): 42530.992
- Recovered Duty (kW): 42530.992
- Residual Hot Utility (kW): 33200.832
- Residual Cold Utility (kW): 48525.726
- Candidates: 2
- Direct Matches: 1
- Indirect Matches: 1
- HTM Inventory (m3): 3.936
- Header Pressure (bar): 14.50
- Pair Score: 0.000
- Control Factor: 2.500

Selected Utility Train

Table 251. Selected Utility Train

Columns: Exchanger; Island; Cluster; Header; Topology; Service; Hot Unit; Cold Unit; Recovered Duty (kW); Medium

Row 1
- Exchanger: HX-01
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Cluster: -
- Header: 1
- Topology: shared HTM island network
- Service: R-rough to D-rough heat recovery
- Hot Unit: R-rough
- Cold Unit: D-rough
- Recovered Duty (kW): 34024.794
- Medium: Dowtherm A

Row 2
- Exchanger: HX-02
- Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
- Cluster: -
- Header: 1
- Topology: shared HTM island network
- Service: D-rough to E-rough heat recovery
- Hot Unit: D-rough
- Cold Unit: E-rough
- Recovered Duty (kW): 8506.198
- Medium: direct

Utility Train Packages

Utility Train Package Summary View

Table 252. Utility Train Package Summary View

Columns: Equipment; Role; Family; Type; Service; Design Temp (C); Design Pressure (bar); Duty (kW); Area (m2)

Row 1
- Equipment: HX-01
- Role: exchanger
- Family: process_exchange
- Type: HTM loop exchanger
- Service: R-rough to D-rough heat recovery
- Design Temp (C): 213.0
- Design Pressure (bar): 11.00
- Duty (kW): 34024.794
- Area (m2): 9049.147

Row 2

Equipment: HX-01-CTRL
Role: controls
Family: process_exchange
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Design Temp (C): 203.0
Design Pressure (bar): 9.00
Duty (kW): 0.000
Area (m2): 0.000

Row 3
Equipment: HX-01-HDR
Role: header
Family: process_exchange
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Design Temp (C): 205.0
Design Pressure (bar): 14.50
Duty (kW): 34024.794
Area (m2): 0.000

Row 4
Equipment: HX-01-PMP
Role: circulation
Family: process_exchange
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Design Temp (C): 205.0
Design Pressure (bar): 14.50
Duty (kW): 0.000
Area (m2): 0.000

Row 5
Equipment: HX-01-EXP
Role: expansion
Family: process_exchange
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Design Temp (C): 210.0
Design Pressure (bar): 6.00
Duty (kW): 0.000
Area (m2): 0.000

Row 6
Equipment: HX-01-RV
Role: relief
Family: process_exchange
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Design Temp (C): 213.0
Design Pressure (bar): 13.50
Duty (kW): 0.000
Area (m2): 0.000

Row 7
Equipment: HX-02
Role: exchanger
Family: process_exchange
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Design Temp (C): 162.0
Design Pressure (bar): 6.50
Duty (kW): 8506.198
Area (m2): 506.321

Row 8
Equipment: HX-02-CTRL
Role: controls
Family: process_exchange
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Design Temp (C): 158.0
Design Pressure (bar): 5.00
Duty (kW): 0.000

Area (m2): 0.000

Utility Train Package Hydraulic View

Table 253. Utility Train Package Hydraulic View

Columns: Equipment; Header; Volume (m3); Power (kW); Flow (m3/h); LMTD (K);
Phase Load (kg/h)
Row 1
Equipment: HX-01
Header: 1
Volume (m3): 542.949
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 8.000
Phase Load (kg/h): 0.000
Row 2
Equipment: HX-01-CTRL
Header: 1
Volume (m3): 0.150
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Phase Load (kg/h): 0.000
Row 3
Equipment: HX-01-HDR
Header: 1
Volume (m3): 0.851
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Phase Load (kg/h): 0.000
Row 4
Equipment: HX-01-PMP
Header: 1
Volume (m3): 0.250
Power (kW): 15.466
Flow (m3/h): 1818.329
LMTD (K): 0.000
Phase Load (kg/h): 0.000
Row 5
Equipment: HX-01-EXP
Header: 1
Volume (m3): 2.835
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Phase Load (kg/h): 0.000
Row 6
Equipment: HX-01-RV
Header: 1
Volume (m3): 1.309
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Phase Load (kg/h): 0.000
Row 7
Equipment: HX-02
Header: 1
Volume (m3): 30.379
Power (kW): 0.000
Flow (m3/h): 111.924
LMTD (K): 30.000
Phase Load (kg/h): 0.000
Row 8
Equipment: HX-02-CTRL
Header: 1
Volume (m3): 0.150
Power (kW): 0.000
Flow (m3/h): 111.924
LMTD (K): 0.000

Phase Load (kg/h): 0.000

Table 254. Utility Train Package Hydraulic View

Columns: Island; Cluster; Header; Step; Equipment; Role; Family; Type; Service; Design Temp (C); Design Pressure (bar); Volume (m3); Duty (kW); Power (kW); Flow (m3/h); LMTD (K); Area (m2); Phase Load (kg/h)

Row 1
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01
Role: exchanger
Family: process_exchange
Type: HTM loop exchanger
Service: R-rough to D-rough heat recovery
Design Temp (C): 213.0
Design Pressure (bar): 11.00
Volume (m3): 542.949
Duty (kW): 34024.794
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 8.000
Area (m2): 9049.147
Phase Load (kg/h): 0.000

Row 2
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01-CTRL
Role: controls
Family: process_exchange
Type: Utility control package
Service: R-rough to D-rough heat recovery control valves, instrumentation, and bypass station
Design Temp (C): 203.0
Design Pressure (bar): 9.00
Volume (m3): 0.150
Duty (kW): 0.000
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Row 3
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01-HDR
Role: header
Family: process_exchange
Type: Shared HTM header manifold
Service: R-rough to D-rough heat recovery network header and isolation package
Design Temp (C): 205.0
Design Pressure (bar): 14.50
Volume (m3): 0.851
Duty (kW): 34024.794
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Row 4
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01-PMP

Role: circulation
Family: process_exchange
Type: HTM circulation skid
Service: R-rough to D-rough heat recovery circulation loop
Design Temp (C): 205.0
Design Pressure (bar): 14.50
Volume (m3): 0.250
Duty (kW): 0.000
Power (kW): 15.466
Flow (m3/h): 1818.329
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Row 5
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01-EXP
Role: expansion
Family: process_exchange
Type: HTM expansion tank
Service: R-rough to D-rough heat recovery HTM expansion and inventory hold-up
Design Temp (C): 210.0
Design Pressure (bar): 6.00
Volume (m3): 2.835
Duty (kW): 0.000
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Row 6
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_01
Equipment: HX-01-RV
Role: relief
Family: process_exchange
Type: HTM relief package
Service: R-rough to D-rough heat recovery thermal relief and collection package
Design Temp (C): 213.0
Design Pressure (bar): 13.50
Volume (m3): 1.309
Duty (kW): 0.000
Power (kW): 0.000
Flow (m3/h): 1818.329
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Row 7
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_02
Equipment: HX-02
Role: exchanger
Family: process_exchange
Type: Heat integration exchanger
Service: D-rough to E-rough heat recovery
Design Temp (C): 162.0
Design Pressure (bar): 6.50
Volume (m3): 30.379
Duty (kW): 8506.198
Power (kW): 0.000
Flow (m3/h): 111.924
LMTD (K): 30.000
Area (m2): 506.321
Phase Load (kg/h): 0.000

Row 8

Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Cluster: -
Header: 1
Step: omega_catalytic_pinch_htm__shared_htm_step_02
Equipment: HX-02-CTRL
Role: controls
Family: process_exchange
Type: Utility control package
Service: D-rough to E-rough heat recovery control valves, instrumentation, and bypass station
Design Temp (C): 158.0
Design Pressure (bar): 5.00
Volume (m3): 0.150
Duty (kW): 0.000
Power (kW): 0.000
Flow (m3/h): 111.924
LMTD (K): 0.000
Area (m2): 0.000
Phase Load (kg/h): 0.000

Operating Mode Decision

Table 255. Operating Mode Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: continuous | Description: Continuous operation | Feasible: yes | Score: 81.01 | Rejected Reasons: -

Control Architecture Decision

Table 256. Control Architecture Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: basic_regulatory | Description: Basic regulatory loops with alarms | Feasible: yes | Score: 62.00 | Rejected Reasons: -
Row 2: Candidate: cascade_override | Description: Cascade temperature/pressure control with override protection | Feasible: yes | Score: 84.00 | Rejected Reasons: -
Row 3: Candidate: sis_augmented | Description: Regulatory control plus SIS/interlocks on critical nodes | Feasible: yes | Score: 78.00 | Rejected Reasons: -

Site Decision

Table 257. Site Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: Dahej | Description: Dahej, Gujarat | Feasible: yes | Score: 94.00 | Rejected Reasons: -
Row 2: Candidate: Jamnagar | Description: Jamnagar, Gujarat | Feasible: yes | Score: 89.50 | Rejected Reasons: -
Row 3: Candidate: Paradip | Description: Paradip, Odisha | Feasible: yes | Score: 78.50 | Rejected Reasons: -

Route Decision

Table 258. Route Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: eo_hydration | Description: Ethylene oxide hydration (Liquid Hydration Train) with Pinch + HTM loop | Feasible: yes | Score: 39.95 | Rejected Reasons: dehydration_duty; glycol_split_vle_basis
Row 2: Candidate: omega_catalytic | Description: OMEGA catalytic route via ethylene carbonate (Liquid Hydration Train) with Pinch + HTM loop | Feasible: yes | Score: 60.60 | Rejected Reasons: dehydration_duty; glycol_split_vle_basis
Row 3: Candidate: chlorohydrin | Description: Ethylene chlorohydrin

hydrolysis (Chlorinated Hydrolysis Train) with No recovery | Feasible: no | Score: -1.00 | Rejected Reasons: Route generates chloride-heavy waste and is not preferred for India deployment under the current policy basis.; chloride_waste_burden; brine_handling_penalty

Utility Basis Decision

Table 259. Utility Basis Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: mp_steam_15bar | Description: 15 bar medium-pressure steam and conventional utility header | Feasible: yes | Score: 70.05 | Rejected Reasons: -
Row 2: Candidate: hp_steam_20bar | Description: 20 bar high-pressure steam basis with integrated recovery support | Feasible: yes | Score: 90.95 | Rejected Reasons: -
Row 3: Candidate: htm_assisted_25bar | Description: 25 bar steam header with hotter utility backbone for HTM-backed recovery | Feasible: yes | Score: 84.65 | Rejected Reasons: -

Economic Basis Decision

Table 260. Economic Basis Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: selected_integrated_base | Description: Selected India base case at Dahej with Pinch + HTM loop | Feasible: yes | Score: 59.43 | Rejected Reasons: Gross margin is non-positive under this basis and needs analyst review.
Row 2: Candidate: no_recovery_counterfactual | Description: Same site and route basis without selected heat-integration savings | Feasible: yes | Score: 86.20 | Rejected Reasons: Gross margin is non-positive under this basis and needs analyst review.
Row 3: Candidate: conservative_case | Description: Selected architecture under conservative India scenario multipliers | Feasible: yes | Score: 42.73 | Rejected Reasons: Gross margin is non-positive under this basis and needs analyst review.

Capacity Case

Table 261. Capacity Case

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: 150000_tpa | Description: 150,000 TPA plant basis | Feasible: yes | Score: 72.10 | Rejected Reasons: -
Row 2: Candidate: 200000_tpa | Description: 200,000 TPA plant basis | Feasible: yes | Score: 93.10 | Rejected Reasons: -
Row 3: Candidate: 250000_tpa | Description: 250,000 TPA plant basis | Feasible: yes | Score: 76.20 | Rejected Reasons: -

Property Method Decision

Table 262. Property Method Decision

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: direct_public_package | Description: Use directly cited public component properties as the primary basis. | Feasible: no | Score: 16.00 | Rejected Reasons: Property coverage still depends materially on estimated values.
Row 2: Candidate: hybrid_cited_plus_library | Description: Use cited public anchors with library-backed calculation and benchmark seeds for missing constants. | Feasible: yes | Score: 62.00 | Rejected Reasons: -
Row 3: Candidate: estimation_dominant | Description: Use estimation-heavy property basis with conservative assumptions. | Feasible: yes | Score: 40.00 | Rejected Reasons: -
Row 4: Candidate: analogy_basis | Description: Use close-compound analogy

as a provisional property basis. | Feasible: yes | Score: 10.00 | Rejected Reasons: -

Thermo Method

Table 263. Thermo Method

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: direct_public_data | Description: Use direct public property and reaction data where available. | Feasible: yes | Score: 91.10 | Rejected Reasons: -
Row 2: Candidate: correlation_interpolation | Description: Use public correlations and interpolation around cited anchors. | Feasible: yes | Score: 88.10 | Rejected Reasons: -
Row 3: Candidate: estimation_method | Description: Use accepted estimation methods with conservative bounds. | Feasible: yes | Score: 65.90 | Rejected Reasons: -
Row 4: Candidate: analogy_basis | Description: Use close-compound analogy as a provisional design basis. | Feasible: yes | Score: 53.50 | Rejected Reasons: -

Kinetics Method

Table 264. Kinetics Method

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: cited_rate_law | Description: Use cited industrial or literature rate law directly. | Feasible: yes | Score: 92.00 | Rejected Reasons: -
Row 2: Candidate: arrhenius_fit | Description: Use Arrhenius-style fitting around cited rate data. | Feasible: yes | Score: 88.50 | Rejected Reasons: -
Row 3: Candidate: apparent_order_fit | Description: Use apparent-order fitting from conversion and residence-time basis. | Feasible: yes | Score: 72.50 | Rejected Reasons: -
Row 4: Candidate: conservative_analogy | Description: Use close-route analogy with conservative residence-time allowance. | Feasible: yes | Score: 56.50 | Rejected Reasons: -

Exchanger Choice

Table 265. Exchanger Choice

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: shell_and_tube | Description: Shell-and-tube exchanger basis | Feasible: yes | Score: 88.00 | Rejected Reasons: -
Row 2: Candidate: plate_exchanger | Description: Plate exchanger basis | Feasible: yes | Score: 70.00 | Rejected Reasons: -
Row 3: Candidate: air_cooler | Description: Air cooler basis | Feasible: yes | Score: 68.00 | Rejected Reasons: -
Row 4: Candidate: kettle_reboiler | Description: Kettle/thermosyphon reboiler basis | Feasible: yes | Score: 58.00 | Rejected Reasons: -

Storage Choice

Table 266. Storage Choice

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: vertical_tank_farm | Description: Vertical atmospheric tank farm | Feasible: yes | Score: 100.00 | Rejected Reasons: -
Row 2: Candidate: pressure_bullets | Description: Pressurized bullet storage | Feasible: yes | Score: 48.00 | Rejected Reasons: -
Row 3: Candidate: silo_hopper | Description: Silo and hopper storage | Feasible: yes | Score: 44.00 | Rejected Reasons: -

Moc Choice

Table 267. Moc Choice

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: carbon_steel | Description: Carbon steel construction | Feasible: yes | Score: 100.00 | Rejected Reasons: -
Row 2: Candidate: ss316l | Description: SS316L construction | Feasible: yes | Score: 98.00 | Rejected Reasons: -
Row 3: Candidate: alloy_steel_converter_service | Description: Alloy steel converter service | Feasible: yes | Score: 52.00 | Rejected Reasons: -
Row 4: Candidate: rubber_lined_cs | Description: Rubber-lined carbon steel | Feasible: yes | Score: 58.00 | Rejected Reasons: -

Procurement Basis

Table 268. Procurement Basis

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: domestic_cluster_procurement | Description: Domestic cluster-led procurement | Feasible: yes | Score: 92.00 | Rejected Reasons: -
Row 2: Candidate: mixed_import_domestic | Description: Mixed import and domestic procurement | Feasible: yes | Score: 84.00 | Rejected Reasons: -
Row 3: Candidate: import_heavy | Description: Import-heavy procurement | Feasible: yes | Score: 68.00 | Rejected Reasons: -

Logistics Basis

Table 269. Logistics Basis

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: coastal_cluster_dispatch | Description: Coastal cluster dispatch basis | Feasible: yes | Score: 92.00 | Rejected Reasons: -
Row 2: Candidate: rail_road_inland | Description: Rail-road inland dispatch basis | Feasible: yes | Score: 78.00 | Rejected Reasons: -

Financing Basis

Table 270. Financing Basis

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: debt_equity_70_30 | Description: 70:30 debt-equity basis | Feasible: yes | Score: 32.25 | Rejected Reasons: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 2: Candidate: debt_equity_60_40 | Description: 60:40 debt-equity basis | Feasible: yes | Score: 26.42 | Rejected Reasons: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 3: Candidate: conservative_50_50 | Description: 50:50 conservative financing basis | Feasible: yes | Score: 17.01 | Rejected Reasons: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening

threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.

Layout Choice

Table 271. Layout Choice

Columns: Candidate; Description; Feasible; Score; Rejected Reasons
Row 1: Candidate: compact_process_core | Description: Compact process core with short pipe runs | Feasible: yes | Score: 78.00 | Rejected Reasons: -
Row 2: Candidate: segregated_hazard_zones | Description: Segregated hazard-zone layout | Feasible: yes | Score: 92.00 | Rejected Reasons: -
Row 3: Candidate: solids_block_layout | Description: Block layout with solids handling and truck access priority | Feasible: yes | Score: 58.00 | Rejected Reasons: -

HAZOP Node Register

Table 272. HAZOP Node Register

Columns: Node; Parameter; Guide Word; Safeguards
Row 1: Node: R-101 | Parameter: Temperature | Guide Word: More | Safeguards: TIC-R-101; PIC-R-101; FRC-R-101
Row 2: Node: PU-201 | Parameter: Pressure | Guide Word: Less | Safeguards: LIC-PU-201; PIC-PU-201; FIC-PU-201
Row 3: Node: E-101 | Parameter: Heat transfer | Guide Word: Less | Safeguards: TIC-R-101; PIC-R-101
Row 4: Node: TK-301 | Parameter: Level | Guide Word: More | Safeguards: LIC-TK-301; PIC-TK-301
Row 5: Node: HX-01 | Parameter: Heat transfer | Guide Word: Less | Safeguards: TIC-R-101; PIC-R-101
Row 6: Node: HX-01-EXP | Parameter: Level | Guide Word: More | Safeguards: LIC-HX-01-EXP; PIC-HX-01-EXP
Row 7: Node: HX-02 | Parameter: Heat transfer | Guide Word: Less | Safeguards: TIC-R-101; PIC-R-101

Rejected Alternative Log

Table 273. Rejected Alternative Log

Columns: Decision; Candidate; Reasons
Row 1: Decision: route_selection | Candidate: eo_hydration | Reasons: dehydration_duty; glycol_split_vle_basis
Row 2: Decision: route_selection | Candidate: omega_catalytic | Reasons: dehydration_duty; glycol_split_vle_basis
Row 3: Decision: route_selection | Candidate: chlorohydrin | Reasons: Route generates chloride-heavy waste and is not preferred for India deployment under the current policy basis.; chloride_waste_burden; brine_handling_penalty
Row 4: Decision: economic_basis | Candidate: selected_integrated_base | Reasons: Gross margin is non-positive under this basis and needs analyst review.
Row 5: Decision: economic_basis | Candidate: no_recovery_counterfactual | Reasons: Gross margin is non-positive under this basis and needs analyst review.
Row 6: Decision: economic_basis | Candidate: conservative_case | Reasons: Gross margin is non-positive under this basis and needs analyst review.
Row 7: Decision: property_method_decision | Candidate: direct_public_package | Reasons: Property coverage still depends materially on estimated values.
Row 8: Decision: financing_basis | Candidate: debt_equity_70_30 | Reasons: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 9: Decision: financing_basis | Candidate: debt_equity_60_40 | Reasons:

Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.
Row 10: Decision: financing_basis | Candidate: conservative_50_50 | Reasons: Minimum DSCR is below the lender screening threshold.; Average DSCR is below the lender screening threshold.; LLCR is below the lender screening threshold.; PLCR is below the lender screening threshold.; conservative scenario minimum DSCR is below the lender screening threshold.; conservative scenario average DSCR is below the lender screening threshold.; conservative scenario LLCR is below the lender screening threshold.; conservative scenario PLCR is below the lender screening threshold.

Economic Scenario Model

Table 274. Economic Scenario Model

Columns: Scenario; Revenue (INR/y); Transport/Service (INR/y); Utility-Island Burden (INR/y); Opex (INR/y); Gross Margin (INR/y)
Row 1: Scenario: base | Revenue (INR/y): 12,600,000,000.00 | Transport/Service (INR/y): 14,583,091.39 | Utility-Island Burden (INR/y): 3,146,650,249.39 | Opex (INR/y): 13,273,756,525.48 | Gross Margin (INR/y): -673,756,525.48
Row 2: Scenario: conservative | Revenue (INR/y): 11,970,000,000.00 | Transport/Service (INR/y): 16,167,728.07 | Utility-Island Burden (INR/y): 3,791,561,480.32 | Opex (INR/y): 14,912,994,804.01 | Gross Margin (INR/y): -2,942,994,804.01
Row 3: Scenario: upside | Revenue (INR/y): 13,104,000,000.00 | Transport/Service (INR/y): 14,088,928.39 | Utility-Island Burden (INR/y): 2,902,794,468.22 | Opex (INR/y): 12,731,575,600.67 | Gross Margin (INR/y): 372,424,399.33

Table 275. Economic Scenario Model

Columns: Scenario; Utility-Island Service (INR/y); Utility-Island Replacement (INR/y); Packing (INR/y); Classifier (INR/y); Filter Media (INR/y); Dryer Exhaust (INR/y)
Row 1
- Scenario: base
- Utility-Island Service (INR/y): 885,675.97
- Utility-Island Replacement (INR/y): 13,697,415.43
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Row 2
- Scenario: conservative
- Utility-Island Service (INR/y): 1,001,404.29
- Utility-Island Replacement (INR/y): 15,166,323.78
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Row 3
- Scenario: upside
- Utility-Island Service (INR/y): 847,060.49
- Utility-Island Replacement (INR/y): 13,241,867.90
- Packing (INR/y): 0.00
- Classifier (INR/y): 0.00
- Filter Media (INR/y): 0.00
- Dryer Exhaust (INR/y): 0.00

Table 276. Economic Scenario Model

Columns: Scenario; Island; Capex Burden (INR); Allocated Utility (INR/y); Service (INR/y); Replacement (INR/y); Operating Burden (INR/y)
Row 1

Scenario: base
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Capex Burden (INR): 648,415,477.00
Allocated Utility (INR/y): 3,132,067,158.00
Service (INR/y): 885,675.97
Replacement (INR/y): 13,697,415.43
Operating Burden (INR/y): 3,146,650,249.39
Row 2
Scenario: conservative
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Capex Burden (INR): 717,951,435.39
Allocated Utility (INR/y): 3,775,393,752.25
Service (INR/y): 1,001,404.29
Replacement (INR/y): 15,166,323.78
Operating Burden (INR/y): 3,791,561,480.32
Row 3
Scenario: upside
Island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01
Capex Burden (INR): 626,850,527.87
Allocated Utility (INR/y): 2,888,705,539.82
Service (INR/y): 847,060.49
Replacement (INR/y): 13,241,867.90
Operating Burden (INR/y): 2,902,794,468.22

Reactor Design Basis

Table 277. Reactor Design Basis

Columns: Field; Value
Row 1: Field: Selected type | Value: Tubular Plug Flow Hydrator
Row 2: Field: Governing equations | Value: V = tau * volumetric_flow; k = A * exp(-Ea/RT); Da = k * tau; Q = U * A * LMTD; Nu = 0.023 * Re^0.8 * Pr^0.4
Row 3: Field: Operating envelope | Value: temperature_c=255.0; pressure_bar=20.00; cooling_medium=Dowtherm A / cooling water; utility_topology=shared HTM island network (shared_htm) with 1 utility islands; utility_architecture_family=shared_htm; coupled_service_basis=standalone utility service; selected_utility_islands=none; selected_header_levels=none; selected_cluster_ids=none; catalyst_name=CO2 catalytic loop; catalyst_inventory_kg=4914758.026; catalyst_cycle_days=270.0; catalyst_regeneration_days=5.0; catalyst_whsv_1_hr=0.0055

Column Hydraulics

Table 278. Column Hydraulics

Columns: Field; Value
Row 1: Field: Flooding fraction | Value: 0.538
Row 2: Field: Tray spacing (m) | Value: 0.450
Row 3: Field: Downcomer area fraction | Value: 0.140
Row 4: Field: Vapor velocity (m/s) | Value: 0.188
Row 5: Field: Allowable vapor velocity (m/s) | Value: 0.350
Row 6: Field: Capacity factor (m/s) | Value: 350000.000
Row 7: Field: Active area (m2) | Value: 1.520
Row 8: Field: Liquid load (m3/h) | Value: 6090.003
Row 9: Field: Pressure drop per stage (kPa) | Value: 0.282

Heat Exchanger Thermal Design

Table 279. Heat Exchanger Thermal Design

Columns: Field; Value
Row 1: Field: Configuration | Value: Shell And Tube
Row 2: Field: Equations | Value: Q = U * A * LMTD; A = Q / (U * LMTD); m_phase = Q / lambda
Row 3: Field: Thermal inputs | Value: heat_load_kw=34024.794; lmtd_k=8.000; u_w_m2_k=470.000; package_family=process_exchange; circulation_flow_m3_hr=1818.329; phase_change_load_kg_hr=0.000; package_holdup_m3=542.949; utility_topology=shared HTM island network (shared_htm) with 1 utility islands; utility_architecture_family=shared_htm; selected_train_step_id=omega_catalytic_pinch_htm__shared_htm_step_01;

selected_island_id=omega_catalytic_pinch_htm__shared_htm_shared_htm_01;
selected_header_level=1; selected_cluster_id=none;
allocated_recovered_duty_target_kw=42530.992;
boiling_side_coefficient_w_m2_k=1200.000;
condensing_side_coefficient_w_m2_k=1200.000
Row 4: Field: Package basis | Value: package_family=process_exchange;
selected_package_roles=circulation, controls, exchanger, expansion,
header, relief; selected_package_items=omega_catalytic_pinch_htm__shared_h
tm_step_01_exchanger,
omega_catalytic_pinch_htm__shared_htm_step_01_controls,
omega_catalytic_pinch_htm__shared_htm_step_01_header,
omega_catalytic_pinch_htm__shared_htm_step_01_circulation,
omega_catalytic_pinch_htm__shared_htm_step_01_expansion,
omega_catalytic_pinch_htm__shared_htm_step_01_relief

Pump Design

Table 280. Pump Design

Columns: Field; Value
Row 1: Field: Pump ID | Value: TK-301_pump
Row 2: Field: Service | Value: Ethylene Glycol storage via vertical tank
farm transfer
Row 3: Field: Flow (m3/h) | Value: 957.690
Row 4: Field: Head (m) | Value: 52.000
Row 5: Field: Power (kW) | Value: 159.360

Mechanical Design Basis

Table 281. Mechanical Design Basis

Columns: Field; Value
Row 1: Field: Code basis | Value: ASME-style screening logic
Row 2: Field: Design pressure basis | Value: Selected equipment design
pressure with feasibility-study margin
Row 3: Field: Design temperature basis | Value: Selected equipment design
temperature with preliminary utility envelope
Row 4: Field: Corrosion allowance (mm) | Value: 3.00
Row 5: Field: Support design basis | Value: Support screening includes
vertical, piping, wind, seismic, and thermal-growth load components by
equipment family.
Row 6: Field: Load case basis | Value: Tall process equipment uses
operating + wind + seismic + nozzle loads; headers use rack piping +
thermal expansion; skids use dead load + piping reaction.
Row 7: Field: Foundation basis | Value: Support variants differentiate
skirt, saddle, leg, rack, and skid foundations with screening footprint
and anchor-group logic.
Row 8: Field: Nozzle load basis | Value: Nozzle schedules include
reinforcement family, local shell interaction, orientation, projection,
and screening piping load allocation.
Row 9: Field: Connection rating basis | Value: Nozzle and header
connection classes follow design-pressure screening via ASME/PN envelope
mapping.
Row 10: Field: Access platform basis | Value: Columns, absorbers,
reactors, and exchanger services above grade include screening access-
platform area allowances.

Tax and Depreciation Basis

Depreciation uses a 12-year straight-line screening basis on total CAPEX of INR 72,959,132,925.40; corporate tax assumed at 25%.

Source Extracts

Table 282. Source Extracts

Columns: Source ID; Kind; Domain; Title; Geography; Location
Row 1: Source ID: user_doc_1 | Kind: user_document | Domain: technical |
Title: EG technical basis | Geography: global | Location: /Users/ayzaman/.
gemini/antigravity/scratch/AoC/examples/eg_sources/technical_basis.md
Row 2: Source ID: user_doc_2 | Kind: user_document | Domain: site | Title:
India site note | Geography: India | Location: /Users/ayzaman/.gemini/anti
gravity/scratch/AoC/examples/eg_sources/india_site_note.md

Row 3: Source ID: india_sheet_1 | Kind: market | Domain: economics | Title: India Market | Geography: India | Location: /Users/ayzaman/.gemini/antigravity/scratch/AoC/examples/eg_sources/india_market.csv
Row 4: Source ID: eg_handbook | Kind: handbook | Domain: technical | Title: Ethylene glycol handbook properties | Geography: global | Location: https://example.com/eg/properties
Row 5: Source ID: eg_kinetics | Kind: literature | Domain: technical | Title: Ethylene oxide hydration kinetics note | Geography: global | Location: https://example.com/eg/kinetics
Row 6: Source ID: eg_market_india | Kind: market | Domain: economics | Title: India MEG market note | Geography: India | Location: https://example.com/india/meg-market
Row 7: Source ID: eg_sites_india | Kind: company_report | Domain: site | Title: India petrochemical cluster siting note | Geography: India | Location: https://example.com/india/sites
Row 8: Source ID: eg_utilities_gujarat | Kind: utility | Domain: utilities | Title: India utility tariff note | Geography: Gujarat | Location: https://example.com/india/utilities
Row 9: Source ID: eg_safety | Kind: sds | Domain: safety | Title: Ethylene oxide and ethylene glycol SDS note | Geography: global | Location: https://example.com/eg/safety

Reaction-System Calculation Traces

Table 283. Reaction-System Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Byproduct selectivity gap | Formula: Gap = 1 - S | Inputs: S=0.9900 | Result: 0.0100 fraction of converted feed | Notes: Gap is allocated explicitly across the reaction-network closure artifact before stream solving.

Stream-Balance Calculation Traces

Table 284. Stream-Balance Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Main reaction extent | Formula: extent = product_kmol / nu_product | Inputs: product_kmol=406.839460; nu_product=1.000 | Result: 406.839460 kmol/h | Notes:
Row 2: Trace: Feed extent with conversion/selectivity | Formula: feed_extent = (product_kmol / nu_product) / (selectivity * conversion) | Inputs: selectivity=0.9900; conversion=0.9600 | Result: 428.071822 kmol/h | Notes:
Row 3: Trace: Byproduct closure allocation | Formula: Residual byproduct mass is allocated across explicit byproduct-closure estimates | Inputs: closure_status=estimated; estimate_count=1; residual_mass_kg_hr=253.021 | Result: 1 estimates items | Notes: This replaces the older implicit heavy-ends fallback whenever the reaction system carries a byproduct-closure artifact.
Row 4: Trace: Multi-component recycle solution | Formula: Solve recycle per reactant using target_total, consumed, recovery, and purge fractions | Inputs: Ethylene oxide=fresh=417.223; recycle=10.849; total=428.072; Water=fresh=420.411; recycle=29.064; total=449.475 | Result: 2 components solved components | Notes: This is the first generic convergence slice: reactants are solved through the same recycle/purge loop instead of family-specific inline math.
Row 5: Trace: Unit-sequence expansion | Formula: stream_count = feeds + mixed feed + reactor + separation + recycle + product/waste | Inputs: family=generic | Result: 10 streams | Notes:
Row 6: Trace: Fresh-feed to external-out closure | Formula: closure = | Inputs: fresh_feed - external_out | Result: / fresh_feed | Notes: fresh_feed=25952.368; external_out=25952.368 | Field 6: 0.000000 % | Field 7:

Energy-Balance Calculation Traces

Table 285. Energy-Balance Calculation Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Feed preheat duty | Formula: Q = m * Cp * dT / 3600 | Inputs: m=25952.368 kg/h; dT=215.0 K | Result: 4532.727 kW | Notes:
Row 2: Trace: Reaction duty | Formula: Q = | Inputs: n * dH | Result: / 3600 | Notes: n=402.364225 kmol/h; dH=-68.000 kJ/mol | Field 6: 7600.213 kW | Field 7:
Row 3: Trace: Unitwise duty expansion | Formula: duty_count = number of explicit unit duties in the solved energy envelope | Inputs: | Result: 3 duties | Notes: The generic energy solver now emits duty rows for each major process section rather than a single family-level total.
Row 4: Trace: Thermal packet expansion | Formula: packet_count = number of unitwise thermal packets carried into utility and equipment design | Inputs: | Result: 3 packets | Notes: These packets preserve unit-level hot and cold thermal interfaces downstream.
Row 5: Trace: Packet-derived exchanger candidates | Formula: candidate_count = number of packet-to-packet heat-recovery candidates | Inputs: | Result: 2 candidates | Notes: These candidates seed utility-architecture selection and detailed exchanger sizing.
Row 6: Trace: Composition-driven thermal basis | Formula: Cp and latent-duty basis are derived from solved unit composition states and mixture-property packages | Inputs: feed_state=feed_prep_composition_state; primary_state=primary_flash_composition_state; purification_state=purification_composition_state; feed_cp=2.816; product_cp=2.452; mixture_packages=5 | Result: generic family | Notes: The energy solver now consumes mixture-property packages first and only retains heuristics as a compatibility fallback.

Reactor Design Traces

Table 286. Reactor Design Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Reactor holdup | Formula: V = Qv * tau * factor | Inputs: Qv=22884.886; tau=1.000 | Result: 22884.886 m3 | Notes:
Row 2: Trace: Reactor design volume | Formula: Vd = V * design_factor | Inputs: V=22884.886; factor=1.180 | Result: 27004.165 m3 | Notes:
Row 3: Trace: Kinetics-coupled sizing basis | Formula: k = A*exp(-Ea/RT); tau_design from kinetic space time, route time, and solved conversion target | Inputs: Ea_kJ_mol=58.000; A_1_hr=1.200e+07; T_K=513.15; k_1_hr=14.9632; tau_kin_hr=0.6000; X_target=0.9600 | Result: 1.000 h | Notes: Stage 3 reactor design now ties residence time to the selected kinetic basis instead of only a fixed route heuristic.
Row 4: Trace: Reactor-side Reynolds number | Formula: Re = rho * v * Dh / mu | Inputs: rho=1.2; v=0.200; Dh=0.032; mu=0.01304 | Result: 1000.0 - | Notes:
Row 5: Trace: Reactor-side Nusselt number | Formula: Nu = 0.023 Re^0.8 Pr^0.4 | Inputs: Re=1000.0; Pr=122.14 | Result: 39.49 - | Notes:
Row 6: Trace: Reactor property-package basis | Formula: Pr = Cp * mu / k using mixture-property package values when available | Inputs: Cp=2.448; mu=0.013037; k=0.2613; mixture_package=reactor_mixture_properties | Result: 122.14 - | Notes:
Row 7: Trace: Solved reactor packet basis | Formula: reactor packet -> inlet mass and thermal duty | Inputs: packet_inlet_mass_kg_hr=26953.864; thermal_packet=R-101_thermal_packet; composition_state=reactor_composition_state | Result: 7600.213 kW | Notes: Reactor sizing now reads the solved reactor unit packet, composition state, and thermal packet before falling back to route-level heuristics.
Row 8: Trace: Reactor utility-train coupling | Formula: Integrated reactor duty = sum(selected train steps tied to reactor source/sink) | Inputs: selected_steps=none; topology=shared HTM island network (shared_htm) with 1 utility islands; architecture_family=shared_htm; selected_islands=none; header_levels=none; allocated_target_kw=0.000 | Result: 0.000 kW | Notes: This captures recovered reactor duty exported to or imported from the selected utility train.
Row 9: Trace: Reactor integrated heat-transfer basis | Formula: A_int = Q_int / (U * LMTD_int) | Inputs: Q_int=0.0; U=850.0; LMTD_int=0.00 | Result: 0.000 m2 | Notes: Integrated heat-transfer area is derived from the selected train-step thermal driving force, not only aggregate recovered duty.
Row 10: Trace: Reactor thermal stability basis | Formula: DeltaTad = Q/(m*Cp); margin = (Qcap-Qduty)/Qduty | Inputs: Qduty=7600.213; m_feed=26953.864; Cp_kJ_kg_K=2.448; DeltaTad=414.729; Qcap=8512.239 |

Result: 0.1200 fraction margin | Notes: This is a screening runaway and heat-removal check to keep reactor selection tied to thermal controllability.
Row 11: Trace: Catalyst service basis | Formula: mcat = V*(1-eps)*rhobulk ; WHSV = mfeed/mcat | Inputs: catalyst=CO2 catalytic loop; void_fraction=0.350; inventory_kg=4914758.026; WHSV_1_hr=0.0055; cycle_days=270.0 | Result: 4914758.026 kg | Notes: Catalyst inventory and cycle are screening values used to connect reactor selection with service-life assumptions.

Column Design Traces

Table 287. Column Design Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Process-unit family | Formula: service = selected separation family | Inputs: | Result: Distillation and purification train | Notes:
Row 2: Trace: Separation thermodynamics basis | Formula: alpha = sqrt(alpha_top * alpha_bottom) from component K-values when a VLE basis exists | Inputs: light_key=Water; heavy_key=Ethylene Glycol; alpha_top=25.9091; alpha_bottom=14.3988; method=ideal_raoult_missing_bip_fallback | Result: 19.3148 - | Notes: Column volatility basis now prefers the separation-thermodynamics artifact built from Antoine / Clausius-Clapeyron K-values when available.
Row 3: Trace: Minimum theoretical stages | Formula: Nmin = log[(xD,LK/xB,LK)*(xB,HK/xD,HK)] / log(alpha) | Inputs: xD,LK=0.9850; xB,LK=0.0040; xD,HK=0.0150; xB,HK=0.9990; alpha=19.315 | Result: 3.278 stages | Notes:
Row 4: Trace: Minimum reflux proxy | Formula: Rmin = f(alpha, q, xD,LK) | Inputs: alpha=19.315; q=1.106; xD,LK=0.9850 | Result: 0.150 - | Notes: This is a screening Underwood-style reflux estimate built from the solved property basis and separation severity.
Row 5: Trace: Feed and volatility section basis | Formula: q from Cp and DeltaT; section volatility anchored to top/bottom alpha estimates | Inputs: q=1.106; alpha_top=25.909; alpha_bottom=14.399; murphree_efficiency=0.634 | Result: 1.106 - | Notes: Stage 2 separation design keeps the feed-quality and section-volatility basis explicit instead of burying it inside the reflux proxy.
Row 6: Trace: Actual stage proxy | Formula: Nactual = g(Nmin, R/Rmin, tray efficiency) | Inputs: Nmin=3.278; R=0.200; Rmin=0.150; tray_eff=0.660 | Result: 7.000 actual stages | Notes:
Row 7: Trace: Rectifying and stripping section basis | Formula: Nrect + Nstrip = Ntheoretical ; section loads derived from reflux and bottoms circulation | Inputs: feed_stage=4; Ntheoretical=4.278; Nrect=2.444; Nstrip=1.833; Vrect=6352.735; Vstrip=5683.010 | Result: 4.278 stages | Notes: This makes the section-level column basis explicit before deeper M9-style tray-by-tray work.
Row 8: Trace: Equivalent diameter | Formula: D = sqrt(4*Aactive/[pi*(1-Adc)]) | Inputs: Aactive=1.105; Adc=0.140 | Result: 1.500 m | Notes:
Row 9: Trace: Column hydraulic capacity basis | Formula: uallow = C * sqrt[(rhoL-rhoV)/rhoV]; uflood = usuperficial/uallow | Inputs: C=0.300; rhoL=1.158; rhoV=6.005; usuperficial=0.188 | Result: 0.538 - | Notes:
Row 10: Trace: Process-unit property-package basis | Formula: Hydraulics proxy uses density, viscosity, and Cp from the mixture-property package | Inputs: density=1.158; viscosity=0.014479; Cp=2.452; mixture_package=purification_mixture_properties | Result: 1.158 kg/m3 | Notes:
Row 11: Trace: Solved process-unit packet basis | Formula: packet basis -> outlet throughput and matched thermal packets | Inputs: process_packet=purification_unit_packet; heating_packet=fallback; cooling_packet=fallback; composition_state=purification_composition_state | Result: 3179.805 kW | Notes: Process-unit sizing now prefers solved separation packet, composition state, and thermal packet before route-level utility heuristics.
Row 12: Trace: Process-unit utility-train coupling | Formula: Integrated reboiler duty = sum(selected train steps tied to process-unit cold sinks | Inputs: heating_steps=none; condenser_steps=none; topology=shared HTM island network (shared_htm) with 1 utility islands; architecture_family=shared_htm; selected_islands=none; header_levels=none | Result: 0.000 kW | Notes: This captures recovered duty delivered into the reboiler/process unit and heat recovered from condenser-side streams.

Row 13: Trace: Reboiler package basis | Formula: Vboil = Qreb / lambda; circulation from selected train package when available | Inputs: package=fallback; package_type=generic; phase_change_load_kg_hr=6187.729; circulation_ratio=3.000; selected_islands=none; target_recovered_kw=0.000; cluster_ids=none | Result: 3179.805 kW | Notes:
Row 14: Trace: Condenser package basis | Formula: mcond = Qcond / lambda; circulation from selected train package when available | Inputs: package=fallback; package_type=generic; phase_change_load_kg_hr=7049.642; circulation_flow_m3_hr=6090.003; selected_islands=none; target_recovered_kw=0.000; cluster_ids=none | Result: 4112.291 kW | Notes:
Row 15: Trace: Integrated reboiler heat-transfer basis | Formula: A_reb,int = Q_reb,int / (U_reb * LMTD_reb,int) | Inputs: Q_reb,int=0.0; U_reb=780.0; LMTD_reb,int=0.00 | Result: 0.000 m2 | Notes: Integrated reboiler area is derived from selected train steps that feed the process-unit heating sink.
Row 16: Trace: Condenser-side recovery basis | Formula: A_cond,rec = Q_cond,rec / (U_cond * LMTD_cond,rec) | Inputs: Q_cond,rec=0.0; U_cond=700.0; LMTD_cond,rec=0.00 | Result: 0.000 m2 | Notes: Recovered condenser duty is tied to selected train steps sourced from the process-unit hot side.
Row 17: Trace: Solved equilibrium packet basis | Formula: selected separation packets provide the active phase-equilibrium model and split basis for this process-unit family | Inputs: packet_ids=purification_separation_packet; models=heuristic; parameter_ids=none; fallback=no | Result: heuristic_split | Notes: This basis is consumed directly from the solved separator packets rather than being inferred from chapter prose.

Heat-Exchanger Design Traces

Table 288. Heat-Exchanger Design Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Exchanger area | Formula: A = Q/(U*dTlm) | Inputs: Q=34024794.0; U=470.0; dTlm=8.0 | Result: 9049.147 m2 | Notes:
Row 2: Trace: Solved exchanger packet basis | Formula: packet basis -> selected thermal packet | Inputs: thermal_packet=fallback; selected_train_step=omega_catalytic_pinch_htm__shared_htm_step_01; architecture_family=shared_htm; selected_island=omega_catalytic_pinch_htm__shared_htm_shared_htm_01 | Result: 34024.794 kW | Notes: Exchanger sizing now prefers the selected utility-train step and solved thermal packet before aggregate-duty fallback.
Row 3: Trace: Utility package exchanger basis | Formula: Selected exchanger package -> family, area, LMTD, circulation, phase load | Inputs: package_family=process_exchange; exchanger_package=omega_catalytic_pinch_htm__shared_htm_step_01_exchanger; circulation_flow_m3_hr=1818.329; phase_change_load_kg_hr=0.000; header_level=1; cluster_id=none; target_recovered_kw=42530.992 | Result: 542.949 m3 | Notes: Reboiler/condenser package sizing now reads the selected utility package item before applying generic exchanger fallback.

Storage Design Traces

Table 289. Storage Design Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Storage inventory-day basis | Formula: inventory days = (dispatch + operating stock + restart buffer) * turnaround buffer factor | Inputs: dispatch_buffer_days=9.400; operating_stock_days=2.500; restart_buffer_days=1.500; turnaround_buffer_factor=1.050 | Result: 14.070 days | Notes: Operations planning now drives finished-product storage instead of a fixed weekly heuristic.
Row 2: Trace: Storage volume | Formula: V = throughput * days / density | Inputs: days=14.1 | Result: 7661.521 m3 | Notes:

Mechanical Design Traces

Table 290. Mechanical Design Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=2.000 MPa; D=17.895 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 157.134 mm | Notes:
Row 2: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=27004.165 m3; duty=7600.213 kW | Result: 9336.1 mm | Notes:
Row 3: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=280488.792 kN; piping=240.402 kN; wind=1152.822 kN; seismic=46748.132 kN | Result: 328630.147 kN | Notes:
Row 4: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=53.685 m; dT=230.0 C | Result: 148.170 mm | Notes:
Row 5: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=9336.1 mm; P=20.00 bar | Result: 1941901.983 mm2 | Notes:
Row 6: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=integral repad; d_nozzle=9336.1 mm; D_shell=17.895 m | Result: 2.359 factor | Notes:
Row 7: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=20.00 bar | Result: 26.000 bar | Notes:
Row 8: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=cylindrical skirt with anchor chair; platform_area=61.737 m2; clearance=1.80 m | Result: 656.205 m2 | Notes:
Row 9: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.200 MPa; D=1.651 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 4.409 mm | Notes:
Row 10: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=21.206 m3; duty=3179.805 kW | Result: 640.2 mm | Notes:
Row 11: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=270.030 kN; piping=16.706 kN; wind=9.813 kN; seismic=45.005 kN | Result: 341.554 kN | Notes:
Row 12: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=4.953 m; dT=115.0 C | Result: 6.835 mm | Notes:
Row 13: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=640.2 mm; P=2.00 bar | Result: 13316.919 mm2 | Notes:
Row 14: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=integral repad; d_nozzle=640.2 mm; D_shell=1.651 m | Result: 1.848 factor | Notes:
Row 15: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=2.00 bar | Result: 3.000 bar | Notes:
Row 16: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=cylindrical skirt with anchor chair; platform_area=5.696 m2; clearance=1.80 m | Result: 6.620 m2 | Notes:
Row 17: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.300 MPa; D=10.104 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 15.940 mm | Notes:
Row 18: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=4860.750 m3; duty=0.000 kW | Result: 3917.0 mm | Notes:
Row 19: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=50477.341 kN; piping=98.976 kN; wind=168.445 kN; seismic=4673.828 kN | Result: 55418.590 kN | Notes:
Row 20: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=30.312 m; dT=60.0 C | Result: 21.824 mm | Notes:
Row 21: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=3917.0 mm; P=3.00 bar | Result: 65806.429 mm2 | Notes:
Row 22: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=shell excess area; d_nozzle=3917.0 mm; D_shell=10.104 m | Result: 1.743 factor | Notes:

Row 23: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=3.00 bar | Result: 4.000 bar | Notes:
Row 24: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=grouted base frame; platform_area=0.000 m2; clearance=1.20 m | Result: 12.125 m2 | Notes:
Row 25: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.800 MPa; D=5.356 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 21.338 mm | Notes:
Row 26: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=723.932 m3; duty=34024.794 kW | Result: 2021.0 mm | Notes:
Row 27: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=7966.118 kN; piping=53.324 kN; wind=47.328 kN; seismic=737.603 kN | Result: 8804.374 kN | Notes:
Row 28: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=16.067 m; dT=155.0 C | Result: 29.885 mm | Notes:
Row 29: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=2021.0 mm; P=8.00 bar | Result: 90539.654 mm2 | Notes:
Row 30: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=shell excess area; d_nozzle=2021.0 mm; D_shell=5.356 m | Result: 1.799 factor | Notes:
Row 31: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=8.00 bar | Result: 10.400 bar | Notes:
Row 32: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=dual saddle; platform_area=3.000 m2; clearance=1.50 m | Result: 47.328 m2 | Notes:
Row 33: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.120 MPa; D=12.211 m; S=120.0 MPa; J=0.85; CA=2.0 mm | Result: 9.188 mm | Notes:
Row 34: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=8580.903 m3; duty=0.000 kW | Result: 5177.3 mm | Notes:
Row 35: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=81830.296 kN; piping=129.853 kN; wind=313.146 kN; seismic=7576.879 kN | Result: 89850.175 kN | Notes:
Row 36: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=36.634 m; dT=20.0 C | Result: 12.456 mm | Notes:
Row 37: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=5177.3 mm; P=1.20 bar | Result: 34791.617 mm2 | Notes:
Row 38: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=shell excess area; d_nozzle=5177.3 mm; D_shell=12.211 m | Result: 1.781 factor | Notes:
Row 39: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=1.20 bar | Result: 2.200 bar | Notes:
Row 40: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=dual saddle; platform_area=0.000 m2; clearance=1.20 m | Result: 246.043 m2 | Notes:
Row 41: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=1.100 MPa; D=4.866 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 25.945 mm | Notes:
Row 42: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=542.949 m3; duty=34024.794 kW | Result: 1822.7 mm | Notes:
Row 43: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=6086.989 kN; piping=71.656 kN; wind=39.069 kN; seismic=563.610 kN | Result: 6761.323 kN | Notes:
Row 44: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=14.598 m; dT=188.0 C | Result: 32.933 mm | Notes:
Row 45: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=1822.7 mm; P=11.00 bar | Result: 152378.966 mm2 | Notes:
Row 46: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=set-on repad; d_nozzle=1822.7 mm; D_shell=4.866 m | Result: 1.839 factor | Notes:

Row 47: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=11.00 bar | Result: 14.300 bar | Notes:
Row 48: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=dual saddle; platform_area=3.000 m2; clearance=1.50 m | Result: 39.069 m2 | Notes:
Row 49: Trace: Skid or frame support basis | Formula: t_support = max(screening minimum, service family basis) | Inputs: service_family=Utility control package | Result: 12.000 mm | Notes:
Row 50: Trace: Connection basis | Formula: d_conn = f(duty, service family) | Inputs: duty=0.000 kW | Result: 25.0 mm | Notes:
Row 51: Trace: Support load basis | Formula: W = static load proxy + thermal allowance | Inputs: volume=0.150 m3; duty=0.000 kW | Result: 6.000 kN | Notes:
Row 52: Trace: Connection class | Formula: Class = f(design pressure) | Inputs: P=9.00 bar | Result: ASME 150 class | Notes:
Row 53: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=panel frame; platform_area=0.000 m2 | Result: 1.200 m2 | Notes:
Row 54: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=1.450 MPa; D=0.900 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 8.604 mm | Notes:
Row 55: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=0.851 m3; duty=34024.794 kW | Result: 417.0 mm | Notes:
Row 56: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=642.056 kN; piping=22.476 kN; wind=2.066 kN; seismic=71.340 kN | Result: 737.937 kN | Notes:
Row 57: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=2.700 m; dT=180.0 C | Result: 5.832 mm | Notes:
Row 58: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=417.0 mm; P=14.50 bar | Result: 45956.111 mm2 | Notes:
Row 59: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=set-on repad; d_nozzle=417.0 mm; D_shell=0.900 m | Result: 2.132 factor | Notes:
Row 60: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=14.50 bar | Result: 18.850 bar | Notes:
Row 61: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=guided rack shoe; platform_area=0.000 m2; clearance=1.20 m | Result: 2.430 m2 | Notes:
Row 62: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=1.450 MPa; D=0.900 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 8.604 mm | Notes:
Row 63: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=0.250 m3; duty=15.466 kW | Result: 117.3 mm | Notes:
Row 64: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=11.589 kN; piping=11.612 kN; wind=1.500 kN; seismic=1.059 kN | Result: 25.760 kN | Notes:
Row 65: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=2.700 m; dT=180.0 C | Result: 5.832 mm | Notes:
Row 66: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=117.3 mm; P=14.50 bar | Result: 12929.954 mm2 | Notes:
Row 67: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=set-on repad; d_nozzle=117.3 mm; D_shell=0.900 m | Result: 1.452 factor | Notes:
Row 68: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=14.50 bar | Result: 18.850 bar | Notes:
Row 69: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=grouted base frame; platform_area=0.000 m2; clearance=1.20 m | Result: 1.200 m2 | Notes:
Row 70: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.600 MPa; D=0.900 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 5.309 mm | Notes:
Row 71: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty,

equipment family) | Inputs: volume=2.835 m3; duty=0.000 kW | Result: 175.1 mm | Notes:
Row 72: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=35.673 kN; piping=9.393 kN; wind=1.701 kN; seismic=3.303 kN | Result: 50.069 kN | Notes:
Row 73: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=2.700 m; dT=185.0 C | Result: 5.994 mm | Notes:
Row 74: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=175.1 mm; P=6.00 bar | Result: 7984.835 mm2 | Notes:
Row 75: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=set-on repad; d_nozzle=175.1 mm; D_shell=0.900 m | Result: 1.440 factor | Notes:
Row 76: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=6.00 bar | Result: 7.800 bar | Notes:
Row 77: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=four-leg support; platform_area=0.000 m2; clearance=1.20 m | Result: 1.600 m2 | Notes:
Row 78: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=1.350 MPa; D=0.900 m; S=138.0 MPa; J=0.85; CA=3.0 mm | Result: 8.215 mm | Notes:
Row 79: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=1.309 m3; duty=0.000 kW | Result: 145.4 mm | Notes:
Row 80: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=20.428 kN; piping=12.123 kN; wind=1.500 kN; seismic=1.892 kN | Result: 35.943 kN | Notes:
Row 81: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=2.700 m; dT=188.0 C | Result: 6.091 mm | Notes:
Row 82: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=145.4 mm; P=13.50 bar | Result: 14920.743 mm2 | Notes:
Row 83: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=set-on repad; d_nozzle=145.4 mm; D_shell=0.900 m | Result: 1.493 factor | Notes:
Row 84: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=13.50 bar | Result: 17.550 bar | Notes:
Row 85: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=grouted base frame; platform_area=0.000 m2; clearance=1.20 m | Result: 1.200 m2 | Notes:
Row 86: Trace: Shell thickness | Formula: t = P*D / (2*S*J - 1.2*P) + CA | Inputs: P=0.650 MPa; D=1.861 m; S=120.0 MPa; J=0.85; CA=3.0 mm | Result: 8.953 mm | Notes:
Row 87: Trace: Nozzle basis | Formula: d_nozzle = f(volume, duty, equipment family) | Inputs: volume=30.379 m3; duty=8506.198 kW | Result: 613.7 mm | Notes:
Row 88: Trace: Support load basis | Formula: W_support = vertical + piping + wind + seismic screening loads | Inputs: vertical=434.303 kN; piping=17.618 kN; wind=5.715 kN; seismic=40.213 kN | Result: 497.849 kN | Notes:
Row 89: Trace: Thermal growth | Formula: delta = alpha * L * dT | Inputs: L=5.583 m; dT=137.0 C | Result: 13.004 mm | Notes:
Row 90: Trace: Nozzle reinforcement area | Formula: A = f(d_nozzle, design pressure, equipment family) | Inputs: d_nozzle=613.7 mm; P=6.50 bar | Result: 22339.297 mm2 | Notes:
Row 91: Trace: Local shell / nozzle interaction | Formula: Interaction = f(nozzle-to-shell ratio, pressure, equipment family) | Inputs: reinforcement_family=shell excess area; d_nozzle=613.7 mm; D_shell=1.861 m | Result: 1.691 factor | Notes:
Row 92: Trace: Hydrotest pressure | Formula: P_hydro = max(1.3*P_design, P_design + 1 bar) | Inputs: P_design=6.50 bar | Result: 8.450 bar | Notes:
Row 93: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=dual saddle; platform_area=3.000 m2; clearance=1.50 m | Result: 5.715 m2 | Notes:
Row 94: Trace: Skid or frame support basis | Formula: t_support = max(screening minimum, service family basis) | Inputs: service_family=Utility control package | Result: 12.000 mm | Notes:
Row 95: Trace: Connection basis | Formula: d_conn = f(duty, service

family) | Inputs: duty=0.000 kW | Result: 25.0 mm | Notes:
Row 96: Trace: Support load basis | Formula: W = static load proxy + thermal allowance | Inputs: volume=0.150 m3; duty=0.000 kW | Result: 6.000 kN | Notes:
Row 97: Trace: Connection class | Formula: Class = f(design pressure) | Inputs: P=5.00 bar | Result: ASME 150 class | Notes:
Row 98: Trace: Foundation / access basis | Formula: Footprint = f(support family, diameter, access area) | Inputs: support_variant=panel frame; platform_area=0.000 m2 | Result: 1.200 m2 | Notes:

Utility Basis Traces

Table 291. Utility Basis Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Selected steam tariff basis | Formula: Steam cost = 1.80 INR/kg | Inputs: | Result: 1.80 INR/kg | Notes:
Row 2: Trace: Selected cooling-water tariff basis | Formula: Cooling-water cost = 8.00 INR/m3 | Inputs: | Result: 8.00 INR/m3 | Notes:
Row 3: Trace: Selected power tariff basis | Formula: Power cost = 8.50 INR/kWh | Inputs: | Result: 8.50 INR/kWh | Notes:

Project-Cost Traces

Table 292. Project-Cost Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Total CAPEX | Formula: CAPEX = direct + indirect + contingency | Inputs: direct=54447114123.44; indirect=13067307389.62; contingency=5444711412.34 | Result: 72959132925.40 INR | Notes:
Row 2: Trace: Procurement timing basis | Formula: Procurement timing uses package-family lead times and import content to distribute CAPEX draws across advance, fabrication, delivery, erection, and retention milestones. | Inputs: profile=domestic_cluster_timing; construction_months=19; imported_fraction=0.140; long_lead_fraction=0.254; package_families=5; import_duty=353373129.56; advance_fraction=0.150; progress_fraction=0.750; retention_fraction=0.100 | Result: 72959132925.43 INR | Notes:
Row 3: Trace: Import-duty burden by equipment class | Formula: Total import duty = sum(bare equipment cost * import content fraction * import-duty fraction) across procurement package families. | Inputs: package_families=general_fabrication, instrumented_skid_package, pressure_vessel_package, tankage_and_balance_of_plant, thermal_exchange_package; imported_fraction=0.140 | Result: 353373129.56 INR | Notes:
Row 4: Trace: Annual OPEX | Formula: OPEX = RM + utilities + labor + maintenance + overheads | Inputs: RM=7510532251.62; utilities=3132067158.00; labor=156000000.00 | Result: 13281189728.49 INR/y | Notes:
Row 5: Trace: Selected utility-train CAPEX | Formula: CAPEX = sum(installed cost of selected utility-train items) | Inputs: | Result: 478104852.53 INR | Notes:
Row 6: Trace: Transport-limited utility penalties | Formula: Penalty utilities = absorber hydraulic electricity + solids auxiliaries + dryer endpoint steam | Inputs: steam_load=71672.82; power_load=28732.70 | Result: 3132067158.00 INR/y | Notes:
Row 7: Trace: Utility island economic basis | Formula: Island economics allocate selected utility-train CAPEX and utility burden across selected utility islands, then add island-specific service costs. | Inputs: selected_islands=1; allocated_utility_cost=3132067158.00; island_service_cost=2952253.22; island_replacement_cost=15494813.83 | Result: 733501670.81 INR | Notes:
Row 8: Trace: Absorber packing replacement basis | Formula: Annual packing replacement = packing inventory * replacement fraction / replacement cycle | Inputs: packing_family=; specific_area=0.00; wetting_ratio=0.000; pressure_drop=0.000; flood_margin=0.000 | Result: 0.00 INR/y | Notes:
Row 9: Trace: Classifier recurring service basis | Formula: Classifier service = f(slurry circulation, cut size, classification sharpness) | Inputs: slurry_rate=0.000; cut_size=0.000; classified_fraction=0.000 | Result: 0.00 INR/y | Notes:
Row 10: Trace: Filter media replacement basis | Formula: Filter media

replacement = f(filter area, cake resistance, medium resistance) | Inputs: filter_area=0.000; alpha_cake=0.000e+00; Rm=0.000e+00 | Result: 0.00 INR/y | Notes:
Row 11: Trace: Dryer exhaust treatment basis | Formula: Dryer exhaust treatment = f(dry-air flow, humidity lift, exhaust saturation) | Inputs: dry_air=0.000; humidity_lift=0.0000; exhaust_sat=0.000 | Result: 0.00 INR/y | Notes:
Row 12: Trace: Recurring transport/service submodel total | Formula: Transport/service = packing replacement + classifier service + filter media replacement + dryer exhaust treatment | Inputs: packing=0.00; classifier=0.00; filter=0.00; dryer=0.00; utility_islands=2952253.22; utility_island_replacement=15494813.83 | Result: 18447067.05 INR/y | Notes:
Row 13: Trace: Availability policy basis | Formula: Route-family outage policy defines minor outages, turnaround window, startup loss, and turnaround interval | Inputs: policy=continuous_liquid_organic_train; minor_days=5.0; major_days=13.0; startup_loss_days=3.5; cycle_years=4 | Result: 768606535.25 INR event | Notes:

Working-Capital Traces

Table 293. Working-Capital Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Working capital | Formula: WC = RM inventory + FG inventory + receivables + cash buffer + restart loss inventory + outage buffer - payables | Inputs: rm_inventory=421309994.11; fg_inventory=376273972.60; receivables=1104657534.25; cash_buffer=254707748.22; restart_loss_inventory=1034880.00; outage_buffer_inventory=95515405.58; payables=493843216.54 | Result: 1759656318.22 INR | Notes: Operations-planning buffers now feed directly into working capital.
Row 2: Trace: Procurement-linked peak working-capital basis | Formula: Peak WC = startup inventory bridge + partial receivables/product stock + cash buffer + restart/outage buffers - partial payables credit, with the uplift timed by the procurement and construction schedule. | Inputs: timing_factor=1.197; precommissioning_days=13.40; precommissioning_month=17.95; peak_month=18.15; payables_realization_fraction=0.641; precommissioning_inventory=275688431.02 | Result: 1759656318.22 INR | Notes: Construction timing now shifts the highest working-capital exposure into the commissioning window.

Financial Traces

Table 294. Financial Traces

Columns: Trace; Formula; Inputs; Result; Notes
Row 1: Trace: Annual cashflow | Formula: Cashflow = PBT - tax + depreciation | Inputs: | Result: 1.00 INR/y | Notes:
Row 2: Trace: Payback | Formula: Payback uses cumulative yearwise cash accrual including replacement and turnaround events | Inputs: | Result: 76261521411.190 y | Notes:
Row 3: Trace: Recurring maintenance timing basis | Formula: Yearwise OPEX includes discrete packing replacement, utility-island replacement, and turnaround events instead of a flat annualized maintenance burden | Inputs: packing_cycle_years=0.00; utility_island_cycle_years=omega_catalytic_pinch_htm__shared_htm_shared_htm_01:4.26; turnaround_cycle_years=4 | Result: 768606535.25 INR event | Notes:
Row 4: Trace: Construction funding timing basis | Formula: Construction funding uses staged procurement draws, procurement-linked peak working capital, and interest during construction before operations start. | Inputs: procurement_profile=domestic_cluster_timing; construction_months=19; peak_working_capital=1,759,656,318.22; peak_working_capital_month=18.15; debt_fraction=0.700; interest_rate=0.1050; draws=advance_release@m6.7:INR 13,583,398,188; advance_release@m10.1:INR 336,687,371; advance_release@m11.2:INR 14,904,585; advance_release@m11.2:INR 1,119,309; fabrication_progress@m11.8:INR 31,241,815,833; advance_release@m13.5:INR 369,916,194; fabrication_progress@m13.6:INR 785,603,867; fabrication_progress@m14.3:INR 34,777,365; fabrication_progress@m14.3:INR

2,611,721; fabrication_progress@m15.5:INR 1,479,664,777; delivery_release@m16.0:INR 12,904,228,279; delivery_release@m16.6:INR 374,097,079; delivery_release@m16.7:INR 16,560,650; delivery_release@m16.7:INR 1,243,677; delivery_release@m16.7:INR 616,526,990; site_erection@m17.3:INR 3,395,849,547; site_erection@m17.5:INR 187,048,540; site_erection@m17.6:INR 8,280,325; site_erection@m17.6:INR 621,838; site_erection@m17.7:INR 308,263,495; performance_retention@m19.0:INR 8,280,325; performance_retention@m19.0:INR 621,838; performance_retention@m19.0:INR 6,791,699,094; performance_retention@m19.0:INR 308,263,495; performance_retention@m19.0:INR 187,048,540 | Result: 1542732167.57 INR | Notes:
Row 5: Trace: Debt-service coverage basis | Formula: DSCR = CFADS / debt service, where CFADS = revenue - opex - tax and debt service = principal + interest. | Inputs: minimum_dscr=0.050; average_dscr=0.050; debt_years=5 | Result: 0.050 ratio | Notes:
Row 6: Trace: Lender-style coverage basis | Formula: LLCR = PV(loan-life CFADS) / opening debt and PLCR = PV(project-life CFADS) / opening debt, discounted at the debt screening rate. | Inputs: opening_debt=52614125215.35; debt_years=5; loan_life_cfads_pv=0.00; project_life_cfads_pv=0.00; discount_rate=0.1050 | Result: LLCR=0.000; PLCR=0.000 ratio | Notes:

Major Assumptions

- Mock EG product profile uses seeded but realistic property values.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Capacity case scored deterministically from benchmark throughput and market basis.
- Mock EG route survey uses seeded industrial route families.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Site choice is ranked deterministically from scored India cluster attributes and configured preferences.
- Mock EG route survey uses seeded industrial route families.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Mock EG route survey uses seeded industrial route families.
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- Mock EG route survey uses seeded industrial route families.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Operating-mode decision uses operations-planning buffers, restart losses, and campaign burden instead of a fixed throughput-only rule.
- Rough alternative duties are used for early process architecture and heat-integration ranking, not final equipment design.
- Heat-integration cases are route-specific and feed the final route choice.
- Mock EG route survey uses seeded industrial route families.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.

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- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock EG route survey uses seeded industrial route families.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Chemistry-family adapter 'continuous_liquid_organic_train' selected from archetype 'organic_liquid_distillation_flash' and top-ranked route descriptors.
- Mock EG route survey uses seeded industrial route families.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Mock EG product profile uses seeded but realistic property values.
- Property-gap resolution applies a single hierarchy from sourced values to conservative assumptions.
- Chemistry-family adapter 'continuous_liquid_organic_train' selected from archetype 'organic_liquid_distillation_flash' and top-ranked route descriptors.
- Sparse-data policy is generated from the chemistry-family adapter and current property-engine coverage.
- Detailed design stages may block even when screening solvers still allow explicit heuristic fallback.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Mock kinetic parameters are seeded values for workflow verification.
- Seeded BFD used for deterministic stage execution.
- Seeded BFD used for deterministic stage execution.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Generic material-balance engine uses a sequence-based flowsheet solver with explicit feed preparation, reaction, staged separations, recycle, and purge handling.
- Sequence family inferred as generic from route separations and expanded into the stage path: feed_handling, reaction, primary_recovery, recycle_recovery_carbonate_loop_cleanup, purification.
- Sectioned flowsheet topology is now emitted explicitly with 5 solved sections and stream roles for feed, recycle, side-draw, product, vent, and waste paths.
- Unitwise composition propagation now aggregates solved inlet and outlet component state for each major unit before energy and equipment sizing.
- Ethylene glycol cases retain strong water recycle inside the generic sequence solver to reflect dilution-intensive hydration practice.
- Mock kinetic parameters are seeded values for workflow verification.

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- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock kinetic parameters are seeded values for workflow verification.
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- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
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- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.

- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
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- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
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- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.

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- Route family liquid_hydration_train drives rough utility/CAPEX screening.
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- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
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- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
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- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: exchanger.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: controls.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Rotating and control-package items use support-frame screening rather than vessel shell-thickness equations.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: header.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.

- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: circulation.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: expansion.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: relief.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: exchanger.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Shell thickness uses allowable-stress and joint-efficiency screening logic with corrosion allowance.
- Cluster id: none.
- Header level: 1.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Utility island: omega_catalytic_pinch_htm__shared_htm_shared_htm_01.
- Utility train package role: controls.
- Mechanical calculations remain feasibility-study level and are not a substitute for code-stamped detailed design.
- Rotating and control-package items use support-frame screening rather than vessel shell-thickness equations.
- Mechanical chapter now uses deterministic screening calculations for shell thickness, nozzles, support load cases, foundation/access basis, connection classes, piping class basis, and equipment-wise material of construction basis with alternates and service rationale.
- Inspection and maintainability basis remains feasibility-level screening, but it now follows service class, support variant, and access provisions equipment by equipment.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family

adapter preferences.

- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
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- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
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- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
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- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
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- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis ranking is deterministic and depends on residual utility demand after the selected heat-integration case.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
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- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family

adapter preferences, and unit-operation family screening.

- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package

power.

- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.

- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Waste treatment remains conceptual until detailed composition data is available.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.

- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct

stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.

- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-

propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same

assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Flowsheet graph nodes considered: 7.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.

- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
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- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-

propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
- Storage selector scored alternatives from product phase liquid and chemistry-family adapter preferences.
- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
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- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
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- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility basis is selected for Dahej using route-level recovery and India tariff realism.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Mock India market assessment uses seeded price points normalized to 2025 INR.
- Route family liquid_hydration_train drives rough utility/CAPEX screening.
- Heat-integration ranking uses annual savings minus annualized CAPEX and fixed operability/safety penalties.
- Utility loads are net of selected heat-integration case omega_catalytic_pinch_htm with 47256.7 kW recovered duty.
- Utility loads include 2 selected train steps with 42531.0 kW packet-derived recovery basis.
- Utility auxiliary power includes 8 selected package items with 15.5 kW explicit package power.
- Cooling water rise fixed at 10 K and steam latent heat at 2200 kJ/kg for preliminary utility sizing.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Seeded BFD used for deterministic stage execution.
- Mock kinetic parameters are seeded values for workflow verification.
- Residence time basis retained at 1.00 h.
- Byproduct closure uses declared trace allocation because no explicit byproduct stoichiometry is available.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Reactor selector scored alternatives from route type, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Mock kinetic parameters are seeded values for workflow verification.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.

- Separation selector scored alternatives from route metadata, process archetype, chemistry-family adapter preferences, and unit-operation family screening.
- Property Engine v1 uses cited product-profile values first, then benchmark public seeds, then optional library-backed calculations, then analogy, then block.
- Library-backed values are treated as estimated unless a cited source also anchors the property basis.
- Binary interaction parameters are only used when a cited/public source adapter resolves the specific pair; otherwise the solver falls back to ideal gamma=1.0.
- Henry-law constants and solubility curves are only used when the property engine resolves a cited/public basis for the active pair; otherwise separators fall back to heuristic splits with explicit warnings.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Mock thermodynamic values are seeded for deterministic test runs and must be replaced by cited values in production use.
- Generic energy solver inferred generic duty structure from route metadata and composition-propagated unit state.
- Exchanger selector scored alternatives from gross unit duties.
- Mock EG product profile uses seeded but realistic property values.
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- Operations planning basis combines route-family campaign behavior, startup loss, and dispatch buffering so storage, working capital, and financial timing use the same assumptions.
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- Economics v2 ties CAPEX and OPEX to the selected procurement and logistics basis.
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- Procurement-linked working-capital uplift uses construction months 19, imported-equipment fraction 0.140, and long-lead fraction 0.254.
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- Financial model v2 includes procurement-timed funding, procurement-linked peak working capital, interest during construction, depreciation, interest, tax, DSCR, LLCR, and PLCR screening.
- Economic basis compares the integrated base case against no-recovery and conservative counterfactuals.

- Mock India site selection uses seeded cluster scoring rather than full geospatial analysis.
- Executive summary compiled from deterministic workflow artifacts.
- Conclusion reflects preliminary feasibility-level economics and design assumptions.